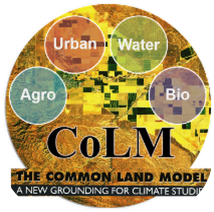

CoLM Desktop

用 户 手 册

面向 CoLM202X 站点、区域与全球空间模拟的桌面工作台

适用版本	v0.3.0-beta.10（测试版）
编写日期	2026-11-13
适用平台	macOS · Windows · Linux
截图来源	真实窗口截图：GUI 全页面、区域/流域、AI 助手等 48 张
内容构成	23 章 + 3 个附录（964 字段 · 1369 参数 · CLI 68 组）

测试版：功能可能变化；正式科研使用前请独立核验结果。

CoLM Desktop 用户手册

适用版本： `0.3.0-beta.10`（测试版）
平台： macOS（Apple Silicon / Intel）、Windows x86_64、Linux x86_64
手册截图： 2026-11-07/08 在 macOS 上由真实窗口采集；示例项目为 `~/Desktop/Github/CoLM-Desktop/tmp/manual-project`（CN-Cng 示例算例，缩到 2008-01-01 → 2008-01-03 便于演示）。
附录： CLI 命令参考 · 配置字段总表（964 项） · 参数库索引（1369 项）

> **警告：** 当前是 **Beta** 版本。请保留原始数据与算例备份；正式科研使用前独立核验结果。

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交付物

本手册同时提供三种可读格式，内容一致：

文件	用途
`README.md`	源文件（Markdown，适合 GitHub 浏览与继续编辑）
`CoLM-Desktop-用户手册.html`	单文件 HTML（含封面，图片已内嵌，可直接浏览器打开/打印）
`CoLM-Desktop-用户手册.pdf`	PDF（含封面、全部附录与 48 张标注截图，适合分发/打印）

文件	用途
`CoLM-Desktop-用户手册.md`	合并版 Markdown（正文 + 三个附录，便于全文搜索）
`images/`	48 张真实窗口截图（含编号标注，JPEG）
`cover.html` / `cover-preview.jpg`	封面设计稿（HTML 可视化 / 图片预览）
`appendices/`	CLI 命令、964 项配置字段、1369 项参数库三个附录

> 截图里的红色圆圈数字就是操作顺序；图下方文字给出每个编号对应控件。示例运行环境：macOS + 本机 Rust 引擎，示例项目缩到 3 天以控制演示时间。

1. 安装与启动

1.1 系统要求

项目	要求
操作系统	macOS 12+（Apple Silicon / Intel）、Windows 10/11 x86_64、主流 Linux x86_64
磁盘	安装包约 0.5–1 GB；每个官方模型修订的源码快照约 126 MB；OpenBench 环境约 400 MB（可选）
内存	站点算例 4 GB 起；空间/全球算例按网格规模准备 32 GB 以上
网络	下载安装包与首次获取模型修订需要；OpenBench、AI 助手、远程服务器按需
本机依赖	**不需要**自行安装 Rust、Fortran、MPI、NetCDF；安装包已带 Rust 引擎与 `colm-cli`

1.2 安装

发行版安装包（推荐）

从 GitHub Releases 下载对应平台：

平台	文件	安装方式
macOS Apple Silicon	`CoLM-Desktop_*_aarch64.dmg`	打开 DMG，把 CoLM Desktop 拖进"应用程序"
macOS Intel	`CoLM-Desktop_*_x64.dmg`	同上
Windows x86_64	`.msi` 或 `.exe`	双击安装
Linux x86_64	`.AppImage` / `.deb` / `.rpm`	AppImage 加执行权限直接运行；deb/rpm 用系统包管理器安装

首次启动会创建应用数据目录（见第 21 章），并自动登记当前官方模型修订（约 126 MB 源码快照）。

从源码运行（开发/自定义构建）

```
# 1) 前端依赖
cd gui && npm ci

# 2) 首次构建 Rust 内核（站点默认预设）
cd .. && ./oracle/scripts/build_kernel.sh default

# 3) 开发模式启动桌面端（Tauri v2）
cd gui/src-tauri && cargo tauri dev
```

```
# 4) 只构建命令行（引擎 workspace）
cargo build --release -p colm-cli    # 产物 target/release/colm-cli
```

> 桌面端的完整打包命令、Windows/Linux 交叉构建与签名说明见仓库 `README.md` 的"从源码构建"一节。安装包内已包含 `colm-cli` 与 `mksrfdata-rs/mkinidata-rs/colm-rs` 四个程序。

1.3 第一次启动

启动后进入**首页**（图 1）。左下角显示版本号与"测试版"标记；右侧是 AI 助手面板（默认展开，可点右上角"AI 助手"收起）。首次启动请先完成两件事：

1. **确认模型版本**：默认使用"官方版"（首页"模型版本"卡片）。切换入口见第 15 章。
2. **配置数据路径**（可选）：如果你有自己的项目目录、地表数据目录、Runtime 与强迫场目录，可到"资源 → 默认数据路径"登记；新建模拟时会自动填值。

---

2. 10 分钟快速开始

下面用内置 CN-Cng 草地示例，从"打开项目"到"看到结果"走一遍。全过程本机运行、不依赖服务器。

1. **打开项目**：首页点"打开项目"（图 1 的 ④），或直接点"最近的项目"里的 `manual-project`。首次使用可点"打开示例 → CN-Cng"。
2. **打开算例**：项目页列出项目中的算例（图 4），点 `CN-Cng`。
3. **核对五步设置**（图 6—图 11）：算例页顶部有 ①运行环境与保存位置 → ②范围与网格 → ③数据与时段 → ④模型结构与方案 → ⑤检查与运行。示例算例已经配好，可以直接从第 1 步点"下一步"到最后。
4. **开始运行**：在第 5 步底部点"开始运行"（图 13）。本机 Rust 引擎会依次执行：
   - `mksrfdata`：制作地表数据；
   - `mkinidata`：制作初始场；
   - `colm`：主模拟。 3 天示例在普通笔记本上通常 **10 秒—1 分钟** 完成；日志逐阶段显示（图 14）。
5. **查看结果**：点右上角"查看结果"或左侧"结果"（图 18）：可以看到评估总览、平均偏差、RMSE、有效配对和时序/观测评估页签。
6. **查看历史**：左侧"历史"（图 23）会记录这次运行，可导出 `CSV` / `JSON`。
7. **再跑一个批量**：返回项目页 →"批量运行"（图 16）：勾选两个站点 →"开始运行"，两个算例并发跑完（图 17）。

下面是本次演示实际跑出的日志（节选）：

```
——— 制作地表数据（mksrfdata）———
Successful in surface data making.
  mksrfdata  ok
——— 制作初始场（mkinidata）———
CoLM Initialization Execution Completed
  mkinidata  ok
——— 模拟（colm）———
colm-rs: 96 step(s) on patch(es) [0]; wrote .../out/CN-Cng/restart/2008-003-00000/...
colm-rs: 1 history file(s)
CoLM Execution Completed.
  colm       ok
```

---

3. 界面总览

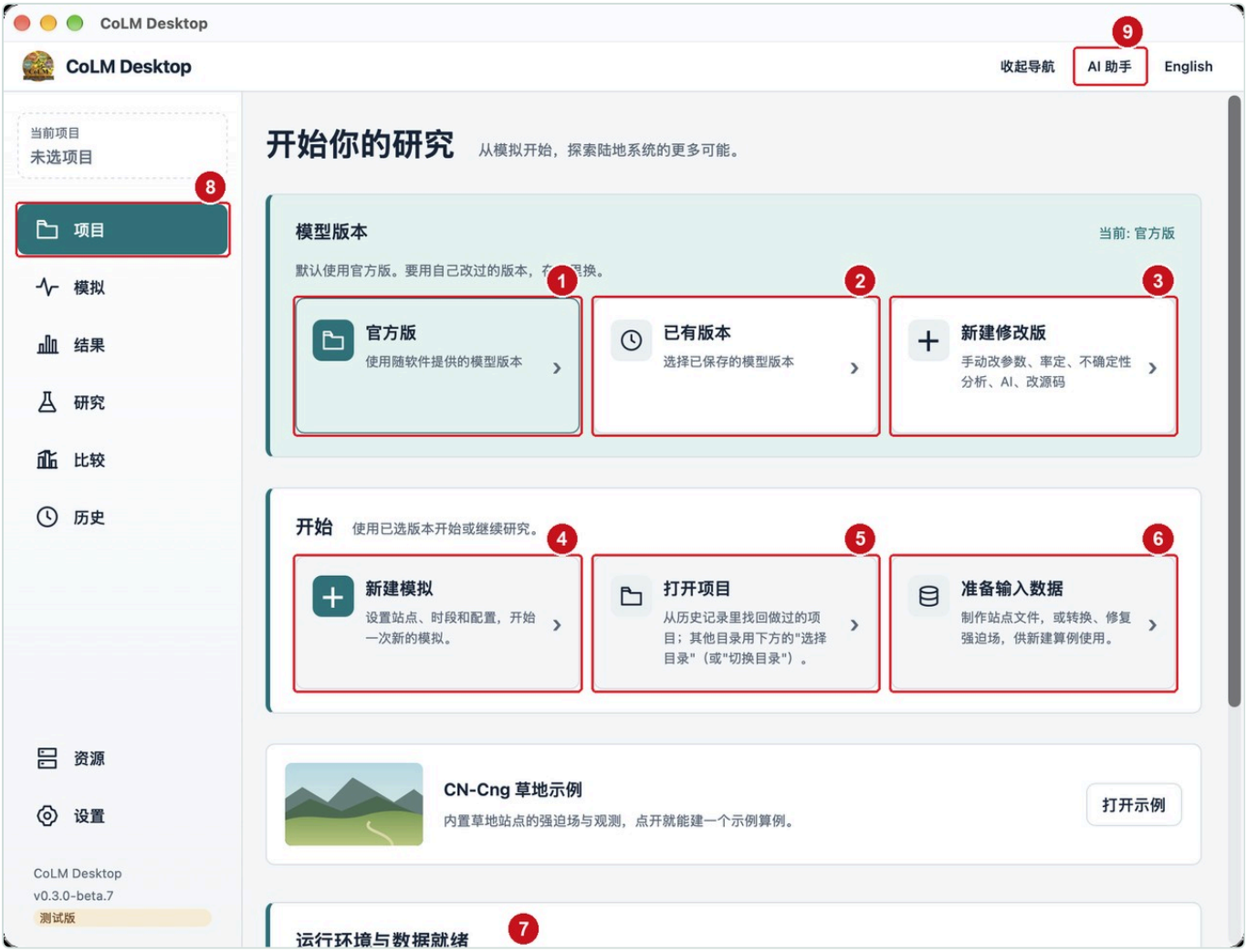

图 1 标注： ① 模型版本卡片（官方版/已有版本/新建修改版） ② "已有版本"入口 ③ "新建模拟" ④ "打开项目" ⑤ "准备输入数据" ⑥ 最近项目 ⑦ 左侧主导航 ⑧ AI 助手开关 ⑨ 语言切换。

界面分为三块：

- **顶部栏**：软件名、当前项目、AI 助手开关、语言（中文/English）。
- **左侧导航**：

导航	作用
项目	首页/当前项目：模型版本、新建模拟、打开项目、准备输入数据
模拟	当前算例的五步设置与运行
结果	运行结果评估、图表、导出
研究	参数修改、率定、不确定性、AI 研究、改源码
比较	OpenBench 跨模拟评估
历史	全部操作记录、筛选与导出
资源	计算服务器、本机环境、默认数据路径
设置	界面语言/外观、助手入口、模型缓存、帮助

- **主面板**：随导航切换；页面内的长表单支持滚动，底部通常固定"上一步/下一步"或"开始运行"。

操作顺序的读法：每张截图的红色圆圈数字就是建议点击顺序；图下方文字给出对应关系。多步页面按"①→②→…"操作即可。

3.1 English 界面

软件内置中英文两套完整界面。点右上角 `English` 即可切换，所有页面、按钮、错误信息都会同步切换；再点 `中文` 切回。

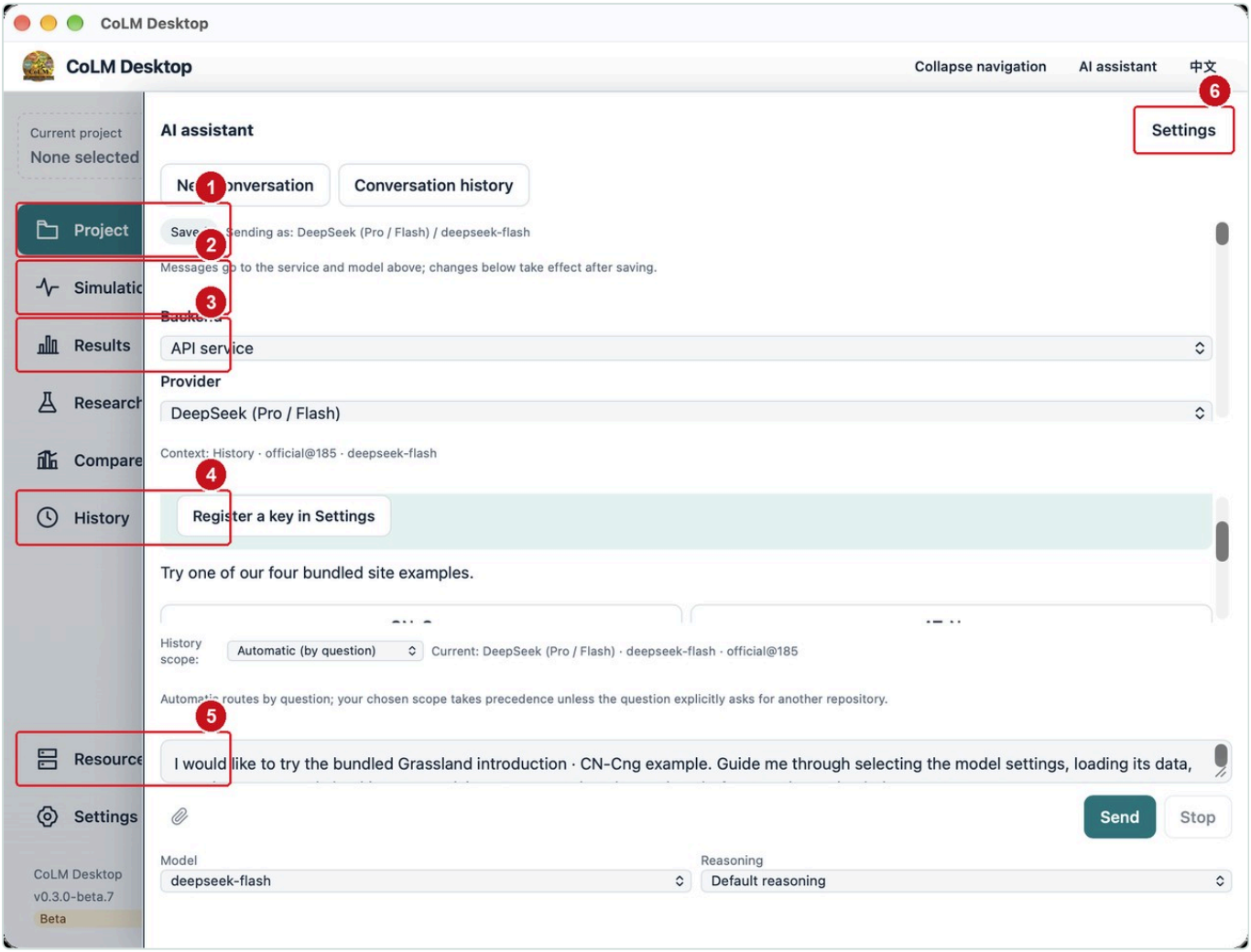

图 2 标注： ① Project ② Simulation ③ Results ④ History ⑤ Resources ⑥ Settings。

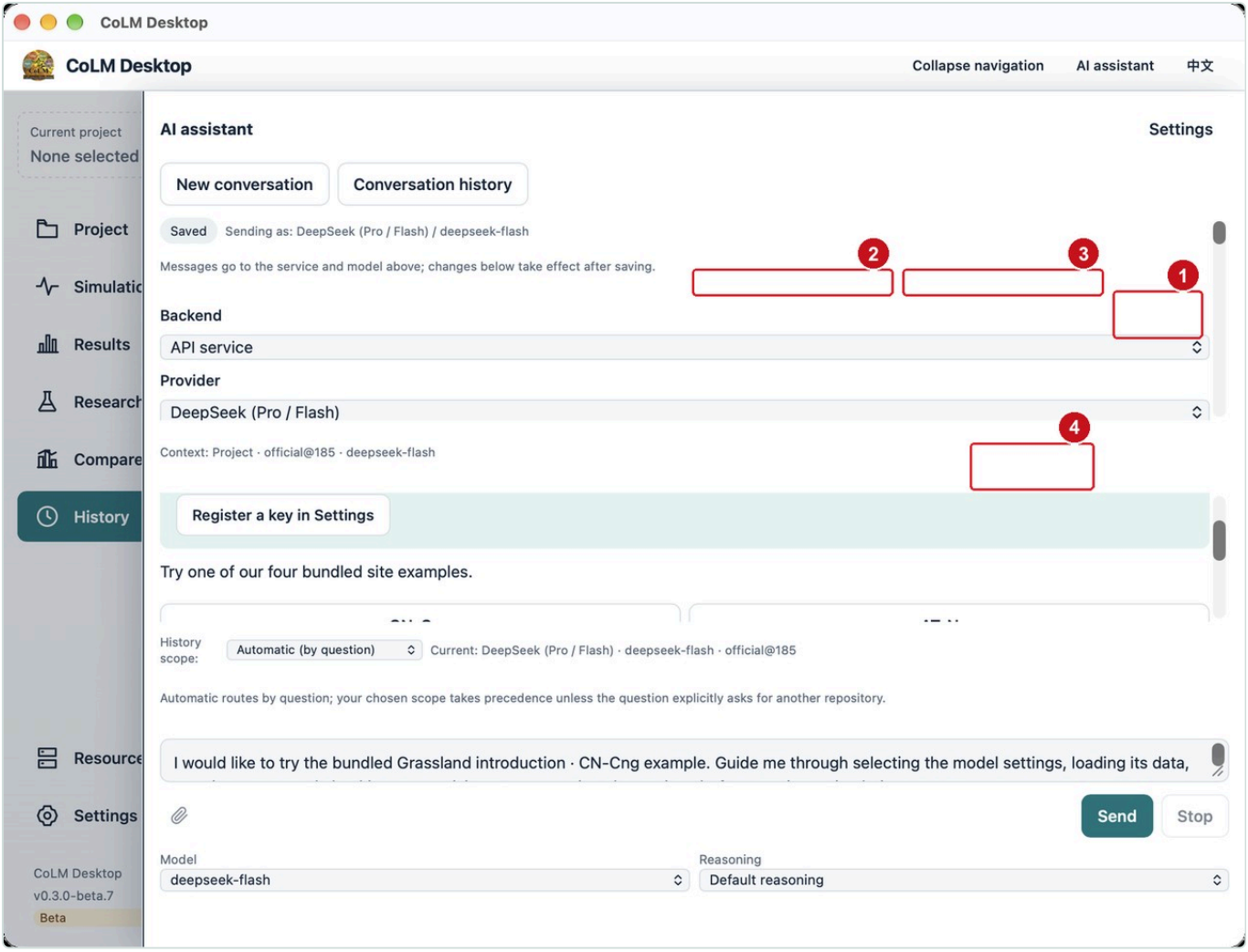

图 3 标注： 英文界面的历史页（筛选、导出、记录卡片与中文版一一对应）。

4. 项目与算例

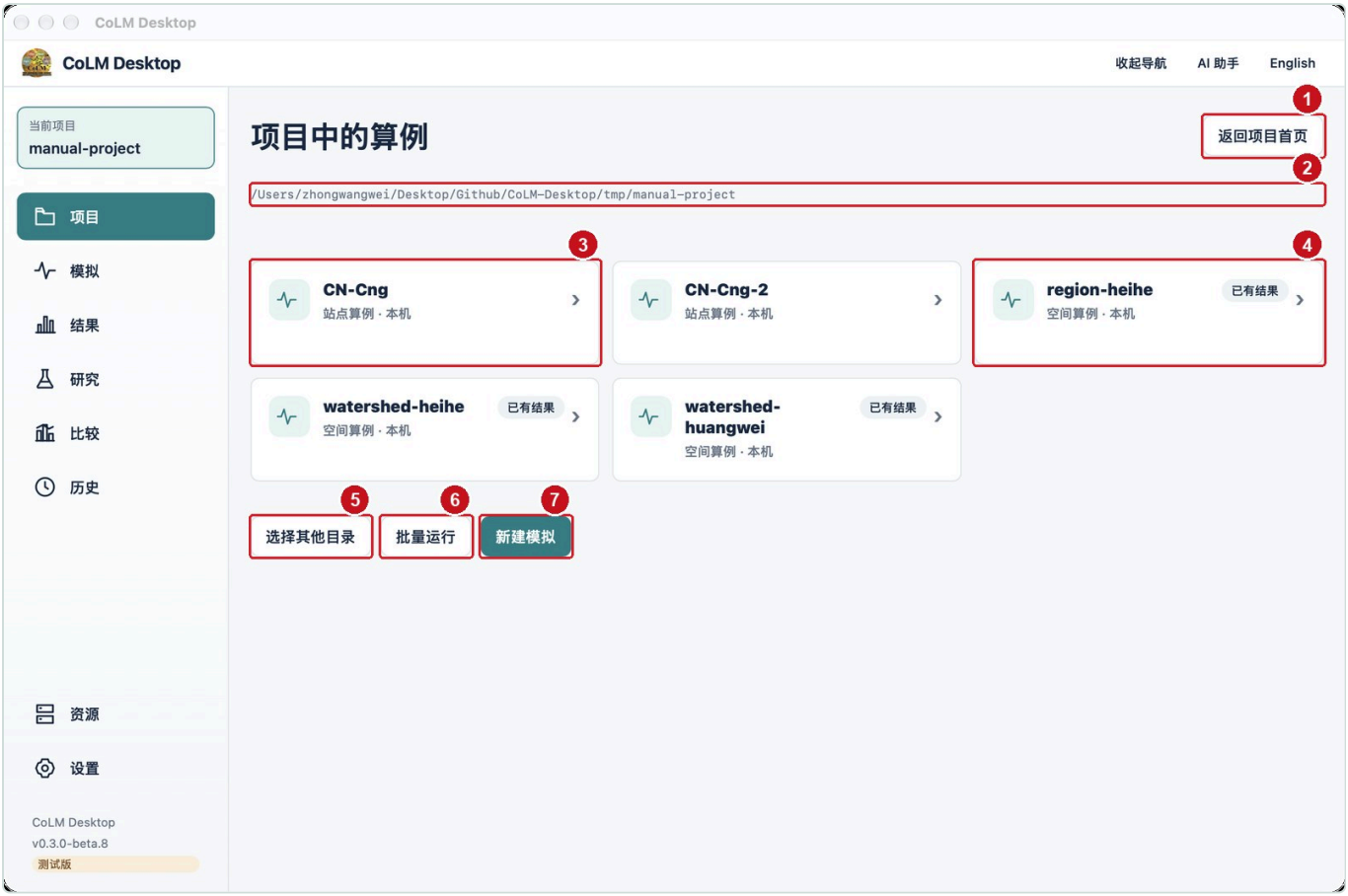

图 4 标注： ① 返回项目首页 ② 项目路径 ③ 站点算例卡片（点开进入五步设置） ④ 空间算例卡片 ⑤ 选择其他目录 ⑥ 批量运行 ⑦ 新建模拟。

- **项目 = 一个目录**，目录下每个子文件夹是一个算例（含 `case.nml`、`forcing.nml`、`site.nc/mesh.nc` 等）。
- 点算例卡片进入第 6 章。
- "选择其他目录"可打开任意位置的项目；打开后会出现在首页"最近的项目"。
- 项目页和首页的"打开项目"都优先从**历史记录**找回做过的项目；项目页另有"从历史项目打开"，按项目聚合历史、显示算例/研究数量，选中后直接打开。
- 历史里没有的新目录才用"选择目录/切换目录"走系统文件夹选择器。
- "新建模拟"进入第 5 章。
- "批量运行"进入第 8 章。

> 当前项目会显示在左侧导航顶部与页面标题；切换项目请回首页"打开项目"或项目页"选择其他目录"。

5. 新建模拟向导

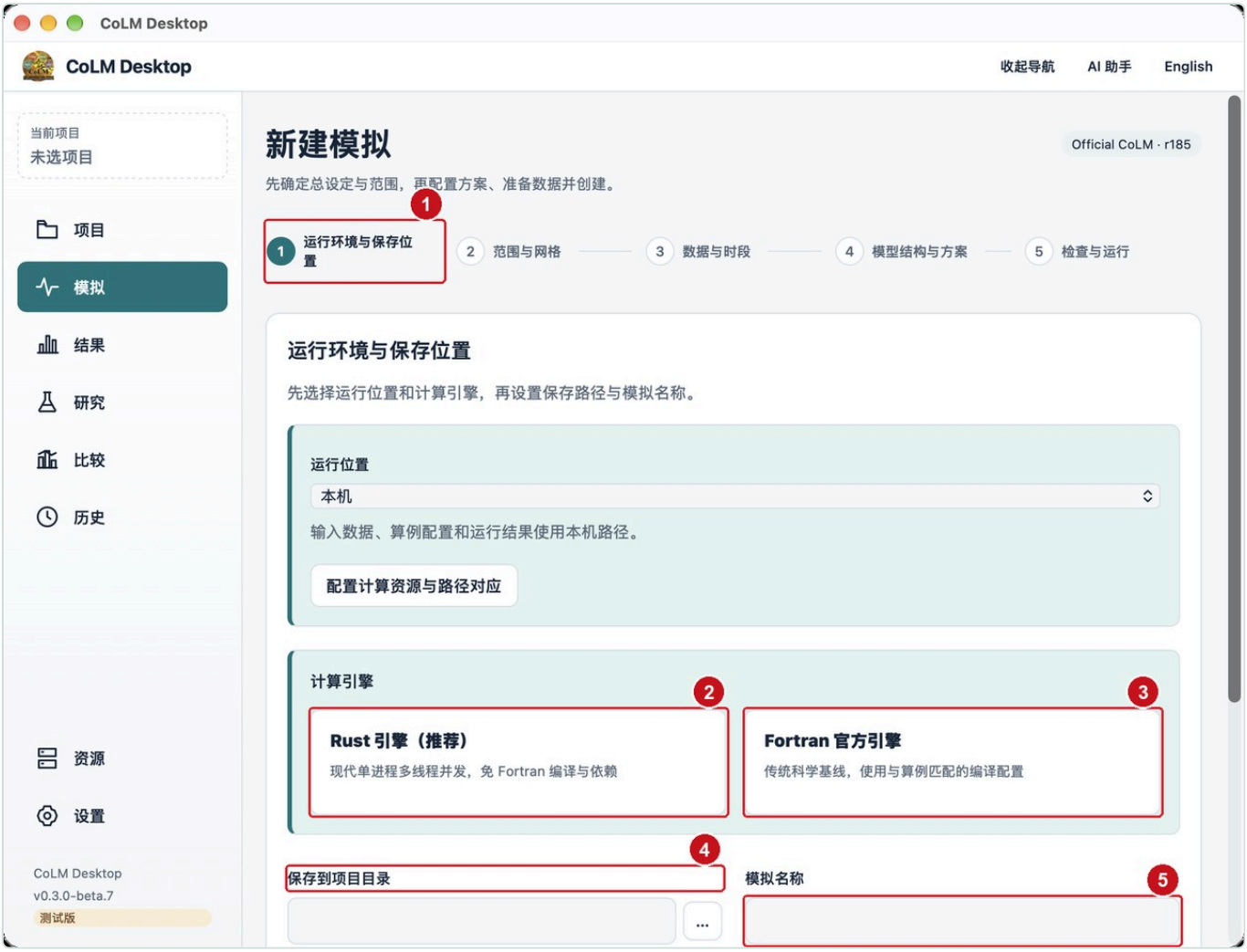

图 5 标注： ① 五步向导 ② Rust 引擎（推荐） ③ Fortran 官方引擎 ④ “保存到项目目录” ⑤ 模拟名称（模拟保存目录与下一步在下方）。

新建模拟与编辑已有算例共用同一套五步结构，但**基础配置（网格类型、引擎、尺寸）在创建时固定**，已有算例不能再改（可在第 1 步看到相应提示）。

5.1 第 1 步：运行环境与保存位置

字段	说明
运行位置	"本机"，或服务器（先在"资源"里添加服务器）。服务器运行见第 13 章
计算引擎	**Rust 引擎（推荐）**：所有网格、无需 Fortran/MPI 环境；**Fortran 官方引擎**：传统科学基线，需要匹配的编译配置（本机需构建内核；服务器上需远程内核）
保存到项目目录	勾选后算例保存在当前项目下；取消可手选目录
模拟名称	必填；不能含路径分隔符
模拟保存目录	由项目目录 + 模拟名称自动组合；可手动改
类型/网格	站点、经纬度网格、非结构网格、流域网格等（由后续步骤或入口决定）

5.2 第 2–5 步

- **第 2 步 范围与网格**：站点选经纬度；空间算例选网格类型、范围、分辨率/非结构文件。

- **第 3 步 数据与时段**：模拟起止日期、预热、站点/地表/Runtime/强迫场路径与强迫参数。
- **第 4 步 模型结构与方案**：地类方案、土壤水力、变饱和流、河湖汇流、LUC、BGC、城市、作物、材料过程等开关，以及"完整参数目录"。其中 USGS 地类方案已退休、未全面测试，不建议新研究使用。
- **第 5 步 检查与运行**：时间步长/输出设置、变量选择、编译配置、提交阶段，最后"开始运行"。

各字段的完整清单见附录 A：配置字段总表（964 项，按页面分组）。

6. 算例五步设置

打开已有算例后，顶部显示五个步骤；已完成的步骤带 ✓。下面按步骤说明重点。所有图都是同一示例算例 `CN-Cng`。

6.1 第 1 步：运行环境与保存位置

图 6 标注： ① 五步导航 ② 运行位置 ③ 计算引擎 ④ 算例路径 ⑤ 输出目录 ⑥ 更改路径 ⑦ 下一步。

要点：

- 运行位置默认"本机"；切到服务器后，下方会显示服务器负载、分区与作业状态。
- 计算引擎在创建时确定；已有算例如需换引擎，请新建模拟。
- 算例路径与输出目录用于定位配置/结果；模型版本运行会在算例目录下 `runs/<操作号>/` 生成独立运行副本。

6.2 第 2 步：范围与网格

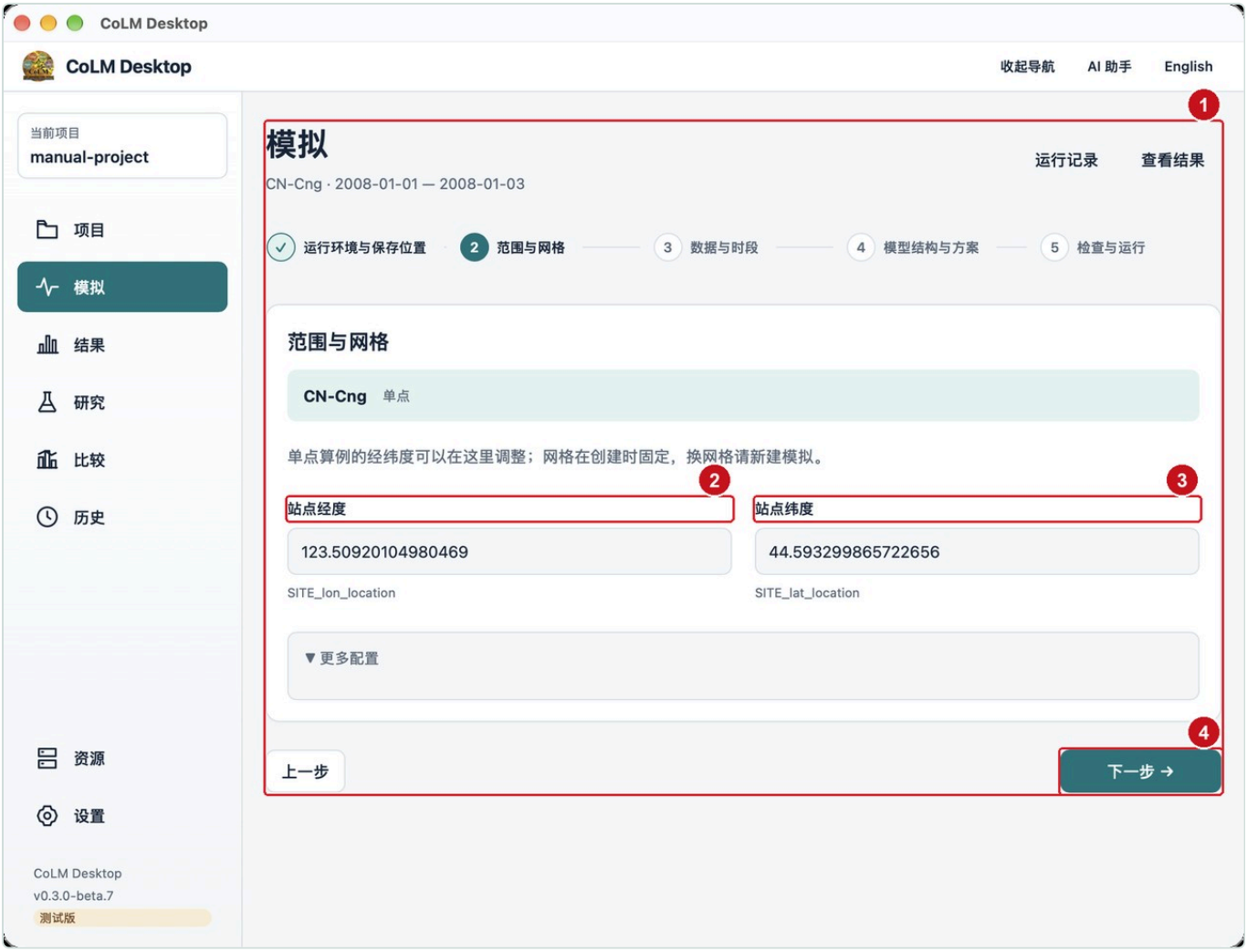

图 7 标注： ① 单点说明 ② 站点经度 ③ 站点纬度 ④ 下一步。

- 站点算例可在这里调整经纬度；**网格在创建时固定**，换网格需新建模拟。
- 空间算例的第 2 步显示范围/网格字段，并在创建后锁定（只能查看）；若需更换网格请新建模拟。

6.3 第 3 步：数据与时段

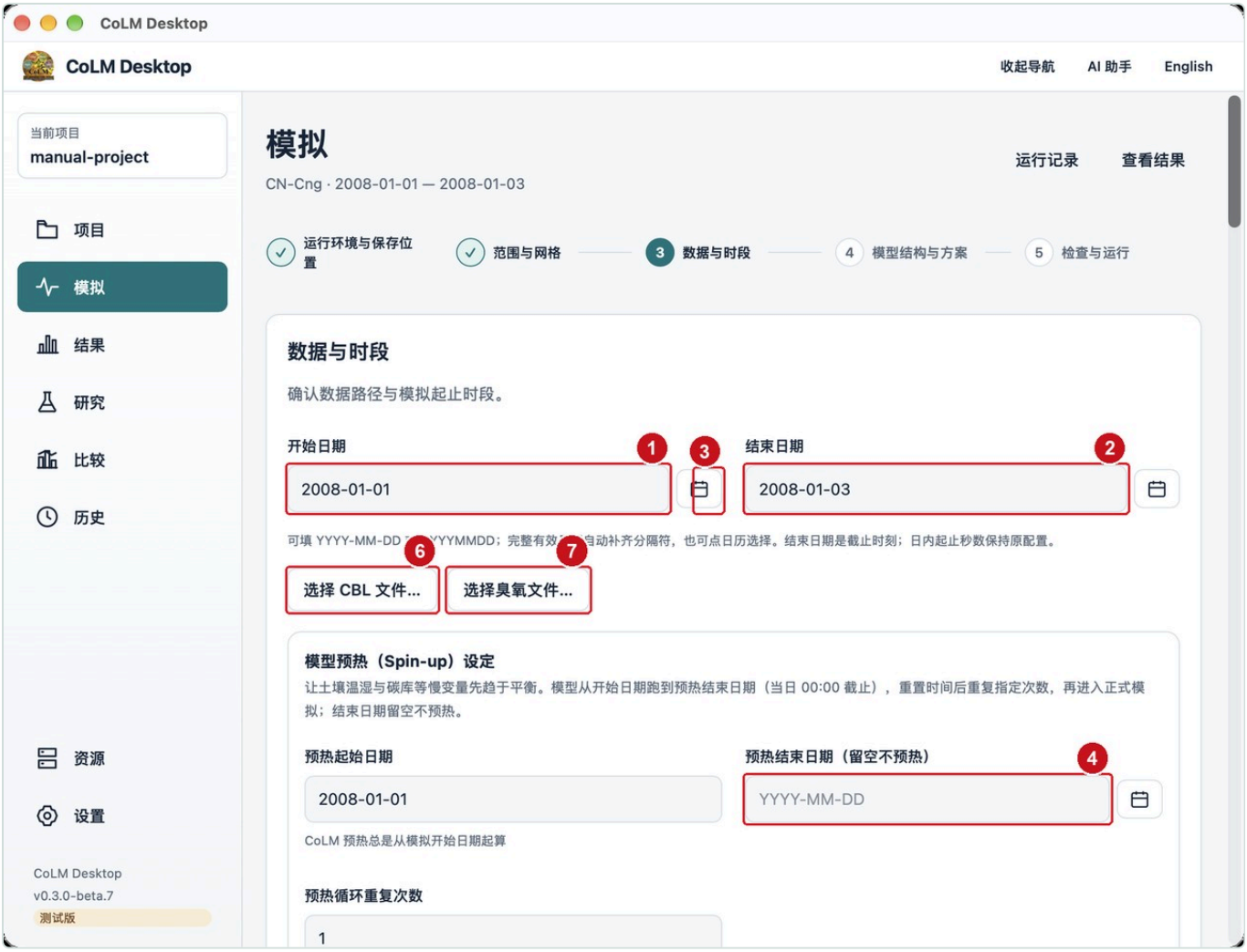

图 8 标注： ① 开始日期 ② 结束日期 ③ 日历选择 ④ 预热结束日期 ⑥ 选择 CBL 文件 ⑦ 选择臭氧文件（⑤“预热循环重复次数”在同一区域下方）。

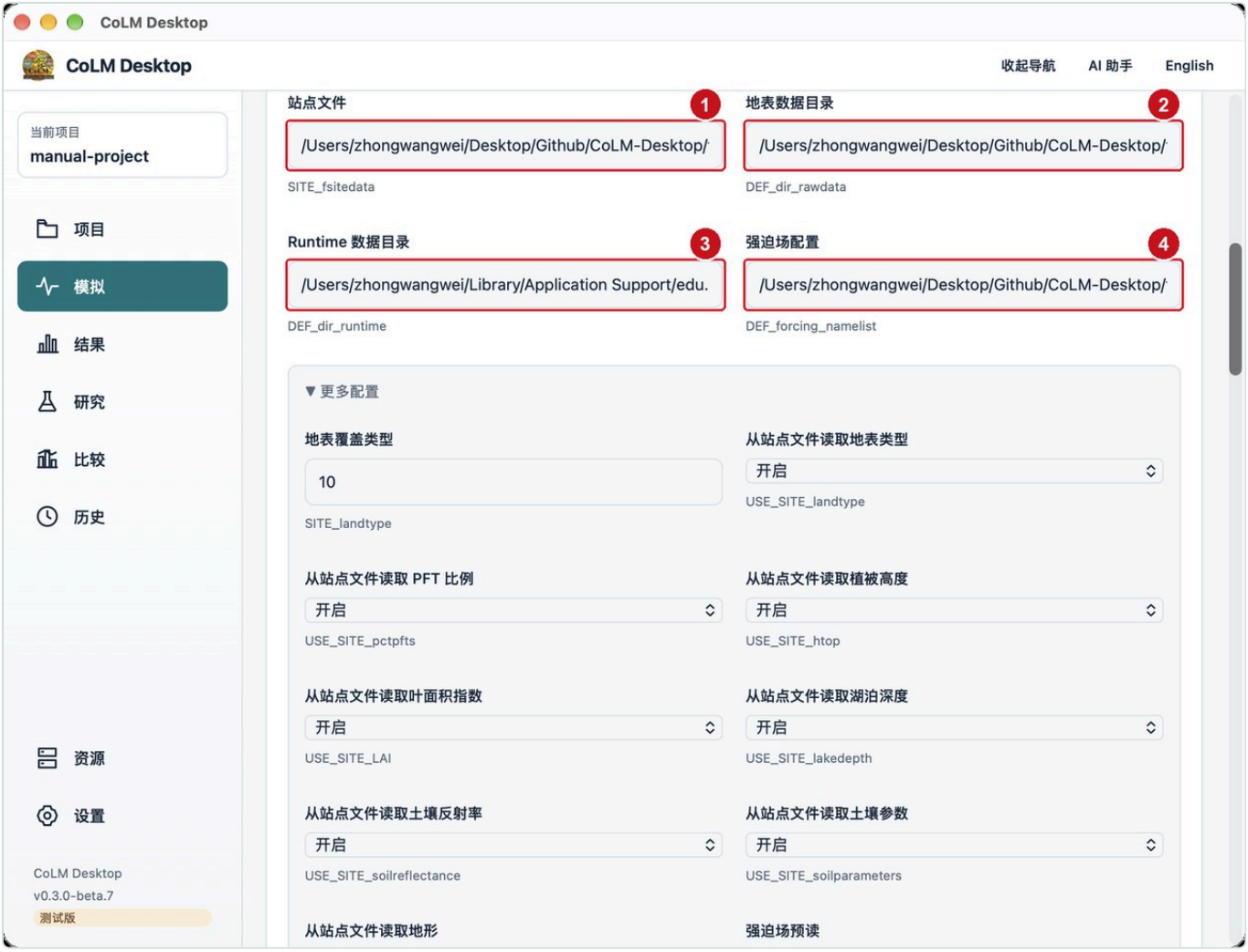

图 9 标注： ① 站点文件 ② 地表数据目录 ③ Runtime 数据目录 ④ 强迫场配置（其下的强迫场目录/数据集在同一卡片内）。

关键字段：

字段	说明
开始/结束日期	支持 YYYY-MM-DD 或 YYYYMMDD；结束日期是截止时刻。结束必须晚于开始
预热起始日期	CoLM 总是从模拟开始日期起算，只读
预热结束日期	留空 = 不预热；模型先跑预热，重置后重复指定次数再进入正式模拟
站点文件	`site.nc`，单站地表数据
地表数据目录	RawData 目录（空间算例或需要重做地表时）；
Runtime 数据目录	模型运行所需附加数据
强迫场配置	`forcing.nml` 路径
强迫场数据目录/数据集	强迫场根目录与数据集名（如 JRA3Q、CMFD、POINT）；可选值来自客户端模板注册表，不是随便填
观测高度方式/高度	强迫场风速/气温/湿度观测高度
数据含缺测值	是否声明缺测值属性与名称
各变量时间间隔/偏移	强迫场变量时间步长与偏移（8 个变量，逗号分隔）

> 强迫场 `dataset` 必须写引擎实际识别的标识（如 `CMFD`、`ERA5`、`CMIP6`、`POINT`），不是模板文件名（`ERA5_LEddy`、`MPI-ESM1-2-HR_ssp585` 是模板名，会被拒绝）。

6.4 第 4 步：模型结构与方案

图 10 标注： ① 地表覆盖方案（USGS 已标记“已退休 · 未全面测试”，不建议新研究使用；土壤水力方案在同页下方继续）。

- **土壤水力方案**：Campbell 或 van Genuchten；van Genuchten 必须开启变饱和流。
- **过程开关**：变饱和流、河湖汇流、土地覆盖变化、BGC、城市、作物、材料过程等。是否可用取决于当前内核编译宏与算例尺度（例如 SinglePoint 不支持土地覆盖变化）。
- **参数卡片**：次网格与植被、土壤方案、水热过程、生态与生地化、其他模型选项；点"完整参数目录"可搜索全部 1369 个参数。
- 参数修改只写入当前算例的覆盖，不派生新版本；要形成可复用版本，请用"研究 → 新建修改版"并冻结。

6.5 第 5 步：检查与运行

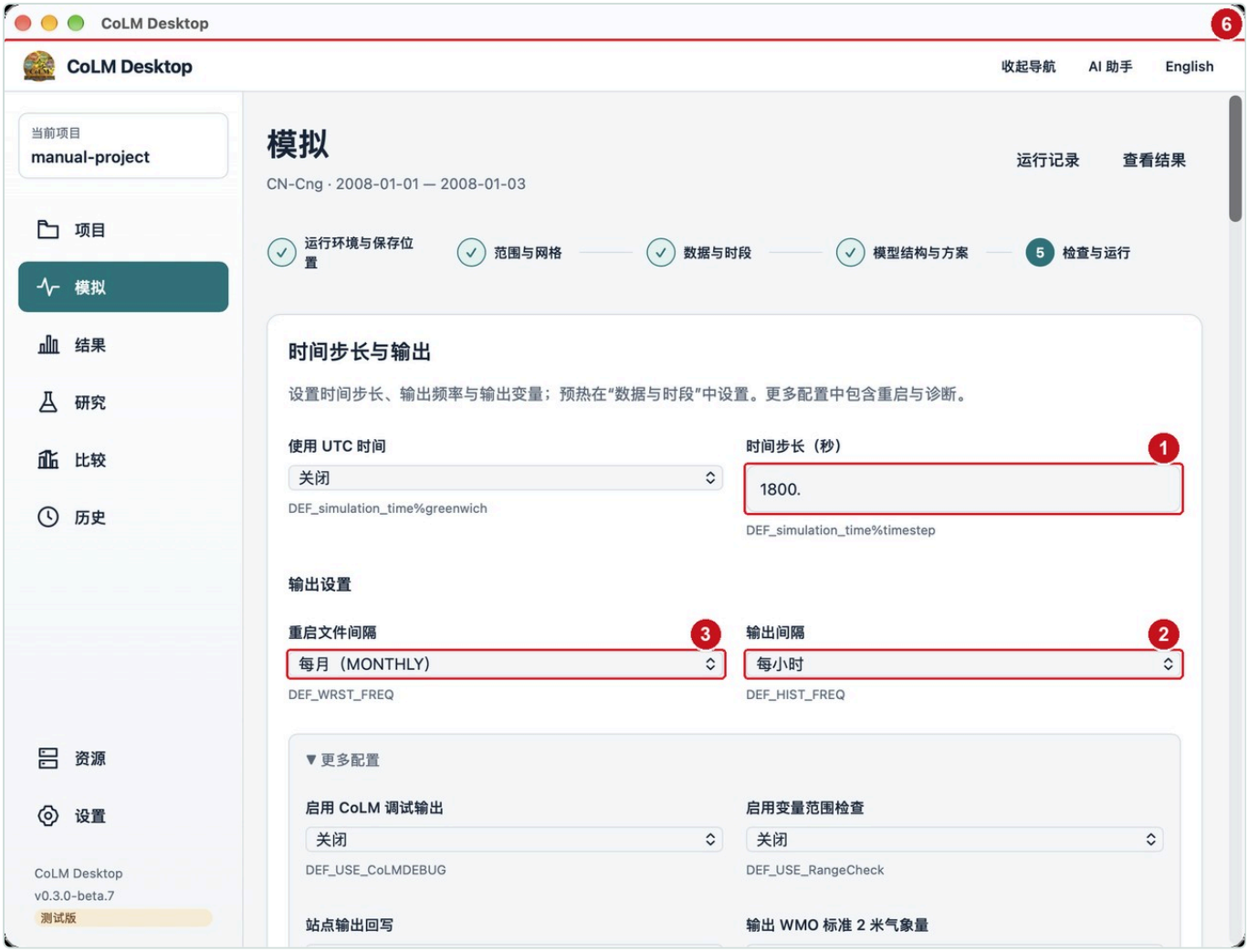

图 11 标注： ① 时间步长 ② 输出间隔 ③ 重启文件间隔 ④ 推荐核心量 ⑤ 一键全开启 ⑥ 全部变量（变量表在下方）。

图 12 标注： ① 检查并运行 ② 本次提交设置 ③ 运行位置 ④ 编译配置（⑤"本次提交阶段"在同一卡片下方）。

图 13 标注： ① 记录其他文件指纹 ② NML 配置预览 ③ 上一步 ④ 开始运行。

要点：

- 输出变量可在"推荐核心量（39）"与"全部变量"之间切换；一键全开/全清。
- **提交阶段**：全部（地表→初始化→主模拟）/ 仅地表 / 仅初始化 / 仅主模拟；切换阶段会自动标记需要重跑的阶段。
- "记录其他文件指纹"默认关闭；开启后会校验输入/程序/输出内容哈希，更严格但更慢。
- 点"开始运行"前会检查输入文件；缺文件会直接列出。
- 运行过程中可"停止"；关闭窗口前如有运行中任务会先询问。

7. 运行算例

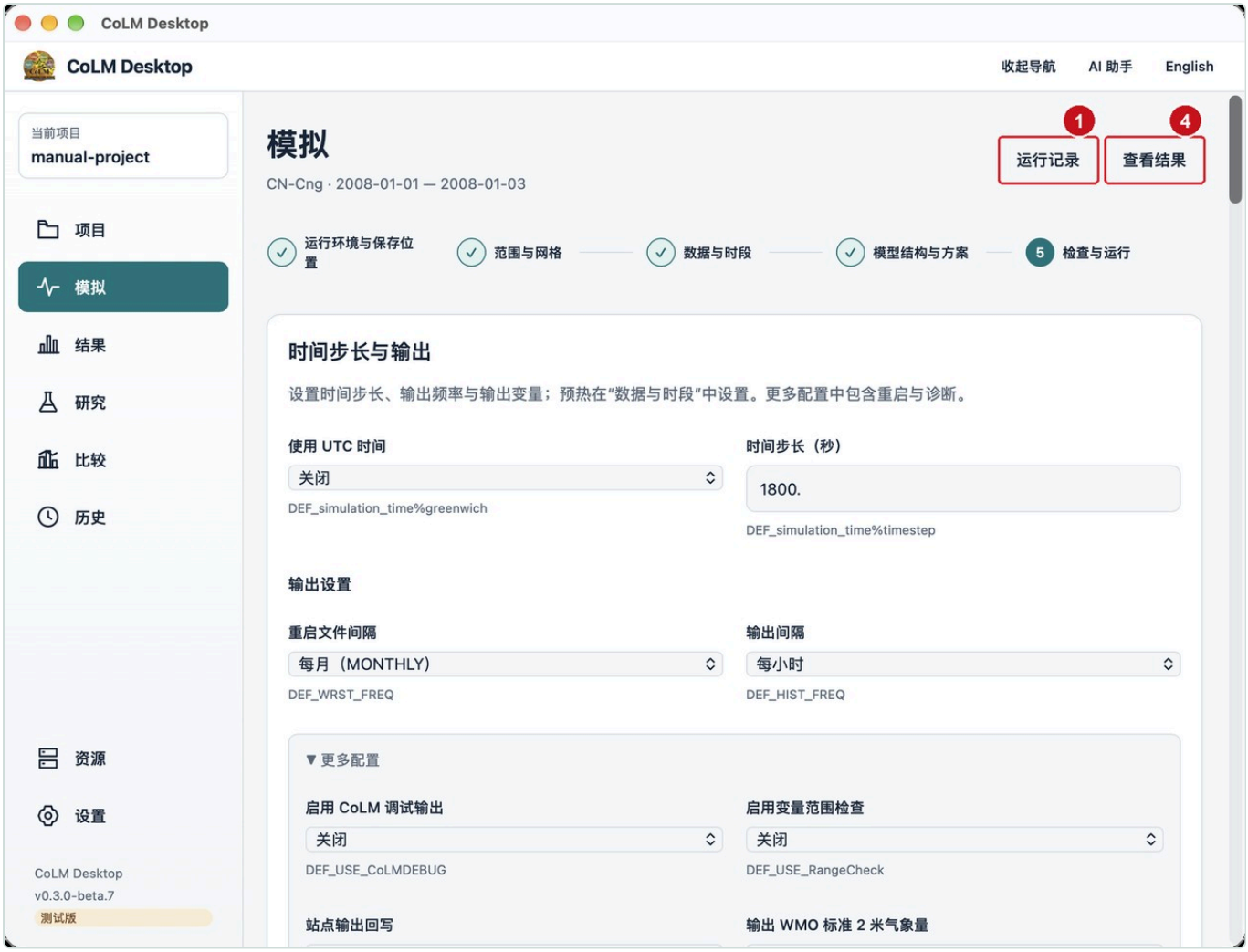

图 14 标注： ① 运行记录 ② 运行状态/完成信息 ③ 查看完整日志 ④ 查看结果。

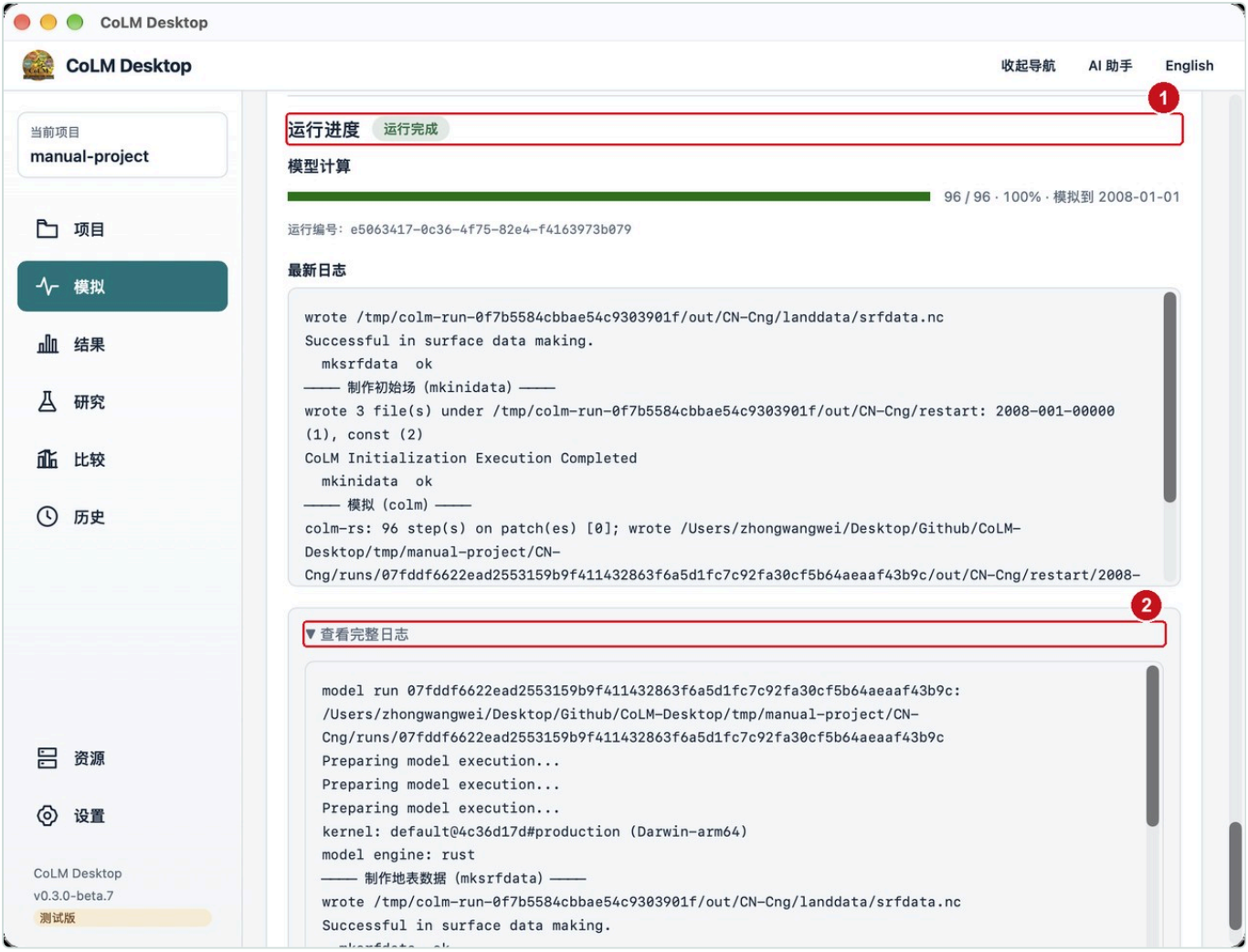

图 15 标注： ① 运行进度 ② 完整日志。

- 本机运行：Rust 引擎直接执行三个阶段；日志按阶段分段。
- 远程运行：提交后每约 10 秒核对一次状态；可在"运行进度"查看阶段、日志末尾、服务器作业详情，并可停止/取回结果。
- 失败时展开日志定位；常见原因见第 22 章。

本次演示实际运行的输出目录（节选）：

```
tmp/manual-project/CN-Cng/runs/07fddf66…/
├── case.nml            # 物化后的运行配置
├── forcing.nml
├── model-run.json      # 运行清单（版本、输入哈希、产物路径）
└── out/CN-Cng/
    ├── landdata/srfdata.nc
    ├── restart/2008-003-00000/…restart….nc
    └── history/CN-Cng_hist_2008-01.nc
```

8. 批量运行

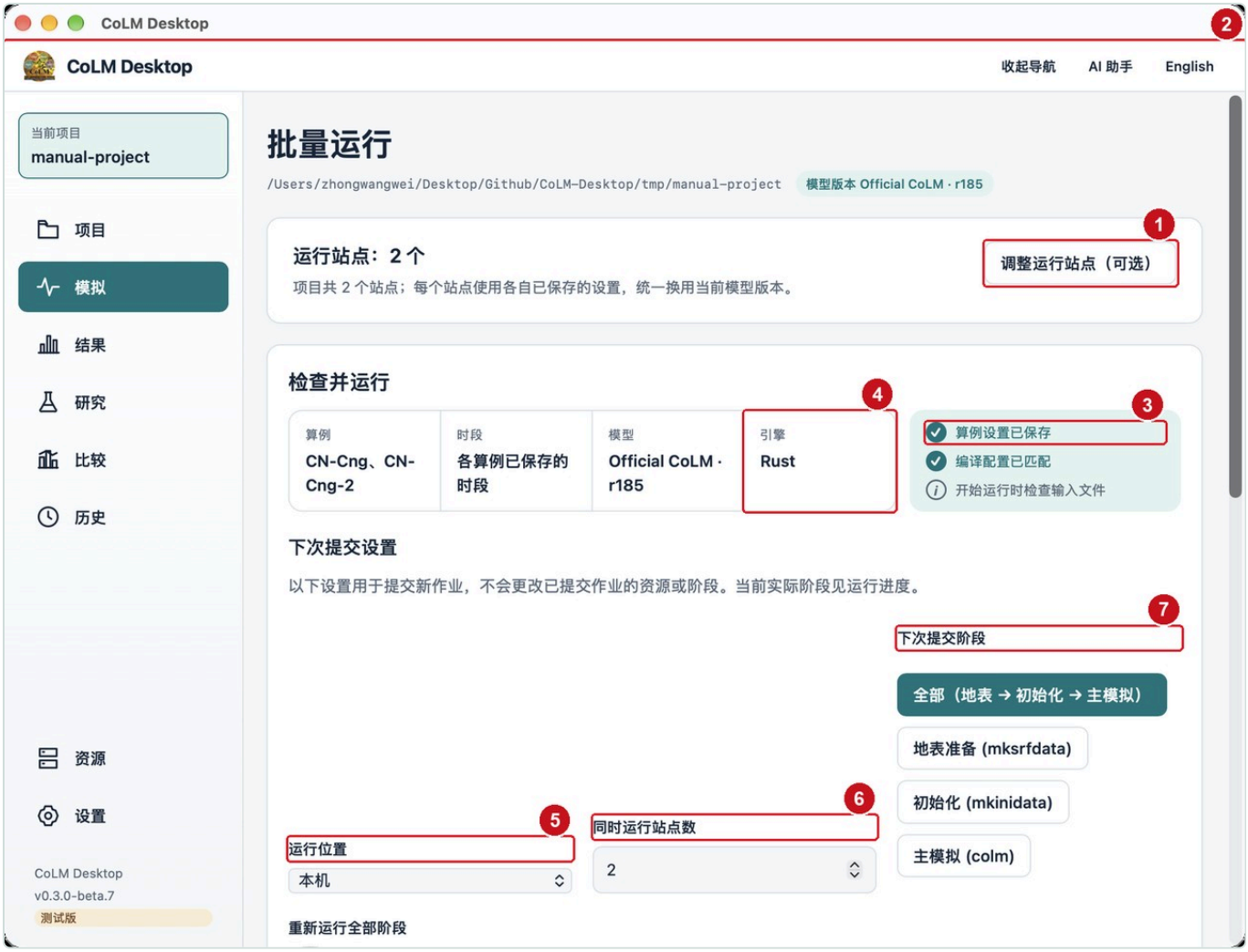

图 16 标注： ① 调整运行站点（可选） ② 检查并运行 ③ 算例与时段摘要 ④ 模型与引擎 ⑤ 运行位置 ⑥ 同时运行站点数 ⑦ 提交阶段。

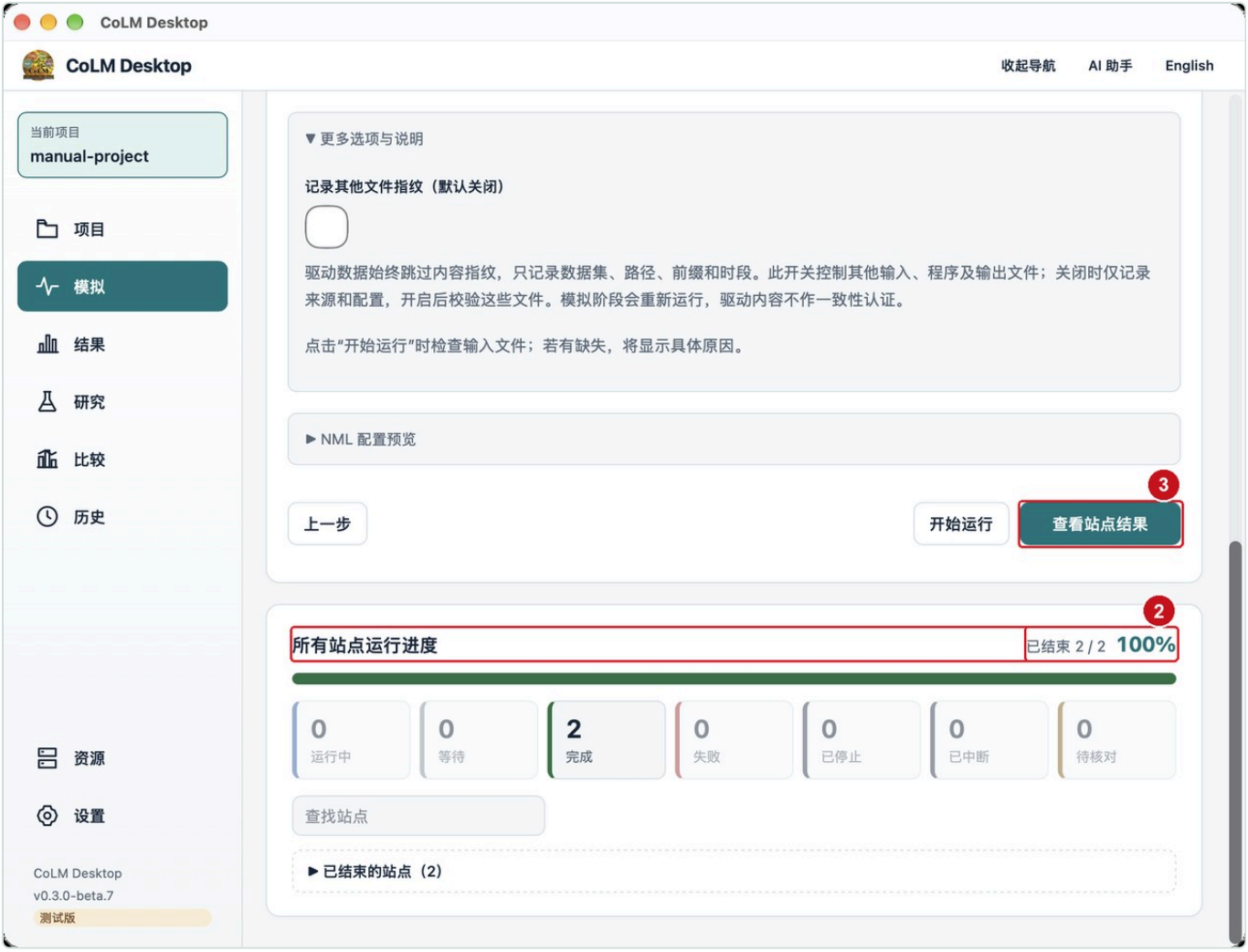

图 17 标注： ① 进度汇总 ② 完成/失败计数 ③ 查看站点结果 ④ 只选失败站点重跑。

操作顺序：

1. 项目页点"批量运行"；
2. 在"调整运行站点"里勾选要运行的站点（默认全选；可按未完成/失败筛选）；
3. 选择运行位置（本机/服务器）、模型版本、引擎与提交阶段；
4. 点"开始运行"；
5. 查看每个站点的进度；失败的站点可"只选失败或已停止的"重跑；
6. 完成后点"查看站点结果"或左侧"结果"查看。

打包与并发：

- 本机：同时运行站点数 决定并发进程数，单点每站固定 1 线程。
- 服务器（Slurm/LSF/PBS）：站点批量默认**按节点打包**——同一批站点进入服务器队列，每个节点作业一次占满节点核数，站点数多时自动多开节点；不再"一个站点占一个节点"。
- Rust 空间算例的并行预算（节点×每节点核数）与站点打包互斥，界面只显示适用的那一种。

本次演示的批量结果：**2 个站点全部完成，0 失败**（与图 17 一致）。

9. 结果页

图 18 标注： ① 结果与评估标题 ② 切换运行记录 ③ 分析范围 ④ 结果/时序/观测/闭合四个页签 ⑤ 导出完整 CSV ⑥ 返回模拟。

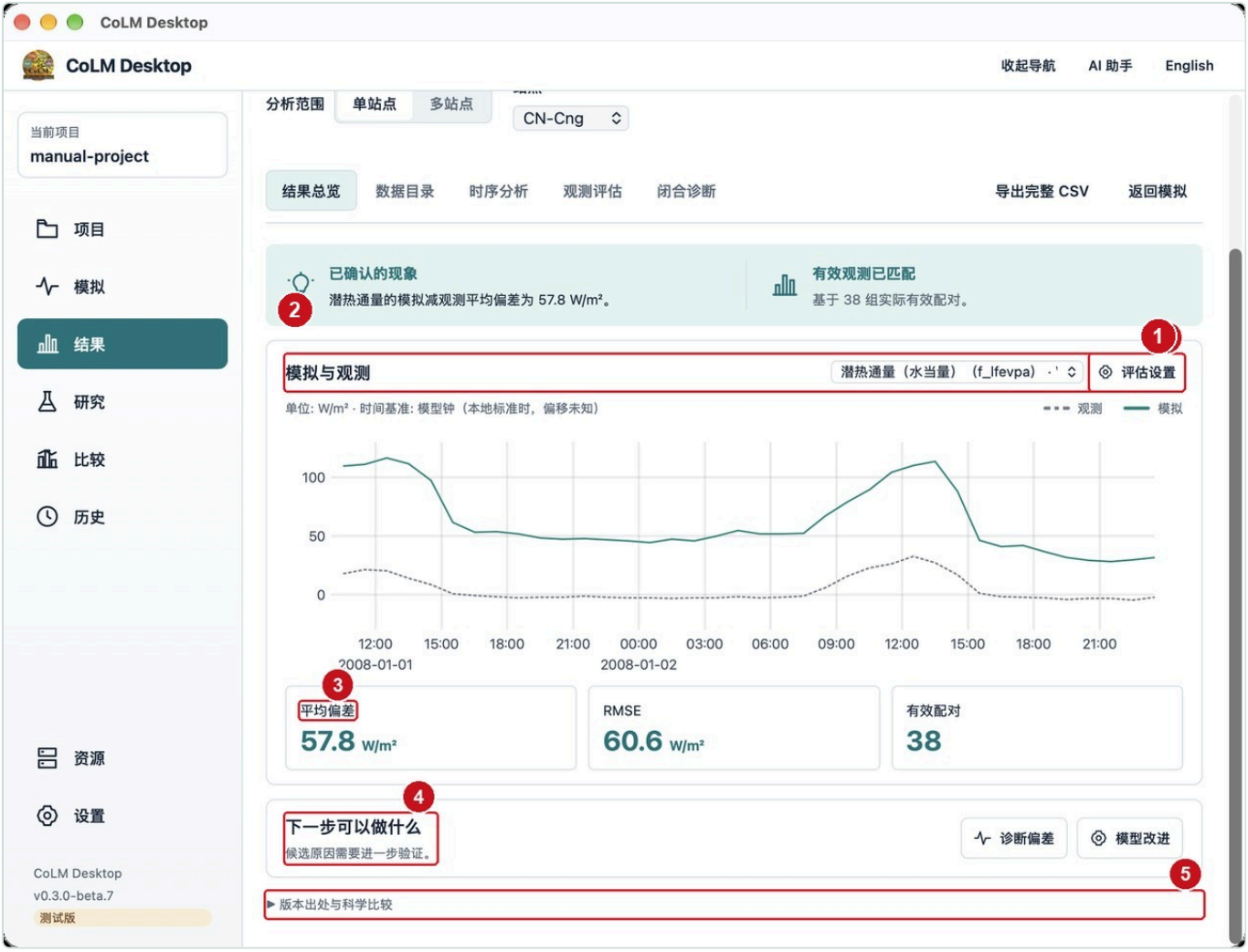

图 19 标注： ① 评估设置 ② 已确认的现象与有效观测匹配 ③ 平均偏差/RMSE/有效配对 ④ 候选原因 ⑤ 版本出处与科学比较。

- **分析范围**：单站点 / 多站点（多站点需同一批运行记录）。
- **页签**：结果总览、时序分析、观测评估、闭合诊断。
- **指标**：平均偏差、RMSE、有效配对数量、观测覆盖；时间基准按运行设置（模型钟/本地标准时）。
- **导出完整 CSV**：导出当前分析范围的时序数据。
- **版本出处**：显示运行使用的模型版本/修订与科学比较入口（跨版本对照需在"比较"页用 OpenBench）。

10. 研究页

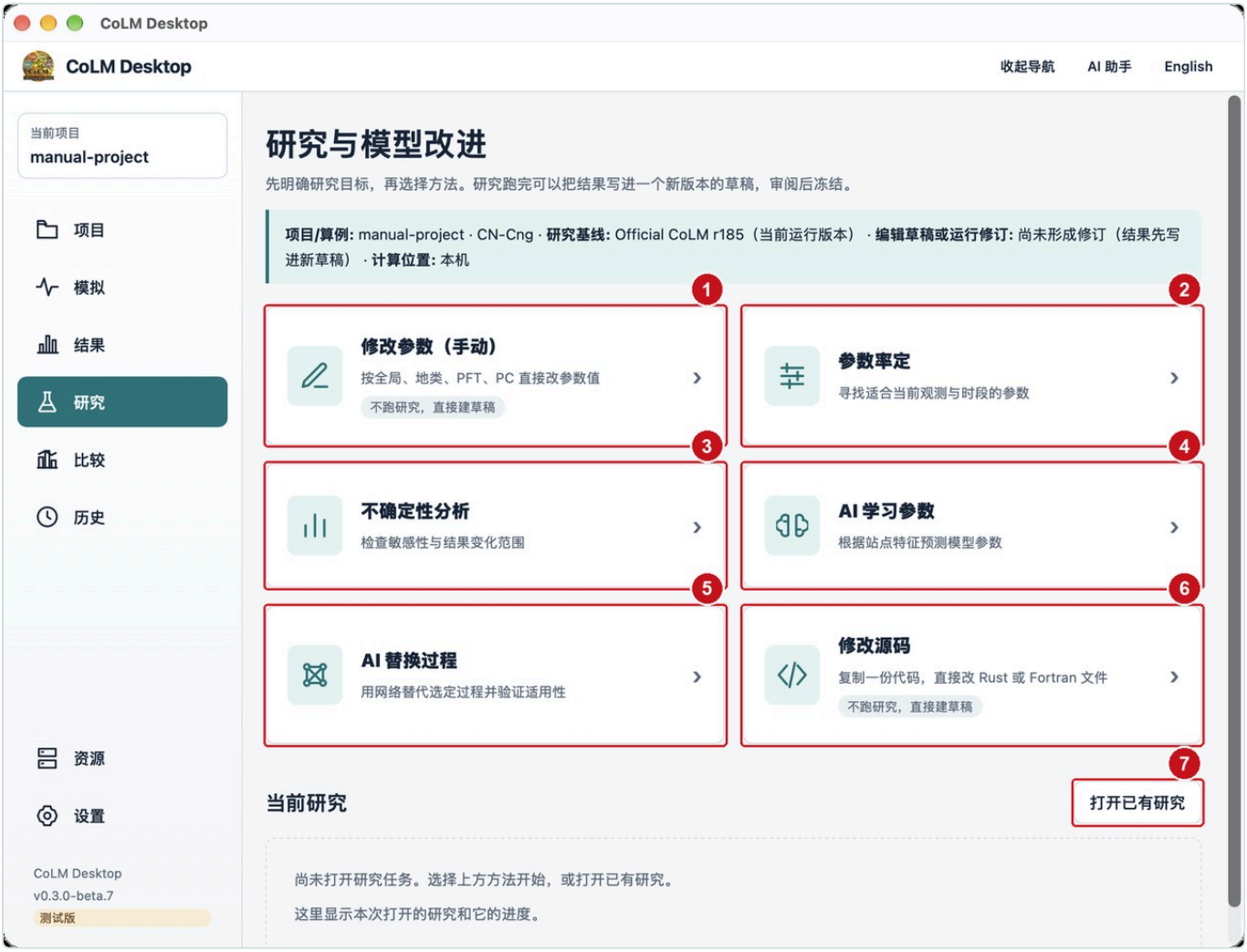

图 20 标注： ① 修改参数（手动） ② 参数率定 ③ 不确定性分析 ④ AI 学习参数 ⑤ AI 替换过程 ⑥ 修改源码 ⑦ 打开已有研究。

方法	用途
修改参数（手动）	直接改全局/地类/PFT/PC 参数；不跑研究，直接建草稿
参数率定	用观测寻找适合当前时段与算例的参数（支持共享参数跨站点验证）
不确定性分析	敏感性分析与结果变化范围
AI 学习参数	根据站点特征预测参数
AI 替换过程	用网络替代选定过程并验证适用性
修改源码	复制源码改 Rust/Fortran，编译与测试后形成修订

研究跑完后，结果先写进新版本的**草稿**，审阅后冻结为模型修订；结果不比默认参数好时拒绝写回。任务进度、收敛、观测约束程度都显示在"当前研究"区域。

> **Rust 固定引擎**：研究页不再提供计算引擎、服务器前处理引擎或编译配置下拉；研究固定使用 Rust，内核由所选算例的网格/地类/作物设置自动匹配。没有匹配内核时会直接提示，不会静默改用其他内核。已有的旧研究仍按它冻结的引擎与内核运行。
>
> **CROP**：Rust 的单点和空间作物算例都不再要求 CROP 编译内核；单点由站点文件的作物数组识别，空间由“作物过程”开关（正的播种日）识别，CLI 会自动加 `--crop`。Fortran 引擎仍需真正的 CROP 内核。

USGS：地类方案中的 USGS 已标记为“已退休 · 未全面测试”，保留给已有 USGS 算例和兼容性对照，不建议新研究使用。

11. 比较页（OpenBench）

图 21 标注： ① 安装、重装与维护（展开后有一键安装；环境状态/本次评估在该节上下方）。

- OpenBench 安装在软件自己的数据目录（约 400 MB），不碰系统 Python，不与你自己的 OpenBench 混用；
- 首次使用：展开"安装、重装与维护" → 一键安装（uv → 专用 Python 3.12 → OpenBench 及依赖），需要联网；
- 安装完成后在"本次评估"选择模拟、参考数据、指标/评分、时间尺度（可到年/月气候态），可本机或服务器运行；日志实时显示、可停止，只取回报告；
- 支持登记模型变量、非标准模拟与多参考数据；评估报告在结果页/比较页查看。

图 22 标注： ②"安装、重装与维护"折叠区 ④ 安装位置（打开目录）；环境状态与一键安装在同页上下方。

> 安装过程会先装 uv（约 20 MB，校验后使用），再装专用 Python 3.12，最后装 OpenBench 及依赖（约 400 MB，需要联网）。三个状态全部显示"已安装"后，"本次评估"区域才会出现配置表单。

12. 历史页

图 23 标注： ① 搜索 ② 类型筛选 ③ 状态筛选 ④ 查询 ⑤ 更多筛选 ⑥ 导出 CSV ⑦ 项目记录 ⑧ 清空历史记录。

- 记录每一次运行、研究、配置修改与远程操作；点击记录查看依据与后续操作；
- 筛选可按类型、状态、算例、服务器等组合；
- 导出 CSV / JSON（全局历史导出为**一行一操作**，不会把批量运行折叠掉）；
- 清空历史可选择是否同时删除本机算例文件夹与未确认结束的记录（危险操作会二次确认）。

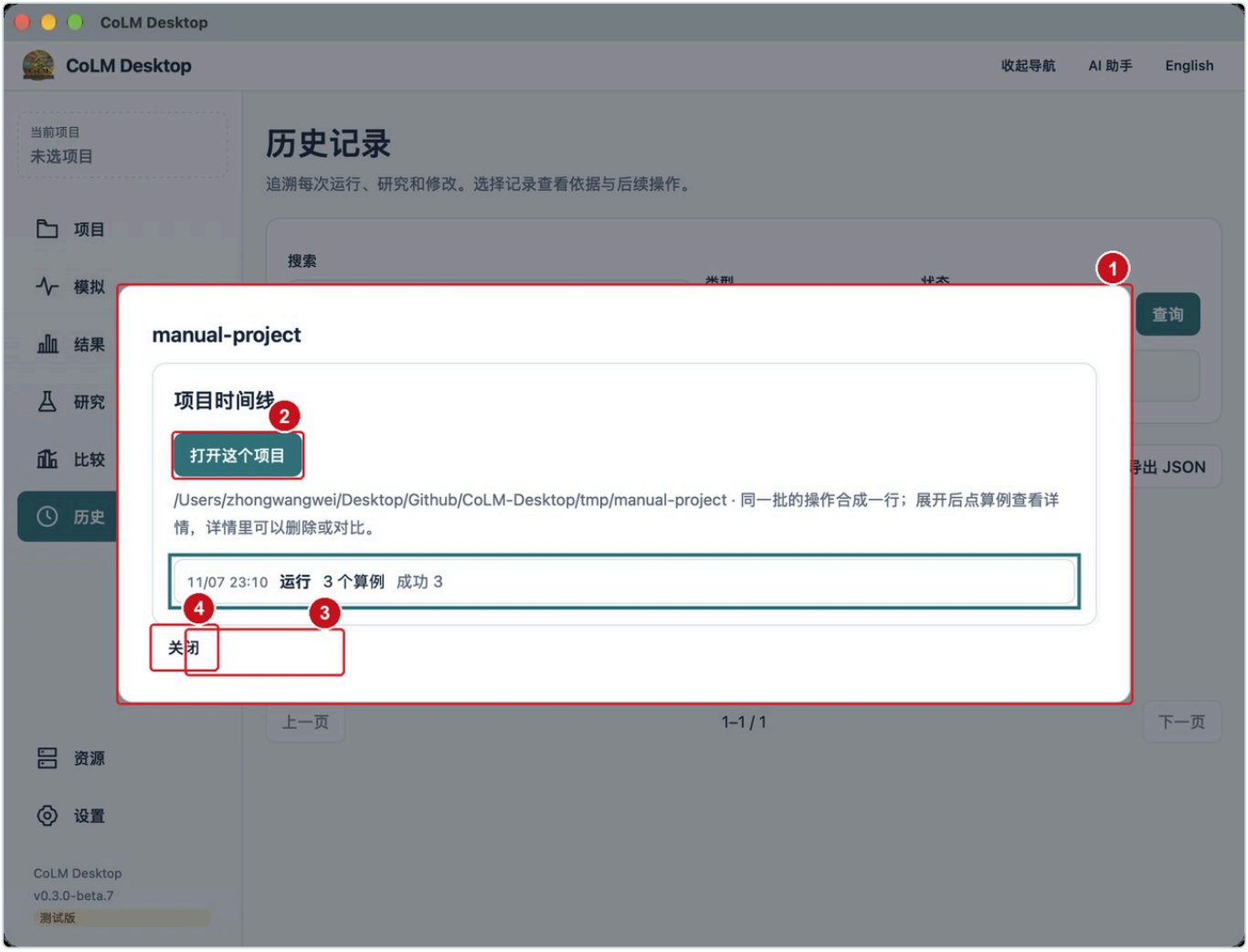

图 24 标注： ① 项目时间线 ② 打开这个项目 ③ 算例记录 ④ 关闭。同一批（一次批量运行）的操作合并成一行；展开后可点算例查看详情、删除或对比。

13. 资源页与远程服务器

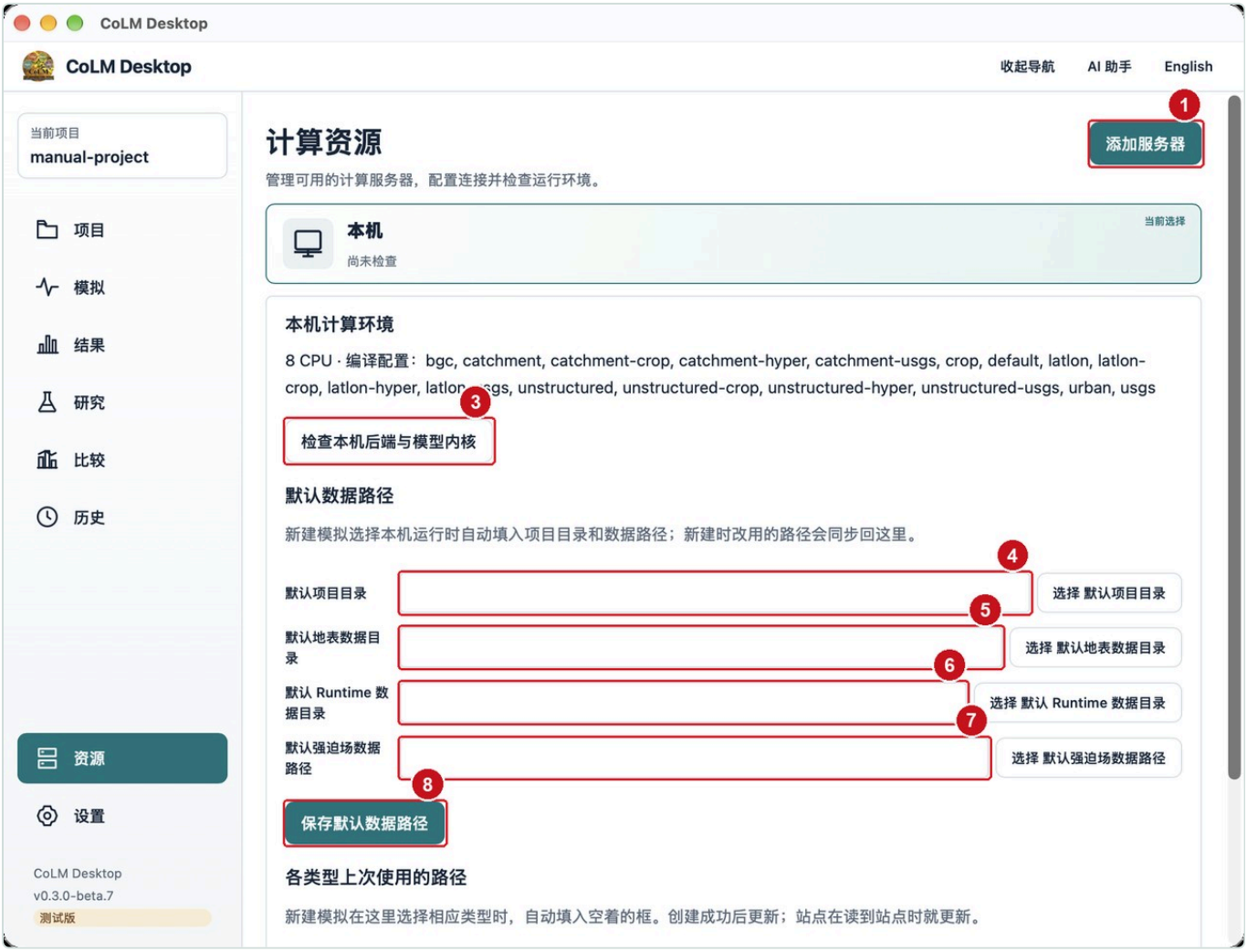

图 25 标注： ① 添加服务器 ② 检查本机后端与模型内核 ③ 默认项目目录 ④ 默认地表数据目录 ⑤ 默认 Runtime 数据目录 ⑥ 默认强迫场数据路径 ⑦ 保存默认数据路径（本机服务器卡片在页面中部）。

13.1 本机

- 显示 CPU 核数、可用编译配置（内核预设）；
- "检查本机后端与模型内核"会验证 `colm-cli`、sidecar 与内核清单；
- 默认数据路径用于新建模拟时自动填值；新建后改用的路径会同步回这里。

13.2 添加远程服务器（SSH）

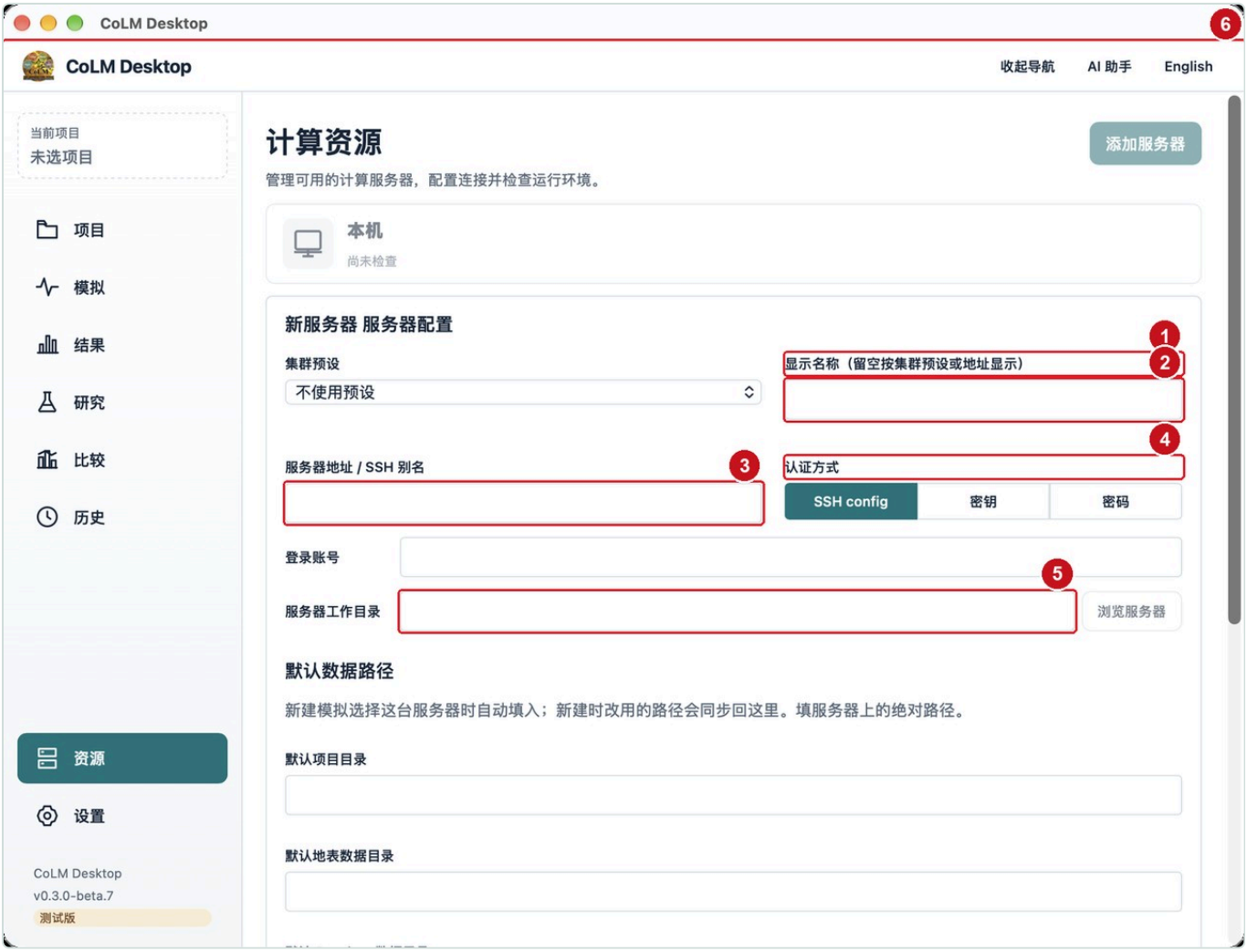

图 26 标注： ① 集群预设 ② 显示名称 ③ 服务器地址/SSH 别名 ④ 认证方式（config/密钥/密码） ⑤ 服务器工作目录 ⑥ 默认数据路径（可折叠区）。保存前请先点"检查连接"，全部检查项通过后再保存。

"添加服务器"后填写：

字段	说明
显示名称	便于识别这条服务器连接的名称
SSH 别名/主机	`~/.ssh/config` 里的 Host，或 `user@host`；land 集群当前使用登录节点别名 `land`
认证方式	SSH config、密钥、密码或密钥与密码；按当前用户的服务器登录要求选择
工作根目录	服务器上的 CoLM-Desktop 根目录（引擎、内核、作业、算例副本都在此）
调度器	`auto` / `bare` / `slurm` / `lsf` / `pbs`；分区/队列、账号、时限
节点与核数	设置节点数量和每节点核数，用于站点打包与并行预算
环境准备（env script）	在服务器每次远程命令前执行，例如 `module load ...` / `source /etc/profile.d/modules.sh; module load ...`。编 Fortran 内核/运行服务器作业都依赖它
目录映射（map）	本机路径前缀 = 服务器路径前缀，用于把本地数据目录对应到服务器（如 PLUMBER2s/Sitedata）
环境变量/指令	额外 `--directive`（Slurm `#SBATCH` 行）、内存等

需要私钥口令或登录密码时，应用会弹出输入框。按提示输入后按 Enter 或点击“继续”，当前操作会接着执行。经过跳板机或需要多项认证时，按每次弹窗的提示依次输入。

默认数据路径可以点击“选择目录”浏览服务器，强迫场也可以选择单个文件。选择集群预设或成功检查连接后，空着的路径会先填入预设值，项目目录可采用已填写的服务器工作目录。已有自定义路径会保留，填入后可继续修改和检查。

Slurm 用户先点击“检查连接”，再到“高级设置”中的“分区 / 队列”下拉框选择分区。LSF 用户在同一位置填写队列名称。节点数量和每节点核数决定申请的计算资源，调度器据此分配计算节点。需要指定具体节点时，可在“调度指令（每行一条）”中填写 Slurm 的 `--nodelist=节点名` 或 LSF 的 `-m "节点名"`，提交时由调度器核验节点和权限。

> 关键规则：
>
> 1. **外部数据目录必须 map**（强迫场、rawdata、Runtime、Sitedata）。目录不会被自动上传；普通文件（如算例目录内的 `site.nc`、`forcing.nml`）在物化运行时由 CLI staging 自动带走。
> 2. Fortran 引擎需要先在服务器上构建内核（"资源 → 服务器 → 准备环境/内核"），并设置正确的 `env script`（gfortran + MPI + NetCDF-Fortran）。
> 3. Rust 引擎首次远程运行会在服务器上构建 Rust 引擎（登录节点编好），需要 Rust/Cargo/CMake；安装包支持离线自举，也可提前 `remote-engine-prepare`。

13.3 服务器运行流程

1. 资源页添加服务器 → 检查连接与目录；
2. 算例第 1 步把"运行位置"切到该服务器；
3. 第 5 步选编译配置/引擎、提交阶段 → "开始运行"；
4. 运行进度里查看服务器作业状态、日志、节点/分区；可"核对服务器状态"、取消、取回结果；
5. 远程结果只取回所需变量/月份（可压缩），本机"结果"页打开。

常用远程 CLI 示例见第 18.6 节。

14. 设置页

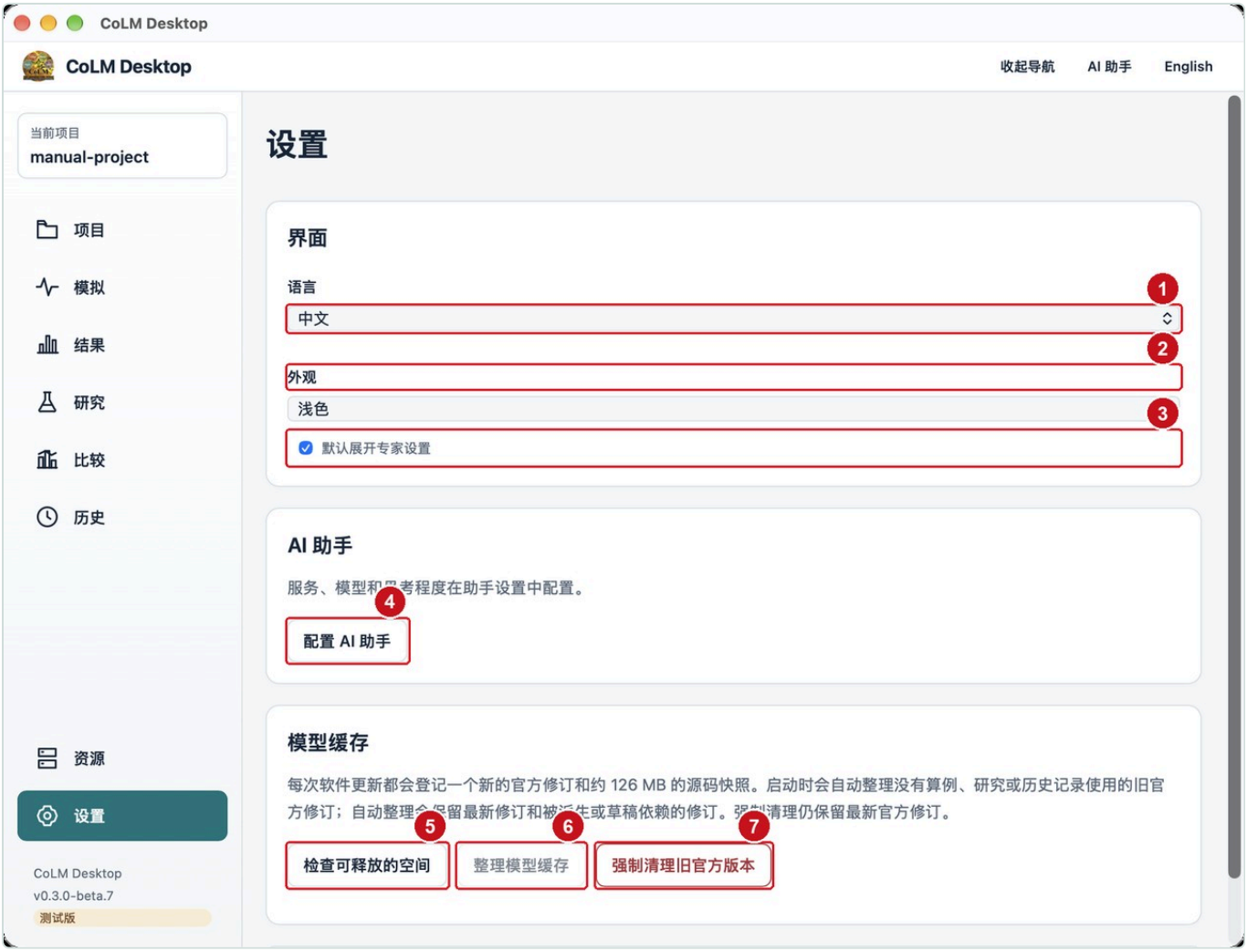

图 27 标注： ① 语言 ② 外观 ③ 默认展开专家设置 ④ 配置 AI 助手 ⑤ 检查可释放空间 ⑥ 整理模型缓存 ⑦ 强制清理旧官方版本（帮助在页面下方）。

- **界面**：语言（中文/English）、外观（浅色/深色/跟随系统）、默认展开专家设置。
- **AI 助手**：跳到助手设置（服务地址、模型、API Key、联网搜索、审批方式等），见第 17 章。
- **模型缓存**：每次软件更新登记新的官方修订与约 126 MB 源码快照；启动时会自动整理没有被算例/研究/历史使用的旧官方修订（保留最新与被依赖的修订）。"检查可释放空间"先统计，"整理模型缓存"按策略清理，"强制清理"需二次确认。
- **帮助**：关于、版本、文档入口。

图 28 标注： ① 配置 AI 助手 ② 模型缓存说明 ③ 检查可释放空间 ④ 整理模型缓存 ⑤ 强制清理旧官方版本 ⑥ 帮助。

15. 模型版本

图 29 标注： ① "选择已有版本"对话框 ② 官方版修订卡片 ③ 清理旧的官方版本 ④ 关闭。

首页"模型版本"卡片有三种入口：

入口	说明
官方版	随软件提供、不可修改；每个版本对应一个冻结修订与源码快照
已有版本	列出所有已保存修订（含自己冻结的修改版、旧官方修订），可切换/删除；删除前会检查绑定
新建修改版	进入"研究"：手动改参数、率定、不确定性、AI、改源码；研究结果经草稿审阅后**冻结**为新修订

注意事项：

- 运行算例时选择的版本会绑定到算例（`model-binding.json`），重启后保持；
- 修改版与官方版可同时存在，结果页/比较页按版本比较；
- 旧官方修订被自动/手动清理后，绑定它的算例会提示重新绑定；
- 清理/删除是不可逆操作，界面会二次确认并说明项目/研究/历史文件保留。

16. 准备输入数据

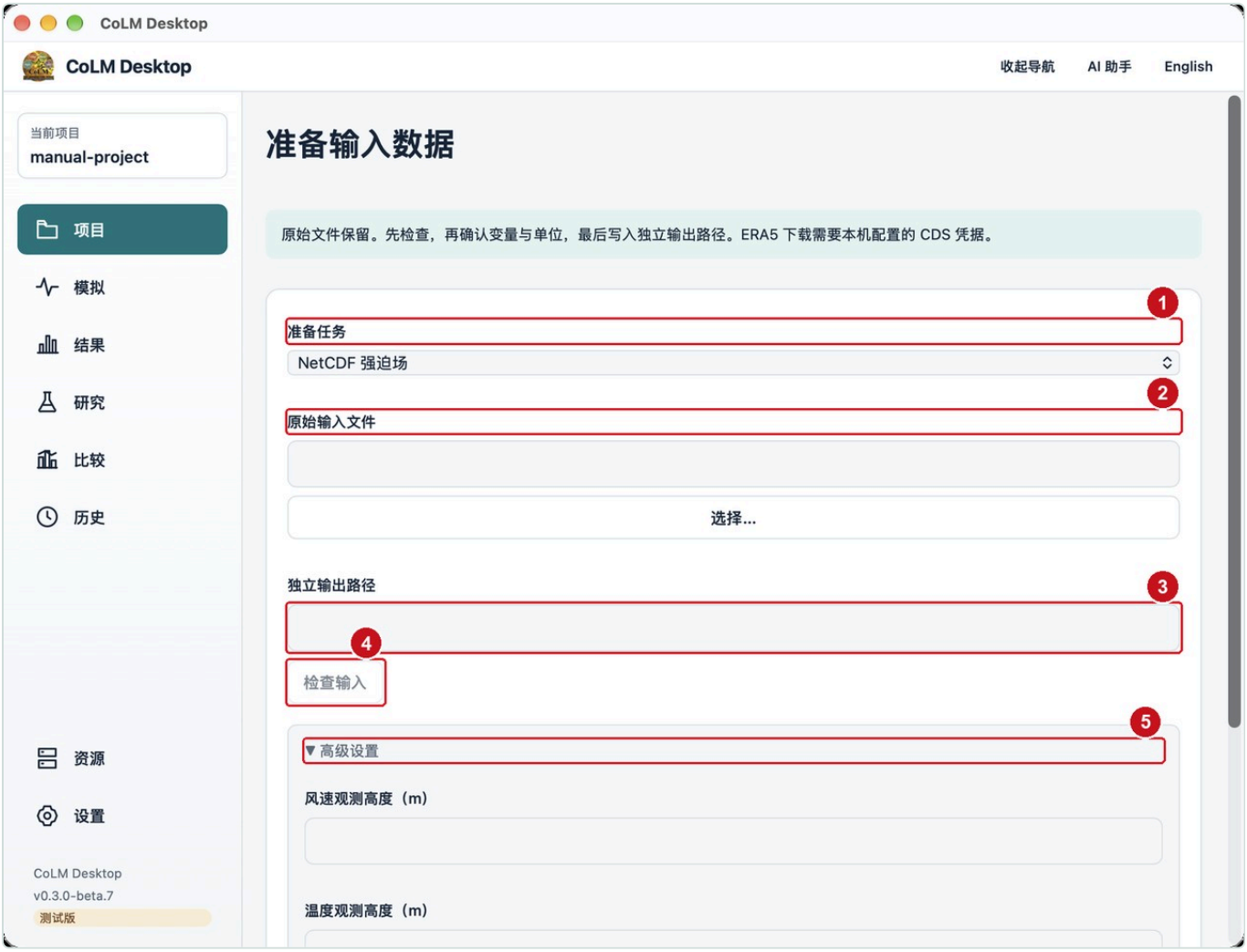

图 30 标注： ① 准备任务 ② 原始输入文件 ③ 独立输出路径 ④ 检查输入 ⑤ 高级设置。

用途：制作站点文件、转换/修复强迫场，供新建算例使用。典型流程：

1. **选择准备任务**：站点建站 / 强迫场转换 / 强迫场缺测修复；
2. **选择原始输入文件**（CSV/TXT/NetCDF 等；ERA5 下载需本机配置 CDS 凭据）；
3. **设置独立输出路径**：原始文件不改，产物写到独立路径；
4. **检查输入**：自动识别分隔符、列名、单位、时间范围与站点；
5. **高级设置**：观测高度、单位换算、缺测处理、变量映射等；
6. **生成数据**：写入产物；
7. **用于新算例** 或 **安装到数据集目录并扫描**：安装到 `Sitedata/`、`Forcing/` 后自动调用 `scan_sites` 刷新数据集，页面显示识别到的站点数量。站点与强迫场词干不同会提示无法配对。

产物页会显示 `site_kind/mode`、`from_lookup` 计数与 warning；rawdata 为零栅格时给出显式警告而不是静默回退。

图 31 标注： ① 准备任务 ② 原始输入文件 ③ 独立输出路径 ④ 检查输入 ⑤ 高级设置（展开后含观测高度、变量映射等）。

17. AI 助手

图 32 标注： ① 新对话 ② 会话历史 ③ 上下文范围（项目/算例/模拟/结果/研究） ④ 内置示例站点 ⑤ 输入框 ⑥ 发送/停止 ⑦ 助手设置。

图 33 标注： ① 运行后端 ② 服务商 ③ 模型选择 ④ 历史范围 ⑤ 未登记 API Key 提示；联网搜索与文档检索等高级项在设置页下方（需滚动查看）。

17.1 首次配置

1. 打开助手设置 → 填写服务地址（如 `https://api.deepseek.com`）、模型名、API Key；
2. 选择 API 格式（chat completions / responses 等）、思考程度、超时；
3. 如使用联网搜索/数据外发，勾选确认外发目标；
4. 保存。若助手正在运行，保存后会自动重配；重配失败会提示"设置已保存，但助手重配失败……下一次发送会自动重试"，不会丢设置。

17.2 使用

- 助手**理解当前页面上下文**（项目/算例/模拟/结果/研究），能读取页面合同并执行登记的按钮操作；
- 发送前可修改问题；支持附件（PDF/DOCX/TXT）；
- 工具调用会显示"完成/失败"卡片，可展开查看参数与错误；
- **危险操作（删除、覆盖、清空等）仍需人工确认**；审批方式可在设置里选；
- 页面上的按钮都登记为助手可执行动作；配置字段受页面合同约束——例如空间算例第 2 步的范围字段在助手工具中也是只读（`ui_fill` 会返回 `cannot be changed here`），避免绕过界面约束；
- 会话历史按项目保存，可恢复；上下文接近窗口上限时会提示并折叠较早工具结果。

17.3 文件与检索

- 可附加本机 PDF/DOCX/文本文档；
- 联网搜索使用 DeepSeek Search API（需 Key）；
- 文档检索可接本机 Xerj 节点（默认关闭；可用户自己运行，也可由应用托管固定版本）；
- 模型/代码问题会路由官方 CoLM 与 CoLM-Desktop 仓库检索，回答带出处；不编造提交/PR。

---

18. 命令行 `colm-cli`

`colm-cli` 与桌面端使用同一套引擎/算例/模型注册表。安装包内路径（macOS）：

```
/Applications/CoLM Desktop.app/Contents/MacOS/colm-cli
# 或源码构建产物
target/release/colm-cli
```

查看全部命令：

```
colm-cli --help
```

> 完整命令索引与 `--help` 原文见附录 C：CLI 命令参考（68 组用法、.--help 全文）。

18.1 通用约定

- 大多数命令支持 `--json 1` 输出机器可读结果；
- 模型相关命令可带 `--model-root <目录>`（默认 `~/Library/Application Support/CoLM-Desktop/models` 或对应平台目录）与 `--model-id/--model-revision`；
- 远程命令共享 `--host`、`--root`、`--scheduler`、`--partition`、`--walltime`、`--env-script` 等选项；
- 错误信息同时给中英文时，脚本可按固定文本匹配；退出码非 0 表示失败。

18.2 站点与强迫场

```
# 扫描站点目录（--quick 跳过强迫场读取）
colm-cli scan --dir "$HOME/Library/Application Support/edu.sysu.colm.desktop/examples/Sitedata" --quick
        1

# 从 CSV/TXT 建站点文件（带 rawdata 可抽取土壤等字段）
colm-cli site-new --out ./CN-Demo_site.nc --lon 113.5 --lat 23.5 \
  --mode igbp --rawdata /path/to/rawdata --json 1

# 强迫场目录探测/修复
colm-cli forcing-directory-probe --help   # 见 --help 中 remote/forcing 组
```

实际运行示例（本手册验证时执行）：

```
$ colm-cli scan --dir ".../examples/Sitedata" --quick 1 --out sites.json
$ head -c 300 sites.json
[
  {
    "name": "AT-Neu",
    "site_file": ".../Sitedata/AT-Neu_2010-2012_FLUXNET-CH4_site.nc",
    "met_file": ".../Forcing/AT-Neu_2010-2012_FLUXNET-CH4_Met.nc",
```

18.3 算例创建与运行

```
# 运行一个已有算例（本机 Rust 引擎；--kernel 指向已构建内核）
colm-cli run ./my-project/CN-Cng --kernel ./kernels/default --stream 1

# 指定阶段/重跑
colm-cli run ./my-project/CN-Cng --kernel ./kernels/default --stage mksrfdata

# 查看历史变量目录
colm-cli history-catalog ./my-project/CN-Cng

# 导出/裁剪历史
colm-cli history-subset ./my-project/CN-Cng --out ./subset --vars f_fevpa,f_fsena --from 2008-01 --to 2
        008-01
```

本次演示实际运行的 `history-catalog` 输出：

```
{"files":1,"steps":48,"start":1199147400,"end":1199316600,
 "variables":[{"name":"f_alb","units":"-","long_name":"averaged albedo","kind":"category", ...}]}
```

18.4 模型版本

```
colm-cli model-list --model-root "$HOME/Library/Application Support/CoLM-Desktop/models" --json 1
colm-cli model-get --model-root ... --model-id official --model-revision 185
colm-cli model-freeze --model-root ... --model-id <draft-id> --asset-root ./assets
```

修改版运行：`--model-version <名称>` 或 `--model-id/--model-revision`；运行会物化到算例目录的 `runs/<操作号>/`，结果与运行清单（`model-run.json`）一起保存，可追溯。

18.5 评估/研究/历史

```
colm-cli evaluation-plan <case> --obs <Flux.nc> --kernel <kernel-dir>
colm-cli study-params
colm-cli history-subset ...
colm-cli model-changes --model-root ... --version <id> --revision <n>
```

18.6 远程服务器与集群

```
# 构建远程 Fortran 内核（服务器需 gfortran + MPI + NetCDF-Fortran）
colm-cli remote-kernel --host my-cluster --root /path/colm-desktop --preset default \
  --env-script 'source /etc/profile.d/modules.sh; module load gcc mpi netcdf' \
  --scheduler slurm --partition debug --walltime 00:30:00

# 远程运行（Rust 引擎，默认；需本机内核目录提供宏清单）
colm-cli remote-run ./my-project/CN-Cng --host my-cluster --root /path/colm-desktop \
  --kernel ./kernels/default --map /local/data=/server/data --upload-unmapped 1

# 远程 Fortran 引擎
colm-cli remote-run ... --engine fortran --preset default --remote-kernel auto

# 站点批量打包（同一批共享 --pack-batch 值）
colm-cli remote-run ./site1 --host my-cluster --root ... --kernel ... --pack-batch <batch-id> --cores-p
        er-node 64
colm-cli remote-run ./site2 --host my-cluster --root ... --kernel ... --pack-batch <batch-id> --cores-p
        er-node 64
```

```
# 查询/取消/取回
colm-cli remote-status ./site1 --host my-cluster --root ...
colm-cli remote-cancel ./site1 --host my-cluster --root ...
colm-cli remote-fetch ./site1 --host my-cluster --root ... --vars f_fevpa,f_fsena
```

- 远程作业的 SSH 凭据可由桌面端以 stdin 注入（`COLM_SSH_AUTH_STDIN=1`）；命令行直接使用时按 `~/.ssh/config` 连接。
- 外部大数据目录用 `--map`；算例内部普通文件（site.nc 等）会随 staging 自动上传。
- 打包批量：同一批站点进同一队列，载体作业按节点槽位并发；每站 1 rank/1 线程。

18.7 常用诊断

```
colm-cli remote-probe --host my-cluster
colm-cli remote-kernels --host my-cluster --root /path/colm-desktop
colm-cli remote-engines --host my-cluster --root /path/colm-desktop
colm-cli remote-load --host my-cluster
colm-cli verify-completed-outputs ./my-project/CN-Cng
```

19. 区域与流域算例（Heihe 实测）

本章的例子都在同一台机器上实际跑过：数据来自 `/Volumes/Data`（Heihe 边界、JRA3Q 强迫、CoLMrawdata、CoLMruntime 单元汇流数据），时间窗口 2003-02-01 → 2003-02-04。

19.1 项目里同时放站点/区域/流域算例

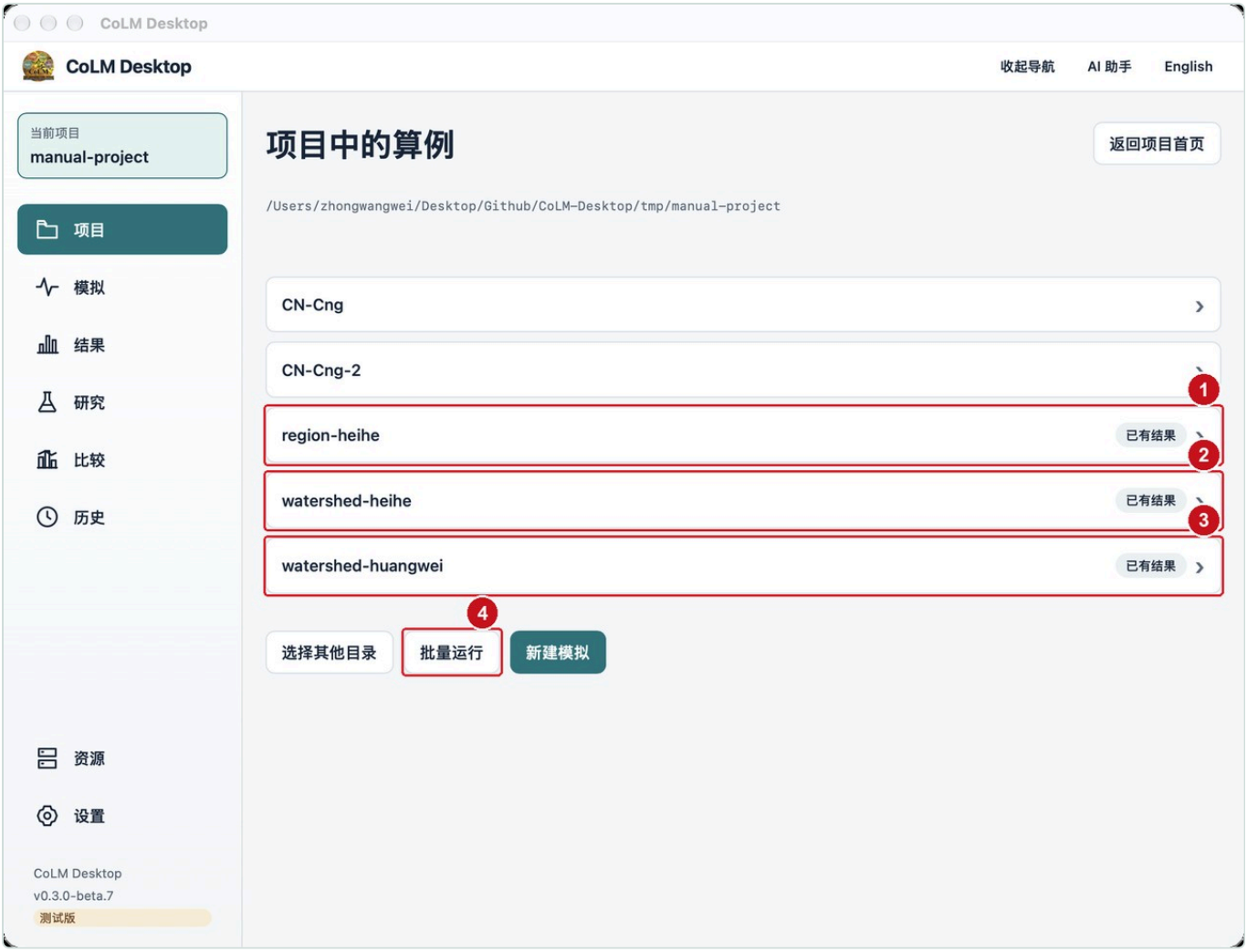

图 34 标注： ① 区域算例 `region-heihe` ② 流域算例 `watershed-heihe`（huahai 边界） ③ 流域算例 `watershed-huangwei`（huangwei 边界） ④ 批量运行入口。算例卡片的"已有结果"表示该算例已有运行输出。

19.2 区域算例（latlon 经纬度网格）

区域算例使用 0.25° 经纬度网格，范围 **96.25—104.25°E、37.75—43.25°N**（Heihe 流域+周边），强迫场 JRA3Q，开启区域单元汇流（河湖流）。

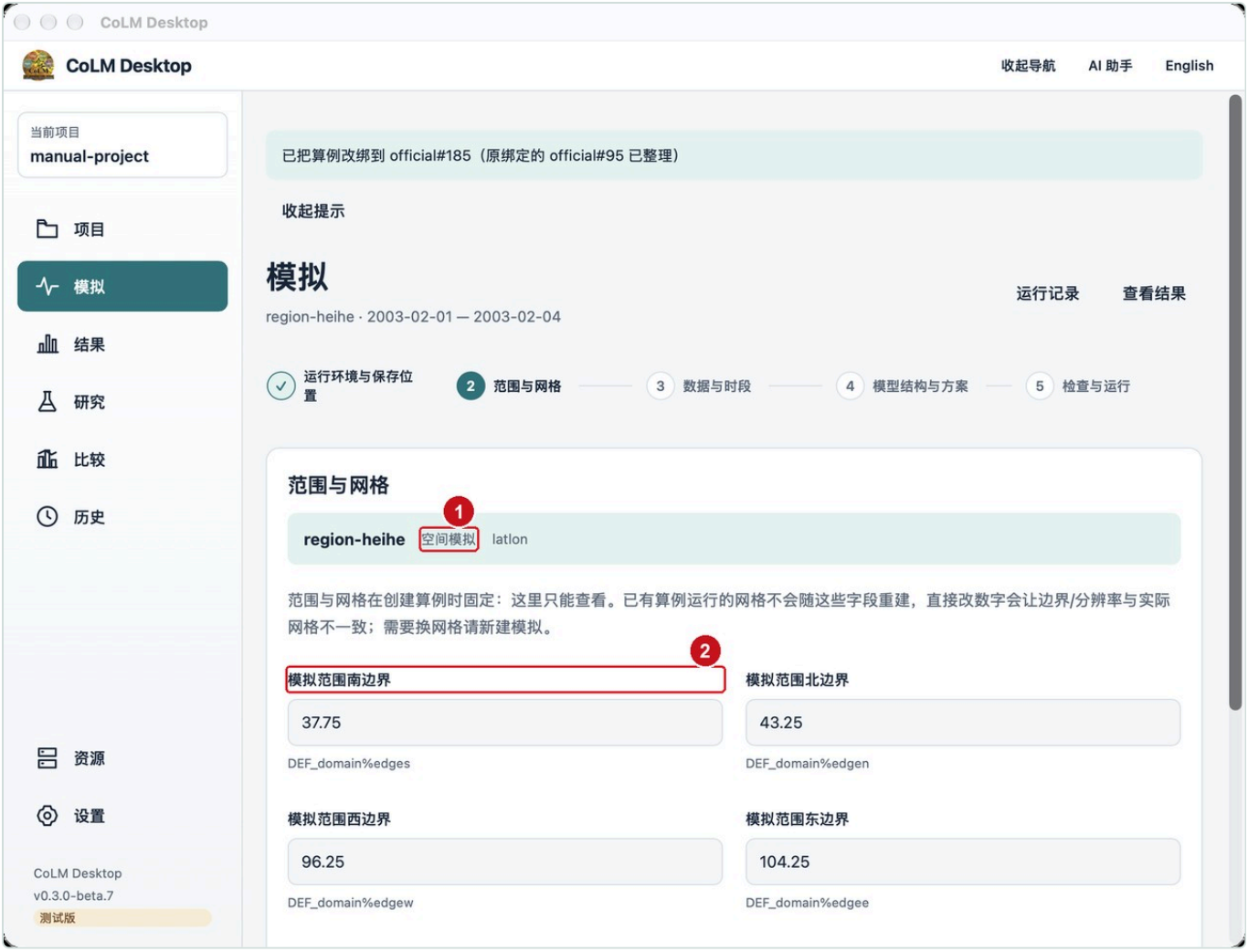

图 35 标注： ① 空间模拟说明（网格创建后固定） ② 模拟范围/网格分辨率字段（只读，换网格需新建模拟）。

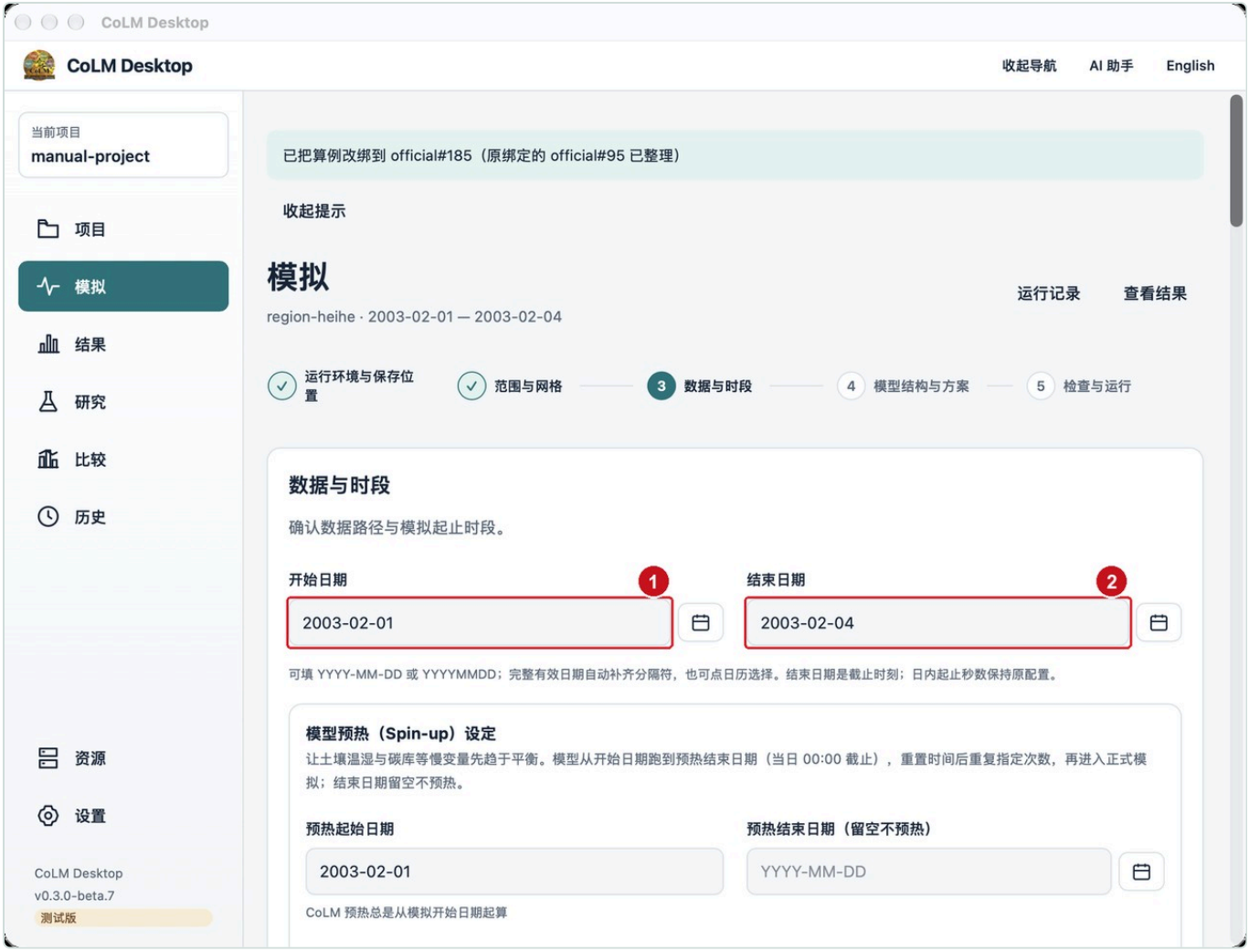

图 36 标注： ① 开始日期 ② 结束日期（空间算例同样支持预热与日内秒）。

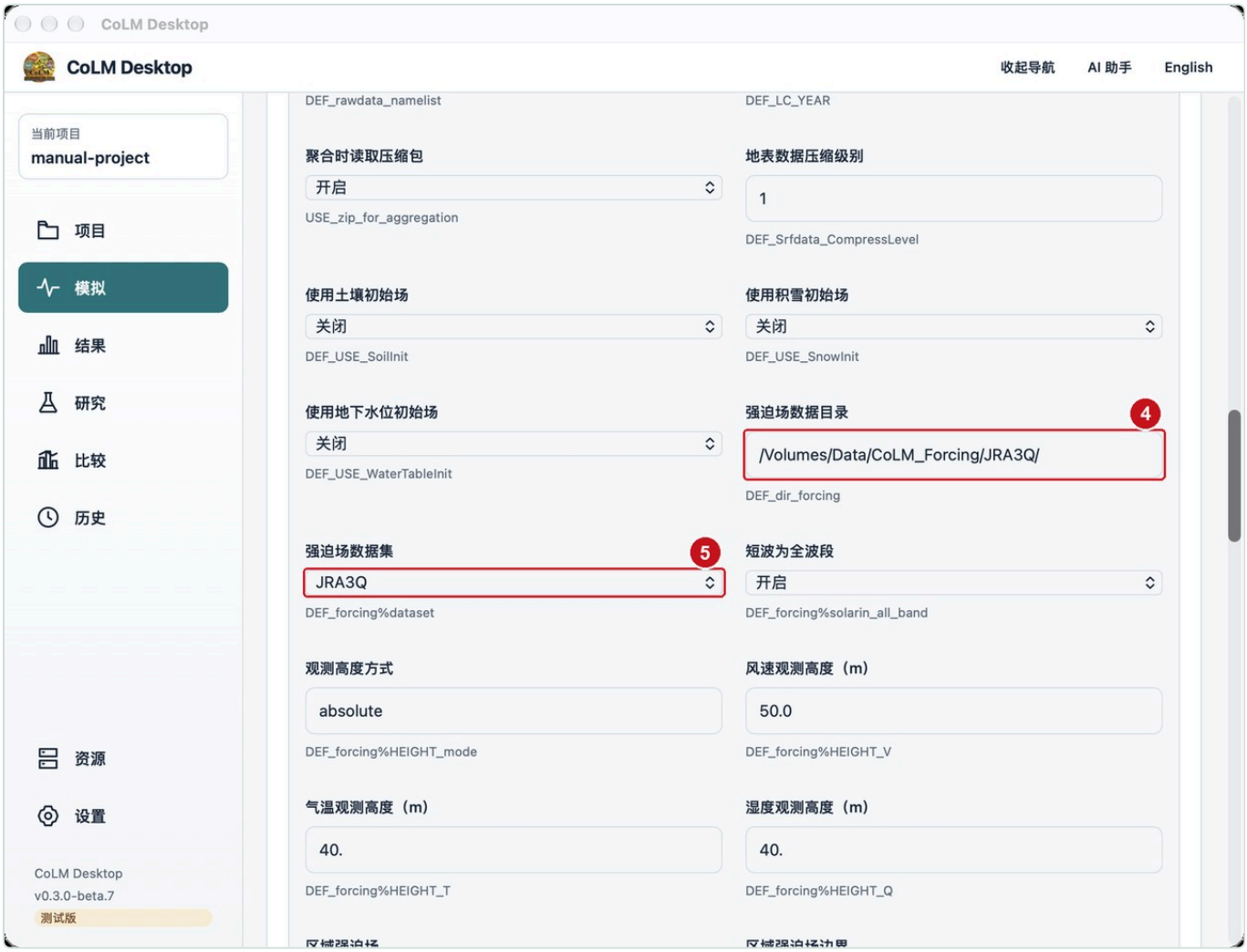

图 37 标注： ① 强迫场配置 ② 强迫场数据目录（JRA3Q） ③ 强迫场数据集。

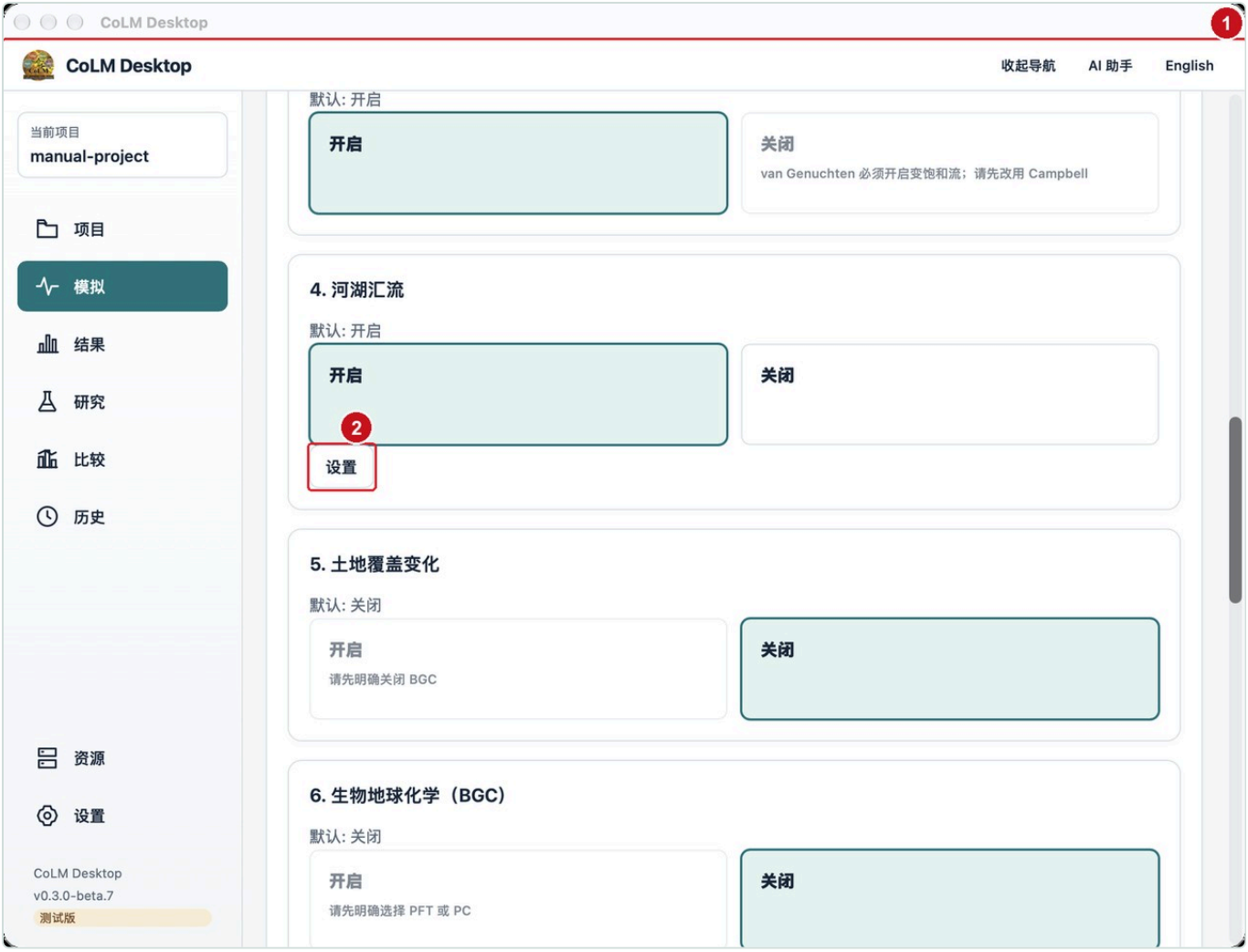

图 38 标注： ① "4. 河湖汇流"过程 ② 该过程右侧的"设置"。

图 39 标注： ① 河湖流与水库设置对话框 ② 单元流域河网文件（`grid_routing_data_15min.nc`） ③ 河湖汇流最大子步长等参数。区域单元汇流开关、水库调度方案、估算河道深度等都在这里；改完点"保存设置"。

图 40 标注： ① 编译配置（区域网格需要 `latlon` 内核） ② 开始运行。

实际运行结果（CLI 实跑）：

```
kernel: latlon@4c36d17d#production (Darwin-arm64)
model engine: rust
=== colm-stage colm ok ===
  colm       ok
EXIT:0
```

- 96 个时间步（3 天，3600 秒步长），输出 `out/region-heihe/history/`、`restart/`；区域网格由 `DEF_GRIDBASED_lon_res/lat_res=0.25` 自动构建（无需 `DEF_file_mesh`）。
- 河湖流：latlon 内核编译了 `GridRiverLakeFlow`，运行时读取 2.0 GB 的 15min 单元汇流数据。

19.3 流域算例（shapefile 边界 → unstructured 网格）

流域算例用 `/Volumes/Data/Data/Heihe-shp/heihe_boundary_with_huahai.shp` 生成非结构网格（0.5°，113 个活跃单元），边界内 267 个 LCT patch，保留河湖流开关。

```
# 1) 由流域边界生成非结构网格（0.5° = 720×360 全球网格中落在边界内的单元）
colm-cli mesh-new --out watershed-heihe/mesh.nc --nlon 720 --nlat 360 \
  --grid-kind unstructured --shp /Volumes/Data/Data/Heihe-shp/heihe_boundary_with_huahai.shp

# 2) 建空间算例（JRA3Q 强迫、本机 rawdata/runtime、3 天）
colm-cli spatial-new --grid-kind unstructured --mesh watershed-heihe/mesh.nc \
  --out watershed-heihe --forcing watershed-heihe/forcing.nml \
  --rawdata /Volumes/Data/CoLMrawdata --runtime /Volumes/Data/CoLMruntime \
```

```
  --start 2003-02-01 --end 2003-02-04 --timestep 3600 \
  --name watershed-heihe --mode igbp
```

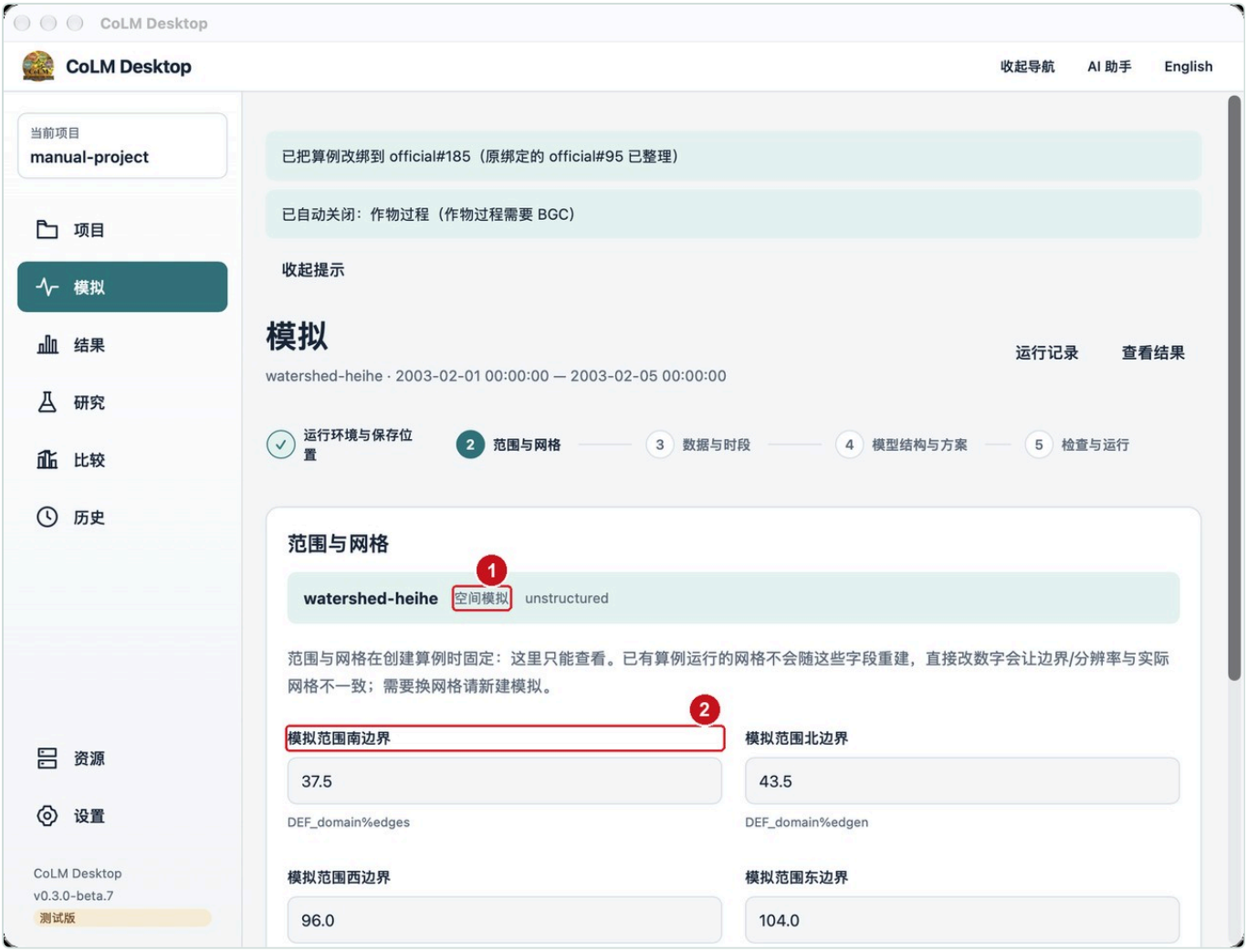

图 41 标注： ① 空间模拟说明 ② 模拟范围（由边界文件决定，创建后锁定）。

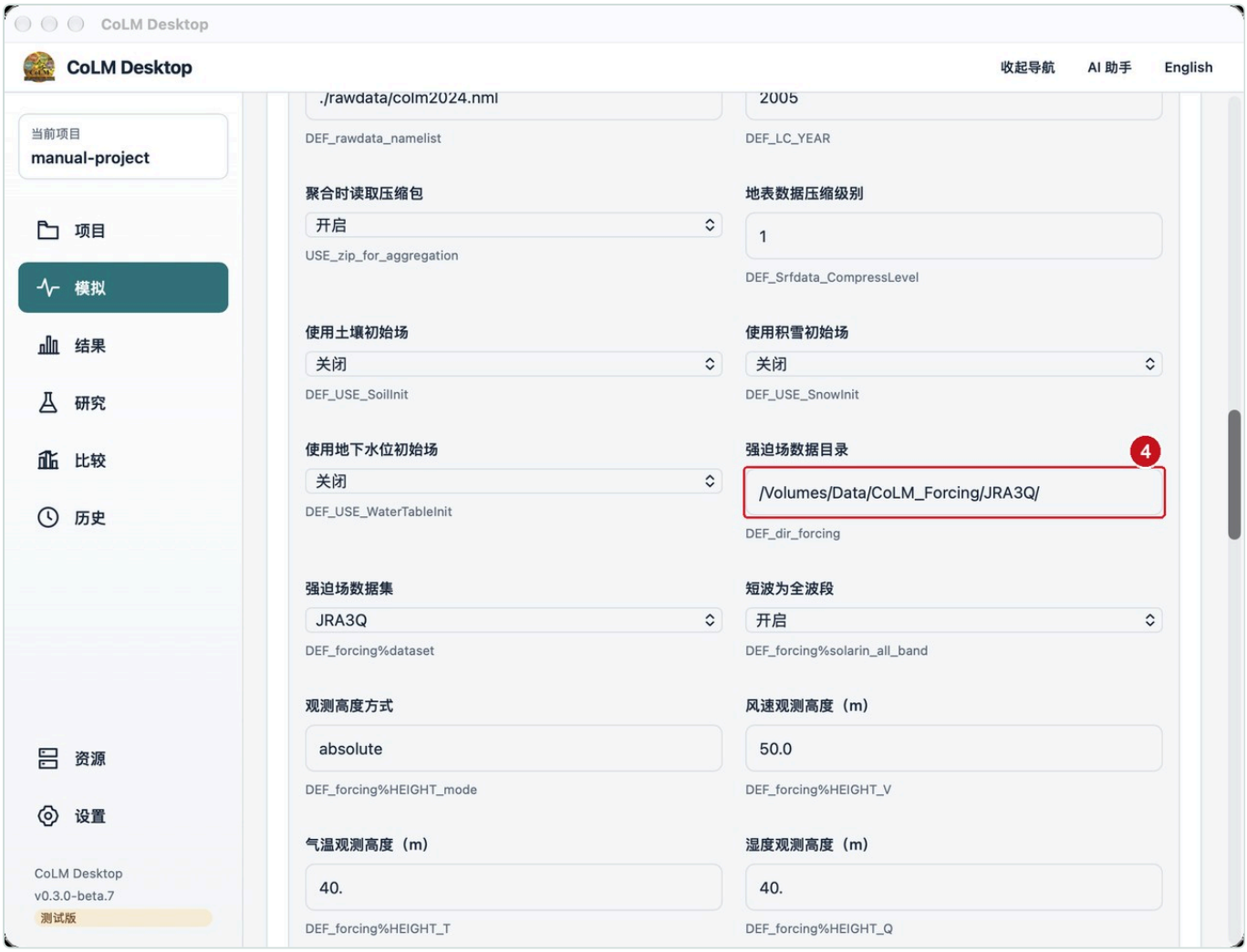

图 42 标注： ① 强迫场配置/目录（JRA3Q）；② 地表数据与 Runtime 目录同区域算例。

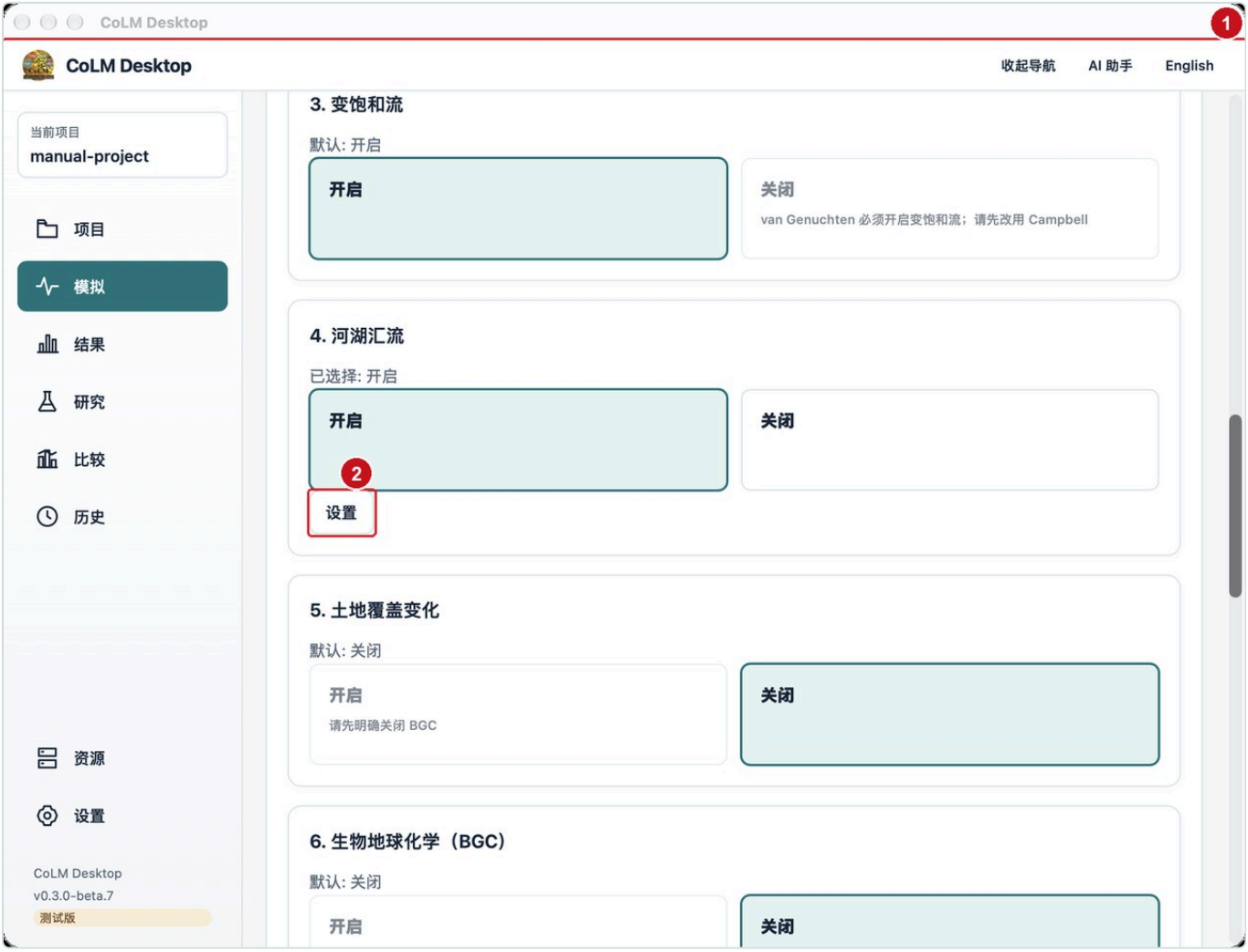

图 43 标注： ① "4. 河湖汇流" ② "设置"。

图 44 标注： ① 河湖流与水库设置 ② 单元流域河网文件（本机 15min 数据）。

图 45 标注： ① 编译配置（unstructured 内核） ② 开始运行。

实际运行结果（GUI 实跑）：

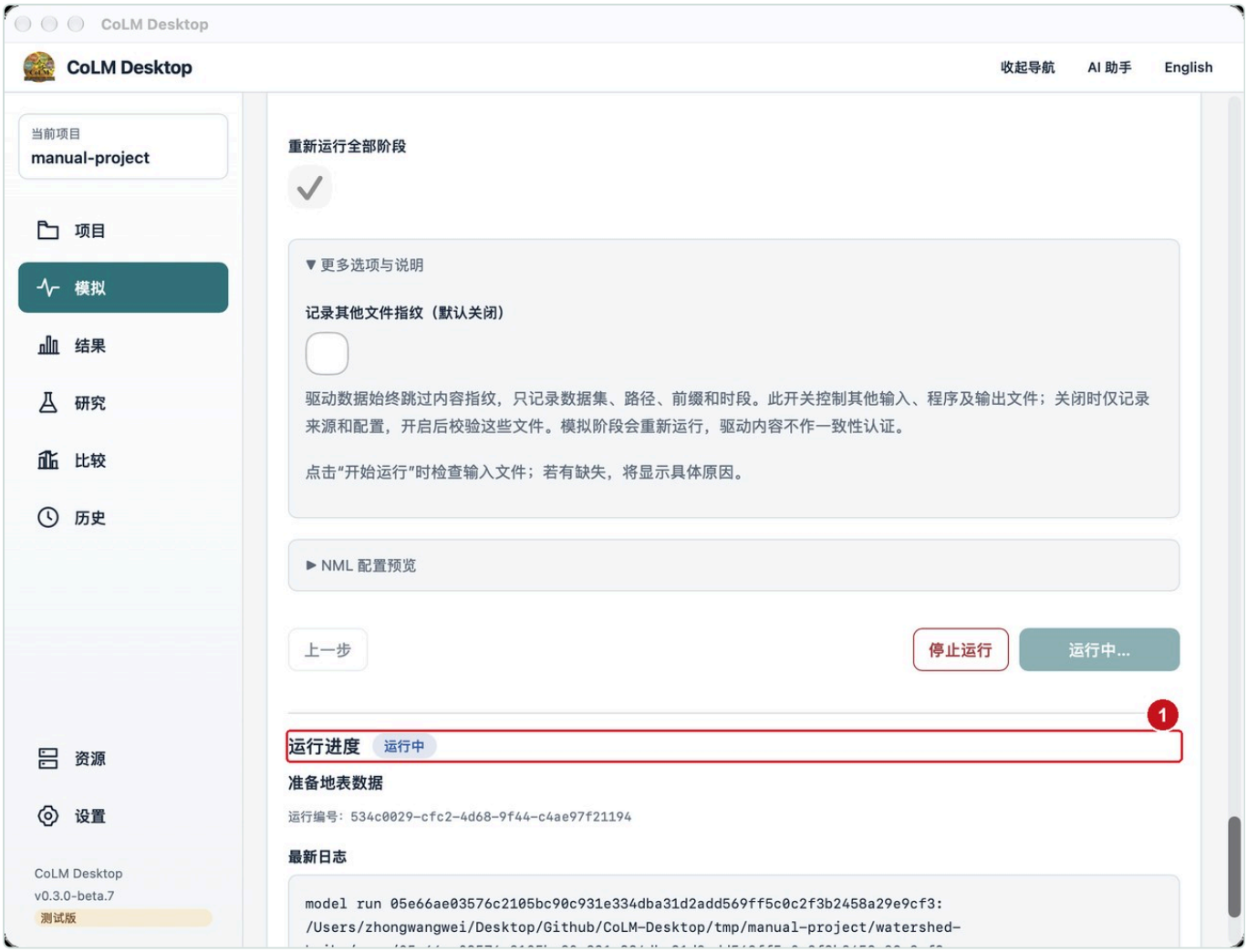

图 46 标注： ① 运行进度（mksrfdata → mkinidata → colm 三阶段）。

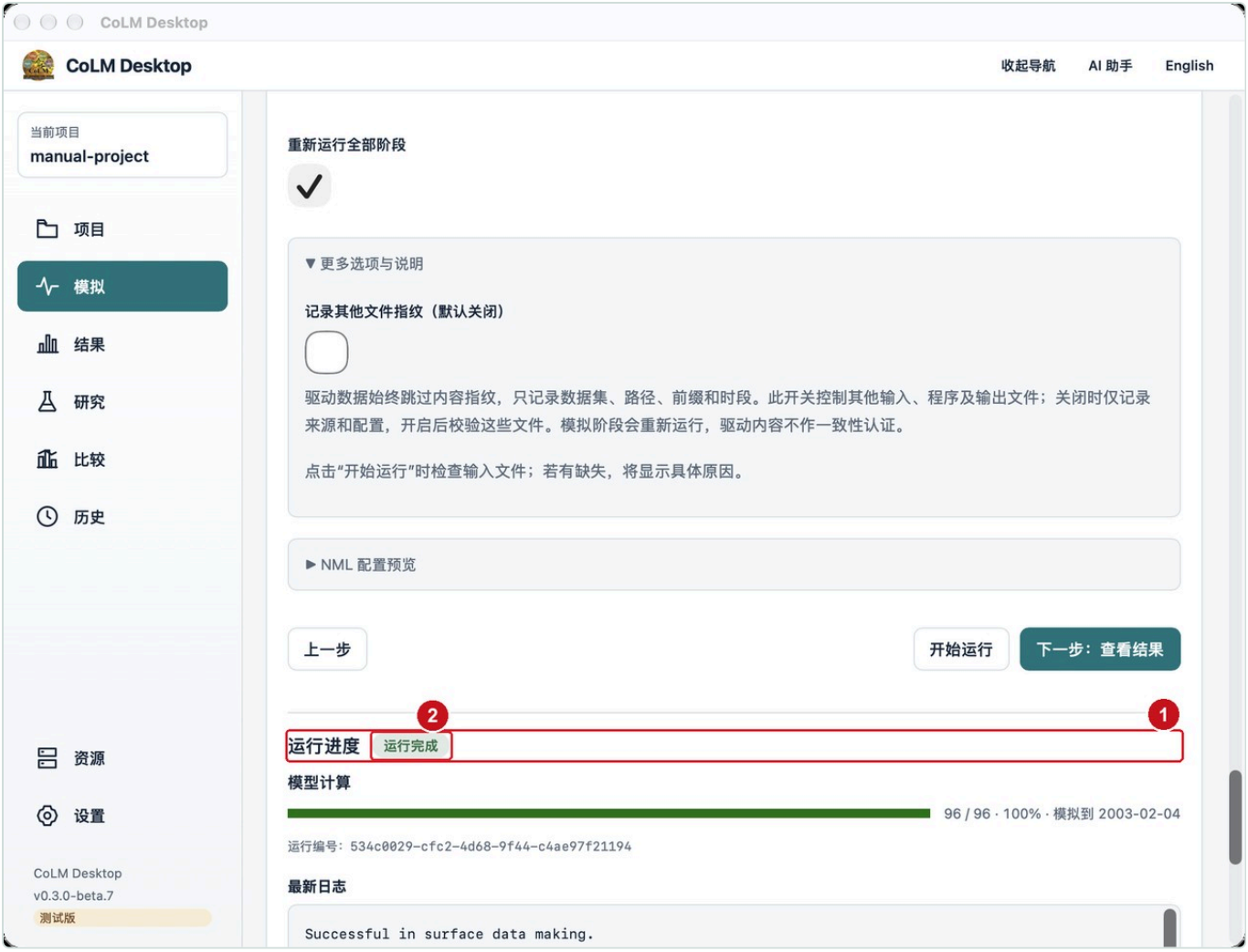

图 47 标注： ① 运行进度 ② 完成状态。

```
wrote 113 spatial land elements and 267 LCT patches
Regional unit-catchment network: 2084 unit catchments in 77 river systems,
  written to .../riverlake/unitcatchment_regional.nc
colm-rs: 96 step(s) on 267 spatial patch(es) in 4 block(s)
CoLM Execution Completed.
=== colm-stage colm ok ===
```

第二个流域边界（`heihe_huangwei.shp`） 也用同样流程实跑：0.5° 网格取到 **59 个活跃单元**（177 个 spatial patch），区域河网 **1758 个单元流域、24 个河系**，3 天 96 步，`EXIT 0`：

```
Regional unit-catchment network: 1758 unit catchments in 24 river systems
colm-rs: 96 step(s) on 177 spatial patch(es) in 4 block(s)
CoLM Execution Completed.
=== colm-stage colm ok ===
EXIT:0
```

图 48 标注： ① 结果与评估 ② 切换运行记录 ③ 分析范围 ④ 结果/时序/观测/闭合页签。空间算例的"模拟与观测"显示**区域平均（按纬度加权）**，下方"空间分布"可逐帧看格点（示例：潜热通量水当量 `f_lfevpa`）。

19.4 区域 vs 流域：怎么选

	区域（latlon）	流域（unstructured / catchment）
网格	规则经纬度网格，`DEF_GRIDBASED_lon_res/lat_res`	边界文件生成的非结构网格（或用户提供的 catchment mesh）
典型用途	区域气候/陆面模拟、与格点观测比较	按流域边界做水文/汇流、河湖流
河湖流	`GridRiverLakeFlow` + 单元汇流文件（latlon 内核已编入）	内核含 `GridRiverLakeFlow`，需 `DEF_UnitCatchment_file` +（流域）`unitcatchment_regional`
本例规模	0.25°，32×22 单元	0.5°，113 单元 / 267 LCT patch
结果展示	区域平均 + 空间分布	区域平均 + 空间分布（按网格）

数据准备要点：

- 区域强迫：JRA3Q 目录需含 `tmp2m/tmp2m_YYYY_MM.nc` 等 8 个变量子目录；`forcing.nml` 的 `DEF_forcing%startyr/endyr` 覆盖模拟年份；
- Heihe 边界 shapefile 必须与 `.shx/.dbf/.prj` 同目录；
- `DEF_UnitCatchment_file` 指向 `CoLMruntime/unitcatchment/grid_routing_data_15min.nc`（区域）或用户自己的 unit catchment mesh；
- 空间算例不支持"参数调优/不确定性分析"（界面会自动提示）。

> 说明：本章"区域"用 CLI 实跑验证（EXIT 0），"流域"用 GUI 实跑并截图；两者使用同一套 Rust 引擎与本地内核。

20. 配置参数总览

界面里所有可编辑字段来自 CoLM namelist schema，共 **964 项**；参数库（研究/助手用）共 **1369 项**。

19.1 按页面的参数分组

页面/分组	主要字段
运行环境与保存位置	`DEF_CASE_NAME`、算例/输出目录、引擎、版本绑定
范围与网格	`SITE_lon_location`、`SITE_lat_location`、网格/范围/分辨率字段
数据与时段	`DEF_simulation_time%*`、`DEF_dir_rawdata`、`DEF_dir_runtime`、`DEF_forcing_namelist`、`DEF_dir_forcing`、`DEF_forcing%*`
模型结构与方案	地类方案、土壤水力、过程开关、参数覆盖、参数卡片
检查与运行	`DEF_simulation_time%timestep`、输出/重启频率、历史变量、提交阶段、指纹开关
运行/配置（高级）	并行、调试、诊断、数据同化等

19.2 字段表怎么查

- 完整字段表：附录 A（按页面分组，含类型、默认值、可选值、依赖宏、派生标记与说明）；
- 参数库：附录 B（按分类，含参数 ID、原始键、默认值、作用域、可见性、源码位置）。

常用字段速查：

字段	含义	备注
`DEF_CASE_NAME`	算例名	字母/数字/-/_，不能有路径分隔符
`DEF_simulation_time%start_year/month/day/sec`	模拟开始	结束必须晚于开始
`DEF_simulation_time%timestep`	时间步长（秒）	单点常用 1800/3600
`DEF_HIST_FREQ`	历史输出间隔	支持步/时/日/月/年
`DEF_WRST_FREQ`	重启文件间隔	用于续跑
`SITE_fsitedata`	站点文件路径	站点算例必需
`DEF_dir_forcing` / `DEF_forcing%dataset`	强迫场目录/数据集标识	标识必须是引擎实际认识的集合（CMFD/ERA5/CMIP6/POINT 等）
`DEF_dir_rawdata`	地表数据/原始数据目录	空间算例或重建地表时需要
`DEF_dir_runtime`	Runtime 数据目录	部分过程/资产需要
`DEF_USE_*`	过程开关	是否可用取决于内核编译宏与算例尺度

21. 文件与目录布局

```
# 项目
<project>/
├── <case>/
│   ├── case.nml                # 算例配置
│   ├── forcing.nml             # 强迫场配置
│   ├── site.nc / mesh.nc       # 站点或网格输入
│   ├── model-binding.json      # 绑定的模型版本
│   ├── .colm-desktop.json      # 算例界面配置（引擎、运行位置等）
│   ├── runs/<操作号>/          # 模型版本运行的物化副本与结果
│   └── out/                    # 直接运行的默认输出
└── ...

# 应用数据目录（macOS 示例）
~/Library/Application Support/CoLM-Desktop/
├── models/                     # 模型注册表：versions/、sources/、drafts/、runs/
├── history.sqlite3             # 历史台账
├── openbench/                  # OpenBench 私有环境（可选）
└── live-runs/                  # 运行中/残留标记

~/Library/Application Support/edu.sysu.colm.desktop/
├── remote.json                 # 远程服务器配置
├── assistant.json / assistant-keys.json  # 助手设置与 Key
├── examples/                   # 内置示例（Sitedata/Forcing/Observation/Runtime）
└── recent.json                 # 最近项目与界面偏好
```

不同平台目录不同：Windows 用 `%APPDATA%\CoLM-Desktop` 等，Linux 用 `~/.local/share/CoLM-Desktop`；桌面端会自动解析。命令行默认模型根目录与历史数据库跟随同一应用数据目录。

22. 故障排查

现象	可能原因	处理
启动后看不到项目/算例	项目目录选择不对	首页"打开项目 → 选择目录"，或检查 `recent.json`
新建模拟提示无权/路径无效	目标目录不存在/不可写	换目录；检查磁盘权限
运行时缺 `site.nc/forcing.nml`	输入路径被移动	第 3 步重新选择；或修复强迫场配置引用
`dataset` 被拒绝	填了模板名而非引擎标识	用 `CMFD/ERA5/CMIP6/POINT` 等实际标识；见 6.3
Fortran 引擎提示编译配置缺失	本机/服务器没有匹配内核	资源页构建内核（本机）或 `remote-kernel`（服务器）
远程运行报 "no server path"	外部目录未 map	资源→服务器→目录映射；目录不会自动上传
远程运行报 "host key unknown/changed"	首次连接或主机密钥变化	终端先 `ssh <别名>` 确认并接受密钥
远程 Fortran 内核构建失败	环境脚本没载入 gfortran/MPI/NetCDF-Fortran	检查 `--env-script`；登录节点可编译，计算节点可能缺开发包
结果页没有数据	没取回结果或变量/月份裁剪为空	运行进度里"取回结果"；检查分析范围
历史导出少了站点	用了折叠视图	历史导出默认一行一操作；批量运行会展开
助手无法发送	未登记 API Key/服务不可达	助手设置填地址+Key；检查网络

现象	可能原因	处理
助手提示"重配失败"	保存成功但启动校验失败（如外发确认缺失）	按提示补全外发确认/Key；设置已保存，下次发送重试
空间算例第 2 步字段不可改	网格创建后固定	新建模拟；助手也不能绕过（契约只读）
模型版本被清理后算例报错	旧官方修订被自动/手动清理	版本对话框重新选一个可用修订并迁移

更多开发/构建问题见仓库 `docs/implementation-verification.md`、`docs/design.md` 与 GitHub Issues。

23. 附录

- 附录 A：配置字段总表（964 项）
- 附录 B：参数库索引（1369 项）
- 附录 C：CLI 命令参考（含完整 `--help`）

本手册的示例与证据

示例	状态	证据
单站 CN-Cng 本机 Rust 运行（3 天）	实际运行完成	图 14/15；日志显示 mksrfdata/mkinidata/colm 全 ok；输出 `history/CN-Cng_hist_2008-01.nc`
2 站点批量运行	实际运行完成（2/2，0 失败）	图 16/17
Heihe 区域算例（latlon 0.25°，JRA3Q）	CLI 实跑 EXIT 0（96 步）	图 35—40；`tmp/manual/data/region-run.log`
Heihe 流域算例（huahai 边界→unstructured，0.5°）	GUI 实跑完成：113 空间单元/267 LCT patch、2084 单元流域/77 河系、96 步 EXIT ok	图 41—48
Heihe 流域算例（huangwei 边界→unstructured，0.5°）	CLI 实跑 EXIT 0：59 活跃单元/177 patch、96 步	`tmp/manual/data/huangwei-run.log`
CLI `scan`	实际执行	输出 AT-Neu/CN-Cng 等 4 个示例站点 JSON
CLI `site-new`	实际执行	生成 `CN-Demo_site.nc` 与 JSON 字段摘要
CLI `history-catalog`	实际执行	输出 48 步、变量目录 JSON
真实远程 Slurm（Rust/Fortran、pack、ranks=1）	已在 tianhe 实测	见 `docs/implementation-verification.md` 2026-11-07 验收章节

手册生成日期：2026-11-13；软件版本 0.3.0-beta.10。

附录 A：配置字段总表（964 项）

本表由 `xtask parameter-audit` 的 schema 字段表生成；类型与默认值使用 namelist 展示格式，覆盖桌面端当前 schema。默认值来自当前目录，并非冻结模型或运行算例的有效值。文件 列表示字段写进哪个 namelist（`nl_colm` = case.nml，`nl_colm_forcing` = forcing.nml，`nl_colm_history` = history.nml，其余为材料 / 同位素等扩展 namelist）。派生 为“是”的字段由程序覆盖，界面上应视为只读。依赖宏 表示该字段只有在对应编译宏开启的内核里才有意义。

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明

初始场

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
1	DEF_USE_CN_INIT	初始场	nl_colm	logical	.false.				
2	DEF_USE_SnowInit	初始场	nl_colm	logical	.false.				
3	DEF_USE_SoilInit	初始场	nl_colm	logical	.false.				
4	DEF_USE_WaterTableInit	初始场	nl_colm	logical	.false.				
5	DEF_file_SnowInit	初始场	nl_colm	character(len=256)	null				
6	DEF_file_SoilInit	初始场	nl_colm	character(len=256)	null				
7	DEF_file_WaterTable	初始场	nl_colm	character(len=256)	null				
8	DEF_file_cn_init	初始场	nl_colm	character(len=256)	null				

地表数据

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
9	DEF_USE_ESACCI	地表数据	``	logical	.false.			是	
10	DEF_USE_GLC30	地表数据	``	logical	.false.			是	
11	DEF_USE_IGBP	地表数据	``	logical	.false.			是	
12	DEF_USE_USGS	地表数据	``	logical	.false.			是	
13	DEF_FAST_PC	地表数据	nl_colm	logical	.true.				
14	DEF_HighResSoil	地表数据	nl_colm	logical	.true.				
15	DEF_HighResUrban_albedo	地表数据	nl_colm	character(len=256)	null				
16	DEF_HighResVeg	地表数据	nl_colm	logical	.true.				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
17	DEF_LAI_CHANGE_YEARLY	地表数据	nl_colm	logical	.true.				
18	DEF_LAI_END_YEAR	地表数据	nl_colm	integer	2020				
19	DEF_LAI_MONTHLY	地表数据	nl_colm	logical	.true.				
20	DEF_LAI_START_YEAR	地表数据	nl_colm	integer	2000				
21	DEF_LANDONLY	地表数据	nl_colm	logical	.true.				
22	DEF_LULCC_SCHEME	地表数据	nl_colm	integer	1	1, 2			
23	DEF_PC_CROP_SPLIT	地表数据	nl_colm	logical	.true.				
24	DEF_SOIL_REFL_SCHEME	地表数据	nl_colm	integer	2	1, 2			
25	DEF_SOLO_PFT	地表数据	nl_colm	logical	.false.				
26	DEF_SUBGRID_SCHEME	地表数据	nl_colm	character(len=256)	LCT				
27	DEF_Srfdata_CompressLevel	地表数据	nl_colm	integer	1				
28	DEF_USE_DOMINANT_PATCHTYPE	地表数据	nl_colm	logical	.false.				
29	DEF_USE_LCT	地表数据	nl_colm	logical	.true.				
30	DEF_USE_LULCC	地表数据	nl_colm	logical	.false.				
31	DEF_USE_PC	地表数据	nl_colm	logical	.false.				
32	DEF_USE_PFT	地表数据	nl_colm	logical	.false.				
33	DEF_USE_SOILPAR_UPS_FIT	地表数据	nl_colm	logical	.true.				
34	DEF_dir_existing_srfdata	地表数据	nl_colm	character(len=256)	path/to/landdata				
35	DEF_dir_rawdata	地表数据	nl_colm	character(len=256)	path/to/rawdata/				
36	DEF_rawdata_namelist	地表数据	nl_colm	character(len=256)	./rawdata/colm2024.nml				
37	USE_srfdata_from_3D_gridded_data	地表数据	nl_colm	logical	.false.				
38	USE_srfdata_from_larger_region	地表数据	nl_colm	logical	.false.				
39	USE_zip_for_aggregation	地表数据	nl_colm	logical	.true.				

城市

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
40	DEF_URBAN_BEM	城市	nl_colm	logical	.true.				
41	DEF_URBAN_LUCY	城市	nl_colm	logical	.true.				
42	DEF_URBAN_ONLY	城市	nl_colm	logical	.false.				
43	DEF_URBAN_RUN	城市	nl_colm	logical	.false.				
44	DEF_URBAN_TREE	城市	nl_colm	logical	.true.				
45	DEF_URBAN_WATER	城市	nl_colm	logical	.true.				
46	DEF_URBAN_type_scheme	城市	nl_colm	integer	2	1, 2			
47	DEF_USE_CANYON_HWR	城市	nl_colm	logical	.true.				
48	USE_SITE_urban_ecology	城市	nl_colm	logical	.true.				
49	USE_SITE_urban_geometry	城市	nl_colm	logical	.true.				
50	USE_SITE_urban_human	城市	nl_colm	logical	.true.				
51	USE_SITE_urban_radiation	城市	nl_colm	logical	.true.				
52	USE_SITE_urban_thermal	城市	nl_colm	logical	.false.				

强迫场

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
53	DEF_DS_HiresTopographyDataDir	强迫场	nl_colm	character(len=256)	null				
54	DEF_DS_LONGWAVE_LAPSE_RATE	强迫场	nl_colm	real	0.032				Glacier longwave lapse rate [W m−2 m−1]
55	DEF_DS_LONGWAVE_LIMIT	强迫场	nl_colm	real	0.5				Relative longwave correction limit [0,1]
56	DEF_DS_SHORTWAVE_LIMIT	强迫场	nl_colm	real	0.5				Full−mode relative shortwave correction limit [0,1]
57	DEF_DS_SHORTWAVE_SIMPLE_LIMIT	强迫场	nl_colm	real	0.2				Simple−mode relative shortwave correction limit [0,1]
58	DEF_DS_TEMP_LAPSE_RATE	强迫场	nl_colm	real	0.006				Temperature lapse rate [K m−1]
59	DEF_DS_longwave_adjust_scheme	强迫场	nl_colm	character(len=5)	II	I, II			

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
60	DEF_DS_precipitation_adjust_scheme	强迫场	nl_colm	character(len=5)	I	I, II, III			
61	DEF_Forcing_Interp_Method	强迫场	nl_colm	character(len=20)	arealweight	arealweight, bilinear			'arealweight' (default) or 'bilinear'
62	DEF_USE_CBL_HEIGHT	强迫场	nl_colm	logical	.false.				
63	DEF_USE_ClimForcing_for_Spinup	强迫场	nl_colm	logical	.false.				
64	DEF_USE_Forcing_Downscaling	强迫场	nl_colm	logical	.false.				
65	DEF_USE_Forcing_Downscaling_Simple	强迫场	nl_colm	logical	.false.				
66	DEF_dir_forcing	强迫场	nl_colm_forcing	character(len=256)	path/to/forcing/data				
67	DEF_forcing%CBL_dtime	强迫场	nl_colm_forcing	integer	21600				
68	DEF_forcing%CBL_fprefix	强迫场	nl_colm_forcing	character(len=256)	TPHWL6Hrly/clmforc.cruncep.V4.c2011.0.5d.TPQWL.				
69	DEF_forcing%CBL_offset	强迫场	nl_colm_forcing	integer	10800				
70	DEF_forcing%CBL_tintalgo	强迫场	nl_colm_forcing	character(len=256)	linear				
71	DEF_forcing%CBL_vname	强迫场	nl_colm_forcing	character(len=256)	blh				
72	DEF_forcing%HEIGHT_Q	强迫场	nl_colm_forcing	real	50.				
73	DEF_forcing%HEIGHT_T	强迫场	nl_colm_forcing	real	50.				
74	DEF_forcing%HEIGHT_V	强迫场	nl_colm_forcing	real	100.0				
75	DEF_forcing%HEIGHT_mode	强迫场	nl_colm_forcing	character(len=256)	absolute				
76	DEF_forcing%NVAR	强迫场	nl_colm_forcing	integer	8				variable number of forcing data

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
77	DEF_forcing%data2d	强迫场	nl_colm_forcing	logical	.true.				data in 2 dimension (lon, lat)
78	DEF_forcing%dataset	强迫场	nl_colm_forcing	character(len=256)	CRUNCEP	CPL7, CRUNCEPV4, CRUNCEPV7, GSWP2, GSWP3, POINT, PRINCETON, QIAN			
79	DEF_forcing%dim2d	强迫场	nl_colm_forcing	logical	.true.				lat/lon value in 2 dimension (lon, lat)
80	DEF_forcing%dtime	强迫场	nl_colm_forcing	integer	(/21600,21600,21600,21600,0,21600,21600,21600/)				
81	DEF_forcing%endmo	强迫场	nl_colm_forcing	integer	12				end month of forcing data
82	DEF_forcing%endyr	强迫场	nl_colm_forcing	integer	2003				end year of forcing data
83	DEF_forcing%fprefix	强迫场	nl_colm_forcing	character(len=256)	(/ 'TPHWL6Hrly/clmforc.cruncep.V4.c2011.0.5d.TPQWL.', 'TPHWL6Hrly/clmforc.cruncep.V4.c2011.0.5d.TPQWL.', 'TPHWL6Hrly/clmforc.cruncep.V4.c2011.0.5d.TPQWL.', 'Precip6Hrly/clmforc.cruncep.V4.c2011.0.5d.Prec.', 'NULL ', 'TPHWL6Hrly/clmforc.cruncep.V4.c2011.0.5d.TPQWL.', 'Solar6Hrly/clmforc.cruncep.V4.c2011.0.5d.Solr. ',				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
					'TPHWL6Hrly/clmforc.cruncep.V4.c2011.0.5d.TPQWL.' /)				
84	DEF_forcing%groupby	强迫场	nl_colm_forcing	character(len=256)	month	day, month, year			file grouped by year/month
85	DEF_forcing%has_missing_value	强迫场	nl_colm_forcing	logical	.false.				
86	DEF_forcing%hightdim	强迫场	nl_colm_forcing	logical	.false.				have "z" dimension
87	DEF_forcing%latname	强迫场	nl_colm_forcing	character(len=256)	LATIXY				dimension name of latitude
88	DEF_forcing%leapyear	强迫场	nl_colm_forcing	logical	.false.				leapyear calendar
89	DEF_forcing%lonname	强迫场	nl_colm_forcing	character(len=256)	LONGXY				dimension name of longitude
90	DEF_forcing%missing_value_name	强迫场	nl_colm_forcing	character(len=256)	missing_value				
91	DEF_forcing%nlands	强迫场	nl_colm_forcing	integer	1				land grid number in 1d
92	DEF_forcing%offset	强迫场	nl_colm_forcing	integer	(/10800,10800,10800,10800,0,10800,0,10800/)				
93	DEF_forcing%regbnd	强迫场	nl_colm_forcing	real	(/-90.0, 90.0, -180.0, 180.0/)				
94	DEF_forcing%regional	强迫场	nl_colm_forcing	logical	.false.				
95	DEF_forcing%solarin_all_band	强迫场	nl_colm_forcing	logical	.true.				
96	DEF_forcing%startmo	强迫场	nl_colm_forcing	integer	1				start month of forcing data
97	DEF_forcing%startyr	强迫场	nl_colm_forcing	integer	2000				start year of forcing data
98	DEF_forcing%timelog	强迫场	nl_colm_forc	character(len=256)	(/ 'instant','instant				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
			ing		','instant','forward', 'NULL','instant','forward','forward' /)				
99	DEF_forcing%tintalgo	强迫场	nl_colm_forcing	character(len=256)	(/ 'linear','linear','linear','nearest', 'NULL','linear','coszen','linear '/)				
100	DEF_forcing%vname	强迫场	nl_colm_forcing	character(len=256)	(/ 'TBOT','QBOT','PSRF','PRECTmms', 'NULL','WIND','FSDS','FLDS '/)				

数据同化

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
101	DEF_DA_ENS_NUM	数据同化	nl_colm	integer	20		DataAssimilation		
102	DEF_DA_ENS_SM	数据同化	nl_colm	logical	.false.		DataAssimilation		
103	DEF_DA_RTM_diel	数据同化	nl_colm	integer	0	0, 1, 2, 3	DataAssimilation		
104	DEF_DA_RTM_rough	数据同化	nl_colm	integer	0	0, 1, 2, 3	DataAssimilation		
105	DEF_DA_SM	数据同化	nl_colm	logical	.false.		DataAssimilation		
106	DEF_DA_SM_FY	数据同化	nl_colm	logical	.false.		DataAssimilation		
107	DEF_DA_SM_SMAP	数据同化	nl_colm	logical	.false.		DataAssimilation		
108	DEF_DA_SM_SYNOP	数据同化	nl_colm	logical	.false.		DataAssimilation		
109	DEF_DA_TWS	数据同化	nl_colm	logical	.false.		DataAssimilation		
110	DEF_DA_TWS_GRACE	数据同化	nl_colm	logical	.false.				
111	DEF_DA_obsdir	数据同化	nl_colm	character(len=256)	null		DataAssimilation		

文件与目录

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
112	DEF_dir_landdata	文件与目录	``	character(len=256)	path/to/landdata			是	
113	DEF_dir_runtime	文件与目录	nl_colm	character(len=256)	path/to/runtime/				
114	DEF_forcing_namelist	文件与目录	nl_colm	character(len=256)	null				

时间与预热

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
115	DEF_simulation_time%end_day	时间与预热	nl_colm	integer	1				
116	DEF_simulation_time%end_month	时间与预热	nl_colm	integer	1				
117	DEF_simulation_time%end_sec	时间与预热	nl_colm	integer	0				
118	DEF_simulation_time%end_year	时间与预热	nl_colm	integer	2003				
119	DEF_simulation_time%greenwich	时间与预热	nl_colm	logical	.true.				
120	DEF_simulation_time%spinup_day	时间与预热	nl_colm	integer	1				
121	DEF_simulation_time%spinup_month	时间与预热	nl_colm	integer	1				
122	DEF_simulation_time%spinup_repeat	时间与预热	nl_colm	integer	1				
123	DEF_simulation_time%spinup_sec	时间与预热	nl_colm	integer	0				
124	DEF_simulation_time%spinup_year	时间与预热	nl_colm	integer	0				
125	DEF_simulation_time%start_day	时间与预热	nl_colm	integer	1				
126	DEF_simulation_time%start_month	时间与预热	nl_colm	integer	1				
127	DEF_simulation_time%start_sec	时间与预热	nl_colm	integer	0				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
128	DEF_simulation_time%start_year	时间与预热	nl_colm	integer	2000				
129	DEF_simulation_time%timestep	时间与预热	nl_colm	real	1800.				

材料

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
130	DEF_USE_NH4_LEACHING	材料	``	logical	.false.			是	
131	DEF_wetland_finundation_scheme	材料	``	integer	1	1, 2, 3, 4, 5, 6, 7		是	
132	DEF_file_GIEMS	材料	nl_colm	character(len=256)	null				
133	DEF_ISO_CANOPY_EQUILIBRATION	材料	nl_colm_isotope_parameter	real	0.0				
134	DEF_ISO_CG_RELHUM_MAX	材料	nl_colm_isotope_parameter	real	0.99				
135	DEF_ISO_ICE_SUPERSAT_SLOPE	材料	nl_colm_isotope_parameter	real	0.003				
136	DEF_ISO_INIT_DELTA_H2_18O	材料	nl_colm_isotope_parameter	real	−10.				
137	DEF_ISO_INIT_DELTA_HD0	材料	nl_colm_isotope_parameter	real	−70.				
138	DEF_ISO_KINETIC_SCHEME	材料	nl_colm_isotope_parameter	character(len=16)	MERLIVAT1978	CAPPA2003, MERLIVAT1978			
139	DEF_ISO_NSS_LEAF_PATH_LENGTH	材料	nl_colm_isotope_parameter	real	0.01				
140	DEF_ISO_NSS_LEAF_RB	材料	nl_colm_isot	real	100.				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
			ope_parameter						
141	DEF_ISO_NSS_LEAF_WATER_PER_LAI	材料	nl_colm_isotope_parameter	real	0.12				
142	DEF_ISO_OPEN_WATER_KINETIC	材料	nl_colm_isotope_parameter	character(len=16)	MJ79	EXPONENT, MJ79			
143	DEF_ISO_REF_RATIO_H2_18O	材料	nl_colm_isotope_parameter	real	2.0052e-3				
144	DEF_ISO_REF_RATIO_HDO	材料	nl_colm_isotope_parameter	real	1.5576e-4				
145	DEF_ISO_SNOWMELT_EQUILIBRATION	材料	nl_colm_isotope_parameter	real	0.0				
146	DEF_ISO_SOIL_DIFFUSION	材料	nl_colm_isotope_parameter	logical	.true.				
147	DEF_ISO_SOIL_KINETIC	材料	nl_colm_isotope_parameter	character(len=16)	RESISTANCE	EXPONENT, RESISTANCE			
148	DEF_ISO_SOIL_VAPOR_DIFFUSION	材料	nl_colm_isotope_parameter	logical	.true.				
149	DEF_ISO_SUBL_SKIN_MM	材料	nl_colm_isotope_parameter	real	5.0				
150	DEF_ISO_USE_FRACTIONATION	材料	nl_colm_isotope_parameter	logical	.false.				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
151	DEF_ISOTOPE_PARAM_FILE	材料	nl_colm_material	character(len=256)	null				
152	DEF_ISOTOPE_SPECIES	材料	nl_colm_material	character(len=256)	H2_18O,HDO				
153	DEF_METHANE_PARAM_FILE	材料	nl_colm_material	character(len=256)	null				
154	DEF_MT_AQUIFER_MIXING_WATER_MM	材料	nl_colm_material	real	-1.				
155	DEF_MT_BALANCE_ABORT_NBAD	材料	nl_colm_material	integer	0				
156	DEF_MT_FORCING_MEMORY_PRECIP	材料	nl_colm_material	real	2.e-5				
157	DEF_MT_FORCING_MIN_PRECIP	材料	nl_colm_material	real	5.e-6				
158	DEF_MT_LULCC_ABORT_NBAD	材料	nl_colm_material	integer	0				
159	DEF_MT_N2O_EF5_NH4	材料	nl_colm_material	real	-1.				
160	DEF_MT_N2O_EF5_NO3	材料	nl_colm_material	real	-1.				
161	DEF_MT_RESID_ABORT_NBAD	材料	nl_colm_material	integer	0				
162	DEF_MT_SOIL_INIT_FILE	材料	nl_colm_material	character(len=256)	null				
163	DEF_MT_SOIL_INIT_VARS	材料	nl_colm_material	character(len=256)	``				
164	DEF_MT_USE_SOIL_INIT	材料	nl_colm_material	logical	.false.				
165	DEF_NH4_LEACHING	材料	nl_colm_material	character(len=16)	auto				
166	DEF_SEDIMENT_PARAM_FILE	材料	nl_colm_material	character(len=256)	null				
167	DEF_SOLUTE_NAMES	材料	nl_colm_mate	character(len=256)	``				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
			rial						
168	DEF_SOLUTE_PARAM_FILES	材料	nl_colm_material	character(len=2048)	null				
169	DEF_USE_ISOTOPE	材料	nl_colm_material	logical	.false.				
170	DEF_USE_METHANE	材料	nl_colm_material	logical	.false.				
171	DEF_USE_SEDIMENT	材料	nl_colm_material	logical	.false.				

水热过程

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
172	DEF_CheckEquilibrium	水热过程	nl_colm	logical	.false.				
173	DEF_Interception_scheme	水热过程	nl_colm	integer	1	1, 2, 3, 4, 5, 6, 7, 8			
174	DEF_Optimize_Baseflow	水热过程	nl_colm	logical	.false.				
175	DEF_RSS_SCHEME	水热过程	nl_colm	integer	1	0, 1, 2, 3, 4, 5, 6			
176	DEF_RS_CROWN_STRUCTURE	水热过程	nl_colm	logical	.false.				
177	DEF_Runoff_SCHEME	水热过程	nl_colm	integer	3	0, 1, 2, 3			
178	DEF_SPLIT_SOILSNOW	水热过程	nl_colm	logical	.false.				
179	DEF_THERMAL_CONDUCTIVITY_SCHEME	水热过程	nl_colm	integer	4	1, 2, 3, 4, 5, 6, 7, 8			Options for soil thermal conductivity schemes
180	DEF_TOPMOD_method	水热过程	nl_colm	integer	0	0, 1, 2			
181	DEF_TUNING_CAPR	水热过程	nl_colm	real	0.34				First-layer-to-surface temperature tuning factor [-]
182	DEF_TUNING_CNFAC	水热过程	nl_colm	real	0.5				Crank-Nicolson weighting factor [0,1]
183	DEF_TUNING_PONDMX	水热过程	nl_colm	real	10.0				Maximum surface ponding depth [mm]
184	DEF_TUNING_SIMPLE_VIC_DS	水热过程	nl_colm	real	0.061				Simple VIC baseflow fraction [-]
185	DEF_TUNING_SIMPLE_VIC_WS	水热过程	nl_colm	real	0.646				Simple VIC baseflow threshold fraction [-]
186	DEF_TUNING_SMPMAX	水热过程	nl_colm	real	-1.5e5				Simple VIC wilting matric potential [mm]

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
187	DEF_TUNING_SMPMAX_HR	水热过程	nl_colm	real	-2.e2				Heterotrophic-respiration soil-potential upper limit [mm]
188	DEF_TUNING_SMPMIN	水热过程	nl_colm	real	-1.e8				Minimum allowed soil matric potential [mm]
189	DEF_TUNING_SMPMIN_HR	水热过程	nl_colm	real	-2.e5				Heterotrophic-respiration soil-potential lower limit [mm]
190	DEF_TUNING_SNOW_COVER_EXPONENT	水热过程	nl_colm	real	1.0				Snow-cover density exponent [-]
191	DEF_TUNING_SOIL_ICE_IMPEDANCE	水热过程	nl_colm	real	6.0				Frozen-soil hydraulic impedance exponent [-]
192	DEF_TUNING_SSI	水热过程	nl_colm	real	0.033				Irreducible snow-water saturation fraction [0,1]
193	DEF_TUNING_TOPMOD_DECAY	水热过程	nl_colm	real	2.0				TOPMODEL water-table decay coefficient [m-1]
194	DEF_TUNING_WETWATMAX	水热过程	nl_colm	real	200.0				Maximum wetland surface-water storage [mm]
195	DEF_TUNING_WIMP	水热过程	nl_colm	real	0.05				Impermeable-soil porosity threshold [0,1)
196	DEF_TUNING_ZLND	水热过程	nl_colm	real	0.01				Soil aerodynamic roughness length [m]
197	DEF_TUNING_ZSNO	水热过程	nl_colm	real	0.0024				Snow aerodynamic roughness length [m]
198	DEF_USE_BEDROCK	水热过程	nl_colm	logical	.false.				
199	DEF_USE_Campbell_SOIL_MODEL	水热过程	nl_colm	logical	.false.				
200	DEF_USE_Dynamic_Lake	水热过程	nl_colm	logical	.false.				Dynamic Lake model
201	DEF_USE_Dynamic_Wetland	水热过程	nl_colm	logical	.false.				Dynamic wetland model
202	DEF_USE_SUPERCOOL_WATER	水热过程	nl_colm	logical	.true.				supercooled soil water scheme, Niu & Yang (2006)
203	DEF_USE_VariablySaturatedFlow	水热过程	nl_colm	logical	.true.				
204	DEF_VIC_OPT	水热过程	nl_colm	logical	.false.				
205	DEF_file_VIC_OPT	水热过程	nl_colm	character(len=256)	null				
206	DEF_file_VIC_para	水热过程	nl_colm	character(len=256)	null				
207	DEF_precip_phase_discrimination_scheme	水热过程	nl_colm	character(len=5)	I	I, II, III			

河道与水库

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
208	DEF_CaMa_FloodFeedback	河道与水库	nl_colm	logical	.false.		CaMa_Flood		
209	DEF_CaMa_Namelist	河道与水库	nl_colm	character(len=256)	null		CaMa_Flood		
210	DEF_CaMa_Restart_file	河道与水库	nl_colm	character(len=256)	null		CaMa_Flood		
211	DEF_CaMa_StrictDomain	河道与水库	nl_colm	logical	.false.		CaMa_Flood		
212	DEF_ElementNeighbour_file	河道与水库	nl_colm	character(len=256)	null				
213	DEF_GRIDBASED_ROUTING_MAX_DT	河道与水库	nl_colm	real	3600.		GridRiver LakeFlow		
214	DEF_GRIDBASED_ROUTING_MOMENTUM_DT_LIMIT	河道与水库	nl_colm	logical	.false.		GridRiver LakeFlow		
215	DEF_GRIDBASED_ROUTING_STORAGE_DT_LIMIT	河道与水库	nl_colm	logical	.true.		GridRiver LakeFlow		
216	DEF_GridRiverLake_FloodFeedback	河道与水库	nl_colm	logical	.false.				
217	DEF_GridRiverLake_FloodInfiltMax	河道与水库	nl_colm	real	5.				
218	DEF_ReservoirPara_file	河道与水库	nl_colm	character(len=256)	null		GridRiver LakeFlow		
219	DEF_Reservoir_Method	河道与水库	nl_colm	integer	0	0, 1, 2			
220	DEF_USE_BIFURCATION	河道与水库	nl_colm	logical	.false.				
221	DEF_USE_EstimatedRiverDepth	河道与水库	nl_colm	logical	.true.		CatchLateralFlow		
222	DEF_USE_GridRiverLakeFlow	河道与水库	nl_colm	logical	.true.				
223	DEF_USE_LEVEE	河道与水库	nl_colm	logical	.false.		GridRiver LakeFlow		
224	DEF_UnitCatchment_file	河道与水库	nl_colm	character(len=256)	null	, NONE, NULL			
225	DEF_UnitCatchment_regional	河道与水库	nl_colm	logical	.false.				

生态与生地化

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
226	DEF_USE_CROP	生态与生地化	``	logical	.false.			是	
227	DEF_Aerosol_Clim	生态与生地化	nl_colm	logical	.false.				
228	DEF_Aerosol_Readin	生态与生地化	nl_colm	logical	.true.				
229	DEF_BALL_BERRY_BINTER	生态与生地化	nl_colm	real	-1.0				Ball–Berry intercept; valid override >= 0
230	DEF_BALL_BERRY_GRADM	生态与生地化	nl_colm	real	-1.0				Ball–Berry slope; valid override > 1.6
231	DEF_FERT_SOURCE	生态与生地化	nl_colm	integer	1	1, 2	CROP		Fertilisation data source
232	DEF_IRRIGATION_ALLOCATION	生态与生地化	nl_colm	integer	1	1, 2, 3	CROP		irrigation allocated method
233	DEF_LC_BETA	生态与生地化	nl_colm	real	-1.e36				
234	DEF_LC_BINTER	生态与生地化	nl_colm	real	-1.e36				
235	DEF_LC_C3C4	生态与生地化	nl_colm	integer	-1				
236	DEF_LC_CHIL	生态与生地化	nl_colm	real	-1.e36				
237	DEF_LC_CK	生态与生地化	nl_colm	real	-1.e36				
238	DEF_LC_D50	生态与生地化	nl_colm	real	-1.e36				
239	DEF_LC_DISPLAR	生态与生地化	nl_colm	real	-1.e36				
240	DEF_LC_EFFCON	生态与生地化	nl_colm	real	-1.e36				
241	DEF_LC_EXTKN	生态与生地化	nl_colm	real	-1.e36				
242	DEF_LC_FVEG0	生态与生地化	nl_colm	real	-1.e36				
243	DEF_LC_G0	生态与生地化	nl_colm	real	-1.e36				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
244	DEF_LC_G1	生态与生地化	nl_colm	real	-1.e36				
245	DEF_LC_GRADM	生态与生地化	nl_colm	real	-1.e36				
246	DEF_LC_HBOT0	生态与生地化	nl_colm	real	-1.e36				
247	DEF_LC_HHTI	生态与生地化	nl_colm	real	-1.e36				
248	DEF_LC_HLTI	生态与生地化	nl_colm	real	-1.e36				
249	DEF_LC_HTOP0	生态与生地化	nl_colm	real	-1.e36				
250	DEF_LC_KMAX_ROOT	生态与生地化	nl_colm	real	-1.e36				
251	DEF_LC_KMAX_SHA	生态与生地化	nl_colm	real	-1.e36				
252	DEF_LC_KMAX_SUN	生态与生地化	nl_colm	real	-1.e36				
253	DEF_LC_KMAX_XYL	生态与生地化	nl_colm	real	-1.e36				
254	DEF_LC_LAMBDA	生态与生地化	nl_colm	real	-1.e36				
255	DEF_LC_PSI50_ROOT	生态与生地化	nl_colm	real	-1.e36				
256	DEF_LC_PSI50_SHA	生态与生地化	nl_colm	real	-1.e36				
257	DEF_LC_PSI50_SUN	生态与生地化	nl_colm	real	-1.e36				
258	DEF_LC_PSI50_XYL	生态与生地化	nl_colm	real	-1.e36				
259	DEF_LC_RESPCP	生态与生地化	nl_colm	real	-1.e36				
260	DEF_LC_RHOL_NIR	生态与生地化	nl_colm	real	-1.e36				
261	DEF_LC_RHOL_VIS	生态与生地化	nl_colm	real	-1.e36				
262	DEF_LC_RHOS_NIR	生态与生地化	nl_colm	real	-1.e36				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
263	DEF_LC_RHOS_VIS	生态与生地化	nl_colm	real	-1.e36				
264	DEF_LC_SAI0	生态与生地化	nl_colm	real	-1.e36				
265	DEF_LC_SHTI	生态与生地化	nl_colm	real	-1.e36				
266	DEF_LC_SLTI	生态与生地化	nl_colm	real	-1.e36				
267	DEF_LC_SQRTDI	生态与生地化	nl_colm	real	-1.e36				
268	DEF_LC_TAUL_NIR	生态与生地化	nl_colm	real	-1.e36				
269	DEF_LC_TAUL_VIS	生态与生地化	nl_colm	real	-1.e36				
270	DEF_LC_TAUS_NIR	生态与生地化	nl_colm	real	-1.e36				
271	DEF_LC_TAUS_VIS	生态与生地化	nl_colm	real	-1.e36				
272	DEF_LC_TRDA	生态与生地化	nl_colm	real	-1.e36				
273	DEF_LC_TRDM	生态与生地化	nl_colm	real	-1.e36				
274	DEF_LC_TROP	生态与生地化	nl_colm	real	-1.e36				
275	DEF_LC_VMAX25	生态与生地化	nl_colm	real	-1.e36				
276	DEF_LC_YEAR	生态与生地化	nl_colm	integer	2005				
277	DEF_LC_Z0MR	生态与生地化	nl_colm	real	-1.e36				
278	DEF_MEDLYN_G0	生态与生地化	nl_colm	real	-1.0				Medlyn intercept; valid override >= 0
279	DEF_MEDLYN_G1	生态与生地化	nl_colm	real	-1.0				Medlyn slope; valid override >= 0
280	DEF_NDEP_FREQUENCY	生态与生地化	nl_colm	integer	1	1, 2			
281	DEF_OZONE_KO3	生态与生地化	nl_colm	real	1.51				Ozone stomatal–resistance coefficient [–]

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
282	DEF_PH_CROOT_LATERAL_LENGTH	生态与生地化	nl_colm	real	0.25				Coarse-root lateral length [m]
283	DEF_PH_FROOT_CARBON	生态与生地化	nl_colm	real	288.392056287006				Fine-root carbon stock [g C m-2]
284	DEF_PH_FROOT_LEAF	生态与生地化	nl_colm	real	1.5				Fine-root to leaf-area allocation factor [-]
285	DEF_PH_KRMAX	生态与生地化	nl_colm	real	3.981071705534969e-009				Maximum radial-root conductance coefficient
286	DEF_PH_K_AXS	生态与生地化	nl_colm	real	2.0e-1				Root axial hydraulic-conductivity coefficient
287	DEF_PH_ROOT_DENSITY	生态与生地化	nl_colm	real	310000.				Root tissue density [g biomass m-3]
288	DEF_PH_ROOT_RADIUS	生态与生地化	nl_colm	real	2.9e-4				Fine-root radius [m]
289	DEF_PROSPECT	生态与生地化	nl_colm	logical	.false.				
290	DEF_RSTFAC	生态与生地化	nl_colm	integer	1	1, 2			root resistance factors option
291	DEF_SSP	生态与生地化	nl_colm	character(len=256)	off	126, 245, 370, 585, off			CMIP6 future CO2 path; off holds the last observed value.
292	DEF_TUNING_CROP_PLANTING_DAY	生态与生地化	nl_colm	real	0.0		CROP		Single-point planting day; 0 reads runtime data [day of year]
293	DEF_TUNING_CSOILC	生态与生地化	nl_colm	real	0.004				Soil drag coefficient below canopy [-]
294	DEF_TUNING_DEWMX	生态与生地化	nl_colm	real	0.1				Maximum canopy dew storage [mm]
295	DEF_TUNING_IRRIGATION_DURATION_SEC	生态与生地化	nl_colm	real	14400.0		CROP		Irrigation application duration [s]
296	DEF_TUNING_IRRIGATION_MAX_CPHASE	生态与生地化	nl_colm	real	4.0		CROP		Exclusive last irrigated crop phase [-]
297	DEF_TUNING_IRRIGATION_MAX_DEPTH	生态与生地化	nl_colm	real	1.0		CROP		Maximum irrigated soil depth [m]
298	DEF_TUNING_IRRIGATION_MIN_CPHASE	生态与生地化	nl_colm	real	1.0		CROP		First irrigated crop phase [-]
299	DEF_TUNING_IRRIGATION_PONDMX	生态与生地化	nl_colm	real	100.0				Paddy maximum ponding depth [mm]
300	DEF_TUNING_IRRIGATION_START_SEC	生态与生地化	nl_colm	real	21600.0		CROP		Local irrigation start second [s]

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
301	DEF_TUNING_IRRIGATION_SUPPLY_FRACTION	生态与生地化	nl_colm	real	1.0		CROP		Irrigation deficit supply fraction [–]
302	DEF_TUNING_IRRIGATION_THRESHOLD_FRACTION	生态与生地化	nl_colm	real	1.0		CROP		Irrigation trigger fraction [–]
303	DEF_TUNING_TRSMX0	生态与生地化	nl_colm	real	2.e-4				Maximum transpiration rate [mm s-1]
304	DEF_USE_BGC	生态与生地化	nl_colm	logical	.false.				
305	DEF_USE_CNSOYFIXN	生态与生地化	nl_colm	logical	.true.				Soy nitrogen fixation
306	DEF_USE_DiagMatrix	生态与生地化	nl_colm	logical	.false.				
307	DEF_USE_FERT	生态与生地化	nl_colm	logical	.true.				Fertilisation on crop
308	DEF_USE_FIRE	生态与生地化	nl_colm	logical	.false.				Fire MODULE
309	DEF_USE_IRRIGATION	生态与生地化	nl_colm	logical	.false.				use irrigation
310	DEF_USE_LAIFEEDBACK	生态与生地化	nl_colm	logical	.false.				
311	DEF_USE_MEDLYNST	生态与生地化	nl_colm	logical	.false.				Medlyn stomata model
312	DEF_USE_NITRIF	生态与生地化	nl_colm	logical	.true.				Nitrification and denitrification switch
313	DEF_USE_NOSTRESSNITROGEN	生态与生地化	nl_colm	logical	.false.				photosynthesis stress option
314	DEF_USE_OZONEDATA	生态与生地化	nl_colm	logical	.false.				
315	DEF_USE_OZONESTRESS	生态与生地化	nl_colm	logical	.false.				
316	DEF_USE_PLANTHYDRAULICS	生态与生地化	nl_colm	logical	.true.				Plant Hydraulics
317	DEF_USE_PN	生态与生地化	nl_colm	logical	.false.				Punctuated nitrogen addition Spin up
318	DEF_USE_SASU	生态与生地化	nl_colm	logical	.false.				Semi-Analytic-Spin-Up
319	DEF_USE_SNICAR	生态与生地化	nl_colm	logical	.false.				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
320	DEF_USE_WUEST	生态与生地化	nl_colm	logical	.true.				WUE stomata model
321	DEF_VEG_SNOW	生态与生地化	nl_colm	logical	.true.				
322	DEF_WUE_LAMBDA	生态与生地化	nl_colm	real	-1.0				WUE marginal water cost; valid override > 0
323	DEF_file_Ozone	生态与生地化	nl_colm	character(len=256)	null				
324	DEF_file_snowaging	生态与生地化	nl_colm	character(len=256)	null				
325	DEF_file_snowoptics	生态与生地化	nl_colm	character(len=256)	null				

站点

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
326	SITE_fsitedata	站点	nl_colm	character(len=256)	null				
327	SITE_landtype	站点	nl_colm	integer	-1				
328	SITE_lat_location	站点	nl_colm	real	-1.e36				
329	SITE_lon_location	站点	nl_colm	real	-1.e36				
330	USE_SITE_ForcingReadAhead	站点	nl_colm	logical	.true.				
331	USE_SITE_LAI	站点	nl_colm	logical	.true.				
332	USE_SITE_dbedrock	站点	nl_colm	logical	.true.				
333	USE_SITE_htop	站点	nl_colm	logical	.true.				
334	USE_SITE_lakedepth	站点	nl_colm	logical	.true.				
335	USE_SITE_landtype	站点	nl_colm	logical	.false.				
336	USE_SITE_pctcrop	站点	nl_colm	logical	.true.				
337	USE_SITE_pctpfts	站点	nl_colm	logical	.true.				
338	USE_SITE_soilparameters	站点	nl_colm	logical	.true.				
339	USE_SITE_soilreflectance	站点	nl_colm	logical	.true.				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
340	USE_SITE_topography	站点	nl_colm	logical	.true.				

算例

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
341	DEF_CASE_NAME	算例	nl_colm	character(len=256)	CASENAME				

网格与并行

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
342	DEF_AverageElementSize	网格与并行	nl_colm	real	-1.				
343	DEF_BlockInfoFile	网格与并行	nl_colm	character(len=256)	null				
344	DEF_CatchmentMesh_data	网格与并行	nl_colm	character(len=256)	path/to/catchment/data				
345	DEF_GRIDBASED_lat_res	网格与并行	nl_colm	real	0.5		GRIDBASED		
346	DEF_GRIDBASED_lon_res	网格与并行	nl_colm	real	0.5		GRIDBASED		
347	DEF_PIO_groupsize	网格与并行	nl_colm	integer	12				
348	DEF_domain%edgee	网格与并行	nl_colm	real	180.0				
349	DEF_domain%edgen	网格与并行	nl_colm	real	90.0				
350	DEF_domain%edges	网格与并行	nl_colm	real	-90.0				
351	DEF_domain%edgew	网格与并行	nl_colm	real	-180.0				
352	DEF_file_mesh	网格与并行	nl_colm	character(len=256)	path/to/mesh/file				
353	DEF_file_mesh_filter	网格与并行	nl_colm	character(len=256)	path/to/mesh/filter				
354	DEF_nIO_eq_nBlock	网格与并行	nl_colm	logical	.false.		USEMPI		
355	DEF_nx_blocks	网格与并行	nl_colm	integer	72				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
356	DEF_ny_blocks	网格与并行	nl_colm	integer	36				

调试与诊断

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
357	DEF_USE_CoLMDEBUG	调试与诊断	nl_colm	logical	.false.				
358	DEF_USE_RangeCheck	调试与诊断	nl_colm	logical	.false.				
359	DEF_USE_SrfdataDiag	调试与诊断	nl_colm	logical	.false.				

输出与重启

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
360	DEF_dir_history	输出与重启	``	character(len=256)	path/to/history			是	
361	DEF_dir_restart	输出与重启	``	character(len=256)	path/to/restart			是	
362	DEF_HISTORY_IN_VECTOR	输出与重启	nl_colm	logical	.false.				
363	DEF_HIST_CompressLevel	输出与重启	nl_colm	integer	1				
364	DEF_HIST_FREQ	输出与重启	nl_colm	character(len=256)	none	DAILY, HOURLY, MONTHLY, TIMESTEP, YEARLY, none			write history file frequency: TIMESTEP/HOURLY/DAILY/MONTHLY/YEARLY
365	DEF_HIST_WriteBack	输出与重启	nl_colm	logical	.false.				
366	DEF_HIST_grid_as_forcing	输出与重启	nl_colm	logical	.false.				
367	DEF_HIST_grid_as_model_mesh	输出与重启	nl_colm	logical	.false.				
368	DEF_HIST_groupby	输出与重启	nl_colm	character(len=256)	MONTH	DAY, MONTH, YEAR			history file in one file: DAY/MONTH/YEAR
369	DEF_HIST_lat_res	输出与重启	nl_colm	real	0.5				
370	DEF_HIST_lon_res	输出与重启	nl_colm	real	0.5				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
371	DEF_HIST_mode	输出与重启	nl_colm	character(len=256)	one	block, one			
372	DEF_HIST_vars_namelist	输出与重启	nl_colm	character(len=256)	null				
373	DEF_HIST_vars_out_default	输出与重启	nl_colm	logical	.true.				
374	DEF_Output_2mWMO	输出与重启	nl_colm	logical	.false.				2m WMO temperature
375	DEF_REST_CompressLevel	输出与重启	nl_colm	integer	1				
376	DEF_WRST_FREQ	输出与重启	nl_colm	character(len=256)	none	DAILY, HOURLY, MONTHLY, TIMESTEP, YEARLY, none			write restart file frequency: TIMESTEP/HOURLY/DAILY/MONTHLY/YEARLY
377	DEF_dir_output	输出与重启	nl_colm	character(len=256)	path/to/output/data		CatchLateralFlow		
378	USE_SITE_HistWriteBack	输出与重启	nl_colm	logical	.true.				

输出变量

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
379	DEF_hist_vars%BD_all	输出变量	nl_colm_history	logical	.true.				
380	DEF_hist_vars%CONC_O2_UNSAT	输出变量	nl_colm_history	logical	.false.				
381	DEF_hist_vars%DA_t_brt_fy3d	输出变量	nl_colm_history	logical	.true.				
382	DEF_hist_vars%DA_t_brt_fy3d_a	输出变量	nl_colm_history	logical	.true.				
383	DEF_hist_vars%DA_t_brt_fy3d_ens	输出变量	nl_colm_history	logical	.true.				
384	DEF_hist_vars%DA_t_brt_fy3d_ens_std	输出变量	nl_colm_history	logical	.true.				
385	DEF_hist_vars%DA_t_brt_smap	输出变量	nl_colm_history	logical	.true.				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
386	DEF_hist_vars%DA_t_brt_smap_a	输出变量	nl_colm_history	logical	.true.				
387	DEF_hist_vars%DA_t_brt_smap_ens	输出变量	nl_colm_history	logical	.true.				
388	DEF_hist_vars%DA_t_brt_smap_ens_std	输出变量	nl_colm_history	logical	.true.				
389	DEF_hist_vars%DA_t_soisno_5cm	输出变量	nl_colm_history	logical	.true.				
390	DEF_hist_vars%DA_t_soisno_5cm_a	输出变量	nl_colm_history	logical	.true.				
391	DEF_hist_vars%DA_t_soisno_5cm_ens_std	输出变量	nl_colm_history	logical	.true.				
392	DEF_hist_vars%DA_t_soisno_ens	输出变量	nl_colm_history	logical	.true.				
393	DEF_hist_vars%DA_wliq_h2osoi_5cm	输出变量	nl_colm_history	logical	.true.				
394	DEF_hist_vars%DA_wliq_h2osoi_5cm_a	输出变量	nl_colm_history	logical	.true.				
395	DEF_hist_vars%DA_wliq_soisno_5cm_ens_std	输出变量	nl_colm_history	logical	.true.				
396	DEF_hist_vars%DA_wliq_soisno_ens	输出变量	nl_colm_history	logical	.true.				
397	DEF_hist_vars%O2_DECOMP_DEPTH_UNSAT	输出变量	nl_colm_history	logical	.false.				
398	DEF_hist_vars%OM_density	输出变量	nl_colm_history	logical	.true.				
399	DEF_hist_vars%abm	输出变量	nl_colm_history	logical	.false.				
400	DEF_hist_vars%alb	输出变量	nl_colm_history	logical	.true.				
401	DEF_hist_vars%alb_hires	输出变量	nl_colm_history	logical	.true.				
402	DEF_hist_vars%ar	输出变量	nl_colm_hist	logical	.true.				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
			ory						
403	DEF_hist_vars%assim	输出变量	nl_colm_history	logical	.true.				
404	DEF_hist_vars%assimsha	输出变量	nl_colm_history	logical	.true.				
405	DEF_hist_vars%assimsun	输出变量	nl_colm_history	logical	.true.				
406	DEF_hist_vars%baf_crop	输出变量	nl_colm_history	logical	.true.				
407	DEF_hist_vars%baf_peatf	输出变量	nl_colm_history	logical	.true.				
408	DEF_hist_vars%bedout	输出变量	nl_colm_history	logical	.true.				
409	DEF_hist_vars%btran2	输出变量	nl_colm_history	logical	.true.				
410	DEF_hist_vars%col_fire_closs	输出变量	nl_colm_history	logical	.true.				
411	DEF_hist_vars%col_fire_nloss	输出变量	nl_colm_history	logical	.true.				
412	DEF_hist_vars%cphase	输出变量	nl_colm_history	logical	.true.				
413	DEF_hist_vars%croppprod1c	输出变量	nl_colm_history	logical	.true.				
414	DEF_hist_vars%croppprod1c_loss	输出变量	nl_colm_history	logical	.true.				
415	DEF_hist_vars%croppprodc_irrigated_cotton	输出变量	nl_colm_history	logical	.true.				
416	DEF_hist_vars%croppprodc_irrigated_rice	输出变量	nl_colm_history	logical	.true.				
417	DEF_hist_vars%croppprodc_irrigated_spwheat	输出变量	nl_colm_history	logical	.true.				
418	DEF_hist_vars%croppprodc_irrigated_sugarcane	输出变量	nl_colm_history	logical	.true.				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
419	DEF_hist_vars%croppro dc_irrigated_temp_cor n	输出变量	nl_col m_hist ory	logical	.true.				
420	DEF_hist_vars%croppro dc_irrigated_temp_soy bean	输出变量	nl_col m_hist ory	logical	.true.				
421	DEF_hist_vars%croppro dc_irrigated_trop_cor n	输出变量	nl_col m_hist ory	logical	.true.				
422	DEF_hist_vars%croppro dc_irrigated_trop_soy bean	输出变量	nl_col m_hist ory	logical	.true.				
423	DEF_hist_vars%croppro dc_irrigated_wtwheat	输出变量	nl_col m_hist ory	logical	.true.				
424	DEF_hist_vars%croppro dc_rainfed_cotton	输出变量	nl_col m_hist ory	logical	.true.				
425	DEF_hist_vars%croppro dc_rainfed_rice	输出变量	nl_col m_hist ory	logical	.true.				
426	DEF_hist_vars%croppro dc_rainfed_spwheat	输出变量	nl_col m_hist ory	logical	.true.				
427	DEF_hist_vars%croppro dc_rainfed_sugarcane	输出变量	nl_col m_hist ory	logical	.true.				
428	DEF_hist_vars%croppro dc_rainfed_temp_corn	输出变量	nl_col m_hist ory	logical	.true.				
429	DEF_hist_vars%croppro dc_rainfed_temp_soybe an	输出变量	nl_col m_hist ory	logical	.true.				
430	DEF_hist_vars%croppro dc_rainfed_trop_corn	输出变量	nl_col m_hist ory	logical	.true.				
431	DEF_hist_vars%croppro dc_rainfed_trop_soybe an	输出变量	nl_col m_hist ory	logical	.true.				
432	DEF_hist_vars%croppro dc_rainfed_wtwheat	输出变量	nl_col m_hist ory	logical	.true.				
433	DEF_hist_vars%croppro dc_unmanagedcrop	输出变量	nl_col m_hist ory	logical	.true.				
434	DEF_hist_vars%cropsee dc_deficit	输出变量	nl_col m_hist ory	logical	.true.				
435	DEF_hist_vars%cwdcCap _vr	输出变量	nl_col m_hist	logical	.false.				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
			ory						
436	DEF_hist_vars%cwdc_vr	输出变量	nl_colm_history	logical	.true.				
437	DEF_hist_vars%cwddecomp	输出变量	nl_colm_history	logical	.true.				
438	DEF_hist_vars%cwdnCap_vr	输出变量	nl_colm_history	logical	.false.				
439	DEF_hist_vars%cwdn_vr	输出变量	nl_colm_history	logical	.true.				
440	DEF_hist_vars%cwdprod	输出变量	nl_colm_history	logical	.true.				
441	DEF_hist_vars%deadcrootc	输出变量	nl_colm_history	logical	.true.				
442	DEF_hist_vars%deadcrootcCap	输出变量	nl_colm_history	logical	.false.				
443	DEF_hist_vars%deadcrootc_storage	输出变量	nl_colm_history	logical	.true.				
444	DEF_hist_vars%deadcrootc_storageCap	输出变量	nl_colm_history	logical	.false.				
445	DEF_hist_vars%deadcrootc_xfer	输出变量	nl_colm_history	logical	.true.				
446	DEF_hist_vars%deadcrootc_xferCap	输出变量	nl_colm_history	logical	.false.				
447	DEF_hist_vars%deadcrootn	输出变量	nl_colm_history	logical	.true.				
448	DEF_hist_vars%deadcrootnCap	输出变量	nl_colm_history	logical	.false.				
449	DEF_hist_vars%deadcrootn_storage	输出变量	nl_colm_history	logical	.true.				
450	DEF_hist_vars%deadcrootn_storageCap	输出变量	nl_colm_history	logical	.false.				
451	DEF_hist_vars%deadcrootn_xfer	输出变量	nl_colm_history	logical	.true.				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
452	DEF_hist_vars%deadcrotn_xferCap	输出变量	nl_colm_history	logical	.false.				
453	DEF_hist_vars%deadstemc	输出变量	nl_colm_history	logical	.true.				
454	DEF_hist_vars%deadstemcCap	输出变量	nl_colm_history	logical	.false.				
455	DEF_hist_vars%deadstemc_storage	输出变量	nl_colm_history	logical	.true.				
456	DEF_hist_vars%deadstemc_storageCap	输出变量	nl_colm_history	logical	.false.				
457	DEF_hist_vars%deadstemc_xfer	输出变量	nl_colm_history	logical	.true.				
458	DEF_hist_vars%deadstemc_xferCap	输出变量	nl_colm_history	logical	.false.				
459	DEF_hist_vars%deadstemn	输出变量	nl_colm_history	logical	.true.				
460	DEF_hist_vars%deadstemnCap	输出变量	nl_colm_history	logical	.false.				
461	DEF_hist_vars%deadstemn_storage	输出变量	nl_colm_history	logical	.true.				
462	DEF_hist_vars%deadstemn_storageCap	输出变量	nl_colm_history	logical	.false.				
463	DEF_hist_vars%deadstemn_xfer	输出变量	nl_colm_history	logical	.true.				
464	DEF_hist_vars%deadstemn_xferCap	输出变量	nl_colm_history	logical	.false.				
465	DEF_hist_vars%discharge	输出变量	nl_colm_history	logical	.true.				
466	DEF_hist_vars%downreg	输出变量	nl_colm_history	logical	.true.				
467	DEF_hist_vars%dz_lake	输出变量	nl_colm_history	logical	.true.				
468	DEF_hist_vars%emis	输出变量	nl_colm_hist	logical	.true.				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
			ory						
469	DEF_hist_vars%etr	输出变量	nl_colm_history	logical	.true.				
470	DEF_hist_vars%etrsha	输出变量	nl_colm_history	logical	.true.				
471	DEF_hist_vars%etrsun	输出变量	nl_colm_history	logical	.true.				
472	DEF_hist_vars%fach	输出变量	nl_colm_history	logical	.true.				
473	DEF_hist_vars%farea_burned	输出变量	nl_colm_history	logical	.true.				
474	DEF_hist_vars%fert_to_sminn	输出变量	nl_colm_history	logical	.true.				
475	DEF_hist_vars%fertnitro_corn	输出变量	nl_colm_history	logical	.true.				
476	DEF_hist_vars%fertnitro_cotton	输出变量	nl_colm_history	logical	.true.				
477	DEF_hist_vars%fertnitro_rice1	输出变量	nl_colm_history	logical	.true.				
478	DEF_hist_vars%fertnitro_rice2	输出变量	nl_colm_history	logical	.true.				
479	DEF_hist_vars%fertnitro_soybean	输出变量	nl_colm_history	logical	.true.				
480	DEF_hist_vars%fertnitro_sugarcane	输出变量	nl_colm_history	logical	.true.				
481	DEF_hist_vars%fertnitro_swheat	输出变量	nl_colm_history	logical	.true.				
482	DEF_hist_vars%fertnitro_wwheat	输出变量	nl_colm_history	logical	.true.				
483	DEF_hist_vars%fevpa	输出变量	nl_colm_history	logical	.true.				
484	DEF_hist_vars%fevpg	输出变量	nl_colm_history	logical	.true.				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
485	DEF_hist_vars%fevpl	输出变量	nl_colm_history	logical	.true.				
486	DEF_hist_vars%fgrnd	输出变量	nl_colm_history	logical	.true.				
487	DEF_hist_vars%fh	输出变量	nl_colm_history	logical	.true.				
488	DEF_hist_vars%fhac	输出变量	nl_colm_history	logical	.true.				
489	DEF_hist_vars%fhah	输出变量	nl_colm_history	logical	.true.				
490	DEF_hist_vars%fldarea	输出变量	nl_colm_history	logical	.true.				
491	DEF_hist_vars%floodarea	输出变量	nl_colm_history	logical	.true.				
492	DEF_hist_vars%floodfrc	输出变量	nl_colm_history	logical	.true.				
493	DEF_hist_vars%fm	输出变量	nl_colm_history	logical	.true.				
494	DEF_hist_vars%fm10m	输出变量	nl_colm_history	logical	.true.				
495	DEF_hist_vars%fpg	输出变量	nl_colm_history	logical	.true.				
496	DEF_hist_vars%fpi	输出变量	nl_colm_history	logical	.true.				
497	DEF_hist_vars%fq	输出变量	nl_colm_history	logical	.true.				
498	DEF_hist_vars%frcsat	输出变量	nl_colm_history	logical	.true.				
499	DEF_hist_vars%frootc	输出变量	nl_colm_history	logical	.true.				
500	DEF_hist_vars%frootcCap	输出变量	nl_colm_history	logical	.false.				
501	DEF_hist_vars%frootc_storage	输出变量	nl_colm_hist	logical	.true.				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
			ory						
502	DEF_hist_vars%frootc_storageCap	输出变量	nl_colm_history	logical	.false.				
503	DEF_hist_vars%frootc_xfer	输出变量	nl_colm_history	logical	.true.				
504	DEF_hist_vars%frootc_xferCap	输出变量	nl_colm_history	logical	.false.				
505	DEF_hist_vars%frootn	输出变量	nl_colm_history	logical	.true.				
506	DEF_hist_vars%frootnCap	输出变量	nl_colm_history	logical	.false.				
507	DEF_hist_vars%frootn_storage	输出变量	nl_colm_history	logical	.true.				
508	DEF_hist_vars%frootn_storageCap	输出变量	nl_colm_history	logical	.false.				
509	DEF_hist_vars%frootn_xfer	输出变量	nl_colm_history	logical	.true.				
510	DEF_hist_vars%frootn_xferCap	输出变量	nl_colm_history	logical	.false.				
511	DEF_hist_vars%fsen_gimp	输出变量	nl_colm_history	logical	.true.				
512	DEF_hist_vars%fsen_gper	输出变量	nl_colm_history	logical	.true.				
513	DEF_hist_vars%fsen_roof	输出变量	nl_colm_history	logical	.true.				
514	DEF_hist_vars%fsen_urbl	输出变量	nl_colm_history	logical	.true.				
515	DEF_hist_vars%fsen_wsha	输出变量	nl_colm_history	logical	.true.				
516	DEF_hist_vars%fsen_wsun	输出变量	nl_colm_history	logical	.true.				
517	DEF_hist_vars%fsena	输出变量	nl_colm_history	logical	.true.				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
518	DEF_hist_vars%fseng	输出变量	nl_colm_history	logical	.true.				
519	DEF_hist_vars%fsenl	输出变量	nl_colm_history	logical	.true.				
520	DEF_hist_vars%fsno	输出变量	nl_colm_history	logical	.true.				
521	DEF_hist_vars%fuelc	输出变量	nl_colm_history	logical	.true.				
522	DEF_hist_vars%fwst	输出变量	nl_colm_history	logical	.true.				
523	DEF_hist_vars%gddmaturity	输出变量	nl_colm_history	logical	.true.				
524	DEF_hist_vars%gddplant	输出变量	nl_colm_history	logical	.true.				
525	DEF_hist_vars%gdp	输出变量	nl_colm_history	logical	.false.				
526	DEF_hist_vars%gpp	输出变量	nl_colm_history	logical	.true.				
527	DEF_hist_vars%gpp_c3arcgrass	输出变量	nl_colm_history	logical	.false.				12
528	DEF_hist_vars%gpp_c3grass	输出变量	nl_colm_history	logical	.false.				13
529	DEF_hist_vars%gpp_c4grass	输出变量	nl_colm_history	logical	.false.				14
530	DEF_hist_vars%gpp_dbfboreal	输出变量	nl_colm_history	logical	.false.				8
531	DEF_hist_vars%gpp_dbftemp	输出变量	nl_colm_history	logical	.false.				7
532	DEF_hist_vars%gpp_dbftrop	输出变量	nl_colm_history	logical	.false.				6
533	DEF_hist_vars%gpp_dbsboreal	输出变量	nl_colm_history	logical	.false.				11
534	DEF_hist_vars%gpp_dbstemp	输出变量	nl_colm_hist	logical	.false.				10

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
			ory						
535	DEF_hist_vars%gpp_dnf boreal	输出变量	nl_col m_hist ory	logical	.false.				3
536	DEF_hist_vars%gpp_ebf temp	输出变量	nl_col m_hist ory	logical	.false.				5
537	DEF_hist_vars%gpp_ebf trop	输出变量	nl_col m_hist ory	logical	.false.				4
538	DEF_hist_vars%gpp_ebs temp	输出变量	nl_col m_hist ory	logical	.false.				9
539	DEF_hist_vars%gpp_enf boreal	输出变量	nl_col m_hist ory	logical	.false.				2
540	DEF_hist_vars%gpp_enf temp	输出变量	nl_col m_hist ory	logical	.false.				1
541	DEF_hist_vars%grainc	输出变量	nl_col m_hist ory	logical	.true.				
542	DEF_hist_vars%grainc_ storage	输出变量	nl_col m_hist ory	logical	.true.				
543	DEF_hist_vars%grainc_ to_cropprodc	输出变量	nl_col m_hist ory	logical	.true.				
544	DEF_hist_vars%grainc_ to_seed	输出变量	nl_col m_hist ory	logical	.true.				
545	DEF_hist_vars%grainc_ xfer	输出变量	nl_col m_hist ory	logical	.true.				
546	DEF_hist_vars%grainn	输出变量	nl_col m_hist ory	logical	.true.				
547	DEF_hist_vars%grainn_ storage	输出变量	nl_col m_hist ory	logical	.true.				
548	DEF_hist_vars%grainn_ xfer	输出变量	nl_col m_hist ory	logical	.true.				
549	DEF_hist_vars%green	输出变量	nl_col m_hist ory	logical	.true.				
550	DEF_hist_vars%groundw ater_demand	输出变量	nl_col m_hist ory	logical	.true.				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
551	DEF_hist_vars%groundwater_supply	输出变量	nl_colm_history	logical	.true.				
552	DEF_hist_vars%gssha	输出变量	nl_colm_history	logical	.true.				
553	DEF_hist_vars%gssun	输出变量	nl_colm_history	logical	.true.				
554	DEF_hist_vars%h2osoi	输出变量	nl_colm_history	logical	.true.				
555	DEF_hist_vars%hdm	输出变量	nl_colm_history	logical	.false.				
556	DEF_hist_vars%hr	输出变量	nl_colm_history	logical	.true.				
557	DEF_hist_vars%hui	输出变量	nl_colm_history	logical	.true.				
558	DEF_hist_vars%huiswheat	输出变量	nl_colm_history	logical	.true.				
559	DEF_hist_vars%irrig_method_corn	输出变量	nl_colm_history	logical	.true.				
560	DEF_hist_vars%irrig_method_cotton	输出变量	nl_colm_history	logical	.true.				
561	DEF_hist_vars%irrig_method_rice1	输出变量	nl_colm_history	logical	.true.				
562	DEF_hist_vars%irrig_method_rice2	输出变量	nl_colm_history	logical	.true.				
563	DEF_hist_vars%irrig_method_soybean	输出变量	nl_colm_history	logical	.true.				
564	DEF_hist_vars%irrig_method_sugarcane	输出变量	nl_colm_history	logical	.true.				
565	DEF_hist_vars%irrig_method_swheat	输出变量	nl_colm_history	logical	.true.				
566	DEF_hist_vars%irrig_method_wwheat	输出变量	nl_colm_history	logical	.true.				
567	DEF_hist_vars%lai	输出变量	nl_colm_hist	logical	.true.				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
			ory						
568	DEF_hist_vars%lai_c3arcgrass	输出变量	nl_colm_history	logical	.false.				12
569	DEF_hist_vars%lai_c3grass	输出变量	nl_colm_history	logical	.false.				13
570	DEF_hist_vars%lai_c4grass	输出变量	nl_colm_history	logical	.false.				14
571	DEF_hist_vars%lai_dbfboreal	输出变量	nl_colm_history	logical	.false.				8
572	DEF_hist_vars%lai_dbftemp	输出变量	nl_colm_history	logical	.false.				7
573	DEF_hist_vars%lai_dbftrop	输出变量	nl_colm_history	logical	.false.				6
574	DEF_hist_vars%lai_dbsboreal	输出变量	nl_colm_history	logical	.false.				11
575	DEF_hist_vars%lai_dbstemp	输出变量	nl_colm_history	logical	.false.				10
576	DEF_hist_vars%lai_dnfboreal	输出变量	nl_colm_history	logical	.false.				3
577	DEF_hist_vars%lai_ebftemp	输出变量	nl_colm_history	logical	.false.				5
578	DEF_hist_vars%lai_ebftrop	输出变量	nl_colm_history	logical	.false.				4
579	DEF_hist_vars%lai_ebstemp	输出变量	nl_colm_history	logical	.false.				9
580	DEF_hist_vars%lai_enfboreal	输出变量	nl_colm_history	logical	.false.				2
581	DEF_hist_vars%lai_enftemp	输出变量	nl_colm_history	logical	.false.				1
582	DEF_hist_vars%laisha	输出变量	nl_colm_history	logical	.true.				
583	DEF_hist_vars%laisun	输出变量	nl_colm_history	logical	.true.				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
584	DEF_hist_vars%lake_deficit	输出变量	nl_colm_history	logical	.true.				
585	DEF_hist_vars%lake_icefrac	输出变量	nl_colm_history	logical	.true.				
586	DEF_hist_vars%ldew	输出变量	nl_colm_history	logical	.true.				
587	DEF_hist_vars%leafc	输出变量	nl_colm_history	logical	.true.				
588	DEF_hist_vars%leafcCap	输出变量	nl_colm_history	logical	.false.				
589	DEF_hist_vars%leafc_c3arcgrass	输出变量	nl_colm_history	logical	.false.				12
590	DEF_hist_vars%leafc_c3grass	输出变量	nl_colm_history	logical	.false.				13
591	DEF_hist_vars%leafc_c4grass	输出变量	nl_colm_history	logical	.false.				14
592	DEF_hist_vars%leafc_dbfboreal	输出变量	nl_colm_history	logical	.false.				8
593	DEF_hist_vars%leafc_dbftemp	输出变量	nl_colm_history	logical	.false.				7
594	DEF_hist_vars%leafc_dbftrop	输出变量	nl_colm_history	logical	.false.				6
595	DEF_hist_vars%leafc_dbsboreal	输出变量	nl_colm_history	logical	.false.				11
596	DEF_hist_vars%leafc_dbstemp	输出变量	nl_colm_history	logical	.false.				10
597	DEF_hist_vars%leafc_dnfboreal	输出变量	nl_colm_history	logical	.false.				3
598	DEF_hist_vars%leafc_ebftemp	输出变量	nl_colm_history	logical	.false.				5
599	DEF_hist_vars%leafc_ebftrop	输出变量	nl_colm_history	logical	.false.				4
600	DEF_hist_vars%leafc_ebstemp	输出变量	nl_colm_hist	logical	.false.				9

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
			ory						
601	DEF_hist_vars%leafc_enfboreal	输出变量	nl_colm_history	logical	.false.				2
602	DEF_hist_vars%leafc_enftemp	输出变量	nl_colm_history	logical	.false.				1
603	DEF_hist_vars%leafc_storage	输出变量	nl_colm_history	logical	.true.				
604	DEF_hist_vars%leafc_storageCap	输出变量	nl_colm_history	logical	.false.				
605	DEF_hist_vars%leafc_xfer	输出变量	nl_colm_history	logical	.true.				
606	DEF_hist_vars%leafc_xferCap	输出变量	nl_colm_history	logical	.false.				
607	DEF_hist_vars%leafn	输出变量	nl_colm_history	logical	.true.				
608	DEF_hist_vars%leafnCap	输出变量	nl_colm_history	logical	.false.				
609	DEF_hist_vars%leafn_storage	输出变量	nl_colm_history	logical	.true.				
610	DEF_hist_vars%leafn_storageCap	输出变量	nl_colm_history	logical	.false.				
611	DEF_hist_vars%leafn_xfer	输出变量	nl_colm_history	logical	.true.				
612	DEF_hist_vars%leafn_xferCap	输出变量	nl_colm_history	logical	.false.				
613	DEF_hist_vars%lfevp_gimp	输出变量	nl_colm_history	logical	.true.				
614	DEF_hist_vars%lfevp_gper	输出变量	nl_colm_history	logical	.true.				
615	DEF_hist_vars%lfevp_roof	输出变量	nl_colm_history	logical	.true.				
616	DEF_hist_vars%lfevp_urbl	输出变量	nl_colm_history	logical	.true.				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
617	DEF_hist_vars%lfevpa	输出变量	nl_colm_history	logical	.true.				
618	DEF_hist_vars%litfire	输出变量	nl_colm_history	logical	.true.				
619	DEF_hist_vars%litr1cCap_vr	输出变量	nl_colm_history	logical	.false.				
620	DEF_hist_vars%litr1c_vr	输出变量	nl_colm_history	logical	.true.				
621	DEF_hist_vars%litr1nCap_vr	输出变量	nl_colm_history	logical	.false.				
622	DEF_hist_vars%litr1n_vr	输出变量	nl_colm_history	logical	.true.				
623	DEF_hist_vars%litr2cCap_vr	输出变量	nl_colm_history	logical	.false.				
624	DEF_hist_vars%litr2c_vr	输出变量	nl_colm_history	logical	.true.				
625	DEF_hist_vars%litr2nCap_vr	输出变量	nl_colm_history	logical	.false.				
626	DEF_hist_vars%litr2n_vr	输出变量	nl_colm_history	logical	.true.				
627	DEF_hist_vars%litr3cCap_vr	输出变量	nl_colm_history	logical	.false.				
628	DEF_hist_vars%litr3c_vr	输出变量	nl_colm_history	logical	.true.				
629	DEF_hist_vars%litr3nCap_vr	输出变量	nl_colm_history	logical	.false.				
630	DEF_hist_vars%litr3n_vr	输出变量	nl_colm_history	logical	.true.				
631	DEF_hist_vars%livecrootc	输出变量	nl_colm_history	logical	.true.				
632	DEF_hist_vars%livecrootcCap	输出变量	nl_colm_history	logical	.false.				
633	DEF_hist_vars%livecrootc_storage	输出变量	nl_colm_hist	logical	.true.				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
			ory						
634	DEF_hist_vars%livecrotc_storageCap	输出变量	nl_colm_history	logical	.false.				
635	DEF_hist_vars%livecrotc_xfer	输出变量	nl_colm_history	logical	.true.				
636	DEF_hist_vars%livecrotc_xferCap	输出变量	nl_colm_history	logical	.false.				
637	DEF_hist_vars%livecrotn	输出变量	nl_colm_history	logical	.true.				
638	DEF_hist_vars%livecrotnCap	输出变量	nl_colm_history	logical	.false.				
639	DEF_hist_vars%livecrotn_storage	输出变量	nl_colm_history	logical	.true.				
640	DEF_hist_vars%livecrotn_storageCap	输出变量	nl_colm_history	logical	.false.				
641	DEF_hist_vars%livecrotn_xfer	输出变量	nl_colm_history	logical	.true.				
642	DEF_hist_vars%livecrotn_xferCap	输出变量	nl_colm_history	logical	.false.				
643	DEF_hist_vars%livestemc	输出变量	nl_colm_history	logical	.true.				
644	DEF_hist_vars%livestemcCap	输出变量	nl_colm_history	logical	.false.				
645	DEF_hist_vars%livestemc_storage	输出变量	nl_colm_history	logical	.true.				
646	DEF_hist_vars%livestemc_storageCap	输出变量	nl_colm_history	logical	.false.				
647	DEF_hist_vars%livestemc_xfer	输出变量	nl_colm_history	logical	.true.				
648	DEF_hist_vars%livestemc_xferCap	输出变量	nl_colm_history	logical	.false.				
649	DEF_hist_vars%livestemn	输出变量	nl_colm_history	logical	.true.				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
650	DEF_hist_vars%livestemnCap	输出变量	nl_colm_history	logical	.false.				
651	DEF_hist_vars%livestemn_storage	输出变量	nl_colm_history	logical	.true.				
652	DEF_hist_vars%livestemn_storageCap	输出变量	nl_colm_history	logical	.false.				
653	DEF_hist_vars%livestemn_xfer	输出变量	nl_colm_history	logical	.true.				
654	DEF_hist_vars%livestemn_xferCap	输出变量	nl_colm_history	logical	.false.				
655	DEF_hist_vars%lnfm	输出变量	nl_colm_history	logical	.false.				
656	DEF_hist_vars%m_cwd_c_to_fire	输出变量	nl_colm_history	logical	.false.				
657	DEF_hist_vars%m_cwd_n_to_fire	输出变量	nl_colm_history	logical	.false.				
658	DEF_hist_vars%m_deadcrootc_storage_to_fire	输出变量	nl_colm_history	logical	.false.				
659	DEF_hist_vars%m_deadcrootc_storage_to_litter_fire	输出变量	nl_colm_history	logical	.false.				
660	DEF_hist_vars%m_deadcrootc_to_fire	输出变量	nl_colm_history	logical	.false.				
661	DEF_hist_vars%m_deadcrootc_to_litter_fire	输出变量	nl_colm_history	logical	.false.				
662	DEF_hist_vars%m_deadcrootc_xfer_to_fire	输出变量	nl_colm_history	logical	.false.				
663	DEF_hist_vars%m_deadcrootc_xfer_to_litter_fire	输出变量	nl_colm_history	logical	.false.				
664	DEF_hist_vars%m_deadcrootn_storage_to_fire	输出变量	nl_colm_history	logical	.false.				
665	DEF_hist_vars%m_deadcrootn_storage_to_litter_fire	输出变量	nl_colm_history	logical	.false.				
666	DEF_hist_vars%m_deadcrootn_to_fire	输出变量	nl_colm_hist	logical	.false.				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
			ory						
667	DEF_hist_vars%m_deadcrootn_to_litter_fire	输出变量	nl_colm_history	logical	.false.				
668	DEF_hist_vars%m_deadcrootn_xfer_to_fire	输出变量	nl_colm_history	logical	.false.				
669	DEF_hist_vars%m_deadcrootn_xfer_to_litter_fire	输出变量	nl_colm_history	logical	.false.				
670	DEF_hist_vars%m_deadstemc_storage_to_fire	输出变量	nl_colm_history	logical	.false.				
671	DEF_hist_vars%m_deadstemc_storage_to_litter_fire	输出变量	nl_colm_history	logical	.false.				
672	DEF_hist_vars%m_deadstemc_to_fire	输出变量	nl_colm_history	logical	.false.				
673	DEF_hist_vars%m_deadstemc_to_litter_fire	输出变量	nl_colm_history	logical	.false.				
674	DEF_hist_vars%m_deadstemc_xfer_to_fire	输出变量	nl_colm_history	logical	.false.				
675	DEF_hist_vars%m_deadstemc_xfer_to_litter_fire	输出变量	nl_colm_history	logical	.false.				
676	DEF_hist_vars%m_deadstemn_storage_to_fire	输出变量	nl_colm_history	logical	.false.				
677	DEF_hist_vars%m_deadstemn_storage_to_litter_fire	输出变量	nl_colm_history	logical	.false.				
678	DEF_hist_vars%m_deadstemn_to_fire	输出变量	nl_colm_history	logical	.false.				
679	DEF_hist_vars%m_deadstemn_to_litter_fire	输出变量	nl_colm_history	logical	.false.				
680	DEF_hist_vars%m_deadstemn_xfer_to_fire	输出变量	nl_colm_history	logical	.false.				
681	DEF_hist_vars%m_deadstemn_xfer_to_litter_fire	输出变量	nl_colm_history	logical	.false.				
682	DEF_hist_vars%m_frootc_storage_to_fire	输出变量	nl_colm_history	logical	.false.				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
683	DEF_hist_vars%m_frootc_storage_to_litter_fire	输出变量	nl_colm_history	logical	.false.				
684	DEF_hist_vars%m_frootc_to_fire	输出变量	nl_colm_history	logical	.false.				
685	DEF_hist_vars%m_frootc_to_litter_fire	输出变量	nl_colm_history	logical	.false.				
686	DEF_hist_vars%m_frootc_xfer_to_fire	输出变量	nl_colm_history	logical	.false.				
687	DEF_hist_vars%m_frootc_xfer_to_litter_fire	输出变量	nl_colm_history	logical	.false.				
688	DEF_hist_vars%m_frootn_storage_to_fire	输出变量	nl_colm_history	logical	.false.				
689	DEF_hist_vars%m_frootn_storage_to_litter_fire	输出变量	nl_colm_history	logical	.false.				
690	DEF_hist_vars%m_frootn_to_fire	输出变量	nl_colm_history	logical	.false.				
691	DEF_hist_vars%m_frootn_to_litter_fire	输出变量	nl_colm_history	logical	.false.				
692	DEF_hist_vars%m_frootn_xfer_to_fire	输出变量	nl_colm_history	logical	.false.				
693	DEF_hist_vars%m_frootn_xfer_to_litter_fire	输出变量	nl_colm_history	logical	.false.				
694	DEF_hist_vars%m_gresp_storage_to_fire	输出变量	nl_colm_history	logical	.false.				
695	DEF_hist_vars%m_gresp_storage_to_litter_fire	输出变量	nl_colm_history	logical	.false.				
696	DEF_hist_vars%m_gresp_xfer_to_fire	输出变量	nl_colm_history	logical	.false.				
697	DEF_hist_vars%m_gresp_xfer_to_litter_fire	输出变量	nl_colm_history	logical	.false.				
698	DEF_hist_vars%m_leafc_storage_to_fire	输出变量	nl_colm_history	logical	.false.				
699	DEF_hist_vars%m_leafc_storage_to_litter_fi	输出变量	nl_colm_hist	logical	.false.				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
	re		ory						
700	DEF_hist_vars%m_leafc_to_fire	输出变量	nl_colm_history	logical	.false.				
701	DEF_hist_vars%m_leafc_to_litter_fire	输出变量	nl_colm_history	logical	.false.				
702	DEF_hist_vars%m_leafc_xfer_to_fire	输出变量	nl_colm_history	logical	.false.				
703	DEF_hist_vars%m_leafc_xfer_to_litter_fire	输出变量	nl_colm_history	logical	.false.				
704	DEF_hist_vars%m_leafn_storage_to_fire	输出变量	nl_colm_history	logical	.false.				
705	DEF_hist_vars%m_leafn_storage_to_litter_fire	输出变量	nl_colm_history	logical	.false.				
706	DEF_hist_vars%m_leafn_to_fire	输出变量	nl_colm_history	logical	.false.				
707	DEF_hist_vars%m_leafn_to_litter_fire	输出变量	nl_colm_history	logical	.false.				
708	DEF_hist_vars%m_leafn_xfer_to_fire	输出变量	nl_colm_history	logical	.false.				
709	DEF_hist_vars%m_leafn_xfer_to_litter_fire	输出变量	nl_colm_history	logical	.false.				
710	DEF_hist_vars%m_litr1_c_to_fire	输出变量	nl_colm_history	logical	.false.				
711	DEF_hist_vars%m_litr1_n_to_fire	输出变量	nl_colm_history	logical	.false.				
712	DEF_hist_vars%m_litr2_c_to_fire	输出变量	nl_colm_history	logical	.false.				
713	DEF_hist_vars%m_litr2_n_to_fire	输出变量	nl_colm_history	logical	.false.				
714	DEF_hist_vars%m_litr3_c_to_fire	输出变量	nl_colm_history	logical	.false.				
715	DEF_hist_vars%m_litr3_n_to_fire	输出变量	nl_colm_history	logical	.false.				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
716	DEF_hist_vars%m_livecrootc_storage_to_fire	输出变量	nl_colm_history	logical	.false.				
717	DEF_hist_vars%m_livecrootc_storage_to_litter_fire	输出变量	nl_colm_history	logical	.false.				
718	DEF_hist_vars%m_livecrootc_to_deadcrootc_fire	输出变量	nl_colm_history	logical	.false.				
719	DEF_hist_vars%m_livecrootc_to_fire	输出变量	nl_colm_history	logical	.false.				
720	DEF_hist_vars%m_livecrootc_to_litter_fire	输出变量	nl_colm_history	logical	.false.				
721	DEF_hist_vars%m_livecrootc_xfer_to_fire	输出变量	nl_colm_history	logical	.false.				
722	DEF_hist_vars%m_livecrootc_xfer_to_litter_fire	输出变量	nl_colm_history	logical	.false.				
723	DEF_hist_vars%m_livecrootn_storage_to_fire	输出变量	nl_colm_history	logical	.false.				
724	DEF_hist_vars%m_livecrootn_storage_to_litter_fire	输出变量	nl_colm_history	logical	.false.				
725	DEF_hist_vars%m_livecrootn_to_deadcrootn_fire	输出变量	nl_colm_history	logical	.false.				
726	DEF_hist_vars%m_livecrootn_to_fire	输出变量	nl_colm_history	logical	.false.				
727	DEF_hist_vars%m_livecrootn_to_litter_fire	输出变量	nl_colm_history	logical	.false.				
728	DEF_hist_vars%m_livecrootn_xfer_to_fire	输出变量	nl_colm_history	logical	.false.				
729	DEF_hist_vars%m_livecrootn_xfer_to_litter_fire	输出变量	nl_colm_history	logical	.false.				
730	DEF_hist_vars%m_livestemc_storage_to_fire	输出变量	nl_colm_history	logical	.false.				
731	DEF_hist_vars%m_livestemc_storage_to_litter_fire	输出变量	nl_colm_history	logical	.false.				
732	DEF_hist_vars%m_livestemc_to_deadstemc_fir	输出变量	nl_colm_hist	logical	.false.				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
	e		ory						
733	DEF_hist_vars%m_livestemc_to_fire	输出变量	nl_colm_history	logical	.false.				
734	DEF_hist_vars%m_livestemc_to_litter_fire	输出变量	nl_colm_history	logical	.false.				
735	DEF_hist_vars%m_livestemc_xfer_to_fire	输出变量	nl_colm_history	logical	.false.				
736	DEF_hist_vars%m_livestemc_xfer_to_litter_fire	输出变量	nl_colm_history	logical	.false.				
737	DEF_hist_vars%m_livestemn_storage_to_fire	输出变量	nl_colm_history	logical	.false.				
738	DEF_hist_vars%m_livestemn_storage_to_litter_fire	输出变量	nl_colm_history	logical	.false.				
739	DEF_hist_vars%m_livestemn_to_deadstemn_fire	输出变量	nl_colm_history	logical	.false.				
740	DEF_hist_vars%m_livestemn_to_fire	输出变量	nl_colm_history	logical	.false.				
741	DEF_hist_vars%m_livestemn_to_litter_fire	输出变量	nl_colm_history	logical	.false.				
742	DEF_hist_vars%m_livestemn_xfer_to_fire	输出变量	nl_colm_history	logical	.false.				
743	DEF_hist_vars%m_livestemn_xfer_to_litter_fire	输出变量	nl_colm_history	logical	.false.				
744	DEF_hist_vars%m_retransn_to_fire	输出变量	nl_colm_history	logical	.false.				
745	DEF_hist_vars%m_retransn_to_litter_fire	输出变量	nl_colm_history	logical	.false.				
746	DEF_hist_vars%manunitro	输出变量	nl_colm_history	logical	.false.				
747	DEF_hist_vars%meta	输出变量	nl_colm_history	logical	.true.				
748	DEF_hist_vars%ndep_to_sminn	输出变量	nl_colm_history	logical	.true.				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
749	DEF_hist_vars%netflw	输出变量	nl_colm_history	logical	.true.				
750	DEF_hist_vars%nfire	输出变量	nl_colm_history	logical	.true.				
751	DEF_hist_vars%npp_c3arcgrass	输出变量	nl_colm_history	logical	.false.				12
752	DEF_hist_vars%npp_c3grass	输出变量	nl_colm_history	logical	.false.				13
753	DEF_hist_vars%npp_c4grass	输出变量	nl_colm_history	logical	.false.				14
754	DEF_hist_vars%npp_dbfboreal	输出变量	nl_colm_history	logical	.false.				8
755	DEF_hist_vars%npp_dbftemp	输出变量	nl_colm_history	logical	.false.				7
756	DEF_hist_vars%npp_dbftrop	输出变量	nl_colm_history	logical	.false.				6
757	DEF_hist_vars%npp_dbsboreal	输出变量	nl_colm_history	logical	.false.				11
758	DEF_hist_vars%npp_dbstemp	输出变量	nl_colm_history	logical	.false.				10
759	DEF_hist_vars%npp_dnfboreal	输出变量	nl_colm_history	logical	.false.				3
760	DEF_hist_vars%npp_ebftemp	输出变量	nl_colm_history	logical	.false.				5
761	DEF_hist_vars%npp_ebftrop	输出变量	nl_colm_history	logical	.false.				4
762	DEF_hist_vars%npp_ebstemp	输出变量	nl_colm_history	logical	.false.				9
763	DEF_hist_vars%npp_enfboreal	输出变量	nl_colm_history	logical	.false.				2
764	DEF_hist_vars%npp_enftemp	输出变量	nl_colm_history	logical	.false.				1
765	DEF_hist_vars%npptoleafc_c3arcgrass	输出变量	nl_colm_hist	logical	.false.				12

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
			ory						
766	DEF_hist_vars%npptoleafc_c3grass	输出变量	nl_colm_history	logical	.false.				13
767	DEF_hist_vars%npptoleafc_c4grass	输出变量	nl_colm_history	logical	.false.				14
768	DEF_hist_vars%npptoleafc_dbfboreal	输出变量	nl_colm_history	logical	.false.				8
769	DEF_hist_vars%npptoleafc_dbftemp	输出变量	nl_colm_history	logical	.false.				7
770	DEF_hist_vars%npptoleafc_dbftrop	输出变量	nl_colm_history	logical	.false.				6
771	DEF_hist_vars%npptoleafc_dbsboreal	输出变量	nl_colm_history	logical	.false.				11
772	DEF_hist_vars%npptoleafc_dbstemp	输出变量	nl_colm_history	logical	.false.				10
773	DEF_hist_vars%npptoleafc_dnfboreal	输出变量	nl_colm_history	logical	.false.				3
774	DEF_hist_vars%npptoleafc_ebftemp	输出变量	nl_colm_history	logical	.false.				5
775	DEF_hist_vars%npptoleafc_ebftrop	输出变量	nl_colm_history	logical	.false.				4
776	DEF_hist_vars%npptoleafc_ebstemp	输出变量	nl_colm_history	logical	.false.				9
777	DEF_hist_vars%npptoleafc_enfboreal	输出变量	nl_colm_history	logical	.false.				2
778	DEF_hist_vars%npptoleafc_enftemp	输出变量	nl_colm_history	logical	.false.				1
779	DEF_hist_vars%o3uptakesha	输出变量	nl_colm_history	logical	.true.				
780	DEF_hist_vars%o3uptakesun	输出变量	nl_colm_history	logical	.true.				
781	DEF_hist_vars%olrg	输出变量	nl_colm_history	logical	.true.				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
782	DEF_hist_vars%pdcorn	输出变量	nl_colm_history	logical	.false.				
783	DEF_hist_vars%pdcotton	输出变量	nl_colm_history	logical	.false.				
784	DEF_hist_vars%pdrice1	输出变量	nl_colm_history	logical	.false.				
785	DEF_hist_vars%pdrice2	输出变量	nl_colm_history	logical	.false.				
786	DEF_hist_vars%pdsoybean	输出变量	nl_colm_history	logical	.false.				
787	DEF_hist_vars%pdsugarcane	输出变量	nl_colm_history	logical	.false.				
788	DEF_hist_vars%pdswheat	输出变量	nl_colm_history	logical	.false.				
789	DEF_hist_vars%pdwwheat	输出变量	nl_colm_history	logical	.false.				
790	DEF_hist_vars%peatf	输出变量	nl_colm_history	logical	.false.				
791	DEF_hist_vars%pft_fire_closs	输出变量	nl_colm_history	logical	.true.				
792	DEF_hist_vars%pft_fire_nloss	输出变量	nl_colm_history	logical	.true.				
793	DEF_hist_vars%plantdate_irrigated_cotton	输出变量	nl_colm_history	logical	.true.				
794	DEF_hist_vars%plantdate_irrigated_rice	输出变量	nl_colm_history	logical	.true.				
795	DEF_hist_vars%plantdate_irrigated_spwheat	输出变量	nl_colm_history	logical	.true.				
796	DEF_hist_vars%plantdate_irrigated_sugarcane	输出变量	nl_colm_history	logical	.true.				
797	DEF_hist_vars%plantdate_irrigated_temp_corn	输出变量	nl_colm_history	logical	.true.				
798	DEF_hist_vars%plantdate_irrigated_temp_soy	输出变量	nl_colm_hist	logical	.true.				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
	bean		ory						
799	DEF_hist_vars%plantdate_irrigated_trop_corn	输出变量	nl_colm_history	logical	.true.				
800	DEF_hist_vars%plantdate_irrigated_trop_soybean	输出变量	nl_colm_history	logical	.true.				
801	DEF_hist_vars%plantdate_irrigated_wtwheat	输出变量	nl_colm_history	logical	.true.				
802	DEF_hist_vars%plantdate_rainfed_cotton	输出变量	nl_colm_history	logical	.true.				
803	DEF_hist_vars%plantdate_rainfed_rice	输出变量	nl_colm_history	logical	.true.				
804	DEF_hist_vars%plantdate_rainfed_spwheat	输出变量	nl_colm_history	logical	.true.				
805	DEF_hist_vars%plantdate_rainfed_sugarcane	输出变量	nl_colm_history	logical	.true.				
806	DEF_hist_vars%plantdate_rainfed_temp_corn	输出变量	nl_colm_history	logical	.true.				
807	DEF_hist_vars%plantdate_rainfed_temp_soybean	输出变量	nl_colm_history	logical	.true.				
808	DEF_hist_vars%plantdate_rainfed_trop_corn	输出变量	nl_colm_history	logical	.true.				
809	DEF_hist_vars%plantdate_rainfed_trop_soybean	输出变量	nl_colm_history	logical	.true.				
810	DEF_hist_vars%plantdate_rainfed_wtwheat	输出变量	nl_colm_history	logical	.true.				
811	DEF_hist_vars%plantdate_unmanagedcrop	输出变量	nl_colm_history	logical	.true.				
812	DEF_hist_vars%qcharge	输出变量	nl_colm_history	logical	.true.				
813	DEF_hist_vars%qdrip	输出变量	nl_colm_history	logical	.true.				
814	DEF_hist_vars%qinfl	输出变量	nl_colm_history	logical	.true.				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
815	DEF_hist_vars%qintr	输出变量	nl_colm_history	logical	.true.				
816	DEF_hist_vars%qlayer	输出变量	nl_colm_history	logical	.true.				
817	DEF_hist_vars%qref	输出变量	nl_colm_history	logical	.true.				
818	DEF_hist_vars%qresv_in	输出变量	nl_colm_history	logical	.true.				
819	DEF_hist_vars%qresv_out	输出变量	nl_colm_history	logical	.true.				
820	DEF_hist_vars%qstar	输出变量	nl_colm_history	logical	.true.				
821	DEF_hist_vars%reflectance_out	输出变量	nl_colm_history	logical	.true.				
822	DEF_hist_vars%reservoirriver_demand	输出变量	nl_colm_history	logical	.true.				
823	DEF_hist_vars%reservoirriver_supply	输出变量	nl_colm_history	logical	.true.				
824	DEF_hist_vars%respc	输出变量	nl_colm_history	logical	.true.				
825	DEF_hist_vars%retrasn	输出变量	nl_colm_history	logical	.true.				
826	DEF_hist_vars%rib	输出变量	nl_colm_history	logical	.true.				
827	DEF_hist_vars%riv_height	输出变量	nl_colm_history	logical	.true.				
828	DEF_hist_vars%riv_veloct	输出变量	nl_colm_history	logical	.true.				
829	DEF_hist_vars%rnet	输出变量	nl_colm_history	logical	.true.				
830	DEF_hist_vars%rnof	输出变量	nl_colm_history	logical	.true.				
831	DEF_hist_vars%rootr	输出变量	nl_colm_hist	logical	.true.				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
			ory						
832	DEF_hist_vars%rss	输出变量	nl_colm_history	logical	.true.				
833	DEF_hist_vars%rstfacsha	输出变量	nl_colm_history	logical	.true.				
834	DEF_hist_vars%rstfacsun	输出变量	nl_colm_history	logical	.true.				
835	DEF_hist_vars%rsub	输出变量	nl_colm_history	logical	.true.				
836	DEF_hist_vars%rsur	输出变量	nl_colm_history	logical	.true.				
837	DEF_hist_vars%rsur_ie	输出变量	nl_colm_history	logical	.true.				
838	DEF_hist_vars%rsur_se	输出变量	nl_colm_history	logical	.true.				
839	DEF_hist_vars%sabg	输出变量	nl_colm_history	logical	.true.				
840	DEF_hist_vars%sabvsha	输出变量	nl_colm_history	logical	.true.				
841	DEF_hist_vars%sabvsun	输出变量	nl_colm_history	logical	.true.				
842	DEF_hist_vars%sai	输出变量	nl_colm_history	logical	.true.				
843	DEF_hist_vars%scv	输出变量	nl_colm_history	logical	.true.				
844	DEF_hist_vars%sedcon	输出变量	nl_colm_history	logical	.true.				
845	DEF_hist_vars%sedinp	输出变量	nl_colm_history	logical	.true.				
846	DEF_hist_vars%sedlayer	输出变量	nl_colm_history	logical	.true.				
847	DEF_hist_vars%sedout	输出变量	nl_colm_history	logical	.true.				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
848	DEF_hist_vars%sensors	输出变量	nl_colm_history	logical	.true.				
849	DEF_hist_vars%shearvel	输出变量	nl_colm_history	logical	.false.				
850	DEF_hist_vars%sigf	输出变量	nl_colm_history	logical	.true.				
851	DEF_hist_vars%sminn_vr	输出变量	nl_colm_history	logical	.true.				
852	DEF_hist_vars%snowdp	输出变量	nl_colm_history	logical	.true.				
853	DEF_hist_vars%soil1cCap_vr	输出变量	nl_colm_history	logical	.false.				
854	DEF_hist_vars%soil1c_vr	输出变量	nl_colm_history	logical	.true.				
855	DEF_hist_vars%soil1nCap_vr	输出变量	nl_colm_history	logical	.false.				
856	DEF_hist_vars%soil1n_vr	输出变量	nl_colm_history	logical	.true.				
857	DEF_hist_vars%soil2cCap_vr	输出变量	nl_colm_history	logical	.false.				
858	DEF_hist_vars%soil2c_vr	输出变量	nl_colm_history	logical	.true.				
859	DEF_hist_vars%soil2nCap_vr	输出变量	nl_colm_history	logical	.false.				
860	DEF_hist_vars%soil2n_vr	输出变量	nl_colm_history	logical	.true.				
861	DEF_hist_vars%soil3cCap_vr	输出变量	nl_colm_history	logical	.false.				
862	DEF_hist_vars%soil3c_vr	输出变量	nl_colm_history	logical	.true.				
863	DEF_hist_vars%soil3nCap_vr	输出变量	nl_colm_history	logical	.false.				
864	DEF_hist_vars%soil3n_vr	输出变量	nl_colm_hist	logical	.true.				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
			ory						
865	DEF_hist_vars%sol_dif_ln_hires	输出变量	nl_colm_history	logical	.true.				
866	DEF_hist_vars%sol_dir_ln_hires	输出变量	nl_colm_history	logical	.true.				
867	DEF_hist_vars%solnd	输出变量	nl_colm_history	logical	.true.				
868	DEF_hist_vars%solndln	输出变量	nl_colm_history	logical	.true.				
869	DEF_hist_vars%solni	输出变量	nl_colm_history	logical	.true.				
870	DEF_hist_vars%solniln	输出变量	nl_colm_history	logical	.true.				
871	DEF_hist_vars%solvd	输出变量	nl_colm_history	logical	.true.				
872	DEF_hist_vars%solvdln	输出变量	nl_colm_history	logical	.true.				
873	DEF_hist_vars%solvi	输出变量	nl_colm_history	logical	.true.				
874	DEF_hist_vars%solviln	输出变量	nl_colm_history	logical	.true.				
875	DEF_hist_vars%somc_fire	输出变量	nl_colm_history	logical	.true.				
876	DEF_hist_vars%somfire	输出变量	nl_colm_history	logical	.true.				
877	DEF_hist_vars%sr	输出变量	nl_colm_history	logical	.true.				
878	DEF_hist_vars%sr_dif_ln_hires	输出变量	nl_colm_history	logical	.true.				
879	DEF_hist_vars%sr_dir_ln_hires	输出变量	nl_colm_history	logical	.true.				
880	DEF_hist_vars%srnd	输出变量	nl_colm_history	logical	.true.				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
881	DEF_hist_vars%srndln	输出变量	nl_colm_history	logical	.true.				
882	DEF_hist_vars%srni	输出变量	nl_colm_history	logical	.true.				
883	DEF_hist_vars%srniln	输出变量	nl_colm_history	logical	.true.				
884	DEF_hist_vars%srvd	输出变量	nl_colm_history	logical	.true.				
885	DEF_hist_vars%srvdln	输出变量	nl_colm_history	logical	.true.				
886	DEF_hist_vars%srvi	输出变量	nl_colm_history	logical	.true.				
887	DEF_hist_vars%srviln	输出变量	nl_colm_history	logical	.true.				
888	DEF_hist_vars%sum_deficit_irrig	输出变量	nl_colm_history	logical	.true.				
889	DEF_hist_vars%sum_irrig	输出变量	nl_colm_history	logical	.true.				
890	DEF_hist_vars%sum_irrig_count	输出变量	nl_colm_history	logical	.true.				
891	DEF_hist_vars%t2m_wmo	输出变量	nl_colm_history	logical	.true.				
892	DEF_hist_vars%t_grnd	输出变量	nl_colm_history	logical	.true.				
893	DEF_hist_vars%t_lake	输出变量	nl_colm_history	logical	.true.				
894	DEF_hist_vars%t_roof	输出变量	nl_colm_history	logical	.true.				
895	DEF_hist_vars%t_room	输出变量	nl_colm_history	logical	.true.				
896	DEF_hist_vars%t_scalar	输出变量	nl_colm_history	logical	.false.				
897	DEF_hist_vars%t_soisno	输出变量	nl_colm_hist	logical	.true.				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
			ory						
898	DEF_hist_vars%t_wall	输出变量	nl_colm_history	logical	.true.				
899	DEF_hist_vars%tafu	输出变量	nl_colm_history	logical	.true.				
900	DEF_hist_vars%taux	输出变量	nl_colm_history	logical	.true.				
901	DEF_hist_vars%tauy	输出变量	nl_colm_history	logical	.true.				
902	DEF_hist_vars%tleaf	输出变量	nl_colm_history	logical	.true.				
903	DEF_hist_vars%totcolc	输出变量	nl_colm_history	logical	.true.				
904	DEF_hist_vars%totcoln	输出变量	nl_colm_history	logical	.true.				
905	DEF_hist_vars%totcwdc	输出变量	nl_colm_history	logical	.true.				
906	DEF_hist_vars%totcwdn	输出变量	nl_colm_history	logical	.true.				
907	DEF_hist_vars%totfire	输出变量	nl_colm_history	logical	.true.				
908	DEF_hist_vars%totlitc	输出变量	nl_colm_history	logical	.true.				
909	DEF_hist_vars%totlitn	输出变量	nl_colm_history	logical	.true.				
910	DEF_hist_vars%totsoiln_vr	输出变量	nl_colm_history	logical	.true.				
911	DEF_hist_vars%totsomc	输出变量	nl_colm_history	logical	.true.				
912	DEF_hist_vars%totsomn	输出变量	nl_colm_history	logical	.true.				
913	DEF_hist_vars%totvegc	输出变量	nl_colm_history	logical	.true.				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
914	DEF_hist_vars%totvegn	输出变量	nl_colm_history	logical	.true.				
915	DEF_hist_vars%trad	输出变量	nl_colm_history	logical	.true.				
916	DEF_hist_vars%transmittance_out	输出变量	nl_colm_history	logical	.true.				
917	DEF_hist_vars%tref	输出变量	nl_colm_history	logical	.true.				
918	DEF_hist_vars%tstar	输出变量	nl_colm_history	logical	.true.				
919	DEF_hist_vars%us10m	输出变量	nl_colm_history	logical	.true.				
920	DEF_hist_vars%ustar	输出变量	nl_colm_history	logical	.true.				
921	DEF_hist_vars%ustar2	输出变量	nl_colm_history	logical	.true.				
922	DEF_hist_vars%vegwp	输出变量	nl_colm_history	logical	.true.				
923	DEF_hist_vars%vehc	输出变量	nl_colm_history	logical	.true.				
924	DEF_hist_vars%veloc_hru	输出变量	nl_colm_history	logical	.true.				
925	DEF_hist_vars%vf	输出变量	nl_colm_history	logical	.true.				
926	DEF_hist_vars%volresv	输出变量	nl_colm_history	logical	.true.				
927	DEF_hist_vars%vs10m	输出变量	nl_colm_history	logical	.true.				
928	DEF_hist_vars%w_scalar	输出变量	nl_colm_history	logical	.false.				
929	DEF_hist_vars%wa	输出变量	nl_colm_history	logical	.true.				
930	DEF_hist_vars%wa_inst	输出变量	nl_colm_hist	logical	.true.				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
			ory						
931	DEF_hist_vars%wat	输出变量	nl_colm_history	logical	.true.				
932	DEF_hist_vars%wat_inst	输出变量	nl_colm_history	logical	.true.				
933	DEF_hist_vars%waterstorage	输出变量	nl_colm_history	logical	.true.				
934	DEF_hist_vars%wdsrf	输出变量	nl_colm_history	logical	.true.				
935	DEF_hist_vars%wdsrf_hru	输出变量	nl_colm_history	logical	.true.				
936	DEF_hist_vars%wdsrf_inst	输出变量	nl_colm_history	logical	.true.				
937	DEF_hist_vars%wetwat	输出变量	nl_colm_history	logical	.true.				
938	DEF_hist_vars%wetwat_inst	输出变量	nl_colm_history	logical	.true.				
939	DEF_hist_vars%wetzwt	输出变量	nl_colm_history	logical	.true.				
940	DEF_hist_vars%wfc	输出变量	nl_colm_history	logical	.true.				
941	DEF_hist_vars%wice_soisno	输出变量	nl_colm_history	logical	.true.				
942	DEF_hist_vars%wliq_soisno	输出变量	nl_colm_history	logical	.true.				
943	DEF_hist_vars%xerr	输出变量	nl_colm_history	logical	.true.				
944	DEF_hist_vars%xsubs_bsn	输出变量	nl_colm_history	logical	.true.				
945	DEF_hist_vars%xsubs_hru	输出变量	nl_colm_history	logical	.true.				
946	DEF_hist_vars%xwsub	输出变量	nl_colm_history	logical	.true.				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
947	DEF_hist_vars%xwsur	输出变量	nl_colm_history	logical	.true.				
948	DEF_hist_vars%xy_frl	输出变量	nl_colm_history	logical	.true.				
949	DEF_hist_vars%xy_hpbl	输出变量	nl_colm_history	logical	.true.				
950	DEF_hist_vars%xy_ozone	输出变量	nl_colm_history	logical	.true.				
951	DEF_hist_vars%xy_pbot	输出变量	nl_colm_history	logical	.true.				
952	DEF_hist_vars%xy_prc	输出变量	nl_colm_history	logical	.true.				
953	DEF_hist_vars%xy_prl	输出变量	nl_colm_history	logical	.true.				
954	DEF_hist_vars%xy_q	输出变量	nl_colm_history	logical	.true.				
955	DEF_hist_vars%xy_rain	输出变量	nl_colm_history	logical	.true.				
956	DEF_hist_vars%xy_snow	输出变量	nl_colm_history	logical	.true.				
957	DEF_hist_vars%xy_solarin	输出变量	nl_colm_history	logical	.true.				
958	DEF_hist_vars%xy_t	输出变量	nl_colm_history	logical	.true.				
959	DEF_hist_vars%xy_us	输出变量	nl_colm_history	logical	.true.				
960	DEF_hist_vars%xy_vs	输出变量	nl_colm_history	logical	.true.				
961	DEF_hist_vars%z0m	输出变量	nl_colm_history	logical	.true.				
962	DEF_hist_vars%zerr	输出变量	nl_colm_history	logical	.true.				
963	DEF_hist_vars%zol	输出变量	nl_colm_hist	logical	.true.				

#	字段	分组	文件	类型	默认值	可选值	依赖宏	派生	说明
			ory						
964	DEF_hist_vars%zwt	输出变量	nl_colm_history	logical	.true.				

附录 B：参数库索引（1369 项）

本表由 `xtask parameter-audit` 的参数目录生成，覆盖模型可调参数（参数值、物理/过程参数、参数与资产来源）。`id` 是研究页与助手工具使用的稳定标识；`scope` 说明参数作用范围；`visibility` 是静态分类（`editable-common` = 常用类），实际可编辑性还取决于引擎、运行选项及写出能力。默认值来自源码声明，实数 kind 后缀已规范化；数组等值仍需对应作用域写入器处理。默认提供方 与 源码位置 指明来源。写入目标 显示 schema 字段的实际文件或引用该文件的键；其余参数由对应作用域的写入器处理。

初始场（8 项）

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
1	case:DEF_USE_CN_INIT	DEF_USE_CN_INIT	DEF USE CN INIT	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:325
2	case:DEF_USE_SnowInit	DEF_USE_SnowInit	DEF USE SnowInit	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:322
3	case:DEF_USE_SoilInit	DEF_USE_SoilInit	DEF USE SoilInit	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:319
4	case:DEF_USE_WaterTableInit	DEF_USE_WaterTableInit	DEF USE WaterTableInit	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:328
5	case:DEF_file_SnowInit	DEF_file_SnowInit	DEF file SnowInit	null		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:323
6	case:DEF_file_SoilInit	DEF_file_SoilInit	DEF file SoilInit	null		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:320
7	case:DEF_file_WaterTable	DEF_file_WaterTable	DEF file WaterTable	null		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:329
8	case:DEF_file_cn_init	DEF_file_cn_init	DEF file cn init	null		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:326

地表数据（31 项）

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
1	case:DEF_FAST_PC	DEF_FAST_PC	DEF FAST PC	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:271
2	case:DEF_HighResSoil	DEF_HighResSoil	DEF HighResSoil	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:463

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
3	case:DEF_HighResUrban_albedo	DEF_HighResUrban_albedo	DEF HighResUrban albedo	null		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:467
4	case:DEF_HighResVeg	DEF_HighResVeg	DEF HighResVeg	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:464
5	case:DEF_LAI_CHANGE_YEARLY	DEF_LAI_CHANGE_YEARLY	DEF LAI CHANGE YEARLY	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:310
6	case:DEF_LAI_END_YEAR	DEF_LAI_END_YEAR	DEF LAI END YEAR	2020		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:301
7	case:DEF_LAI_MONTHLY	DEF_LAI_MONTHLY	DEF LAI MONTHLY	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:306
8	case:DEF_LAI_START_YEAR	DEF_LAI_START_YEAR	DEF LAI START YEAR	2000		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:300
9	case:DEF_LANDONLY	DEF_LANDONLY	DEF LANDONLY	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:277
10	case:DEF_LULCC_SCHEME	DEF_LULCC_SCHEME	DEF LULCC SCHEME	1		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:338
11	case:DEF_PC_CROP_SPLIT	DEF_PC_CROP_SPLIT	DEF PC CROP SPLIT	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:272
12	case:DEF_SOIL_REFL_SCHEME	DEF_SOIL_REFL_SCHEME	DEF SOIL REFL SCHEME	2		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:287
13	case:DEF_SOLO_PFT	DEF_SOLO_PFT	DEF SOLO PFT	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:270
14	case:DEF_SUBGRID_SCHEME	DEF_SUBGRID_SCHEME	DEF SUBGRID SCHEME	LCT		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:275
15	case:DEF_Srfdata_CompressLevel	DEF_Srfdata_CompressLevel	DEF Srfdata CompressLevel	1		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:293
16	case:DEF_USE_DOMINANT_PATCHTYPE	DEF_USE_DOMINANT_PATCHTYPE	DEF USE DOMINANT PATCHTYPE	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:278
17	case:DEF_USE_ESACCI	DEF_USE_ESACCI	DEF USE ESACCI	.false.		case-scalar	read-only-context		MOD_Namelist.F90 declaration	MOD_Namelist.F90:255
18	case:DEF_USE_GLC30	DEF_USE_GLC30	DEF USE GLC30	.false.		case-scalar	read-only-context		MOD_Namelist.F90 declaration	MOD_Namelist.F90:254
19	case:DEF_USE_IGBP	DEF_USE_IGBP	DEF USE IGBP	.false.		case-scalar	read-only-context		MOD_Namelist.F90 declaration	MOD_Namelist.F90:262

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
20	case:DEF_USE_LCT	DEF_USE_LCT	DEF USE LCT	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:267
21	case:DEF_USE_LULCC	DEF_USE_LULCC	DEF USE LULCC	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:344
22	case:DEF_USE_PC	DEF_USE_PC	DEF USE PC	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:269
23	case:DEF_USE_PFT	DEF_USE_PFT	DEF USE PFT	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:268
24	case:DEF_USE_SOILPAR_UPS_FIT	DEF_USE_SOILPAR_UPS_FIT	DEF USE SOILPAR UPS FIT	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:282
25	case:DEF_USE_USGS	DEF_USE_USGS	DEF USE USGS	.false.		case-scalar	read-only-context		MOD_Namelist.F90 declaration	MOD_Namelist.F90:261
26	case:DEF_dir_existing_srfdata	DEF_dir_existing_srfdata	DEF dir existing srfdata	path/to/landdata		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:150
27	case:DEF_dir_rawdata	DEF_dir_rawdata	DEF dir rawdata	path/to/rawdata/		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:127
28	case:DEF_rawdata_namelist	DEF_rawdata_namelist	DEF rawdata namelist	./rawdata/colm2024.nml		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:168
29	case:USE_srfdata_from_3D_gridded_data	USE_srfdata_from_3D_gridded_data	USE srfdata from 3D gridded data	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:153
30	case:USE_srfdata_from_larger_region	USE_srfdata_from_larger_region	USE srfdata from larger region	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:149
31	case:USE_zip_for_aggregation	USE_zip_for_aggregation	USE zip for aggregation	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:290

城市（13 项）

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
1	case:DEF_URBAN_BEM	DEF_URBAN_BEM	DEF URBAN BEM	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:356
2	case:DEF_URBAN_LUCY	DEF_URBAN_LUCY	DEF URBAN LUCY	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:359
3	case:DEF_URBAN_ONLY	DEF_URBAN_ONLY	DEF URBAN ONLY	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:354

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
4	case:DEF_URBAN_RUN	DEF_URBAN_RUN	DEF URBAN RUN	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:355
5	case:DEF_URBAN_TREE	DEF_URBAN_TREE	DEF URBAN TREE	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:357
6	case:DEF_URBAN_WATER	DEF_URBAN_WATER	DEF URBAN WATER	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:358
7	case:DEF_URBAN_type_scheme	DEF_URBAN_type_scheme	DEF URBAN type scheme	2		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:353
8	case:DEF_USE_CANYON_HWR	DEF_USE_CANYON_HWR	DEF USE CANYON HWR	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:360
9	case:USE_SITE_urban_ecology	USE_SITE_urban_ecology	USE SITE urban ecology	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:92
10	case:USE_SITE_urban_geometry	USE_SITE_urban_geometry	USE SITE urban geometry	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:91
11	case:USE_SITE_urban_human	USE_SITE_urban_human	USE SITE urban human	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:95
12	case:USE_SITE_urban_radiation	USE_SITE_urban_radiation	USE SITE urban radiation	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:93
13	case:USE_SITE_urban_thermal	USE_SITE_urban_thermal	USE SITE urban thermal	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:94

强迫场（48 项）

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
1	case:DEF_DS_HiresTopographyDataDir	DEF_DS_HiresTopographyDataDir	DEF DS HiresTopographyDataDir	null		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:792
2	case:DEF_DS_LONGWAVE_LAPSE_RATE	DEF_DS_LONGWAVE_LAPSE_RATE	DEF DS LONGWAVE LAPSE RATE	0.032		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:664
3	case:DEF_DS_LONGWAVE_LIMIT	DEF_DS_LONGWAVE_LIMIT	DEF DS LONGWAVE LIMIT	0.5		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:665
4	case:DEF_DS_SHORTWAVE_LIMIT	DEF_DS_SHORTWAVE_LIMIT	DEF DS SHORTWAVE LIMIT	0.5		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:666
5	case:DEF_DS_SHORTWAVE_SIMPLE_LIMIT	DEF_DS_SHORTWAVE_SIMPLE_LIMIT	DEF DS SHORTWAVE SIMPLE LIMIT	0.2		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:667

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
6	case:DEF_DS_TEMP_LAPSE_RATE	DEF_DS_TEMP_LAPSE_RATE	DEF DS TEMP LAPSE RATE	0.006		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:663
7	case:DEF_DS_longwave_adjust_scheme	DEF_DS_longwave_adjust_scheme	DEF DS longwave adjust scheme	II		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:794
8	case:DEF_DS_precipitation_adjust_scheme	DEF_DS_precipitation_adjust_scheme	DEF DS precipitation adjust scheme	I		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:793
9	case:DEF_Forcing_Interp_Method	DEF_Forcing_Interp_Method	DEF Forcing Interp Method	arealweight		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:788
10	case:DEF_USE_CBL_HEIGHT	DEF_USE_CBL_HEIGHT	DEF USE CBL HEIGHT	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:786
11	case:DEF_USE_ClimForcing_for_Spinup	DEF_USE_ClimForcing_for_Spinup	DEF USE ClimForcing for Spinup	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:796
12	case:DEF_USE_Forcing_Downscaling	DEF_USE_Forcing_Downscaling	DEF USE Forcing Downscaling	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:790
13	case:DEF_USE_Forcing_Downscaling_Simple	DEF_USE_Forcing_Downscaling_Simple	DEF USE Forcing Downscaling Simple	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:791
14	case:DEF_dir_forcing	DEF_dir_forcing	DEF dir forcing	path/to/forcing/data		case-scalar	editable-common	forcing.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:722
15	case:DEF_forcing%CBL_dtime	DEF_forcing%CBL_dtime	DEF forcing CBL dtime	21600		case-scalar	editable-common	forcing.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:779
16	case:DEF_forcing%CBL_fprefix	DEF_forcing%CBL_fprefix	DEF forcing CBL fprefix	TPHWL6Hrly/clmforc.cruncep.V4.c2011.0.5d.TPQWL.		case-scalar	editable-common	forcing.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:776
17	case:DEF_forcing%CBL_offset	DEF_forcing%CBL_offset	DEF forcing CBL offset	10800		case-scalar	editable-common	forcing.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:780
18	case:DEF_forcing%CBL_tintalgo	DEF_forcing%CBL_tintalgo	DEF forcing CBL tintalgo	linear		case-scalar	editable-common	forcing.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:778
19	case:DEF_forcing%CBL_vname	DEF_forcing%CBL_vname	DEF forcing CBL vname	blh		case-scalar	editable-common	forcing.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:777
20	case:DEF_forcing%HEIGHT_Q	DEF_forcing%HEIGHT_Q	DEF forcing HEIGHT Q	50.		case-scalar	editable-common	forcing.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:731
21	case:DEF_forcing%HEIGHT_T	DEF_forcing%HEIGHT_T	DEF forcing HEIGHT T	50.		case-scalar	editable-common	forcing.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:730

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
22	case:DEF_forcing%HEIGHT_V	DEF_forcing%HEIGHT_V	DEF forcing HEIGHT V	100.0		case-scalar	editable-common	forcing.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:729
23	case:DEF_forcing%HEIGHT_mode	DEF_forcing%HEIGHT_mode	DEF forcing HEIGHT mode	absolute		case-scalar	editable-common	forcing.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:728
24	case:DEF_forcing%NVAR	DEF_forcing%NVAR	DEF forcing NVAR	8		case-scalar	editable-common	forcing.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:738
25	case:DEF_forcing%data2d	DEF_forcing%data2d	DEF forcing data2d	.true.		case-scalar	editable-common	forcing.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:748
26	case:DEF_forcing%dataset	DEF_forcing%dataset	DEF forcing dataset	CRUNCEP		case-scalar	editable-common	forcing.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:726
27	case:DEF_forcing%dim2d	DEF_forcing%dim2d	DEF forcing dim2d	.true.		case-scalar	editable-common	forcing.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:750
28	case:DEF_forcing%dtime	DEF_forcing%dtime	DEF forcing dtime	(/21600,21600,21600,21600,0,21600,21600,21600/)		case-scalar	editable-common	forcing.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:743
29	case:DEF_forcing%endmo	DEF_forcing%endmo	DEF forcing endmo	12		case-scalar	editable-common	forcing.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:742
30	case:DEF_forcing%endyr	DEF_forcing%endyr	DEF forcing endyr	2003		case-scalar	editable-common	forcing.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:741
31	case:DEF_forcing%fprefix	DEF_forcing%fprefix	DEF forcing fprefix	(/ 'TPHWL6Hrly/clmforc.cruncep.V4.c2011.0.5d.TPQWL.', 'TPHWL6Hrly/clmforc.cruncep.V4.c2011.0.5d.TPQWL.', 'TPHWL6Hrly/clmforc.cruncep.V4.c2011.0.5d.TPQWL.', 'Precip6Hrly/clmforc.cruncep.V4.c2011.0.5d.Prec.', 'NULL', 'TPHWL6Hrly/cl		case-scalar	editable-common	forcing.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:757

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
				mforc.cruncep.V4.c2011.0.5d.TPQWL.', 'Solar6Hrly/clmforc.cruncep.V4.c2011.0.5d.Solr. ', 'TPHWL6Hrly/clmforc.cruncep.V4.c2011.0.5d.TPQWL.' /)						
32	case:DEF_forcing%groupby	DEF_forcing%groupby	DEF forcing groupby	month		case-scalar	editable-common	forcing.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:755
33	case:DEF_forcing%has_missing_value	DEF_forcing%has_missing_value	DEF forcing has missing value	.false.		case-scalar	editable-common	forcing.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:735
34	case:DEF_forcing%hightdim	DEF_forcing%hightdim	DEF forcing hightdim	.false.		case-scalar	editable-common	forcing.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:749
35	case:DEF_forcing%latname	DEF_forcing%latname	DEF forcing latname	LATIXY		case-scalar	editable-common	forcing.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:752
36	case:DEF_forcing%leapyear	DEF_forcing%leapyear	DEF forcing leapyear	.false.		case-scalar	editable-common	forcing.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:747
37	case:DEF_forcing%lonname	DEF_forcing%lonname	DEF forcing lonname	LONGXY		case-scalar	editable-common	forcing.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:753
38	case:DEF_forcing%missing_value_name	DEF_forcing%missing_value_name	DEF forcing missing value name	missing_value		case-scalar	editable-common	forcing.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:736
39	case:DEF_forcing%nlands	DEF_forcing%nlands	DEF forcing nlands	1		case-scalar	editable-common	forcing.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:745
40	case:DEF_forcing%offset	DEF_forcing%offset	DEF forcing offset	(/10800,10800,10800,10800,0,10800,0,10800/)		case-scalar	editable-common	forcing.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:744
41	case:DEF_forcing%regbnd	DEF_forcing%regbnd	DEF forcing regbnd	(/-90.0, 90.0, -180.0, 180.0/)		case-scalar	editable-common	forcing.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:734
42	case:DEF_forcing%regional	DEF_forcing%regional	DEF forcing regional	.false.		case-scalar	editable-common	forcing.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:733

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
43	case:DEF_forcing%solarin_all_band	DEF_forcing%solarin_all_band	DEF forcing solarin all band	.true.		case-scalar	editable-common	forcing.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:727
44	case:DEF_forcing%startmo	DEF_forcing%startmo	DEF forcing startmo	1		case-scalar	editable-common	forcing.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:740
45	case:DEF_forcing%startyr	DEF_forcing%startyr	DEF forcing startyr	2000		case-scalar	editable-common	forcing.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:739
46	case:DEF_forcing%timelog	DEF_forcing%timelog	DEF forcing timelog	(/ 'instant ','instant ','instant ','forward ', 'NULL ','instant ','forward ','forward ' /)		case-scalar	editable-common	forcing.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:769
47	case:DEF_forcing%tintalgo	DEF_forcing%tintalgo	DEF forcing tintalgo	(/ 'linear ','linear ','linear ','nearest', 'NULL ','linear ','coszen ','linear ' /)		case-scalar	editable-common	forcing.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:772
48	case:DEF_forcing%vname	DEF_forcing%vname	DEF forcing vname	(/ 'TBOT ','QBOT ','PSRF ','PRECTmms', 'NULL ','WIND ','FSDS ','FLDS ' /)		case-scalar	editable-common	forcing.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:766

数据同化（11 项）

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
1	case:DEF_DA_ENS_NUM	DEF_DA_ENS_NUM	DEF DA ENS NUM	20		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:805
2	case:DEF_DA_ENS_SM	DEF_DA_ENS_SM	DEF DA ENS SM	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:806

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
3	case:DEF_DA_RTM_diel	DEF_DA_RTM_diel	DEF DA RTM diel	0		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:810
4	case:DEF_DA_RTM_rough	DEF_DA_RTM_rough	DEF DA RTM rough	0		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:811
5	case:DEF_DA_SM	DEF_DA_SM	DEF DA SM	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:804
6	case:DEF_DA_SM_FY	DEF_DA_SM_FY	DEF DA SM FY	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:808
7	case:DEF_DA_SM_SMAP	DEF_DA_SM_SMAP	DEF DA SM SMAP	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:807
8	case:DEF_DA_SM_SYNOP	DEF_DA_SM_SYNOP	DEF DA SM SYNOP	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:809
9	case:DEF_DA_TWS	DEF_DA_TWS	DEF DA TWS	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:802
10	case:DEF_DA_TWS_GRACE	DEF_DA_TWS_GRACE	DEF DA TWS GRACE	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:803
11	case:DEF_DA_obsdir	DEF_DA_obsdir	DEF DA obsdir	null		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:801

文件与目录（3 项）

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
1	case:DEF_dir_landdata	DEF_dir_landdata	DEF dir landdata	path/to/landdata		case-scalar	read-only-context		MOD_Namelist.F90 declaration	MOD_Namelist.F90:131
2	case:DEF_dir_runtime	DEF_dir_runtime	DEF dir runtime	path/to/runtime/		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:128
3	case:DEF_forcing_namelist	DEF_forcing_namelist	DEF forcing namelist	null		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:720

时间与预热（15 项）

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
1	case:DEF_simulation_time%end_day	DEF_simulation_time%end_day	DEF simulation time end day	1		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:111

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
2	case:DEF_simulation_time%end_month	DEF_simulation_time%end_month	DEF simulation time end month	1		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:110
3	case:DEF_simulation_time%end_sec	DEF_simulation_time%end_sec	DEF simulation time end sec	0		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:112
4	case:DEF_simulation_time%end_year	DEF_simulation_time%end_year	DEF simulation time end year	2003		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:109
5	case:DEF_simulation_time%greenwich	DEF_simulation_time%greenwich	DEF simulation time greenwich	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:104
6	case:DEF_simulation_time%spinup_day	DEF_simulation_time%spinup_day	DEF simulation time spinup day	1		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:115
7	case:DEF_simulation_time%spinup_month	DEF_simulation_time%spinup_month	DEF simulation time spinup month	1		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:114
8	case:DEF_simulation_time%spinup_repeat	DEF_simulation_time%spinup_repeat	DEF simulation time spinup repeat	1		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:117
9	case:DEF_simulation_time%spinup_sec	DEF_simulation_time%spinup_sec	DEF simulation time spinup sec	0		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:116
10	case:DEF_simulation_time%spinup_year	DEF_simulation_time%spinup_year	DEF simulation time spinup year	0		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:113
11	case:DEF_simulation_time%start_day	DEF_simulation_time%start_day	DEF simulation time start day	1		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:107
12	case:DEF_simulation_time%start_month	DEF_simulation_time%start_month	DEF simulation time start month	1		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:106
13	case:DEF_simulation_time%start_sec	DEF_simulation_time%start_sec	DEF simulation time start sec	0		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:108
14	case:DEF_simulation_time%start_year	DEF_simulation_time%start_year	DEF simulation time start year	2000		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:105
15	case:DEF_simulation_time%timestep	DEF_simulation_time%timestep	DEF simulation time timestep	1800.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:118

材料（229 项）

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
1	case:DEF_ISOTOPE_PARAM_FILE	DEF_ISOTOPE_PARAM_FILE	DEF ISOTOPE PARAM FILE	null		case-scalar	editable-common	case.nml	MOD_MT_Namelist.F90 declaration	MOD_MT_Namelist.F90:45

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
2	case:DEF_ISOTOPE_SPECIES	DEF_ISOTOPE_SPECIES	DEF ISOTOPE SPECIES	H2_18O,HDO		case-scalar	editable-common	case.nml	MOD_MT_Namelist.F90 declaration	MOD_MT_Namelist.F90:44
3	case:DEF_ISO_CANOPY_EQUILIBRATION	DEF_ISO_CANOPY_EQUILIBRATION	DEF ISO CANOPY EQUILIBRATION	0.0		case-scalar	editable-expert	DEF_ISOTOPE_PARAM_FILE	MOD_MT_Namelist.F90 declaration	MOD_MT_Namelist.F90:133
4	case:DEF_ISO_CG_RELHUM_MAX	DEF_ISO_CG_RELHUM_MAX	DEF ISO CG RELHUM MAX	0.99		case-scalar	editable-expert	DEF_ISOTOPE_PARAM_FILE	MOD_MT_Namelist.F90 declaration	MOD_MT_Namelist.F90:127
5	case:DEF_ISO_ICE_SUPERSAT_SLOPE	DEF_ISO_ICE_SUPERSAT_SLOPE	DEF ISO ICE SUPERSAT SLOPE	0.003		case-scalar	editable-expert	DEF_ISOTOPE_PARAM_FILE	MOD_MT_Namelist.F90 declaration	MOD_MT_Namelist.F90:126
6	case:DEF_ISO_INIT_DELTA_H2_18O	DEF_ISO_INIT_DELTA_H2_18O	DEF ISO INIT DELTA H2 18O	-10.		case-scalar	editable-expert	DEF_ISOTOPE_PARAM_FILE	MOD_MT_Namelist.F90 declaration	MOD_MT_Namelist.F90:141
7	case:DEF_ISO_INIT_DELTA_HDO	DEF_ISO_INIT_DELTA_HDO	DEF ISO INIT DELTA HDO	-70.		case-scalar	editable-expert	DEF_ISOTOPE_PARAM_FILE	MOD_MT_Namelist.F90 declaration	MOD_MT_Namelist.F90:143
8	case:DEF_ISO_KINETIC_SCHEME	DEF_ISO_KINETIC_SCHEME	DEF ISO KINETIC SCHEME	MERLIVAT1978		case-scalar	editable-common	DEF_ISOTOPE_PARAM_FILE	MOD_MT_Namelist.F90 declaration	MOD_MT_Namelist.F90:125
9	case:DEF_ISO_NSS_LEAF_PATH_LENGTH	DEF_ISO_NSS_LEAF_PATH_LENGTH	DEF ISO NSS LEAF PATH LENGTH	0.01		case-scalar	editable-expert	DEF_ISOTOPE_PARAM_FILE	MOD_MT_Namelist.F90 declaration	MOD_MT_Namelist.F90:136
10	case:DEF_ISO_NSS_LEAF_RB	DEF_ISO_NSS_LEAF_RB	DEF ISO NSS LEAF RB	100.		case-scalar	editable-expert	DEF_ISOTOPE_PARAM_FILE	MOD_MT_Namelist.F90 declaration	MOD_MT_Namelist.F90:137
11	case:DEF_ISO_NSS_LEAF_WATER_PER_LAI	DEF_ISO_NSS_LEAF_WATER_PER_LAI	DEF ISO NSS LEAF WATER PER LAI	0.12		case-scalar	editable-expert	DEF_ISOTOPE_PARAM_FILE	MOD_MT_Namelist.F90 declaration	MOD_MT_Namelist.F90:135
12	case:DEF_ISO_OPEN_WATER_KINETIC	DEF_ISO_OPEN_WATER_KINETIC	DEF ISO OPEN WATER KINETIC	MJ79		case-scalar	editable-common	DEF_ISOTOPE_PARAM_FILE	MOD_MT_Namelist.F90 declaration	MOD_MT_Namelist.F90:128
13	case:DEF_ISO_REF_RATIO_H2_18O	DEF_ISO_REF_RATIO_H2_18O	DEF ISO REF RATIO H2 18O	2.0052e-3		case-scalar	editable-expert	DEF_ISOTOPE_PARAM_FILE	MOD_MT_Namelist.F90 declaration	MOD_MT_Namelist.F90:140

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
14	case:DEF_ISO_REF_RATIO_HDO	DEF_ISO_REF_RATIO_HDO	DEF ISO REF RATIO HDO	1.5576e-4		case-scalar	editable-expert	DEF_ISOTOPE_PARAM_FILE	MOD_MT_Namelist.F90 declaration	MOD_MT_Namelist.F90:142
15	case:DEF_ISO_SNOWMELT_EQUILIBRATION	DEF_ISO_SNOWMELT_EQUILIBRATION	DEF ISO SNOWMELT EQUILIBRATION	0.0		case-scalar	editable-expert	DEF_ISOTOPE_PARAM_FILE	MOD_MT_Namelist.F90 declaration	MOD_MT_Namelist.F90:134
16	case:DEF_ISO_SOIL_DIFFUSION	DEF_ISO_SOIL_DIFFUSION	DEF ISO SOIL DIFFUSION	.true.		case-scalar	editable-common	DEF_ISOTOPE_PARAM_FILE	MOD_MT_Namelist.F90 declaration	MOD_MT_Namelist.F90:131
17	case:DEF_ISO_SOIL_KINETIC	DEF_ISO_SOIL_KINETIC	DEF ISO SOIL KINETIC	RESISTANCE		case-scalar	editable-common	DEF_ISOTOPE_PARAM_FILE	MOD_MT_Namelist.F90 declaration	MOD_MT_Namelist.F90:130
18	case:DEF_ISO_SOIL_VAPOR_DIFFUSION	DEF_ISO_SOIL_VAPOR_DIFFUSION	DEF ISO SOIL VAPOR DIFFUSION	.true.		case-scalar	editable-common	DEF_ISOTOPE_PARAM_FILE	MOD_MT_Namelist.F90 declaration	MOD_MT_Namelist.F90:132
19	case:DEF_ISO_SUBL_SKIN_MM	DEF_ISO_SUBL_SKIN_MM	DEF ISO SUBL SKIN MM	5.0		case-scalar	editable-expert	DEF_ISOTOPE_PARAM_FILE	MOD_MT_Namelist.F90 declaration	MOD_MT_Namelist.F90:129
20	case:DEF_ISO_USE_FRACTIONATION	DEF_ISO_USE_FRACTIONATION	DEF ISO USE FRACTIONATION	.false.		case-scalar	editable-common	DEF_ISOTOPE_PARAM_FILE	MOD_MT_Namelist.F90 declaration	MOD_MT_Namelist.F90:124
21	case:DEF_METHANE_PARAM_FILE	DEF_METHANE_PARAM_FILE	DEF METHANE PARAM FILE	null		case-scalar	editable-common	case.nml	MOD_MT_Namelist.F90 declaration	MOD_MT_Namelist.F90:59
22	case:DEF_MT_AQUIFER_MIXING_WATER_MM	DEF_MT_AQUIFER_MIXING_WATER_MM	DEF MT AQUIFER MIXING WATER MM	-1.		case-scalar	editable-expert	case.nml	MOD_MT_Namelist.F90 declaration	MOD_MT_Namelist.F90:67
23	case:DEF_MT_BALANCE_ABORT_NBAD	DEF_MT_BALANCE_ABORT_NBAD	DEF MT BALANCE ABORT NBAD	0		case-scalar	editable-expert	case.nml	MOD_MT_Namelist.F90 declaration	MOD_MT_Namelist.F90:62
24	case:DEF_MT_FORCING_MEMORY_PRECIP	DEF_MT_FORCING_MEMORY_PRECIP	DEF MT FORCING MEMORY PRECIP	2.e-5		case-scalar	editable-expert	case.nml	MOD_MT_Namelist.F90 declaration	MOD_MT_Namelist.F90:88
25	case:DEF_MT_FORCING_MIN_PRECIP	DEF_MT_FORCING_MIN_PRECIP	DEF MT FORCING MIN PRECIP	5.e-6		case-scalar	editable-expert	case.nml	MOD_MT_Namelist.F90 declaration	MOD_MT_Namelist.F90:78

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
26	case:DEF_MT_LULCC_ABORT_NBAD	DEF_MT_LULCC_ABORT_NBAD	DEF MT LULCC ABORT NBAD	0		case-scalar	editable-expert	case.nml	MOD_MT_Namelist.F90 declaration	MOD_MT_Namelist.F90:64
27	case:DEF_MT_N2O_EF5_NH4	DEF_MT_N2O_EF5_NH4	DEF MT N2O EF5 NH4	-1.		case-scalar	editable-expert	case.nml	MOD_MT_Namelist.F90 declaration	MOD_MT_Namelist.F90:117
28	case:DEF_MT_N2O_EF5_NO3	DEF_MT_N2O_EF5_NO3	DEF MT N2O EF5 NO3	-1.		case-scalar	editable-expert	case.nml	MOD_MT_Namelist.F90 declaration	MOD_MT_Namelist.F90:118
29	case:DEF_MT_RESID_ABORT_NBAD	DEF_MT_RESID_ABORT_NBAD	DEF MT RESID ABORT NBAD	0		case-scalar	editable-expert	case.nml	MOD_MT_Namelist.F90 declaration	MOD_MT_Namelist.F90:63
30	case:DEF_MT_SOIL_INIT_FILE	DEF_MT_SOIL_INIT_FILE	DEF MT SOIL INIT FILE	null		case-scalar	editable-common	case.nml	MOD_MT_Namelist.F90 declaration	MOD_MT_Namelist.F90:120
31	case:DEF_MT_SOIL_INIT_VARS	DEF_MT_SOIL_INIT_VARS	DEF MT SOIL INIT VARS	``		case-scalar	editable-common	case.nml	MOD_MT_Namelist.F90 declaration	MOD_MT_Namelist.F90:121
32	case:DEF_MT_USE_SOIL_INIT	DEF_MT_USE_SOIL_INIT	DEF MT USE SOIL INIT	.false.		case-scalar	editable-common	case.nml	MOD_MT_Namelist.F90 declaration	MOD_MT_Namelist.F90:119
33	case:DEF_NH4_LEACHING	DEF_NH4_LEACHING	DEF NH4 LEACHING	auto		case-scalar	editable-common	case.nml	MOD_MT_Namelist.F90 declaration	MOD_MT_Namelist.F90:51
34	case:DEF_SEDIMENT_PARAM_FILE	DEF_SEDIMENT_PARAM_FILE	DEF SEDIMENT PARAM FILE	null		case-scalar	editable-common	case.nml	MOD_MT_Namelist.F90 declaration	MOD_MT_Namelist.F90:56
35	case:DEF_SOLUTE_NAMES	DEF_SOLUTE_NAMES	DEF SOLUTE NAMES	``		case-scalar	editable-common	case.nml	MOD_MT_Namelist.F90 declaration	MOD_MT_Namelist.F90:46
36	case:DEF_SOLUTE_PARAM_FILES	DEF_SOLUTE_PARAM_FILES	DEF SOLUTE PARAM FILES	null		case-scalar	editable-common	case.nml	MOD_MT_Namelist.F90 declaration	MOD_MT_Namelist.F90:55
37	case:DEF_USE_ISOTOPE	DEF_USE_ISOTOPE	DEF USE ISOTOPE	.false.		case-scalar	editable-common	case.nml	MOD_MT_Namelist.F90 declaration	MOD_MT_Namelist.F90:42

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
38	case:DEF_USE_METHANE	DEF_USE_METHANE	DEF USE METHANE	.false.		case-scalar	editable-common	case.nml	MOD_MT_Namelist.F90 declaration	MOD_MT_Namelist.F90:58
39	case:DEF_USE_NH4_LEACHING	DEF_USE_NH4_LEACHING	DEF USE NH4 LEACHING	.false.		case-scalar	read-only-context		MOD_MT_Namelist.F90 declaration	MOD_MT_Namelist.F90:52
40	case:DEF_USE_SEDIMENT	DEF_USE_SEDIMENT	DEF USE SEDIMENT	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:519
41	case:DEF_file_GIEMS	DEF_file_GIEMS	DEF file GIEMS	null		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:533
42	case:DEF_wetland_finundation_scheme	DEF_wetland_finundation_scheme	DEF wetland finundation scheme	1		case-scalar	read-only-context		MOD_Namelist.F90 declaration	MOD_Namelist.F90:534
43	process:material:DEF_ISO_CANOPY_EQUILIBRATION	DEF_ISO_CANOPY_EQUILIBRATION	DEF ISO CANOPY EQUILIBRATION	0.0		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Namelist.F90:133
44	process:material:DEF_ISO_CG_RELHUM_MAX	DEF_ISO_CG_RELHUM_MAX	DEF ISO CG RELHUM MAX	0.99		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Namelist.F90:127
45	process:material:DEF_ISO_ICE_SUPERSAT_SLOPE	DEF_ISO_ICE_SUPERSAT_SLOPE	DEF ISO ICE SUPERSAT SLOPE	0.003		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Namelist.F90:126
46	process:material:DEF_ISO_INIT_DELTA_H2_18O	DEF_ISO_INIT_DELTA_H2_18O	DEF ISO INIT DELTA H2 18O	-10.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Namelist.F90:141
47	process:material:DEF_ISO_INIT_DELTA_HDO	DEF_ISO_INIT_DELTA_HDO	DEF ISO INIT DELTA HDO	-70.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Namelist.F90:143
48	process:material:DEF_ISO_KINETIC_SCHEME	DEF_ISO_KINETIC_SCHEME	DEF ISO KINETIC SCHEME	'MERLIVAT1978'		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Namelist.F90:125
49	process:material:DEF_ISO_NSS_LEAF_PATH_LENGTH	DEF_ISO_NSS_LEAF_PATH_LENGTH	DEF ISO NSS LEAF PATH LENGTH	0.01		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Namelist.F90:136
50	process:material:DEF_ISO_NSS_LEAF_RB	DEF_ISO_NSS_LEAF_RB	DEF ISO NSS LEAF RB	100.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Namelist.F90:137

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
51	process:material:DEF_ISO_NSS_LEAF_WATER_PER_LAI	DEF_ISO_NSS_LEAF_WATER_PER_LAI	DEF ISO NSS LEAF WATER PER LAI	0.12		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Namelist.F90:135
52	process:material:DEF_ISO_OPEN_WATER_KINETIC	DEF_ISO_OPEN_WATER_KINETIC	DEF ISO OPEN WATER KINETIC	'MJ79'		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Namelist.F90:128
53	process:material:DEF_ISO_REF_RATIO_H2_18O	DEF_ISO_REF_RATIO_H2_18O	DEF ISO REF RATIO H2 18O	2.0052e-3		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Namelist.F90:140
54	process:material:DEF_ISO_REF_RATIO_HDO	DEF_ISO_REF_RATIO_HDO	DEF ISO REF RATIO HDO	1.5576e-4		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Namelist.F90:142
55	process:material:DEF_ISO_SNOWMELT_EQUILIBRATION	DEF_ISO_SNOWMELT_EQUILIBRATION	DEF ISO SNOWMELT EQUILIBRATION	0.0		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Namelist.F90:134
56	process:material:DEF_ISO_SOIL_DIFFUSION	DEF_ISO_SOIL_DIFFUSION	DEF ISO SOIL DIFFUSION	.true.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Namelist.F90:131
57	process:material:DEF_ISO_SOIL_KINETIC	DEF_ISO_SOIL_KINETIC	DEF ISO SOIL KINETIC	'RESISTANCE'		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Namelist.F90:130
58	process:material:DEF_ISO_SOIL_VAPOR_DIFFUSION	DEF_ISO_SOIL_VAPOR_DIFFUSION	DEF ISO SOIL VAPOR DIFFUSION	.true.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Namelist.F90:132
59	process:material:DEF_ISO_SUBL_SKIN_MM	DEF_ISO_SUBL_SKIN_MM	DEF ISO SUBL SKIN MM	5.0		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Namelist.F90:129
60	process:material:DEF_ISO_USE_FRACTIONATION	DEF_ISO_USE_FRACTIONATION	DEF ISO USE FRACTIONATION	.false.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Namelist.F90:124
61	process:material:DEF_SOLUTE%charge	DEF_SOLUTE%charge	DEF SOLUTE charge	0		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Solute.F90:34
62	process:material:DEF_SOLUTE%decay_rate	DEF_SOLUTE%decay_rate	DEF SOLUTE decay rate	0.0		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Solute.F90:30

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
63	process:material:DEF_SOLUTE%init_conc	DEF_SOLUTE%init_conc	DEF SOLUTE init conc	0.0		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Solute.F90:26
64	process:material:DEF_SOLUTE%max_dissolved_conc	DEF_SOLUTE%max_dissolved_conc	DEF SOLUTE max dissolved conc	huge(1.0)		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Solute.F90:29
65	process:material:DEF_SOLUTE%mol_weight	DEF_SOLUTE%mol_weight	DEF SOLUTE mol weight	0.0		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Solute.F90:25
66	process:material:DEF_SOLUTE%n2o_ef5	DEF_SOLUTE%n2o_ef5	DEF SOLUTE n2o ef5	0.0		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Solute.F90:33
67	process:material:DEF_SOLUTE%precip_default_conc	DEF_SOLUTE%precip_default_conc	DEF SOLUTE precip default conc	0.0		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Solute.F90:27
68	process:material:DEF_SOLUTE%unit_kind	DEF_SOLUTE%unit_kind	DEF SOLUTE unit kind	'material_per_water'		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Solute.F90:24
69	process:material:DEF_SOLUTE%vapor_default_conc	DEF_SOLUTE%vapor_default_conc	DEF SOLUTE vapor default conc	0.0		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Solute.F90:28
70	process:material:forcing_dtime	forcing_dtime	forcing dtime	21600		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_ForcingInput.F90:nl_colm_mt_forcing
71	process:material:forcing_fprefix	forcing_fprefix	forcing fprefix	'null'		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_ForcingInput.F90:nl_colm_mt_forcing
72	process:material:forcing_input_mode	forcing_input_mode	forcing input mode	'normalized_over_total'		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_ForcingInput.F90:nl_colm_mt_forcing
73	process:material:forcing_num	forcing_num	forcing num	0		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_ForcingInput.F90:nl_colm_mt_forcing
74	process:material:forcing_offset	forcing_offset	forcing offset	0		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_ForcingInput.F90:nl_colm_mt_forcing

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75	process:material:forcing_role	forcing_role	forcing role	'none'		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_ForcingInput.F90:nl_colm_mt_forcing
76	process:material:forcing_tintalgo	forcing_tintalgo	forcing tintalgo	'linear'		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_ForcingInput.F90:nl_colm_mt_forcing
77	process:material:forcing_vname	forcing_vname	forcing vname	'null'		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_ForcingInput.F90:nl_colm_mt_forcing
78	process:methane:DEF_METHANE%B_init_methanogen	DEF_METHANE%B_init_methanogen	DEF METHANE B init methanogen	1.0		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:239
79	process:methane:DEF_METHANE%B_init_methanotroph	DEF_METHANE%B_init_methanotroph	DEF METHANE B init methanotroph	1.0		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:240
80	process:methane:DEF_METHANE%B_max_fraction_methanogen	DEF_METHANE%B_max_fraction_methanogen	DEF METHANE B max fraction methanogen	-1.0		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:246
81	process:methane:DEF_METHANE%B_max_fraction_methanotroph	DEF_METHANE%B_max_fraction_methanotroph	DEF METHANE B max fraction methanotroph	-1.0		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:247
82	process:methane:DEF_METHANE%B_min_methanogen	DEF_METHANE%B_min_methanogen	DEF METHANE B min methanogen	1.0e-3		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:241
83	process:methane:DEF_METHANE%B_min_methanotroph	DEF_METHANE%B_min_methanotroph	DEF METHANE B min methanotroph	1.0e-3		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:242
84	process:methane:DEF_METHANE%K_inh_O2_methanogen	DEF_METHANE%K_inh_O2_methanogen	DEF METHANE K inh O2 methanogen	1.0e-3		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:255
85	process:methane:DEF_METHANE%K_substrate_methanogen_pool	DEF_METHANE%K_substrate_methanogen_pool	DEF METHANE K substrate methanogen pool	0.04		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:254
86	process:methane:DEF_METHANE%T_ref_microbe	DEF_METHANE%T_ref_microbe	DEF METHANE T ref microbe	298.15		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:262

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
87	process:methane:DEF_METHANE%aere_radius	DEF_METHANE%aere_radius	DEF METHANE aere radius	2.9e-3		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:278
88	process:methane:DEF_METHANE%aereoxid	DEF_METHANE%aereoxid	DEF METHANE aereoxid	0.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:312
89	process:methane:DEF_METHANE%allowlakeprod	DEF_METHANE%allowlakeprod	DEF METHANE allowlakeprod	.false.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:337
90	process:methane:DEF_METHANE%anoxia	DEF_METHANE%anoxia	DEF METHANE anoxia	.true.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:406
91	process:methane:DEF_METHANE%anoxicmicrosites	DEF_METHANE%anoxicmicrosites	DEF METHANE anoxicmicrosites	.false.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:396
92	process:methane:DEF_METHANE%atm_methane	DEF_METHANE%atm_methane	DEF METHANE atm methane	1.7e-6		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:320
93	process:methane:DEF_METHANE%atm_methane_file	DEF_METHANE%atm_methane_file	DEF METHANE atm methane file	'null'		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:322
94	process:methane:DEF_METHANE%atm_methane_file_units	DEF_METHANE%atm_methane_file_units	DEF METHANE atm methane file units	'auto'		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:323
95	process:methane:DEF_METHANE%bgc_anoxia_limits_decomp	DEF_METHANE%bgc_anoxia_limits_decomp	DEF METHANE bgc anoxia limits decomp	.false.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:412
96	process:methane:DEF_METHANE%bubble_f	DEF_METHANE%bubble_f	DEF METHANE bubble f	0.57		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:309
97	process:methane:DEF_METHANE%capthick	DEF_METHANE%capthick	DEF METHANE capthick	100.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:317
98	process:methane:DEF_METHANE%ch4_history_vars	DEF_METHANE%ch4_history_vars	DEF METHANE ch4 history vars	'core'		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:426

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
99	process:methane:DEF_METHANE%cnscalefactor	DEF_METHANE%cnscalefactor	DEF METHANE cnscalefactor	1.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:296
100	process:methane:DEF_METHANE%dormancy_rate_active	DEF_METHANE%dormancy_rate_active	DEF METHANE dormancy rate active	0.1		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:263
101	process:methane:DEF_METHANE%dormancy_rate_revive	DEF_METHANE%dormancy_rate_revive	DEF METHANE dormancy rate revive	1.0		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:264
102	process:methane:DEF_METHANE%dormancy_threshold_methanogen_fO2	DEF_METHANE%dormancy_threshold_methanogen_fO2	DEF METHANE dormancy threshold methanogen fO2	0.1		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:266
103	process:methane:DEF_METHANE%dormancy_threshold_methanogen_fS	DEF_METHANE%dormancy_threshold_methanogen_fS	DEF METHANE dormancy threshold methanogen fS	0.1		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:265
104	process:methane:DEF_METHANE%dormancy_threshold_methanotroph_fO2	DEF_METHANE%dormancy_threshold_methanotroph_fO2	DEF METHANE dormancy threshold methanotroph fO2	0.1		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:268
105	process:methane:DEF_METHANE%dormancy_threshold_methanotroph_fS	DEF_METHANE%dormancy_threshold_methanotroph_fS	DEF METHANE dormancy threshold methanotroph fS	0.1		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:267
106	process:methane:DEF_METHANE%enable_rice_paddy	DEF_METHANE%enable_rice_paddy	DEF METHANE enable rice paddy	.false.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:152
107	process:methane:DEF_METHANE%enable_wetwat_finundated_override	DEF_METHANE%enable_wetwat_finundated_override	DEF METHANE enable wetwat finundated override	.false.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:467
108	process:methane:DEF_METHANE%f_methane	DEF_METHANE%f_methane	DEF METHANE f methane	0.2		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:159
109	process:methane:DEF_METHANE%f_methane_boreal_bog	DEF_METHANE%f_methane_boreal_bog	DEF METHANE f methane boreal bog	0.08		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:178
110	process:methane:DEF_METHANE%f_methane_boreal_fen	DEF_METHANE%f_methane_boreal_fen	DEF METHANE f methane boreal fen	0.12		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:177

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111	process:methane:DEF_METHANE%f_methane_floodplain	DEF_METHANE%f_methane_floodplain	DEF METHANE f methane floodplain	0.10		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:175
112	process:methane:DEF_METHANE%f_methane_rice_paddy	DEF_METHANE%f_methane_rice_paddy	DEF METHANE f methane rice paddy	0.10		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:179
113	process:methane:DEF_METHANE%f_methane_temperate_marsh	DEF_METHANE%f_methane_temperate_marsh	DEF METHANE f methane temperate marsh	0.15		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:176
114	process:methane:DEF_METHANE%f_methane_tropical_floodplain	DEF_METHANE%f_methane_tropical_floodplain	DEF METHANE f methane tropical floodplain	0.07		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:174
115	process:methane:DEF_METHANE%f_methane_tropical_peat	DEF_METHANE%f_methane_tropical_peat	DEF METHANE f methane tropical peat	0.12		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:173
116	process:methane:DEF_METHANE%f_methane_upland_soil	DEF_METHANE%f_methane_upland_soil	DEF METHANE f methane upland soil	0.05		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:180
117	process:methane:DEF_METHANE%gamma_methanogen	DEF_METHANE%gamma_methanogen	DEF METHANE gamma methanogen	0.05		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:250
118	process:methane:DEF_METHANE%gamma_methanotroph	DEF_METHANE%gamma_methanotroph	DEF METHANE gamma methanotroph	0.10		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:251
119	process:methane:DEF_METHANE%gamma_microbial_dormant	DEF_METHANE%gamma_microbial_dormant	DEF METHANE gamma microbial dormant	0.005		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:252
120	process:methane:DEF_METHANE%gamma_microbial_freeze	DEF_METHANE%gamma_microbial_freeze	DEF METHANE gamma microbial freeze	0.001		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:253
121	process:methane:DEF_METHANE%grnd_methane_cond_default	DEF_METHANE%grnd_methane_cond_default	DEF METHANE grnd methane cond default	1.e-6		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:289
122	process:methane:DEF_METHANE%host_water_tolerance	DEF_METHANE%host_water_tolerance	DEF METHANE host water tolerance	0.05		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:449

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123	process:methane:DEF_METHANE%hybrid_soil_threshold	DEF_METHANE%hybrid_soil_threshold	DEF METHANE hybrid soil threshold	0.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:381
124	process:methane:DEF_METHANE%inundation_mode	DEF_METHANE%inundation_mode	DEF METHANE inundation mode	'hybrid'		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:151
125	process:methane:DEF_METHANE%k_m	DEF_METHANE%k_m	DEF METHANE k m	5.e-3		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:222
126	process:methane:DEF_METHANE%k_m_o2	DEF_METHANE%k_m_o2	DEF METHANE k m o2	2.e-2		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:224
127	process:methane:DEF_METHANE%k_m_unsat	DEF_METHANE%k_m_unsat	DEF METHANE k m unsat	5.e-4		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:223
128	process:methane:DEF_METHANE%kappa_m_methanogen	DEF_METHANE%kappa_m_methanogen	DEF METHANE kappa m methanogen	1.0e-2		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:256
129	process:methane:DEF_METHANE%kappa_m_methanotroph	DEF_METHANE%kappa_m_methanotroph	DEF METHANE kappa m methanotroph	1.0e-2		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:257
130	process:methane:DEF_METHANE%lake_decomp_fact	DEF_METHANE%lake_decomp_fact	DEF METHANE lake decomp fact	9.e-11		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:299
131	process:methane:DEF_METHANE%lake_k_m_o2	DEF_METHANE%lake_k_m_o2	DEF METHANE lake k m o2	-1.0		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:230
132	process:methane:DEF_METHANE%lake_liqdiff_scale	DEF_METHANE%lake_liqdiff_scale	DEF METHANE lake liqdiff scale	1.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:285
133	process:methane:DEF_METHANE%lake_o2_liqdiff_scale	DEF_METHANE%lake_o2_liqdiff_scale	DEF METHANE lake o2 liqdiff scale	1.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:286
134	process:methane:DEF_METHANE%lake_oxic_sediment_depth	DEF_METHANE%lake_oxic_sediment_depth	DEF METHANE lake oxic sediment depth	-1.0		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:232

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135	process:methane:DEF_METHANE%lake_oxid_scale	DEF_METHANE%lake_oxid_scale	DEF METHANE lake oxid scale	1.0		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:229
136	process:methane:DEF_METHANE%lake_restart_debug	DEF_METHANE%lake_restart_debug	DEF METHANE lake restart debug	.false.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:458
137	process:methane:DEF_METHANE%lake_restart_debug_end_doy	DEF_METHANE%lake_restart_debug_end_doy	DEF METHANE lake restart debug end doy	3		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:461
138	process:methane:DEF_METHANE%lake_restart_debug_sec	DEF_METHANE%lake_restart_debug_sec	DEF METHANE lake restart debug sec	1800		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:462
139	process:methane:DEF_METHANE%lake_restart_debug_start_doy	DEF_METHANE%lake_restart_debug_start_doy	DEF METHANE lake restart debug start doy	1		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:460
140	process:methane:DEF_METHANE%lake_restart_debug_year	DEF_METHANE%lake_restart_debug_year	DEF METHANE lake restart debug year	1996		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:459
141	process:methane:DEF_METHANE%lake_vmax_methane_oxid	DEF_METHANE%lake_vmax_methane_oxid	DEF METHANE lake vmax methane oxid	-1.0		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:231
142	process:methane:DEF_METHANE%lake_zero_depth_fatal	DEF_METHANE%lake_zero_depth_fatal	DEF METHANE lake zero depth fatal	.false.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:453
143	process:methane:DEF_METHANE%max_microbe_prod_multiplier	DEF_METHANE%max_microbe_prod_multiplier	DEF METHANE max microbe prod multiplier	3.0		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:260
144	process:methane:DEF_METHANE%methane_frzout	DEF_METHANE%methane_frzout	DEF METHANE methane frzout	.false.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:399
145	process:methane:DEF_METHANE%methane_offline	DEF_METHANE%methane_offline	DEF METHANE methane offline	.true.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:387
146	process:methane:DEF_METHANE%methane_rmcnlim	DEF_METHANE%methane_rmcnlim	DEF METHANE methane rmcnlim	.false.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:391

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
147	process:methane:DEF_METHANE%mino2lim	DEF_METHANE%mino2lim	DEF METHANE mino2lim	0.2		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:293
148	process:methane:DEF_METHANE%mu_max_methanogen	DEF_METHANE%mu_max_methanogen	DEF METHANE mu max methanogen	0.2		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:248
149	process:methane:DEF_METHANE%mu_max_methanotroph	DEF_METHANE%mu_max_methanotroph	DEF METHANE mu max methanotroph	0.5		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:249
150	process:methane:DEF_METHANE%nongrassporosratio	DEF_METHANE%nongrassporosratio	DEF METHANE nongrassporosratio	0.333333333333		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:274
151	process:methane:DEF_METHANE%numerical_correction_fatal_threshold	DEF_METHANE%numerical_correction_fatal_threshold	DEF METHANE numerical correction fatal threshold	-1.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:430
152	process:methane:DEF_METHANE%om_frac_sf	DEF_METHANE%om_frac_sf	DEF METHANE om frac sf	1.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:324
153	process:methane:DEF_METHANE%only_wetland	DEF_METHANE%only_wetland	DEF METHANE only wetland	.false.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:153
154	process:methane:DEF_METHANE%oxinhib	DEF_METHANE%oxinhib	DEF METHANE oxinhib	400.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:303
155	process:methane:DEF_METHANE%pHmax	DEF_METHANE%pHmax	DEF METHANE pHmax	9.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:301
156	process:methane:DEF_METHANE%pHmin	DEF_METHANE%pHmin	DEF METHANE pHmin	2.2		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:302
157	process:methane:DEF_METHANE%poros_tiller	DEF_METHANE%poros_tiller	DEF METHANE poros tiller	0.3		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:275
158	process:methane:DEF_METHANE%porosmin	DEF_METHANE%porosmin	DEF METHANE porosmin	0.05		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:277

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
159	process:methane:DEF_METHANE%q10_methane_oxid	DEF_METHANE%q10_methane_oxid	DEF METHANE q10 methane oxid	1.9		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:225
160	process:methane:DEF_METHANE%q10_microbe_growth	DEF_METHANE%q10_microbe_growth	DEF METHANE q10 microbe growth	3.0		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:261
161	process:methane:DEF_METHANE%q10lakebase	DEF_METHANE%q10lakebase	DEF METHANE q10lakebase	298.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:295
162	process:methane:DEF_METHANE%q10methane	DEF_METHANE%q10methane	DEF METHANE q10methane	2.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:158
163	process:methane:DEF_METHANE%q10methane_base	DEF_METHANE%q10methane_base	DEF METHANE q10methane base	295.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:294
164	process:methane:DEF_METHANE%redoxlag	DEF_METHANE%redoxlag	DEF METHANE redoxlag	30.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:298
165	process:methane:DEF_METHANE%redoxlag_boreal_bog	DEF_METHANE%redoxlag_boreal_bog	DEF METHANE redoxlag boreal bog	45.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:198
166	process:methane:DEF_METHANE%redoxlag_boreal_fen	DEF_METHANE%redoxlag_boreal_fen	DEF METHANE redoxlag boreal fen	30.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:197
167	process:methane:DEF_METHANE%redoxlag_rice_paddy	DEF_METHANE%redoxlag_rice_paddy	DEF METHANE redoxlag rice paddy	25.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:199
168	process:methane:DEF_METHANE%redoxlag_temperate_marsh	DEF_METHANE%redoxlag_temperate_marsh	DEF METHANE redoxlag temperate marsh	20.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:196
169	process:methane:DEF_METHANE%redoxlag_tropical_floodplain	DEF_METHANE%redoxlag_tropical_floodplain	DEF METHANE redoxlag tropical floodplain	7.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:195
170	process:methane:DEF_METHANE%redoxlag_tropical_peat	DEF_METHANE%redoxlag_tropical_peat	DEF METHANE redoxlag tropical peat	12.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:194

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
171	process:methane:DEF_METHANE%redoxlag_upland_soil	DEF_METHANE%redoxlag_upland_soil	DEF METHANE redoxlag upland soil	30.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:200
172	process:methane:DEF_METHANE%redoxlag_vertical	DEF_METHANE%redoxlag_vertical	DEF METHANE redoxlag vertical	30.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:300
173	process:methane:DEF_METHANE%replenishlakec	DEF_METHANE%replenishlakec	DEF METHANE replenishlakec	.false.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:383
174	process:methane:DEF_METHANE%rice_drain_window_days	DEF_METHANE%rice_drain_window_days	DEF METHANE rice drain window days	30.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:476
175	process:methane:DEF_METHANE%rice_substrate_boost	DEF_METHANE%rice_substrate_boost	DEF METHANE rice substrate boost	1.0		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:491
176	process:methane:DEF_METHANE%rob	DEF_METHANE%rob	DEF METHANE rob	3.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:279
177	process:methane:DEF_METHANE%satpow	DEF_METHANE%satpow	DEF METHANE satpow	2.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:316
178	process:methane:DEF_METHANE%scale_factor_aere	DEF_METHANE%scale_factor_aere	DEF METHANE scale factor aere	1.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:280
179	process:methane:DEF_METHANE%scale_factor_gasdiff	DEF_METHANE%scale_factor_gasdiff	DEF METHANE scale factor gasdiff	1.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:283
180	process:methane:DEF_METHANE%scale_factor_liqdiff	DEF_METHANE%scale_factor_liqdiff	DEF METHANE scale factor liqdiff	1.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:284
181	process:methane:DEF_METHANE%smp_crit	DEF_METHANE%smp_crit	DEF METHANE smp crit	-2.4e5		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:306
182	process:methane:DEF_METHANE%tiller_C	DEF_METHANE%tiller_C	DEF METHANE tiller C	0.22		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:313

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
183	process:methane:DEF_METHANE%transpirationloss	DEF_METHANE%transpirationloss	DEF METHANE transpirationloss	.true.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:331
184	process:methane:DEF_METHANE%unsat_aere_ratio	DEF_METHANE%unsat_aere_ratio	DEF METHANE unsat aere ratio	0.166666666667		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:276
185	process:methane:DEF_METHANE%use_aereoxid_prog	DEF_METHANE%use_aereoxid_prog	DEF METHANE use aereoxid prog	.true.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:327
186	process:methane:DEF_METHANE%use_biome_f_methane	DEF_METHANE%use_biome_f_methane	DEF METHANE use biome f methane	.false.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:172
187	process:methane:DEF_METHANE%use_biome_redoxlag	DEF_METHANE%use_biome_redoxlag	DEF METHANE use biome redoxlag	.false.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:193
188	process:methane:DEF_METHANE%use_ch4_sif	DEF_METHANE%use_ch4_sif	DEF METHANE use ch4 sif	.true.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:413
189	process:methane:DEF_METHANE%use_microbial_dormancy	DEF_METHANE%use_microbial_dormancy	DEF METHANE use microbial dormancy	.false.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:238
190	process:methane:DEF_METHANE%use_microbial_flux_override	DEF_METHANE%use_microbial_flux_override	DEF METHANE use microbial flux override	.false.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:237
191	process:methane:DEF_METHANE%use_microbial_pools	DEF_METHANE%use_microbial_pools	DEF METHANE use microbial pools	.false.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:236
192	process:methane:DEF_METHANE%use_nitrif_denitrif	DEF_METHANE%use_nitrif_denitrif	DEF METHANE use nitrif denitrif	.true.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:404
193	process:methane:DEF_METHANE%use_routing_for_soil	DEF_METHANE%use_routing_for_soil	DEF METHANE use routing for soil	.false.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:368
194	process:methane:DEF_METHANE%use_spatial_ph	DEF_METHANE%use_spatial_ph	DEF METHANE use spatial ph	.false.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:154

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
195	process:methane:DEF_METHANE%use_transient_atm_methane	DEF_METHANE%use_transient_atm_methane	DEF METHANE use transient atm methane	.false.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:321
196	process:methane:DEF_METHANE%use_vertical_redoxlag	DEF_METHANE%use_vertical_redoxlag	DEF METHANE use vertical redoxlag	.true.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:409
197	process:methane:DEF_METHANE%usephfact	DEF_METHANE%usephfact	DEF METHANE usephfact	.true.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:343
198	process:methane:DEF_METHANE%vgc_max	DEF_METHANE%vgc_max	DEF METHANE vgc max	0.15		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:271
199	process:methane:DEF_METHANE%vmax_methane_oxid	DEF_METHANE%vmax_methane_oxid	DEF METHANE vmax methane oxid	1.25e-5		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:220
200	process:methane:DEF_METHANE%vmax_oxid_unsat	DEF_METHANE%vmax_oxid_unsat	DEF METHANE vmax oxid unsat	1.25e-6		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:221
201	process:methane:DEF_METHANE%wetland_dry_unsat_branch	DEF_METHANE%wetland_dry_unsat_branch	DEF METHANE wetland dry unsat branch	.false.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:472
202	process:methane:DEF_METHANE%write_ch4_history	DEF_METHANE%write_ch4_history	DEF METHANE write ch4 history	.true.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:425
203	process:methane:DEF_METHANE%wtd_inflection	DEF_METHANE%wtd_inflection	DEF METHANE wtd inflection	0.30		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:352
204	process:methane:DEF_METHANE%wtd_inflection_soil	DEF_METHANE%wtd_inflection_soil	DEF METHANE wtd inflection soil	0.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:359
205	process:methane:DEF_METHANE%wtd_steepness	DEF_METHANE%wtd_steepness	DEF METHANE wtd steepness	0.05		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:353
206	process:methane:DEF_METHANE%wtd_steepness_soil	DEF_METHANE%wtd_steepness_soil	DEF METHANE wtd steepness soil	0.3		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:360

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
207	process:methane:DEF_METHANE%z0_methane_prod	DEF_METHANE%z0_methane_prod	DEF METHANE z0 methane prod	0.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:215
208	process:methane:DEF_METHANE_hydrology%pc	DEF_METHANE_hydrology%pc	DEF METHANE hydrology pc	0.4		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:501
209	process:methane:DEF_METHANE_hydrology%slopebeta	DEF_METHANE_hydrology%slopebeta	DEF METHANE hydrology slopebeta	-3.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:499
210	process:methane:DEF_METHANE_hydrology%slopemax	DEF_METHANE_hydrology%slopemax	DEF METHANE hydrology slopemax	0.4		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:500
211	process:methane:DEF_METHANE_hydrology%vdcf	DEF_METHANE_hydrology%vdcf	DEF METHANE hydrology vdcf	2.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Methane_Const.F90:498
212	process:sediment:DEF_SEDIMENT%active_layer_depth	DEF_SEDIMENT%active_layer_depth	DEF SEDIMENT active layer depth	-1.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Sediment.F90:216
213	process:sediment:DEF_SEDIMENT%bed_depth	DEF_SEDIMENT%bed_depth	DEF SEDIMENT bed depth	-1.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Sediment.F90:226
214	process:sediment:DEF_SEDIMENT%cfl_adv	DEF_SEDIMENT%cfl_adv	DEF SEDIMENT cfl adv	-1.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Sediment.F90:224
215	process:sediment:DEF_SEDIMENT%grain_density	DEF_SEDIMENT%grain_density	DEF SEDIMENT grain density	-1.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Sediment.F90:211
216	process:sediment:DEF_SEDIMENT%grain_diameter	DEF_SEDIMENT%grain_diameter	DEF SEDIMENT grain diameter	-1.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Sediment.F90:210
217	process:sediment:DEF_SEDIMENT%ignore_depth_m	DEF_SEDIMENT%ignore_depth_m	DEF SEDIMENT ignore depth m	-1.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Sediment.F90:215
218	process:sediment:DEF_SEDIMENT%max_timestep_s	DEF_SEDIMENT%max_timestep_s	DEF SEDIMENT max timestep s	-1.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Sediment.F90:225

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
219	process:sediment:DEF_SEDIMENT%ndeposit_layers	DEF_SEDIMENT%ndeposit_layers	DEF SEDIMENT ndeposit layers	-1		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Sediment.F90:214
220	process:sediment:DEF_SEDIMENT%nsed	DEF_SEDIMENT%nsed	DEF SEDIMENT nsed	-1		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Sediment.F90:209
221	process:sediment:DEF_SEDIMENT%porosity	DEF_SEDIMENT%porosity	DEF SEDIMENT porosity	-1.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Sediment.F90:213
222	process:sediment:DEF_SEDIMENT%precipitation_exponent	DEF_SEDIMENT%precipitation_exponent	DEF SEDIMENT precipitation exponent	-1.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Sediment.F90:222
223	process:sediment:DEF_SEDIMENT%settling_multiplier	DEF_SEDIMENT%settling_multiplier	DEF SEDIMENT settling multiplier	-1.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Sediment.F90:219
224	process:sediment:DEF_SEDIMENT%slope_exponent	DEF_SEDIMENT%slope_exponent	DEF SEDIMENT slope exponent	-1.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Sediment.F90:221
225	process:sediment:DEF_SEDIMENT%unit_conversion	DEF_SEDIMENT%unit_conversion	DEF SEDIMENT unit conversion	-1.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Sediment.F90:223
226	process:sediment:DEF_SEDIMENT%viscosity	DEF_SEDIMENT%viscosity	DEF SEDIMENT viscosity	-1.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Sediment.F90:217
227	process:sediment:DEF_SEDIMENT%von_karman	DEF_SEDIMENT%von_karman	DEF SEDIMENT von karman	-1.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Sediment.F90:218
228	process:sediment:DEF_SEDIMENT%water_density	DEF_SEDIMENT%water_density	DEF SEDIMENT water density	-1.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Sediment.F90:212
229	process:sediment:DEF_SEDIMENT%yield_coefficient	DEF_SEDIMENT%yield_coefficient	DEF SEDIMENT yield coefficient	-1.		process-file	editable-expert		Fortran process type declaration or initialization	MOD_MT_Sediment.F90:220

水热过程（36 项）

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
1	case:DEF_CheckEquilibrium	DEF_CheckEquilibrium	DEF CheckEquilibrium	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:712
2	case:DEF_Interception_scheme	DEF_Interception_scheme	DEF Interception scheme	1		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:366
3	case:DEF_Optimize_Baseflow	DEF_Optimize_Baseflow	DEF Optimize Baseflow	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:817
4	case:DEF_RSS_SCHEME	DEF_RSS_SCHEME	DEF RSS SCHEME	1		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:389
5	case:DEF_RS_CROWN_STRUCTURE	DEF_RS_CROWN_STRUCTURE	DEF RS CROWN STRUCTURE	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:274
6	case:DEF_Runoff_SCHEME	DEF_Runoff_SCHEME	DEF Runoff SCHEME	3		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:397
7	case:DEF_SPLIT_SOILSNOW	DEF_SPLIT_SOILSNOW	DEF SPLIT SOILSNOW	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:408
8	case:DEF_THERMAL_CONDUCTIVITY_SCHEME	DEF_THERMAL_CONDUCTIVITY_SCHEME	DEF THERMAL CONDUCTIVITY SCHEME	4		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:369
9	case:DEF_TOPMOD_method	DEF_TOPMOD_method	DEF TOPMOD method	0		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:402
10	case:DEF_TUNING_CAPR	DEF_TUNING_CAPR	DEF TUNING CAPR	0.34		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:630
11	case:DEF_TUNING_CNFAC	DEF_TUNING_CNFAC	DEF TUNING CNFAC	0.5		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:631
12	case:DEF_TUNING_PONDMX	DEF_TUNING_PONDMX	DEF TUNING PONDMX	10.0		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:634
13	case:DEF_TUNING_SIMPLE_VIC_DS	DEF_TUNING_SIMPLE_VIC_DS	DEF TUNING SIMPLE VIC DS	0.061		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:643
14	case:DEF_TUNING_SIMPLE_VIC_WS	DEF_TUNING_SIMPLE_VIC_WS	DEF TUNING SIMPLE VIC WS	0.646		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:644
15	case:DEF_TUNING_SMPMAX	DEF_TUNING_SMPMAX	DEF TUNING SMPMAX	-1.5e5		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:635
16	case:DEF_TUNING_SMPMAX_HR	DEF_TUNING_SMPMAX_HR	DEF TUNING SMPMAX HR	-2.e2		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:637

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
17	case:DEF_TUNING_SMPMIN	DEF_TUNING_SMPMIN	DEF TUNING SMPMIN	-1.e8		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:636
18	case:DEF_TUNING_SMPMIN_HR	DEF_TUNING_SMPMIN_HR	DEF TUNING SMPMIN HR	-2.e5		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:638
19	case:DEF_TUNING_SNOW_COVER_EXPONENT	DEF_TUNING_SNOW_COVER_EXPONENT	DEF TUNING SNOW COVER EXPONENT	1.0		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:645
20	case:DEF_TUNING_SOIL_ICE_IMPEDANCE	DEF_TUNING_SOIL_ICE_IMPEDANCE	DEF TUNING SOIL ICE IMPEDANCE	6.0		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:641
21	case:DEF_TUNING_SSI	DEF_TUNING_SSI	DEF TUNING SSI	0.033		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:632
22	case:DEF_TUNING_TOPMOD_DECAY	DEF_TUNING_TOPMOD_DECAY	DEF TUNING TOPMOD DECAY	2.0		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:642
23	case:DEF_TUNING_WETWATMAX	DEF_TUNING_WETWATMAX	DEF TUNING WETWATMAX	200.0		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:640
24	case:DEF_TUNING_WIMP	DEF_TUNING_WIMP	DEF TUNING WIMP	0.05		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:633
25	case:DEF_TUNING_ZLND	DEF_TUNING_ZLND	DEF TUNING ZLND	0.01		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:626
26	case:DEF_TUNING_ZSNO	DEF_TUNING_ZSNO	DEF TUNING ZSNO	0.0024		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:627
27	case:DEF_USE_BEDROCK	DEF_USE_BEDROCK	DEF USE BEDROCK	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:417
28	case:DEF_USE_Campbell_SOIL_MODEL	DEF_USE_Campbell_SOIL_MODEL	DEF USE Campbell SOIL MODEL	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:429
29	case:DEF_USE_Dynamic_Lake	DEF_USE_Dynamic_Lake	DEF USE Dynamic Lake	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:708
30	case:DEF_USE_Dynamic_Wetland	DEF_USE_Dynamic_Wetland	DEF USE Dynamic Wetland	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:710
31	case:DEF_USE_SUPERCOOL_WATER	DEF_USE_SUPERCOOL_WATER	DEF USE SUPERCOOL WATER	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:379
32	case:DEF_USE_VariablySaturatedFlow	DEF_USE_VariablySaturatedFlow	DEF USE VariablySaturatedFlow	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:416
33	case:DEF_VIC_OPT	DEF_VIC_OPT	DEF VIC OPT	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:400

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
34	case:DEF_file_VIC_OPT	DEF_file_VIC_OPT	DEF file VIC OPT	null		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:399
35	case:DEF_file_VIC_para	DEF_file_VIC_para	DEF file VIC para	null		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:398
36	case:DEF_precip_phase_discrimination_scheme	DEF_precip_phase_discrimination_scheme	DEF precip phase discrimination scheme	I		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:537

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#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
1	case:DEF_CaMa_FloodFeedback	DEF_CaMa_FloodFeedback	DEF CaMa FloodFeedback	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:488
2	case:DEF_CaMa_Namelist	DEF_CaMa_Namelist	DEF CaMa Namelist	null		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:486
3	case:DEF_CaMa_Restart_file	DEF_CaMa_Restart_file	DEF CaMa Restart file	null		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:487
4	case:DEF_CaMa_StrictDomain	DEF_CaMa_StrictDomain	DEF CaMa StrictDomain	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:489
5	case:DEF_ElementNeighbour_file	DEF_ElementNeighbour_file	DEF ElementNeighbour file	null		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:496
6	case:DEF_GRIDBASED_ROUTING_MAX_DT	DEF_GRIDBASED_ROUTING_MAX_DT	DEF GRIDBASED ROUTING MAX DT	3600.		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:505
7	case:DEF_GRIDBASED_ROUTING_MOMENTUM_DT_LIMIT	DEF_GRIDBASED_ROUTING_MOMENTUM_DT_LIMIT	DEF GRIDBASED ROUTING MOMENTUM DT LIMIT	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:506
8	case:DEF_GRIDBASED_ROUTING_STORAGE_DT_LIMIT	DEF_GRIDBASED_ROUTING_STORAGE_DT_LIMIT	DEF GRIDBASED ROUTING STORAGE DT LIMIT	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:510
9	case:DEF_GridRiverLake_FloodFeedback	DEF_GridRiverLake_FloodFeedback	DEF GridRiverLake FloodFeedback	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:511
10	case:DEF_GridRiverLake_FloodInfiltMax	DEF_GridRiverLake_FloodInfiltMax	DEF GridRiverLake FloodInfiltMax	5.		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:512
11	case:DEF_ReservoirPara_file	DEF_ReservoirPara_file	DEF ReservoirPara file	null		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:500
12	case:DEF_Reservoir_Method	DEF_Reservoir_Method	DEF Reservoir Method	0		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:504

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
13	case:DEF_USE_BIFURCATION	DEF_USE_BIFURCATION	DEF USE BIFURCATION	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:514
14	case:DEF_USE_EstimatedRiverDepth	DEF_USE_EstimatedRiverDepth	DEF USE EstimatedRiverDepth	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:502
15	case:DEF_USE_GridRiverLakeFlow	DEF_USE_GridRiverLakeFlow	DEF USE GridRiverLakeFlow	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:495
16	case:DEF_USE_LEVEE	DEF_USE_LEVEE	DEF USE LEVEE	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:513
17	case:DEF_UnitCatchment_file	DEF_UnitCatchment_file	DEF UnitCatchment file	null		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:497
18	case:DEF_UnitCatchment_regional	DEF_UnitCatchment_regional	DEF UnitCatchment regional	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:498

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#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
1	case:DEF_Aerosol_Clim	DEF_Aerosol_Clim	DEF Aerosol Clim	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:477
2	case:DEF_Aerosol_Readin	DEF_Aerosol_Readin	DEF Aerosol Readin	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:474
3	case:DEF_BALL_BERRY_BINTER	DEF_BALL_BERRY_BINTER	DEF BALL BERRY BINTER	-1.0		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:552
4	case:DEF_BALL_BERRY_GRADM	DEF_BALL_BERRY_GRADM	DEF BALL BERRY GRADM	-1.0		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:551
5	case:DEF_FERT_SOURCE	DEF_FERT_SOURCE	DEF FERT SOURCE	1		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:702
6	case:DEF_IRRIGATION_ALLOCATION	DEF_IRRIGATION_ALLOCATION	DEF IRRIGATION ALLOCATION	1		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:542
7	case:DEF_LC_YEAR	DEF_LC_YEAR	DEF LC YEAR	2005		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:249
8	case:DEF_MEDLYN_G0	DEF_MEDLYN_G0	DEF MEDLYN G0	-1.0		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:554
9	case:DEF_MEDLYN_G1	DEF_MEDLYN_G1	DEF MEDLYN G1	-1.0		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:553

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
10	case:DEF_NDEP_FREQUENCY	DEF_NDEP_FREQUENCY	DEF NDEP FREQUENCY	1		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:483
11	case:DEF_OZONE_KO3	DEF_OZONE_KO3	DEF OZONE KO3	1.51		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:662
12	case:DEF_PH_CROOT_LATERAL_LENGTH	DEF_PH_CROOT_LATERAL_LENGTH	DEF PH CROOT LATERAL LENGTH	0.25		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:655
13	case:DEF_PH_FROOT_CARBON	DEF_PH_FROOT_CARBON	DEF PH FROOT CARBON	288.392056287006		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:657
14	case:DEF_PH_FROOT_LEAF	DEF_PH_FROOT_LEAF	DEF PH FROOT LEAF	1.5		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:660
15	case:DEF_PH_KRMAX	DEF_PH_KRMAX	DEF PH KRMAX	3.981071705534969e-009		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:661
16	case:DEF_PH_K_AXS	DEF_PH_K_AXS	DEF PH K AXS	2.0e-1		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:656
17	case:DEF_PH_ROOT_DENSITY	DEF_PH_ROOT_DENSITY	DEF PH ROOT DENSITY	310000.		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:659
18	case:DEF_PH_ROOT_RADIUS	DEF_PH_ROOT_RADIUS	DEF PH ROOT RADIUS	2.9e-4		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:658
19	case:DEF_PROSPECT	DEF_PROSPECT	DEF PROSPECT	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:465
20	case:DEF_RSTFAC	DEF_RSTFAC	DEF RSTFAC	1		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:545
21	case:DEF_SSP	DEF_SSP	DEF SSP	off		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:539
22	case:DEF_TUNING_CROP_PLANTING_DAY	DEF_TUNING_CROP_PLANTING_DAY	DEF TUNING CROP PLANTING DAY	0.0		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:654
23	case:DEF_TUNING_CSOILC	DEF_TUNING_CSOILC	DEF TUNING CSOILC	0.004		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:628
24	case:DEF_TUNING_DEWMX	DEF_TUNING_DEWMX	DEF TUNING DEWMX	0.1		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:629
25	case:DEF_TUNING_IRRIGATION_DURATION_SEC	DEF_TUNING_IRRIGATION_DURATION_SEC	DEF TUNING IRRIGATION DURATION SEC	14400.0		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:647
26	case:DEF_TUNING_IRRIGATION_MAX_CPHASE	DEF_TUNING_IRRIGATION_MAX_CPHASE	DEF TUNING IRRIGATION MAX CPHASE	4.0		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:652

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
27	case:DEF_TUNING_IRRIGATION_MAX_DEPTH	DEF_TUNING_IRRIGATION_MAX_DEPTH	DEF TUNING IRRIGATION MAX DEPTH	1.0		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:648
28	case:DEF_TUNING_IRRIGATION_MIN_CPHASE	DEF_TUNING_IRRIGATION_MIN_CPHASE	DEF TUNING IRRIGATION MIN CPHASE	1.0		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:651
29	case:DEF_TUNING_IRRIGATION_PONDMX	DEF_TUNING_IRRIGATION_PONDMX	DEF TUNING IRRIGATION PONDMX	100.0		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:653
30	case:DEF_TUNING_IRRIGATION_START_SEC	DEF_TUNING_IRRIGATION_START_SEC	DEF TUNING IRRIGATION START SEC	21600.0		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:646
31	case:DEF_TUNING_IRRIGATION_SUPPLY_FRACTION	DEF_TUNING_IRRIGATION_SUPPLY_FRACTION	DEF TUNING IRRIGATION SUPPLY FRACTION	1.0		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:650
32	case:DEF_TUNING_IRRIGATION_THRESHOLD_FRACTION	DEF_TUNING_IRRIGATION_THRESHOLD_FRACTION	DEF TUNING IRRIGATION THRESHOLD FRACTION	1.0		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:649
33	case:DEF_TUNING_TRSMX0	DEF_TUNING_TRSMX0	DEF TUNING TRSMX0	2.e-4		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:639
34	case:DEF_USE_BGC	DEF_USE_BGC	DEF USE BGC	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:676
35	case:DEF_USE_CNSOYFIXN	DEF_USE_CNSOYFIXN	DEF USE CNSOYFIXN	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:704
36	case:DEF_USE_CROP	DEF_USE_CROP	DEF USE CROP	.false.		case-scalar	read-only-context		MOD_Namelist.F90 declaration	MOD_Namelist.F90:695
37	case:DEF_USE_DiagMatrix	DEF_USE_DiagMatrix	DEF USE DiagMatrix	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:698
38	case:DEF_USE_FERT	DEF_USE_FERT	DEF USE FERT	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:701
39	case:DEF_USE_FIRE	DEF_USE_FIRE	DEF USE FIRE	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:706
40	case:DEF_USE_IRRIGATION	DEF_USE_IRRIGATION	DEF USE IRRIGATION	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:541
41	case:DEF_USE_LAIFEEDBACK	DEF_USE_LAIFEEDBACK	DEF USE LAIFEEDBACK	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:313
42	case:DEF_USE_MEDLYNST	DEF_USE_MEDLYNST	DEF USE MEDLYNST	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:547
43	case:DEF_USE_NITRIF	DEF_USE_NITRIF	DEF USE NITRIF	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:703

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
44	case:DEF_USE_NOSTRESSNITROGEN	DEF_USE_NOSTRESSNITROGEN	DEF USE NOSTRESSNITROGEN	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:544
45	case:DEF_USE_OZONEDATA	DEF_USE_OZONEDATA	DEF USE OZONEDATA	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:454
46	case:DEF_USE_OZONESTRESS	DEF_USE_OZONESTRESS	DEF USE OZONESTRESS	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:453
47	case:DEF_USE_PLANTHYDRAULICS	DEF_USE_PLANTHYDRAULICS	DEF USE PLANTHYDRAULICS	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:546
48	case:DEF_USE_PN	DEF_USE_PN	DEF USE PN	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:699
49	case:DEF_USE_SASU	DEF_USE_SASU	DEF USE SASU	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:697
50	case:DEF_USE_SNICAR	DEF_USE_SNICAR	DEF USE SNICAR	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:458
51	case:DEF_USE_WUEST	DEF_USE_WUEST	DEF USE WUEST	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:548
52	case:DEF_VEG_SNOW	DEF_VEG_SNOW	DEF VEG SNOW	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:413
53	case:DEF_WUE_LAMBDA	DEF_WUE_LAMBDA	DEF WUE LAMBDA	-1.0		case-scalar	editable-expert	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:555
54	case:DEF_file_Ozone	DEF_file_Ozone	DEF file Ozone	null		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:455
55	case:DEF_file_snowaging	DEF_file_snowaging	DEF file snowaging	null		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:460
56	case:DEF_file_snowoptics	DEF_file_snowoptics	DEF file snowoptics	null		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:459
57	lct:IGBP:DEF_LC_BETA	DEF_LC_BETA	根系分布形状参数	-1.e36	–	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:592
58	lct:IGBP:DEF_LC_BINTER	DEF_LC_BINTER	Ball–Berry 截距	-1.e36	–	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:596

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
59	lct:IGBP:DEF_LC_C3C4	DEF_LC_C3C4	C3/C4 标志	-1	–	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:609
60	lct:IGBP:DEF_LC_CHIL	DEF_LC_CHIL	叶倾角分布参数	-1.e36	–	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:571
61	lct:IGBP:DEF_LC_CK	DEF_LC_CK	脆弱性曲线形状参数	-1.e36	–	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:608
62	lct:IGBP:DEF_LC_D50	DEF_LC_D50	50% 根系深度	-1.e36	cm	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:591
63	lct:IGBP:DEF_LC_DISPLAR	DEF_LC_DISPLAR	零平面位移高度比例	-1.e36	–	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:569
64	lct:IGBP:DEF_LC_EFFCON	DEF_LC_EFFCON	量子效率	-1.e36	–	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:581
65	lct:IGBP:DEF_LC_EXTKN	DEF_LC_EXTKN	叶氮分配系数	-1.e36	–	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:590
66	lct:IGBP:DEF_LC_FVEG0	DEF_LC_FVEG0	植被覆盖度	-1.e36	–	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:566
67	lct:IGBP:DEF_LC_G0	DEF_LC_G0	Medlyn 截距 g0	-1.e36	–	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:598

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
68	lct:IGBP:DEF_LC_G1	DEF_LC_G1	Medlyn 斜率 g1	-1.e36	–	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:597
69	lct:IGBP:DEF_LC_GRADM	DEF_LC_GRADM	Ball–Berry 斜率	-1.e36	–	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:595
70	lct:IGBP:DEF_LC_HBOT0	DEF_LC_HBOT0	冠层底部高度	-1.e36	m	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:565
71	lct:IGBP:DEF_LC_HHTI	DEF_LC_HHTI	高温抑制半响应温度	-1.e36	K	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:588
72	lct:IGBP:DEF_LC_HLTI	DEF_LC_HLTI	低温抑制半响应温度	-1.e36	K	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:589
73	lct:IGBP:DEF_LC_HTOP0	DEF_LC_HTOP0	冠层顶部高度	-1.e36	m	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:564
74	lct:IGBP:DEF_LC_KMAX_ROOT	DEF_LC_KMAX_ROOT	根最大导水率	-1.e36		land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:603
75	lct:IGBP:DEF_LC_KMAX_SHA	DEF_LC_KMAX_SHA	阴叶最大导水率	-1.e36		land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:601
76	lct:IGBP:DEF_LC_KMAX_SUN	DEF_LC_KMAX_SUN	阳叶最大导水率	-1.e36		land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:600

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
77	lct:IGBP:DEF_LC_KMAX_XYL	DEF_LC_KMAX_XYL	木质部最大导水率	-1.e36		land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:602
78	lct:IGBP:DEF_LC_LAMBDA	DEF_LC_LAMBDA	WUE 边际水分成本	-1.e36	mol mol-1	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:599
79	lct:IGBP:DEF_LC_PSI50_ROOT	DEF_LC_PSI50_ROOT	根 50% 失导水势	-1.e36	mmH2O	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:607
80	lct:IGBP:DEF_LC_PSI50_SHA	DEF_LC_PSI50_SHA	阴叶 50% 失导水势	-1.e36	mmH2O	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:605
81	lct:IGBP:DEF_LC_PSI50_SUN	DEF_LC_PSI50_SUN	阳叶 50% 失导水势	-1.e36	mmH2O	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:604
82	lct:IGBP:DEF_LC_PSI50_XYL	DEF_LC_PSI50_XYL	木质部 50% 失导水势	-1.e36	mmH2O	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:606
83	lct:IGBP:DEF_LC_RESPCP	DEF_LC_RESPCP	叶呼吸比例	-1.e36	–	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:582
84	lct:IGBP:DEF_LC_RHOL_NIR	DEF_LC_RHOL_NIR	绿叶近红外反射率	-1.e36	–	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:573
85	lct:IGBP:DEF_LC_RHOL_VIS	DEF_LC_RHOL_VIS	绿叶可见光反射率	-1.e36	–	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:572

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
86	lct:IGBP:DEF_LC_RHOS_NIR	DEF_LC_RHOS_NIR	枯叶近红外反射率	-1.e36	-	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:575
87	lct:IGBP:DEF_LC_RHOS_VIS	DEF_LC_RHOS_VIS	枯叶可见光反射率	-1.e36	-	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:574
88	lct:IGBP:DEF_LC_SAI0	DEF_LC_SAI0	茎面积指数	-1.e36	m2 m-2	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:567
89	lct:IGBP:DEF_LC_SHTI	DEF_LC_SHTI	高温抑制斜率	-1.e36	-	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:583
90	lct:IGBP:DEF_LC_SLTI	DEF_LC_SLTI	低温抑制斜率	-1.e36	-	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:584
91	lct:IGBP:DEF_LC_SQRTDI	DEF_LC_SQRTDI	叶片特征尺寸倒平方根	-1.e36	m-0.5	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:570
92	lct:IGBP:DEF_LC_TAUL_NIR	DEF_LC_TAUL_NIR	绿叶近红外透射率	-1.e36	-	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:577
93	lct:IGBP:DEF_LC_TAUL_VIS	DEF_LC_TAUL_VIS	绿叶可见光透射率	-1.e36	-	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:576
94	lct:IGBP:DEF_LC_TAUS_NIR	DEF_LC_TAUS_NIR	枯叶近红外透射率	-1.e36	-	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:579

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
95	lct:IGBP:DEF_LC_TAUS_VIS	DEF_LC_TAUS_VIS	枯叶可见光透射率	-1.e36	–	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:578
96	lct:IGBP:DEF_LC_TRDA	DEF_LC_TRDA	气孔模型温度系数 A	-1.e36	–	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:585
97	lct:IGBP:DEF_LC_TRDM	DEF_LC_TRDM	气孔模型温度系数 M	-1.e36	K	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:586
98	lct:IGBP:DEF_LC_TROP	DEF_LC_TROP	最适温度	-1.e36	K	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:587
99	lct:IGBP:DEF_LC_VMAX25	DEF_LC_VMAX25	25℃ 最大羧化速率	-1.e36	umol m-2 s-1	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:580
100	lct:IGBP:DEF_LC_Z0MR	DEF_LC_Z0MR	粗糙度长度比例	-1.e36	–	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:568
101	lct:USGS:DEF_LC_BETA	DEF_LC_BETA	根系分布形状参数	-1.e36	–	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:592
102	lct:USGS:DEF_LC_BINTER	DEF_LC_BINTER	Ball–Berry 截距	-1.e36	–	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:596
103	lct:USGS:DEF_LC_C3C4	DEF_LC_C3C4	C3/C4 标志	-1	–	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:609

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
104	lct:USGS:DEF_LC_CHIL	DEF_LC_CHIL	叶倾角分布参数	-1.e36	–	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:571
105	lct:USGS:DEF_LC_CK	DEF_LC_CK	脆弱性曲线形状参数	-1.e36	–	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:608
106	lct:USGS:DEF_LC_D50	DEF_LC_D50	50% 根系深度	-1.e36	cm	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:591
107	lct:USGS:DEF_LC_DISPLAR	DEF_LC_DISPLAR	零平面位移高度比例	-1.e36	–	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:569
108	lct:USGS:DEF_LC_EFFCON	DEF_LC_EFFCON	量子效率	-1.e36	–	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:581
109	lct:USGS:DEF_LC_EXTKN	DEF_LC_EXTKN	叶氮分配系数	-1.e36	–	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:590
110	lct:USGS:DEF_LC_FVEG0	DEF_LC_FVEG0	植被覆盖度	-1.e36	–	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:566
111	lct:USGS:DEF_LC_G0	DEF_LC_G0	Medlyn 截距 g0	-1.e36	–	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:598
112	lct:USGS:DEF_LC_G1	DEF_LC_G1	Medlyn 斜率 g1	-1.e36	–	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:597

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
113	lct:USGS:DEF_LC_GRADM	DEF_LC_GRADM	Ball–Berry 斜率	-1.e36	–	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:595
114	lct:USGS:DEF_LC_HBOT0	DEF_LC_HBOT0	冠层底部高度	-1.e36	m	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:565
115	lct:USGS:DEF_LC_HHTI	DEF_LC_HHTI	高温抑制半响应温度	-1.e36	K	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:588
116	lct:USGS:DEF_LC_HLTI	DEF_LC_HLTI	低温抑制半响应温度	-1.e36	K	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:589
117	lct:USGS:DEF_LC_HTOP0	DEF_LC_HTOP0	冠层顶部高度	-1.e36	m	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:564
118	lct:USGS:DEF_LC_KMAX_ROOT	DEF_LC_KMAX_ROOT	根最大导水率	-1.e36		land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:603
119	lct:USGS:DEF_LC_KMAX_SHA	DEF_LC_KMAX_SHA	阴叶最大导水率	-1.e36		land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:601
120	lct:USGS:DEF_LC_KMAX_SUN	DEF_LC_KMAX_SUN	阳叶最大导水率	-1.e36		land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:600
121	lct:USGS:DEF_LC_KMAX_XYL	DEF_LC_KMAX_XYL	木质部最大导水率	-1.e36		land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:602

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
122	lct:USGS:DEF_LC_LAMBDA	DEF_LC_LAMBDA	WUE 边际水分成本	-1.e36	mol mol-1	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:599
123	lct:USGS:DEF_LC_PSI50_ROOT	DEF_LC_PSI50_ROOT	根 50% 失导水势	-1.e36	mmH2O	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:607
124	lct:USGS:DEF_LC_PSI50_SHA	DEF_LC_PSI50_SHA	阴叶 50% 失导水势	-1.e36	mmH2O	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:605
125	lct:USGS:DEF_LC_PSI50_SUN	DEF_LC_PSI50_SUN	阳叶 50% 失导水势	-1.e36	mmH2O	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:604
126	lct:USGS:DEF_LC_PSI50_XYL	DEF_LC_PSI50_XYL	木质部 50% 失导水势	-1.e36	mmH2O	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:606
127	lct:USGS:DEF_LC_RESPCP	DEF_LC_RESPCP	叶呼吸比例	-1.e36	-	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:582
128	lct:USGS:DEF_LC_RHOL_NIR	DEF_LC_RHOL_NIR	绿叶近红外反射率	-1.e36	-	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:573
129	lct:USGS:DEF_LC_RHOL_VIS	DEF_LC_RHOL_VIS	绿叶可见光反射率	-1.e36	-	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:572
130	lct:USGS:DEF_LC_RHOS_NIR	DEF_LC_RHOS_NIR	枯叶近红外反射率	-1.e36	-	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:575

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
131	lct:USGS:DEF_LC_RHOS_VIS	DEF_LC_RHOS_VIS	枯叶可见光反射率	-1.e36	–	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:574
132	lct:USGS:DEF_LC_SAI0	DEF_LC_SAI0	茎面积指数	-1.e36	m2 m-2	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:567
133	lct:USGS:DEF_LC_SHTI	DEF_LC_SHTI	高温抑制斜率	-1.e36	–	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:583
134	lct:USGS:DEF_LC_SLTI	DEF_LC_SLTI	低温抑制斜率	-1.e36	–	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:584
135	lct:USGS:DEF_LC_SQRTDI	DEF_LC_SQRTDI	叶片特征尺寸倒平方根	-1.e36	m-0.5	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:570
136	lct:USGS:DEF_LC_TAUL_NIR	DEF_LC_TAUL_NIR	绿叶近红外透射率	-1.e36	–	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:577
137	lct:USGS:DEF_LC_TAUL_VIS	DEF_LC_TAUL_VIS	绿叶可见光透射率	-1.e36	–	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:576
138	lct:USGS:DEF_LC_TAUS_NIR	DEF_LC_TAUS_NIR	枯叶近红外透射率	-1.e36	–	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:579
139	lct:USGS:DEF_LC_TAUS_VIS	DEF_LC_TAUS_VIS	枯叶可见光透射率	-1.e36	–	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:578

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
140	lct:USGS:DEF_LC_TRDA	DEF_LC_TRDA	气孔模型温度系数 A	-1.e36	–	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:585
141	lct:USGS:DEF_LC_TRDM	DEF_LC_TRDM	气孔模型温度系数 M	-1.e36	K	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:586
142	lct:USGS:DEF_LC_TROP	DEF_LC_TROP	最适温度	-1.e36	K	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:587
143	lct:USGS:DEF_LC_VMAX25	DEF_LC_VMAX25	25°C 最大羧化速率	-1.e36	umol m-2 s-1	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:580
144	lct:USGS:DEF_LC_Z0MR	DEF_LC_Z0MR	粗糙度长度比例	-1.e36	–	land-cover-class	editable-expert	case.nml	MOD_Const_LC.F90 contextual IGBP/USGS table	MOD_Namelist.F90:568
145	pc-pft:DEF_PFT_ALLCONSS	DEF_PFT_ALLCONSS	茎分配衰减指数	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
146	pc-pft:DEF_PFT_AROOTF	DEF_PFT_AROOTF	最终根分配比例	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
147	pc-pft:DEF_PFT_AROOTI	DEF_PFT_AROOTI	初始根分配比例	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
148	pc-pft:DEF_PFT_ASTEMF	DEF_PFT_ASTEMF	成熟期最低茎分配比例	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
149	pc-pft:DEF_PFT_BASET	DEF_PFT_BASET	积温基准温度	``	°C	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
150	pc-pft:DEF_PFT_BETA	DEF_PFT_BETA	根系分布形状参数	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
151	pc-pft:DEF_PFT_BFACT	DEF_PFT_BFACT	叶片分配曲线形状系数	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
152	pc-pft:DEF_PFT_BINTER	DEF_PFT_BINTER	Ball–Berry 截距	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
153	pc-pft:DEF_PFT_C3C4	DEF_PFT_C3C4	C3/C4 标志	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
154	pc-pft:DEF_PFT_CC_DSTEM	DEF_PFT_CC_DSTEM	死茎燃烧完全度	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
155	pc-pft:DEF_PFT_CC_LEAF	DEF_PFT_CC_LEAF	叶片燃烧完全度	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
156	pc-pft:DEF_PFT_CC_LSTEM	DEF_PFT_CC_LSTEM	活茎燃烧完全度	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
157	pc-pft:DEF_PFT_CC_OTHER	DEF_PFT_CC_OTHER	其他组织燃烧完全度	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
158	pc-pft:DEF_PFT_CHIL	DEF_PFT_CHIL	叶倾角分布参数	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
159	pc-pft:DEF_PFT_CK	DEF_PFT_CK	脆弱性曲线形状参数	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
160	pc-pft:DEF_PFT_CROOT_STEM	DEF_PFT_CROOT_STEM	粗根与茎分配比	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
161	pc-pft:DEF_PFT_D50	DEF_PFT_D50	50%根系深度	``	cm	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
162	pc-pft:DEF_PFT_DEADWDCN	DEF_PFT_DEADWDCN	死木质部碳氮比	``	gC gN-1	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
163	pc-pft:DEF_PFT_EFFCON	DEF_PFT_EFFCON	量子效率	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
164	pc-pft:DEF_PFT_EXTKN	DEF_PFT_EXTKN	叶氮分配系数	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
165	pc-pft:DEF_PFT_FCUR2	DEF_PFT_FCUR2	当前光合产物直接生长比例	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
166	pc-pft:DEF_PFT_FD_PFT	DEF_PFT_FD_PFT	火灾持续时间	``	h	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
167	pc-pft:DEF_PFT_FFROOTCN	DEF_PFT_FFROOTCN	灌浆期细根碳氮比	``	gC gN-1	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
168	pc-pft:DEF_PFT_FLEAFCN	DEF_PFT_FLEAFCN	灌浆期叶片碳氮比	``	gC gN-1	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
169	pc-pft:DEF_PFT_FLEAFI	DEF_PFT_FLEAFI	初始叶片分配比例	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
170	pc-pft:DEF_PFT_FLIVEWD	DEF_PFT_FLIVEWD	活木质部比例	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
171	pc-pft:DEF_PFT_FM_DROOT	DEF_PFT_FM_DROOT	死粗根火致死亡比例	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
172	pc-pft:DEF_PFT_FM_LEAF	DEF_PFT_FM_LEAF	叶片火致死亡比例	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
173	pc-pft:DEF_PFT_FM_LROOT	DEF_PFT_FM_LROOT	活粗根火致死亡比例	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
174	pc-pft:DEF_PFT_FM_LSTEM	DEF_PFT_FM_LSTEM	活茎火致死亡比例	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
175	pc-pft:DEF_PFT_FM_OTHER	DEF_PFT_FM_OTHER	其他组织火致死亡比例	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
176	pc-pft:DEF_PFT_FM_ROOT	DEF_PFT_FM_ROOT	细根火致死亡比例	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
177	pc-pft:DEF_PFT_FROOTCN	DEF_PFT_FROOTCN	细根碳氮比	``	gC gN-1	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
178	pc-pft:DEF_PFT_FROOT_LEAF	DEF_PFT_FROOT_LEAF	细根与叶片分配比	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
179	pc-pft:DEF_PFT_FR_FCEL	DEF_PFT_FR_FCEL	细根凋落物纤维素组分比例	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
180	pc-pft:DEF_PFT_FR_FLAB	DEF_PFT_FR_FLAB	细根凋落物易分解组分比例	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
181	pc-pft:DEF_PFT_FR_FLIG	DEF_PFT_FR_FLIG	细根凋落物木质素组分比例	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
182	pc-pft:DEF_PFT_FSR_PFT	DEF_PFT_FSR_PFT	火灾存活率	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
183	pc-pft:DEF_PFT_FSTEMCN	DEF_PFT_FSTEMCN	灌浆期茎碳氮比	``	gC gN-1	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
184	pc-pft:DEF_PFT_G0	DEF_PFT_G0	Medlyn 截距 g0	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
185	pc-pft:DEF_PFT_G1	DEF_PFT_G1	Medlyn 斜率 g1	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
186	pc-pft:DEF_PFT_GRADM	DEF_PFT_GRADM	Ball–Berry 斜率	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
187	pc-pft:DEF_PFT_GRAINCN	DEF_PFT_GRAINCN	籽粒碳氮比	``	gC gN-1	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
188	pc-pft:DEF_PFT_GRNFILL	DEF_PFT_GRNFILL	灌浆热量指数阈值	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
189	pc-pft:DEF_PFT_GRPERC	DEF_PFT_GRPERC	生长呼吸比例	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
190	pc-pft:DEF_PFT_GRPNOW	DEF_PFT_GRPNOW	生长呼吸即时支付比例	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
191	pc-pft:DEF_PFT_HBOT0	DEF_PFT_HBOT0	冠层底部高度	``	m	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
192	pc-pft:DEF_PFT_HHTI	DEF_PFT_HHTI	高温抑制半响应温度	``	K	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
193	pc-pft:DEF_PFT_HLTI	DEF_PFT_HLTI	低温抑制半响应温度	``	K	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
194	pc-pft:DEF_PFT_HTOP0	DEF_PFT_HTOP0	冠层顶部高度	``	m	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
195	pc-pft:DEF_PFT_KMAX_ROOT	DEF_PFT_KMAX_ROOT	根最大导水率	``		pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
196	pc-pft:DEF_PFT_KMAX_SHA	DEF_PFT_KMAX_SHA	阴叶最大导水率	``		pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
197	pc-pft:DEF_PFT_KMAX_SUN	DEF_PFT_KMAX_SUN	阳叶最大导水率	``		pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
198	pc-pft:DEF_PFT_KMAX_XYL	DEF_PFT_KMAX_XYL	木质部最大导水率	``		pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
199	pc-pft:DEF_PFT_LAIMX	DEF_PFT_LAIMX	最大叶面积指数	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
200	pc-pft:DEF_PFT_LAMBDA	DEF_PFT_LAMBDA	WUE 边际水分成本	``	mol mol−1	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
201	pc-pft:DEF_PFT_LEAFCN	DEF_PFT_LEAFCN	叶片碳氮比	``	gC gN−1	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
202	pc-pft:DEF_PFT_LEAF_LONG	DEF_PFT_LEAF_LONG	叶片寿命	``	yr	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
203	pc-pft:DEF_PFT_LFEMERG	DEF_PFT_LFEMERG	出苗热量指数阈值	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
204	pc-pft:DEF_PFT_LFLITCN	DEF_PFT_LFLITCN	叶凋落物碳氮比	``	gC gN-1	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
205	pc-pft:DEF_PFT_LF_FCEL	DEF_PFT_LF_FCEL	叶凋落物纤维素组分比例	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
206	pc-pft:DEF_PFT_LF_FLAB	DEF_PFT_LF_FLAB	叶凋落物易分解组分比例	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
207	pc-pft:DEF_PFT_LF_FLIG	DEF_PFT_LF_FLIG	叶凋落物木质素组分比例	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
208	pc-pft:DEF_PFT_LIVEWDCN	DEF_PFT_LIVEWDCN	活木质部碳氮比	``	gC gN-1	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
209	pc-pft:DEF_PFT_MANURE	DEF_PFT_MANURE	最大有机肥氮施用量	``	kg N m-2	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
210	pc-pft:DEF_PFT_MXMAT	DEF_PFT_MXMAT	最大成熟天数	``	d	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
211	pc-pft:DEF_PFT_PSI50_ROOT	DEF_PFT_PSI50_ROOT	根50%失导水势	``	mmH2O	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
212	pc-pft:DEF_PFT_PSI50_SHA	DEF_PFT_PSI50_SHA	阴叶50%失导水势	``	mmH2O	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
213	pc-pft:DEF_PFT_PSI50_SUN	DEF_PFT_PSI50_SUN	阳叶50%失导水势	``	mmH2O	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
214	pc-pft:DEF_PFT_PSI50_XYL	DEF_PFT_PSI50_XYL	木质部50%失导水势	``	mmH2O	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
215	pc-pft:DEF_PFT_RHOL_NIR	DEF_PFT_RHOL_NIR	绿叶近红外反射率	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
216	pc-pft:DEF_PFT_RHOL_VIS	DEF_PFT_RHOL_VIS	绿叶可见光反射率	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
217	pc-pft:DEF_PFT_RHOS_NIR	DEF_PFT_RHOS_NIR	枯叶近红外反射率	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
218	pc-pft:DEF_PFT_RHOS_VIS	DEF_PFT_RHOS_VIS	枯叶可见光反射率	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
219	pc-pft:DEF_PFT_SHTI	DEF_PFT_SHTI	高温抑制斜率	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
220	pc-pft:DEF_PFT_SLATOP	DEF_PFT_SLATOP	冠层顶部比叶面积	``	m2 gC-1	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
221	pc-pft:DEF_PFT_SLTI	DEF_PFT_SLTI	低温抑制斜率	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
222	pc-pft:DEF_PFT_SQRTDI	DEF_PFT_SQRTDI	叶片特征尺寸倒平方根	``	m-0.5	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
223	pc-pft:DEF_PFT_STEM_LEAF	DEF_PFT_STEM_LEAF	茎与叶片分配比	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
224	pc-pft:DEF_PFT_TAUL_NIR	DEF_PFT_TAUL_NIR	绿叶近红外透射率	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
225	pc-pft:DEF_PFT_TAUL_VIS	DEF_PFT_TAUL_VIS	绿叶可见光透射率	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
226	pc-pft:DEF_PFT_TAUS_NIR	DEF_PFT_TAUS_NIR	枯叶近红外透射率	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
227	pc-pft:DEF_PFT_TAUS_VIS	DEF_PFT_TAUS_VIS	枯叶可见光透射率	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
228	pc-pft:DEF_PFT_TRDA	DEF_PFT_TRDA	气孔温度系数 A	``	–	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
229	pc-pft:DEF_PFT_TRDM	DEF_PFT_TRDM	气孔温度系数 M	``	K	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
230	pc-pft:DEF_PFT_TROP	DEF_PFT_TROP	最适温度	``	K	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
231	pc-pft:DEF_PFT_VMAX25	DEF_PFT_VMAX25	25°C最大羧化速率	``	umol m-2 s-1	pc-pft-component	editable-expert		MOD_Const_PFT.F90 current PC branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
232	pft:DEF_PFT_ALLCONSS	DEF_PFT_ALLCONSS	茎分配衰减指数	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
233	pft:DEF_PFT_AROOTF	DEF_PFT_AROOTF	最终根分配比例	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
234	pft:DEF_PFT_AROOTI	DEF_PFT_AROOTI	初始根分配比例	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
235	pft:DEF_PFT_ASTEMF	DEF_PFT_ASTEMF	成熟期最低茎分配比例	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
236	pft:DEF_PFT_BASET	DEF_PFT_BASET	积温基准温度	``	°C	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
237	pft:DEF_PFT_BETA	DEF_PFT_BETA	根系分布形状参数	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
238	pft:DEF_PFT_BFACT	DEF_PFT_BFACT	叶片分配曲线形状系数	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
239	pft:DEF_PFT_BINTER	DEF_PFT_BINTER	Ball–Berry 截距	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
240	pft:DEF_PFT_C3C4	DEF_PFT_C3C4	C3/C4 标志	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
241	pft:DEF_PFT_CC_DSTEM	DEF_PFT_CC_DSTEM	死茎燃烧完全度	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
242	pft:DEF_PFT_CC_LEAF	DEF_PFT_CC_LEAF	叶片燃烧完全度	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
243	pft:DEF_PFT_CC_LSTEM	DEF_PFT_CC_LSTEM	活茎燃烧完全度	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
244	pft:DEF_PFT_CC_OTHER	DEF_PFT_CC_OTHER	其他组织燃烧完全度	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
245	pft:DEF_PFT_CHIL	DEF_PFT_CHIL	叶倾角分布参数	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
246	pft:DEF_PFT_CK	DEF_PFT_CK	脆弱性曲线形状参数	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
247	pft:DEF_PFT_CROOT_STEM	DEF_PFT_CROOT_STEM	粗根与茎分配比	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
248	pft:DEF_PFT_D50	DEF_PFT_D50	50%根系深度	``	cm	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
249	pft:DEF_PFT_DEADWDCN	DEF_PFT_DEADWDCN	死木质部碳氮比	``	gC gN-1	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
250	pft:DEF_PFT_EFFCON	DEF_PFT_EFFCON	量子效率	``	-	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
251	pft:DEF_PFT_EXTKN	DEF_PFT_EXTKN	叶氮分配系数	``	-	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
252	pft:DEF_PFT_FCUR2	DEF_PFT_FCUR2	当前光合产物直接生长比例	``	-	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
253	pft:DEF_PFT_FD_PFT	DEF_PFT_FD_PFT	火灾持续时间	``	h	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
254	pft:DEF_PFT_FFROOTCN	DEF_PFT_FFROOTCN	灌浆期细根碳氮比	``	gC gN-1	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
255	pft:DEF_PFT_FLEAFCN	DEF_PFT_FLEAFCN	灌浆期叶片碳氮比	``	gC gN-1	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
256	pft:DEF_PFT_FLEAFI	DEF_PFT_FLEAFI	初始叶片分配比例	``	-	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
257	pft:DEF_PFT_FLIVEWD	DEF_PFT_FLIVEWD	活木质部比例	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
258	pft:DEF_PFT_FM_DROOT	DEF_PFT_FM_DROOT	死粗根火致死亡比例	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
259	pft:DEF_PFT_FM_LEAF	DEF_PFT_FM_LEAF	叶片火致死亡比例	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
260	pft:DEF_PFT_FM_LROOT	DEF_PFT_FM_LROOT	活粗根火致死亡比例	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
261	pft:DEF_PFT_FM_LSTEM	DEF_PFT_FM_LSTEM	活茎火致死亡比例	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
262	pft:DEF_PFT_FM_OTHER	DEF_PFT_FM_OTHER	其他组织火致死亡比例	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
263	pft:DEF_PFT_FM_ROOT	DEF_PFT_FM_ROOT	细根火致死亡比例	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
264	pft:DEF_PFT_FROOTCN	DEF_PFT_FROOTCN	细根碳氮比	``	gC gN-1	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
265	pft:DEF_PFT_FROOT_LEAF	DEF_PFT_FROOT_LEAF	细根与叶片分配比	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
266	pft:DEF_PFT_FR_FCEL	DEF_PFT_FR_FCEL	细根凋落物纤维素组分比例	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
267	pft:DEF_PFT_FR_FLAB	DEF_PFT_FR_FLAB	细根凋落物易分解组分比例	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
268	pft:DEF_PFT_FR_FLIG	DEF_PFT_FR_FLIG	细根凋落物木质素组分比例	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
269	pft:DEF_PFT_FSR_PFT	DEF_PFT_FSR_PFT	火灾存活率	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
270	pft:DEF_PFT_FSTEMCN	DEF_PFT_FSTEMCN	灌浆期茎碳氮比	``	gC gN-1	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
271	pft:DEF_PFT_G0	DEF_PFT_G0	Medlyn 截距 g0	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
272	pft:DEF_PFT_G1	DEF_PFT_G1	Medlyn 斜率 g1	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
273	pft:DEF_PFT_GRADM	DEF_PFT_GRADM	Ball–Berry 斜率	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
274	pft:DEF_PFT_GRAINCN	DEF_PFT_GRAINCN	籽粒碳氮比	``	gC gN-1	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
275	pft:DEF_PFT_GRNFILL	DEF_PFT_GRNFILL	灌浆热量指数阈值	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
276	pft:DEF_PFT_GRPERC	DEF_PFT_GRPERC	生长呼吸比例	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
277	pft:DEF_PFT_GRPNOW	DEF_PFT_GRPNOW	生长呼吸即时支付比例	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
278	pft:DEF_PFT_HBOT0	DEF_PFT_HBOT0	冠层底部高度	``	m	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
279	pft:DEF_PFT_HHTI	DEF_PFT_HHTI	高温抑制半响应温度	``	K	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
280	pft:DEF_PFT_HLTI	DEF_PFT_HLTI	低温抑制半响应温度	``	K	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
281	pft:DEF_PFT_HTOP0	DEF_PFT_HTOP0	冠层顶部高度	``	m	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
282	pft:DEF_PFT_KMAX_ROOT	DEF_PFT_KMAX_ROOT	根最大导水率	``		pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
283	pft:DEF_PFT_KMAX_SHA	DEF_PFT_KMAX_SHA	阴叶最大导水率	``		pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
284	pft:DEF_PFT_KMAX_SUN	DEF_PFT_KMAX_SUN	阳叶最大导水率	``		pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
285	pft:DEF_PFT_KMAX_XYL	DEF_PFT_KMAX_XYL	木质部最大导水率	``		pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
286	pft:DEF_PFT_LAIMX	DEF_PFT_LAIMX	最大叶面积指数	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
287	pft:DEF_PFT_LAMBDA	DEF_PFT_LAMBDA	WUE 边际水分成本	``	mol mol-1	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
288	pft:DEF_PFT_LEAFCN	DEF_PFT_LEAFCN	叶片碳氮比	``	gC gN-1	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
289	pft:DEF_PFT_LEAF_LONG	DEF_PFT_LEAF_LONG	叶片寿命	``	yr	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
290	pft:DEF_PFT_LFEMERG	DEF_PFT_LFEMERG	出苗热量指数阈值	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
291	pft:DEF_PFT_LFLITCN	DEF_PFT_LFLITCN	叶凋落物碳氮比	``	gC gN-1	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
292	pft:DEF_PFT_LF_FCEL	DEF_PFT_LF_FCEL	叶凋落物纤维素组分比例	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
293	pft:DEF_PFT_LF_FLAB	DEF_PFT_LF_FLAB	叶凋落物易分解组分比例	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
294	pft:DEF_PFT_LF_FLIG	DEF_PFT_LF_FLIG	叶凋落物木质素组分比例	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
295	pft:DEF_PFT_LIVEWDCN	DEF_PFT_LIVEWDCN	活木质部碳氮比	``	gC gN-1	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
296	pft:DEF_PFT_MANURE	DEF_PFT_MANURE	最大有机肥氮施用量	``	kg N m-2	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
297	pft:DEF_PFT_MXMAT	DEF_PFT_MXMAT	最大成熟天数	``	d	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
298	pft:DEF_PFT_PSI50_ROOT	DEF_PFT_PSI50_ROOT	根50%失导水势	``	mmH2O	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
299	pft:DEF_PFT_PSI50_SHA	DEF_PFT_PSI50_SHA	阴叶50%失导水势	``	mmH2O	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
300	pft:DEF_PFT_PSI50_SUN	DEF_PFT_PSI50_SUN	阳叶50%失导水势	``	mmH2O	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
301	pft:DEF_PFT_PSI50_XYL	DEF_PFT_PSI50_XYL	木质部50%失导水势	``	mmH2O	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
302	pft:DEF_PFT_RHOL_NIR	DEF_PFT_RHOL_NIR	绿叶近红外反射率	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
303	pft:DEF_PFT_RHOL_VIS	DEF_PFT_RHOL_VIS	绿叶可见光反射率	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
304	pft:DEF_PFT_RHOS_NIR	DEF_PFT_RHOS_NIR	枯叶近红外反射率	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
305	pft:DEF_PFT_RHOS_VIS	DEF_PFT_RHOS_VIS	枯叶可见光反射率	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
306	pft:DEF_PFT_SHTI	DEF_PFT_SHTI	高温抑制斜率	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
307	pft:DEF_PFT_SLATOP	DEF_PFT_SLATOP	冠层顶部比叶面积	``	m2 gC-1	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
308	pft:DEF_PFT_SLTI	DEF_PFT_SLTI	低温抑制斜率	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
309	pft:DEF_PFT_SQRTDI	DEF_PFT_SQRTDI	叶片特征尺寸倒平方根	``	m-0.5	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
310	pft:DEF_PFT_STEM_LEAF	DEF_PFT_STEM_LEAF	茎与叶片分配比	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
311	pft:DEF_PFT_TAUL_NIR	DEF_PFT_TAUL_NIR	绿叶近红外透射率	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
312	pft:DEF_PFT_TAUL_VIS	DEF_PFT_TAUL_VIS	绿叶可见光透射率	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
313	pft:DEF_PFT_TAUS_NIR	DEF_PFT_TAUS_NIR	枯叶近红外透射率	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
314	pft:DEF_PFT_TAUS_VIS	DEF_PFT_TAUS_VIS	枯叶可见光透射率	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
315	pft:DEF_PFT_TRDA	DEF_PFT_TRDA	气孔温度系数 A	``	–	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
316	pft:DEF_PFT_TRDM	DEF_PFT_TRDM	气孔温度系数 M	``	K	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
317	pft:DEF_PFT_TROP	DEF_PFT_TROP	最适温度	``	K	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc
318	pft:DEF_PFT_VMAX25	DEF_PFT_VMAX25	25°C最大羧化速率	``	umol m-2 s-1	pft-type	editable-expert		MOD_Const_PFT.F90 current PFT branch	MOD_Const_PFT.F90 + include/pft_override_fields.inc

站点（15 项）

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
1	case:SITE_fsitedata	SITE_fsitedata	SITE fsitedata	null		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:74

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
2	case:SITE_landtype	SITE_landtype	SITE landtype	-1		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:79
3	case:SITE_lat_location	SITE_lat_location	SITE lat location	-1.e36		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:77
4	case:SITE_lon_location	SITE_lon_location	SITE lon location	-1.e36		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:76
5	case:USE_SITE_ForcingReadAhead	USE_SITE_ForcingReadAhead	USE SITE ForcingReadAhead	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:97
6	case:USE_SITE_LAI	USE_SITE_LAI	USE SITE LAI	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:85
7	case:USE_SITE_dbedrock	USE_SITE_dbedrock	USE SITE dbedrock	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:89
8	case:USE_SITE_htop	USE_SITE_htop	USE SITE htop	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:84
9	case:USE_SITE_lakedepth	USE_SITE_lakedepth	USE SITE lakedepth	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:86
10	case:USE_SITE_landtype	USE_SITE_landtype	USE SITE landtype	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:81
11	case:USE_SITE_pctcrop	USE_SITE_pctcrop	USE SITE pctcrop	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:83
12	case:USE_SITE_pctpfts	USE_SITE_pctpfts	USE SITE pctpfts	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:82
13	case:USE_SITE_soilparameters	USE_SITE_soilparameters	USE SITE soilparameters	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:88
14	case:USE_SITE_soilreflectance	USE_SITE_soilreflectance	USE SITE soilreflectance	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:87
15	case:USE_SITE_topography	USE_SITE_topography	USE SITE topography	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:90

算例（1 项）

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
1	case:DEF_CASE_NAME	DEF_CASE_NAME	DEF CASE NAME	CASENAME		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:23

网格与并行（15 项）

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
1	case:DEF_AverageElementSize	DEF_AverageElementSize	DEF AverageElementSize	-1.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:61
2	case:DEF_BlockInfoFile	DEF_BlockInfoFile	DEF BlockInfoFile	null		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:60
3	case:DEF_CatchmentMesh_data	DEF_CatchmentMesh_data	DEF CatchmentMesh data	path/to/catchment/data		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:143
4	case:DEF_GRIDBASED_lat_res	DEF_GRIDBASED_lat_res	DEF GRIDBASED lat res	0.5		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:141
5	case:DEF_GRIDBASED_lon_res	DEF_GRIDBASED_lon_res	DEF GRIDBASED lon res	0.5		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:140
6	case:DEF_PIO_groupsize	DEF_PIO_groupsize	DEF PIO groupsize	12		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:67
7	case:DEF_domain%edgee	DEF_domain%edgee	DEF domain edgee	180.0		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:46
8	case:DEF_domain%edgen	DEF_domain%edgen	DEF domain edgen	90.0		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:44
9	case:DEF_domain%edges	DEF_domain%edges	DEF domain edges	-90.0		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:43
10	case:DEF_domain%edgew	DEF_domain%edgew	DEF domain edgew	-180.0		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:45
11	case:DEF_file_mesh	DEF_file_mesh	DEF file mesh	path/to/mesh/file		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:139
12	case:DEF_file_mesh_filter	DEF_file_mesh_filter	DEF file mesh filter	path/to/mesh/filter		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:145
13	case:DEF_nIO_eq_nBlock	DEF_nIO_eq_nBlock	DEF nIO eq nBlock	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:68
14	case:DEF_nx_blocks	DEF_nx_blocks	DEF nx blocks	72		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:62
15	case:DEF_ny_blocks	DEF_ny_blocks	DEF ny blocks	36		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:63

调试与诊断（3 项）

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
1	case:DEF_USE_CoLMDEBUG	DEF_USE_CoLMDEBUG	DEF USE CoLMDEBUG	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:34
2	case:DEF_USE_RangeCheck	DEF_USE_RangeCheck	DEF USE RangeCheck	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:35
3	case:DEF_USE_SrfdataDiag	DEF_USE_SrfdataDiag	DEF USE SrfdataDiag	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:36

输出与重启（19 项）

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
1	case:DEF_HISTORY_IN_VECTOR	DEF_HISTORY_IN_VECTOR	DEF HISTORY IN VECTOR	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:823
2	case:DEF_HIST_CompressLevel	DEF_HIST_CompressLevel	DEF HIST CompressLevel	1		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:836
3	case:DEF_HIST_FREQ	DEF_HIST_FREQ	DEF HIST FREQ	none		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:831
4	case:DEF_HIST_WriteBack	DEF_HIST_WriteBack	DEF HIST WriteBack	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:834
5	case:DEF_HIST_grid_as_forcing	DEF_HIST_grid_as_forcing	DEF HIST grid as forcing	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:826
6	case:DEF_HIST_grid_as_model_mesh	DEF_HIST_grid_as_model_mesh	DEF HIST grid as model mesh	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:825
7	case:DEF_HIST_groupby	DEF_HIST_groupby	DEF HIST groupby	MONTH		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:832
8	case:DEF_HIST_lat_res	DEF_HIST_lat_res	DEF HIST lat res	0.5		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:828
9	case:DEF_HIST_lon_res	DEF_HIST_lon_res	DEF HIST lon res	0.5		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:827
10	case:DEF_HIST_mode	DEF_HIST_mode	DEF HIST mode	one		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:833
11	case:DEF_HIST_vars_namelist	DEF_HIST_vars_namelist	DEF HIST vars namelist	null		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:838

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
12	case:DEF_HIST_vars_out_default	DEF_HIST_vars_out_default	DEF HIST vars out default	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:839
13	case:DEF_Output_2mWMO	DEF_Output_2mWMO	DEF Output 2mWMO	.false.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:714
14	case:DEF_REST_CompressLevel	DEF_REST_CompressLevel	DEF REST CompressLevel	1		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:835
15	case:DEF_WRST_FREQ	DEF_WRST_FREQ	DEF WRST FREQ	none		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:830
16	case:DEF_dir_history	DEF_dir_history	DEF dir history	path/to/history		case-scalar	read-only-context		MOD_Namelist.F90 declaration	MOD_Namelist.F90:133
17	case:DEF_dir_output	DEF_dir_output	DEF dir output	path/to/output/data		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:129
18	case:DEF_dir_restart	DEF_dir_restart	DEF dir restart	path/to/restart		case-scalar	read-only-context		MOD_Namelist.F90 declaration	MOD_Namelist.F90:132
19	case:USE_SITE_HistWriteBack	USE_SITE_HistWriteBack	USE SITE HistWriteBack	.true.		case-scalar	editable-common	case.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:96

输出变量（586 项）

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
1	case:DEF_hist_vars%BD_all	DEF_hist_vars%BD_all	DEF hist vars BD all	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1336
2	case:DEF_hist_vars%CONC_O2_UNSAT	DEF_hist_vars%CONC_O2_UNSAT	DEF hist vars CONC O2 UNSAT	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1171
3	case:DEF_hist_vars%DA_t_brt_fy3d	DEF_hist_vars%DA_t_brt_fy3d	DEF hist vars DA t brt fy3d	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1361
4	case:DEF_hist_vars%DA_t_brt_fy3d_a	DEF_hist_vars%DA_t_brt_fy3d_a	DEF hist vars DA t brt fy3d a	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1362
5	case:DEF_hist_vars%DA_t_brt_fy3d_ens	DEF_hist_vars%DA_t_brt_fy3d_ens	DEF hist vars DA t brt fy3d ens	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1363
6	case:DEF_hist_vars%DA_t_brt_fy3d_ens_std	DEF_hist_vars%DA_t_brt_fy3d_ens_std	DEF hist vars DA t brt fy3d ens std	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1364
7	case:DEF_hist_vars%DA_t_brt_smap	DEF_hist_vars%DA_t_brt_smap	DEF hist vars DA t brt smap	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1357

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
8	case:DEF_hist_vars%DA_t_brt_smap_a	DEF_hist_vars%DA_t_brt_smap_a	DEF hist vars DA t brt smap a	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1358
9	case:DEF_hist_vars%DA_t_brt_smap_ens	DEF_hist_vars%DA_t_brt_smap_ens	DEF hist vars DA t brt smap ens	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1359
10	case:DEF_hist_vars%DA_t_brt_smap_ens_std	DEF_hist_vars%DA_t_brt_smap_ens_std	DEF hist vars DA t brt smap ens std	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1360
11	case:DEF_hist_vars%DA_t_soisno_5cm	DEF_hist_vars%DA_t_soisno_5cm	DEF hist vars DA t soisno 5cm	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1351
12	case:DEF_hist_vars%DA_t_soisno_5cm_a	DEF_hist_vars%DA_t_soisno_5cm_a	DEF hist vars DA t soisno 5cm a	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1352
13	case:DEF_hist_vars%DA_t_soisno_5cm_ens_std	DEF_hist_vars%DA_t_soisno_5cm_ens_std	DEF hist vars DA t soisno 5cm ens std	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1356
14	case:DEF_hist_vars%DA_t_soisno_ens	DEF_hist_vars%DA_t_soisno_ens	DEF hist vars DA t soisno ens	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1354
15	case:DEF_hist_vars%DA_wliq_h2osoi_5cm	DEF_hist_vars%DA_wliq_h2osoi_5cm	DEF hist vars DA wliq h2osoi 5cm	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1349
16	case:DEF_hist_vars%DA_wliq_h2osoi_5cm_a	DEF_hist_vars%DA_wliq_h2osoi_5cm_a	DEF hist vars DA wliq h2osoi 5cm a	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1350
17	case:DEF_hist_vars%DA_wliq_soisno_5cm_ens_std	DEF_hist_vars%DA_wliq_soisno_5cm_ens_std	DEF hist vars DA wliq soisno 5cm ens std	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1355
18	case:DEF_hist_vars%DA_wliq_soisno_ens	DEF_hist_vars%DA_wliq_soisno_ens	DEF hist vars DA wliq soisno ens	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1353
19	case:DEF_hist_vars%O2_DECOMP_DEPTH_UNSAT	DEF_hist_vars%O2_DECOMP_DEPTH_UNSAT	DEF hist vars O2 DECOMP DEPTH UNSAT	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1172
20	case:DEF_hist_vars%OM_density	DEF_hist_vars%OM_density	DEF hist vars OM density	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1338
21	case:DEF_hist_vars%abm	DEF_hist_vars%abm	DEF hist vars abm	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1278
22	case:DEF_hist_vars%alb	DEF_hist_vars%alb	DEF hist vars alb	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:909
23	case:DEF_hist_vars%alb_hires	DEF_hist_vars%alb_hires	DEF hist vars alb hires	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:910
24	case:DEF_hist_vars%ar	DEF_hist_vars%ar	DEF hist vars ar	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:995

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
25	case:DEF_hist_vars%assim	DEF_hist_vars%assim	DEF hist vars assim	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:893
26	case:DEF_hist_vars%assimsha	DEF_hist_vars%assimsha	DEF hist vars assimsha	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:943
27	case:DEF_hist_vars%assimsun	DEF_hist_vars%assimsun	DEF hist vars assimsun	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:942
28	case:DEF_hist_vars%baf_crop	DEF_hist_vars%baf_crop	DEF hist vars baf crop	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1175
29	case:DEF_hist_vars%baf_peatf	DEF_hist_vars%baf_peatf	DEF hist vars baf peatf	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1176
30	case:DEF_hist_vars%bedout	DEF_hist_vars%bedout	DEF hist vars bedout	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1449
31	case:DEF_hist_vars%btran2	DEF_hist_vars%btran2	DEF hist vars btran2	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1179
32	case:DEF_hist_vars%col_fire_closs	DEF_hist_vars%col_fire_closs	DEF hist vars col fire closs	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1182
33	case:DEF_hist_vars%col_fire_nloss	DEF_hist_vars%col_fire_nloss	DEF hist vars col fire nloss	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1183
34	case:DEF_hist_vars%cphase	DEF_hist_vars%cphase	DEF hist vars cphase	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1083
35	case:DEF_hist_vars%croppprod1c	DEF_hist_vars%cropprod1c	DEF hist vars cropprod1c	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1088
36	case:DEF_hist_vars%croppprod1c_loss	DEF_hist_vars%cropprod1c_loss	DEF hist vars cropprod1c loss	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1089
37	case:DEF_hist_vars%croppprodc_irrigated_cotton	DEF_hist_vars%cropprodc_irrigated_cotton	DEF hist vars cropprodc irrigated cotton	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1120
38	case:DEF_hist_vars%croppprodc_irrigated_rice	DEF_hist_vars%cropprodc_irrigated_rice	DEF hist vars cropprodc irrigated rice	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1122
39	case:DEF_hist_vars%croppprodc_irrigated_spwheat	DEF_hist_vars%cropprodc_irrigated_spwheat	DEF hist vars cropprodc irrigated spwheat	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1114
40	case:DEF_hist_vars%croppprodc_irrigated_sugarcane	DEF_hist_vars%cropprodc_irrigated_sugarcane	DEF hist vars cropprodc irrigated sugarcane	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1124
41	case:DEF_hist_vars%croppprodc_irrigated_temp_corn	DEF_hist_vars%cropprodc_irrigated_temp_corn	DEF hist vars cropprodc irrigated temp corn	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1112

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
42	case:DEF_hist_vars%croppro dc_irrigated_temp_soybean	DEF_hist_vars%cropprodc_ir rigated_temp_soybean	DEF hist vars cropprodc irrigated temp soybean	.true.		case-scalar	editable-common	history.nml	MOD_Namelist. F90 declaration	MOD_Namelist.F90:1 118
43	case:DEF_hist_vars%croppro dc_irrigated_trop_corn	DEF_hist_vars%cropprodc_ir rigated_trop_corn	DEF hist vars cropprodc irrigated trop corn	.true.		case-scalar	editable-common	history.nml	MOD_Namelist. F90 declaration	MOD_Namelist.F90:1 126
44	case:DEF_hist_vars%croppro dc_irrigated_trop_soybean	DEF_hist_vars%cropprodc_ir rigated_trop_soybean	DEF hist vars cropprodc irrigated trop soybean	.true.		case-scalar	editable-common	history.nml	MOD_Namelist. F90 declaration	MOD_Namelist.F90:1 128
45	case:DEF_hist_vars%croppro dc_irrigated_wtwheat	DEF_hist_vars%cropprodc_ir rigated_wtwheat	DEF hist vars cropprodc irrigated wtwheat	.true.		case-scalar	editable-common	history.nml	MOD_Namelist. F90 declaration	MOD_Namelist.F90:1 116
46	case:DEF_hist_vars%croppro dc_rainfed_cotton	DEF_hist_vars%cropprodc_ra infed_cotton	DEF hist vars cropprodc rainfed cotton	.true.		case-scalar	editable-common	history.nml	MOD_Namelist. F90 declaration	MOD_Namelist.F90:1 119
47	case:DEF_hist_vars%croppro dc_rainfed_rice	DEF_hist_vars%cropprodc_ra infed_rice	DEF hist vars cropprodc rainfed rice	.true.		case-scalar	editable-common	history.nml	MOD_Namelist. F90 declaration	MOD_Namelist.F90:1 121
48	case:DEF_hist_vars%croppro dc_rainfed_spwheat	DEF_hist_vars%cropprodc_ra infed_spwheat	DEF hist vars cropprodc rainfed spwheat	.true.		case-scalar	editable-common	history.nml	MOD_Namelist. F90 declaration	MOD_Namelist.F90:1 113
49	case:DEF_hist_vars%croppro dc_rainfed_sugarcane	DEF_hist_vars%cropprodc_ra infed_sugarcane	DEF hist vars cropprodc rainfed sugarcane	.true.		case-scalar	editable-common	history.nml	MOD_Namelist. F90 declaration	MOD_Namelist.F90:1 123
50	case:DEF_hist_vars%croppro dc_rainfed_temp_corn	DEF_hist_vars%cropprodc_ra infed_temp_corn	DEF hist vars cropprodc rainfed temp corn	.true.		case-scalar	editable-common	history.nml	MOD_Namelist. F90 declaration	MOD_Namelist.F90:1 111
51	case:DEF_hist_vars%croppro dc_rainfed_temp_soybean	DEF_hist_vars%cropprodc_ra infed_temp_soybean	DEF hist vars cropprodc rainfed temp soybean	.true.		case-scalar	editable-common	history.nml	MOD_Namelist. F90 declaration	MOD_Namelist.F90:1 117
52	case:DEF_hist_vars%croppro dc_rainfed_trop_corn	DEF_hist_vars%cropprodc_ra infed_trop_corn	DEF hist vars cropprodc rainfed trop corn	.true.		case-scalar	editable-common	history.nml	MOD_Namelist. F90 declaration	MOD_Namelist.F90:1 125
53	case:DEF_hist_vars%croppro dc_rainfed_trop_soybean	DEF_hist_vars%cropprodc_ra infed_trop_soybean	DEF hist vars cropprodc rainfed trop soybean	.true.		case-scalar	editable-common	history.nml	MOD_Namelist. F90 declaration	MOD_Namelist.F90:1 127
54	case:DEF_hist_vars%croppro dc_rainfed_wtwheat	DEF_hist_vars%cropprodc_ra infed_wtwheat	DEF hist vars cropprodc rainfed wtwheat	.true.		case-scalar	editable-common	history.nml	MOD_Namelist. F90 declaration	MOD_Namelist.F90:1 115
55	case:DEF_hist_vars%croppro dc_unmanagedcrop	DEF_hist_vars%cropprodc_un managedcrop	DEF hist vars cropprodc unmanagedcrop	.true.		case-scalar	editable-common	history.nml	MOD_Namelist. F90 declaration	MOD_Namelist.F90:1 129
56	case:DEF_hist_vars%cropsee dc_deficit	DEF_hist_vars%cropseedc_de ficit	DEF hist vars cropseedc deficit	.true.		case-scalar	editable-common	history.nml	MOD_Namelist. F90 declaration	MOD_Namelist.F90:1 090
57	case:DEF_hist_vars%cwdcCap _vr	DEF_hist_vars%cwdcCap_vr	DEF hist vars cwdcCap vr	.false.		case-scalar	editable-common	history.nml	MOD_Namelist. F90 declaration	MOD_Namelist.F90:1 387
58	case:DEF_hist_vars%cwdc_vr	DEF_hist_vars%cwdc_vr	DEF hist vars cwdc vr	.true.		case-scalar	editable-common	history.nml	MOD_Namelist. F90 declaration	MOD_Namelist.F90:1 372

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
59	case:DEF_hist_vars%cwddecomp	DEF_hist_vars%cwddecomp	DEF hist vars cwddecomp	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:997
60	case:DEF_hist_vars%cwdnCap_vr	DEF_hist_vars%cwdnCap_vr	DEF hist vars cwdnCap vr	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1394
61	case:DEF_hist_vars%cwdn_vr	DEF_hist_vars%cwdn_vr	DEF hist vars cwdn vr	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1379
62	case:DEF_hist_vars%cwdprod	DEF_hist_vars%cwdprod	DEF hist vars cwdprod	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:996
63	case:DEF_hist_vars%deadcrootc	DEF_hist_vars%deadcrootc	DEF hist vars deadcrootc	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:965
64	case:DEF_hist_vars%deadcrootcCap	DEF_hist_vars%deadcrootcCap	DEF hist vars deadcrootcCap	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1299
65	case:DEF_hist_vars%deadcrootc_storage	DEF_hist_vars%deadcrootc_storage	DEF hist vars deadcrootc storage	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:966
66	case:DEF_hist_vars%deadcrootc_storageCap	DEF_hist_vars%deadcrootc_storageCap	DEF hist vars deadcrootc storageCap	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1300
67	case:DEF_hist_vars%deadcrootc_xfer	DEF_hist_vars%deadcrootc_xfer	DEF hist vars deadcrootc xfer	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:967
68	case:DEF_hist_vars%deadcrootc_xferCap	DEF_hist_vars%deadcrootc_xferCap	DEF hist vars deadcrootc xferCap	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1301
69	case:DEF_hist_vars%deadcrootn	DEF_hist_vars%deadcrootn	DEF hist vars deadcrootn	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:986
70	case:DEF_hist_vars%deadcrootnCap	DEF_hist_vars%deadcrootnCap	DEF hist vars deadcrootnCap	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1317
71	case:DEF_hist_vars%deadcrootn_storage	DEF_hist_vars%deadcrootn_storage	DEF hist vars deadcrootn storage	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:987
72	case:DEF_hist_vars%deadcrootn_storageCap	DEF_hist_vars%deadcrootn_storageCap	DEF hist vars deadcrootn storageCap	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1318
73	case:DEF_hist_vars%deadcrootn_xfer	DEF_hist_vars%deadcrootn_xfer	DEF hist vars deadcrootn xfer	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:988
74	case:DEF_hist_vars%deadcrootn_xferCap	DEF_hist_vars%deadcrootn_xferCap	DEF hist vars deadcrootn xferCap	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1319
75	case:DEF_hist_vars%deadstemc	DEF_hist_vars%deadstemc	DEF hist vars deadstemc	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:959

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
76	case:DEF_hist_vars%deadstemcCap	DEF_hist_vars%deadstemcCap	DEF hist vars deadstemcCap	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1293
77	case:DEF_hist_vars%deadstemc_storage	DEF_hist_vars%deadstemc_storage	DEF hist vars deadstemc storage	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:960
78	case:DEF_hist_vars%deadstemc_storageCap	DEF_hist_vars%deadstemc_storageCap	DEF hist vars deadstemc storageCap	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1294
79	case:DEF_hist_vars%deadstemc_xfer	DEF_hist_vars%deadstemc_xfer	DEF hist vars deadstemc xfer	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:961
80	case:DEF_hist_vars%deadstemc_xferCap	DEF_hist_vars%deadstemc_xferCap	DEF hist vars deadstemc xferCap	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1295
81	case:DEF_hist_vars%deadstemn	DEF_hist_vars%deadstemn	DEF hist vars deadstemn	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:980
82	case:DEF_hist_vars%deadstemnCap	DEF_hist_vars%deadstemnCap	DEF hist vars deadstemnCap	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1311
83	case:DEF_hist_vars%deadstemn_storage	DEF_hist_vars%deadstemn_storage	DEF hist vars deadstemn storage	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:981
84	case:DEF_hist_vars%deadstemn_storageCap	DEF_hist_vars%deadstemn_storageCap	DEF hist vars deadstemn storageCap	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1312
85	case:DEF_hist_vars%deadstemn_xfer	DEF_hist_vars%deadstemn_xfer	DEF hist vars deadstemn xfer	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:982
86	case:DEF_hist_vars%deadstemn_xferCap	DEF_hist_vars%deadstemn_xferCap	DEF hist vars deadstemn xferCap	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1313
87	case:DEF_hist_vars%discharge	DEF_hist_vars%discharge	DEF hist vars discharge	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1438
88	case:DEF_hist_vars%downreg	DEF_hist_vars%downreg	DEF hist vars downreg	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:994
89	case:DEF_hist_vars%dz_lake	DEF_hist_vars%dz_lake	DEF hist vars dz lake	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1345
90	case:DEF_hist_vars%emis	DEF_hist_vars%emis	DEF hist vars emis	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:913
91	case:DEF_hist_vars%etr	DEF_hist_vars%etr	DEF hist vars etr	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:866
92	case:DEF_hist_vars%etrsha	DEF_hist_vars%etrsha	DEF hist vars etrsha	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:945

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
93	case:DEF_hist_vars%etrsun	DEF_hist_vars%etrsun	DEF hist vars etrsun	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:944
94	case:DEF_hist_vars%fach	DEF_hist_vars%fach	DEF hist vars fach	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:933
95	case:DEF_hist_vars%farea_burned	DEF_hist_vars%farea_burned	DEF hist vars farea burned	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1174
96	case:DEF_hist_vars%fert_to_sminn	DEF_hist_vars%fert_to_sminn	DEF hist vars fert to sminn	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1132
97	case:DEF_hist_vars%fertnitro_corn	DEF_hist_vars%fertnitro_corn	DEF hist vars fertnitro corn	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1144
98	case:DEF_hist_vars%fertnitro_cotton	DEF_hist_vars%fertnitro_cotton	DEF hist vars fertnitro cotton	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1148
99	case:DEF_hist_vars%fertnitro_rice1	DEF_hist_vars%fertnitro_rice1	DEF hist vars fertnitro rice1	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1149
100	case:DEF_hist_vars%fertnitro_rice2	DEF_hist_vars%fertnitro_rice2	DEF hist vars fertnitro rice2	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1150
101	case:DEF_hist_vars%fertnitro_soybean	DEF_hist_vars%fertnitro_soybean	DEF hist vars fertnitro soybean	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1147
102	case:DEF_hist_vars%fertnitro_sugarcane	DEF_hist_vars%fertnitro_sugarcane	DEF hist vars fertnitro sugarcane	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1151
103	case:DEF_hist_vars%fertnitro_swheat	DEF_hist_vars%fertnitro_swheat	DEF hist vars fertnitro swheat	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1145
104	case:DEF_hist_vars%fertnitro_wwheat	DEF_hist_vars%fertnitro_wwheat	DEF hist vars fertnitro wwheat	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1146
105	case:DEF_hist_vars%fevpa	DEF_hist_vars%fevpa	DEF hist vars fevpa	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:863
106	case:DEF_hist_vars%fevpg	DEF_hist_vars%fevpg	DEF hist vars fevpg	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:868
107	case:DEF_hist_vars%fevpl	DEF_hist_vars%fevpl	DEF hist vars fevpl	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:865
108	case:DEF_hist_vars%fgrnd	DEF_hist_vars%fgrnd	DEF hist vars fgrnd	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:869
109	case:DEF_hist_vars%fh	DEF_hist_vars%fh	DEF hist vars fh	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1405

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
110	case:DEF_hist_vars%fhac	DEF_hist_vars%fhac	DEF hist vars fhac	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:931
111	case:DEF_hist_vars%fhah	DEF_hist_vars%fhah	DEF hist vars fhah	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:934
112	case:DEF_hist_vars%fldarea	DEF_hist_vars%fldarea	DEF hist vars fldarea	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:884
113	case:DEF_hist_vars%floodarea	DEF_hist_vars%floodarea	DEF hist vars floodarea	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1439
114	case:DEF_hist_vars%floodfrc	DEF_hist_vars%floodfrc	DEF hist vars floodfrc	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1440
115	case:DEF_hist_vars%fm	DEF_hist_vars%fm	DEF hist vars fm	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1404
116	case:DEF_hist_vars%fm10m	DEF_hist_vars%fm10m	DEF hist vars fm10m	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1409
117	case:DEF_hist_vars%fpg	DEF_hist_vars%fpg	DEF hist vars fpg	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:999
118	case:DEF_hist_vars%fpi	DEF_hist_vars%fpi	DEF hist vars fpi	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1000
119	case:DEF_hist_vars%fq	DEF_hist_vars%fq	DEF hist vars fq	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1406
120	case:DEF_hist_vars%frcsat	DEF_hist_vars%frcsat	DEF hist vars frcsat	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:902
121	case:DEF_hist_vars%frootc	DEF_hist_vars%frootc	DEF hist vars frootc	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:953
122	case:DEF_hist_vars%frootcCap	DEF_hist_vars%frootcCap	DEF hist vars frootcCap	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1287
123	case:DEF_hist_vars%frootc_storage	DEF_hist_vars%frootc_storage	DEF hist vars frootc storage	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:954
124	case:DEF_hist_vars%frootc_storageCap	DEF_hist_vars%frootc_storageCap	DEF hist vars frootc storageCap	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1288
125	case:DEF_hist_vars%frootc_xfer	DEF_hist_vars%frootc_xfer	DEF hist vars frootc xfer	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:955
126	case:DEF_hist_vars%frootc_xferCap	DEF_hist_vars%frootc_xferCap	DEF hist vars frootc xferCap	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1289

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
127	case:DEF_hist_vars%frootn	DEF_hist_vars%frootn	DEF hist vars frootn	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:974
128	case:DEF_hist_vars%frootnCap	DEF_hist_vars%frootnCap	DEF hist vars frootnCap	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1305
129	case:DEF_hist_vars%frootn_storage	DEF_hist_vars%frootn_storage	DEF hist vars frootn storage	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:975
130	case:DEF_hist_vars%frootn_storageCap	DEF_hist_vars%frootn_storageCap	DEF hist vars frootn storageCap	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1306
131	case:DEF_hist_vars%frootn_xfer	DEF_hist_vars%frootn_xfer	DEF hist vars frootn xfer	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:976
132	case:DEF_hist_vars%frootn_xferCap	DEF_hist_vars%frootn_xferCap	DEF hist vars frootn xferCap	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1307
133	case:DEF_hist_vars%fsen_gimp	DEF_hist_vars%fsen_gimp	DEF hist vars fsen gimp	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:924
134	case:DEF_hist_vars%fsen_gper	DEF_hist_vars%fsen_gper	DEF hist vars fsen gper	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:925
135	case:DEF_hist_vars%fsen_roof	DEF_hist_vars%fsen_roof	DEF hist vars fsen roof	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:921
136	case:DEF_hist_vars%fsen_urbl	DEF_hist_vars%fsen_urbl	DEF hist vars fsen urbl	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:926
137	case:DEF_hist_vars%fsen_wsha	DEF_hist_vars%fsen_wsha	DEF hist vars fsen wsha	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:923
138	case:DEF_hist_vars%fsen_wsun	DEF_hist_vars%fsen_wsun	DEF hist vars fsen wsun	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:922
139	case:DEF_hist_vars%fsena	DEF_hist_vars%fsena	DEF hist vars fsena	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:861
140	case:DEF_hist_vars%fseng	DEF_hist_vars%fseng	DEF hist vars fseng	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:867
141	case:DEF_hist_vars%fsenl	DEF_hist_vars%fsenl	DEF hist vars fsenl	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:864
142	case:DEF_hist_vars%fsno	DEF_hist_vars%fsno	DEF hist vars fsno	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:901
143	case:DEF_hist_vars%fuelc	DEF_hist_vars%fuelc	DEF hist vars fuelc	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1178

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
144	case:DEF_hist_vars%fwst	DEF_hist_vars%fwst	DEF hist vars fwst	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:932
145	case:DEF_hist_vars%gddmaturity	DEF_hist_vars%gddmaturity	DEF hist vars gddmaturity	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1084
146	case:DEF_hist_vars%gddplant	DEF_hist_vars%gddplant	DEF hist vars gddplant	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1085
147	case:DEF_hist_vars%gdp	DEF_hist_vars%gdp	DEF hist vars gdp	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1279
148	case:DEF_hist_vars%gpp	DEF_hist_vars%gpp	DEF hist vars gpp	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:993
149	case:DEF_hist_vars%gpp_c3arcgrass	DEF_hist_vars%gpp_c3arcgrass	DEF hist vars gpp c3arcgrass	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1023
150	case:DEF_hist_vars%gpp_c3grass	DEF_hist_vars%gpp_c3grass	DEF hist vars gpp c3grass	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1024
151	case:DEF_hist_vars%gpp_c4grass	DEF_hist_vars%gpp_c4grass	DEF hist vars gpp c4grass	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1025
152	case:DEF_hist_vars%gpp_dbfboreal	DEF_hist_vars%gpp_dbfboreal	DEF hist vars gpp dbfboreal	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1019
153	case:DEF_hist_vars%gpp_dbftemp	DEF_hist_vars%gpp_dbftemp	DEF hist vars gpp dbftemp	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1018
154	case:DEF_hist_vars%gpp_dbftrop	DEF_hist_vars%gpp_dbftrop	DEF hist vars gpp dbftrop	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1017
155	case:DEF_hist_vars%gpp_dbsboreal	DEF_hist_vars%gpp_dbsboreal	DEF hist vars gpp dbsboreal	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1022
156	case:DEF_hist_vars%gpp_dbstemp	DEF_hist_vars%gpp_dbstemp	DEF hist vars gpp dbstemp	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1021
157	case:DEF_hist_vars%gpp_dnfboreal	DEF_hist_vars%gpp_dnfboreal	DEF hist vars gpp dnfboreal	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1014
158	case:DEF_hist_vars%gpp_ebftemp	DEF_hist_vars%gpp_ebftemp	DEF hist vars gpp ebftemp	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1016
159	case:DEF_hist_vars%gpp_ebftrop	DEF_hist_vars%gpp_ebftrop	DEF hist vars gpp ebftrop	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1015
160	case:DEF_hist_vars%gpp_ebstemp	DEF_hist_vars%gpp_ebstemp	DEF hist vars gpp ebstemp	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1020

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
161	case:DEF_hist_vars%gpp_enfboreal	DEF_hist_vars%gpp_enfboreal	DEF hist vars gpp enfboreal	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1013
162	case:DEF_hist_vars%gpp_enftemp	DEF_hist_vars%gpp_enftemp	DEF hist vars gpp enftemp	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1012
163	case:DEF_hist_vars%grainc	DEF_hist_vars%grainc	DEF hist vars grainc	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:968
164	case:DEF_hist_vars%grainc_storage	DEF_hist_vars%grainc_storage	DEF hist vars grainc storage	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:969
165	case:DEF_hist_vars%grainc_to_cropprodc	DEF_hist_vars%grainc_to_cropprodc	DEF hist vars grainc to cropprodc	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1091
166	case:DEF_hist_vars%grainc_to_seed	DEF_hist_vars%grainc_to_seed	DEF hist vars grainc to seed	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1131
167	case:DEF_hist_vars%grainc_xfer	DEF_hist_vars%grainc_xfer	DEF hist vars grainc xfer	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:970
168	case:DEF_hist_vars%grainn	DEF_hist_vars%grainn	DEF hist vars grainn	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:989
169	case:DEF_hist_vars%grainn_storage	DEF_hist_vars%grainn_storage	DEF hist vars grainn storage	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:990
170	case:DEF_hist_vars%grainn_xfer	DEF_hist_vars%grainn_xfer	DEF hist vars grainn xfer	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:991
171	case:DEF_hist_vars%green	DEF_hist_vars%green	DEF hist vars green	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:904
172	case:DEF_hist_vars%groundwater_demand	DEF_hist_vars%groundwater_demand	DEF hist vars groundwater demand	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1165
173	case:DEF_hist_vars%groundwater_supply	DEF_hist_vars%groundwater_supply	DEF hist vars groundwater supply	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1166
174	case:DEF_hist_vars%gssha	DEF_hist_vars%gssha	DEF hist vars gssha	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1333
175	case:DEF_hist_vars%gssun	DEF_hist_vars%gssun	DEF hist vars gssun	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1332
176	case:DEF_hist_vars%h2osoi	DEF_hist_vars%h2osoi	DEF hist vars h2osoi	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1327
177	case:DEF_hist_vars%hdm	DEF_hist_vars%hdm	DEF hist vars hdm	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1281

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
178	case:DEF_hist_vars%hr	DEF_hist_vars%hr	DEF hist vars hr	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:998
179	case:DEF_hist_vars%hui	DEF_hist_vars%hui	DEF hist vars hui	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1087
180	case:DEF_hist_vars%huiswheat	DEF_hist_vars%huiswheat	DEF hist vars huiswheat	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1134
181	case:DEF_hist_vars%irrig_method_corn	DEF_hist_vars%irrig_method_corn	DEF hist vars irrig method corn	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1152
182	case:DEF_hist_vars%irrig_method_cotton	DEF_hist_vars%irrig_method_cotton	DEF hist vars irrig method cotton	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1156
183	case:DEF_hist_vars%irrig_method_rice1	DEF_hist_vars%irrig_method_rice1	DEF hist vars irrig method rice1	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1157
184	case:DEF_hist_vars%irrig_method_rice2	DEF_hist_vars%irrig_method_rice2	DEF hist vars irrig method rice2	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1158
185	case:DEF_hist_vars%irrig_method_soybean	DEF_hist_vars%irrig_method_soybean	DEF hist vars irrig method soybean	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1155
186	case:DEF_hist_vars%irrig_method_sugarcane	DEF_hist_vars%irrig_method_sugarcane	DEF hist vars irrig method sugarcane	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1159
187	case:DEF_hist_vars%irrig_method_swheat	DEF_hist_vars%irrig_method_swheat	DEF hist vars irrig method swheat	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1153
188	case:DEF_hist_vars%irrig_method_wwheat	DEF_hist_vars%irrig_method_wwheat	DEF hist vars irrig method wwheat	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1154
189	case:DEF_hist_vars%lai	DEF_hist_vars%lai	DEF hist vars lai	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:905
190	case:DEF_hist_vars%lai_c3arcgrass	DEF_hist_vars%lai_c3arcgrass	DEF hist vars lai c3arcgrass	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1051
191	case:DEF_hist_vars%lai_c3grass	DEF_hist_vars%lai_c3grass	DEF hist vars lai c3grass	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1052
192	case:DEF_hist_vars%lai_c4grass	DEF_hist_vars%lai_c4grass	DEF hist vars lai c4grass	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1053
193	case:DEF_hist_vars%lai_dbfboreal	DEF_hist_vars%lai_dbfboreal	DEF hist vars lai dbfboreal	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1047
194	case:DEF_hist_vars%lai_dbftemp	DEF_hist_vars%lai_dbftemp	DEF hist vars lai dbftemp	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1046

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
195	case:DEF_hist_vars%lai_dbftrop	DEF_hist_vars%lai_dbftrop	DEF hist vars lai dbftrop	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1045
196	case:DEF_hist_vars%lai_dbsboreal	DEF_hist_vars%lai_dbsboreal	DEF hist vars lai dbsboreal	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1050
197	case:DEF_hist_vars%lai_dbstemp	DEF_hist_vars%lai_dbstemp	DEF hist vars lai dbstemp	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1049
198	case:DEF_hist_vars%lai_dnfboreal	DEF_hist_vars%lai_dnfboreal	DEF hist vars lai dnfboreal	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1042
199	case:DEF_hist_vars%lai_ebftemp	DEF_hist_vars%lai_ebftemp	DEF hist vars lai ebftemp	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1044
200	case:DEF_hist_vars%lai_ebftrop	DEF_hist_vars%lai_ebftrop	DEF hist vars lai ebftrop	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1043
201	case:DEF_hist_vars%lai_ebstemp	DEF_hist_vars%lai_ebstemp	DEF hist vars lai ebstemp	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1048
202	case:DEF_hist_vars%lai_enfboreal	DEF_hist_vars%lai_enfboreal	DEF hist vars lai enfboreal	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1041
203	case:DEF_hist_vars%lai_enftemp	DEF_hist_vars%lai_enftemp	DEF hist vars lai enftemp	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1040
204	case:DEF_hist_vars%laisha	DEF_hist_vars%laisha	DEF hist vars laisha	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:907
205	case:DEF_hist_vars%laisun	DEF_hist_vars%laisun	DEF hist vars laisun	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:906
206	case:DEF_hist_vars%lake_deficit	DEF_hist_vars%lake_deficit	DEF hist vars lake deficit	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1329
207	case:DEF_hist_vars%lake_icefrac	DEF_hist_vars%lake_icefrac	DEF hist vars lake icefrac	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1347
208	case:DEF_hist_vars%ldew	DEF_hist_vars%ldew	DEF hist vars ldew	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:898
209	case:DEF_hist_vars%leafc	DEF_hist_vars%leafc	DEF hist vars leafc	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:950
210	case:DEF_hist_vars%leafcCap	DEF_hist_vars%leafcCap	DEF hist vars leafcCap	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1284
211	case:DEF_hist_vars%leafc_c3arcgrass	DEF_hist_vars%leafc_c3arcgrass	DEF hist vars leafc c3arcgrass	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1037

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
212	case:DEF_hist_vars%leafc_c3grass	DEF_hist_vars%leafc_c3grass	DEF hist vars leafc c3grass	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1038
213	case:DEF_hist_vars%leafc_c4grass	DEF_hist_vars%leafc_c4grass	DEF hist vars leafc c4grass	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1039
214	case:DEF_hist_vars%leafc_dbfboreal	DEF_hist_vars%leafc_dbfboreal	DEF hist vars leafc dbfboreal	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1033
215	case:DEF_hist_vars%leafc_dbftemp	DEF_hist_vars%leafc_dbftemp	DEF hist vars leafc dbftemp	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1032
216	case:DEF_hist_vars%leafc_dbftrop	DEF_hist_vars%leafc_dbftrop	DEF hist vars leafc dbftrop	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1031
217	case:DEF_hist_vars%leafc_dbsboreal	DEF_hist_vars%leafc_dbsboreal	DEF hist vars leafc dbsboreal	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1036
218	case:DEF_hist_vars%leafc_dbstemp	DEF_hist_vars%leafc_dbstemp	DEF hist vars leafc dbstemp	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1035
219	case:DEF_hist_vars%leafc_dnfboreal	DEF_hist_vars%leafc_dnfboreal	DEF hist vars leafc dnfboreal	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1028
220	case:DEF_hist_vars%leafc_ebftemp	DEF_hist_vars%leafc_ebftemp	DEF hist vars leafc ebftemp	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1030
221	case:DEF_hist_vars%leafc_ebftrop	DEF_hist_vars%leafc_ebftrop	DEF hist vars leafc ebftrop	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1029
222	case:DEF_hist_vars%leafc_ebstemp	DEF_hist_vars%leafc_ebstemp	DEF hist vars leafc ebstemp	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1034
223	case:DEF_hist_vars%leafc_enfboreal	DEF_hist_vars%leafc_enfboreal	DEF hist vars leafc enfboreal	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1027
224	case:DEF_hist_vars%leafc_enftemp	DEF_hist_vars%leafc_enftemp	DEF hist vars leafc enftemp	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1026
225	case:DEF_hist_vars%leafc_storage	DEF_hist_vars%leafc_storage	DEF hist vars leafc storage	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:951
226	case:DEF_hist_vars%leafc_storageCap	DEF_hist_vars%leafc_storageCap	DEF hist vars leafc storageCap	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1285
227	case:DEF_hist_vars%leafc_xfer	DEF_hist_vars%leafc_xfer	DEF hist vars leafc xfer	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:952
228	case:DEF_hist_vars%leafc_xferCap	DEF_hist_vars%leafc_xferCap	DEF hist vars leafc xferCap	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1286

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
229	case:DEF_hist_vars%leafn	DEF_hist_vars%leafn	DEF hist vars leafn	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:971
230	case:DEF_hist_vars%leafnCap	DEF_hist_vars%leafnCap	DEF hist vars leafnCap	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1302
231	case:DEF_hist_vars%leafn_storage	DEF_hist_vars%leafn_storage	DEF hist vars leafn storage	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:972
232	case:DEF_hist_vars%leafn_storageCap	DEF_hist_vars%leafn_storageCap	DEF hist vars leafn storageCap	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1303
233	case:DEF_hist_vars%leafn_xfer	DEF_hist_vars%leafn_xfer	DEF hist vars leafn xfer	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:973
234	case:DEF_hist_vars%leafn_xferCap	DEF_hist_vars%leafn_xferCap	DEF hist vars leafn xferCap	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1304
235	case:DEF_hist_vars%lfevp_gimp	DEF_hist_vars%lfevp_gimp	DEF hist vars lfevp gimp	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:928
236	case:DEF_hist_vars%lfevp_gper	DEF_hist_vars%lfevp_gper	DEF hist vars lfevp gper	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:929
237	case:DEF_hist_vars%lfevp_roof	DEF_hist_vars%lfevp_roof	DEF hist vars lfevp roof	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:927
238	case:DEF_hist_vars%lfevp_urbl	DEF_hist_vars%lfevp_urbl	DEF hist vars lfevp urbl	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:930
239	case:DEF_hist_vars%lfevpa	DEF_hist_vars%lfevpa	DEF hist vars lfevpa	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:862
240	case:DEF_hist_vars%litfire	DEF_hist_vars%litfire	DEF hist vars litfire	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1185
241	case:DEF_hist_vars%litr1cCap_vr	DEF_hist_vars%litr1cCap_vr	DEF hist vars litr1cCap vr	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1381
242	case:DEF_hist_vars%litr1c_vr	DEF_hist_vars%litr1c_vr	DEF hist vars litr1c vr	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1366
243	case:DEF_hist_vars%litr1nCap_vr	DEF_hist_vars%litr1nCap_vr	DEF hist vars litr1nCap vr	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1388
244	case:DEF_hist_vars%litr1n_vr	DEF_hist_vars%litr1n_vr	DEF hist vars litr1n vr	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1373
245	case:DEF_hist_vars%litr2cCap_vr	DEF_hist_vars%litr2cCap_vr	DEF hist vars litr2cCap vr	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1382

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
246	case:DEF_hist_vars%litr2c_vr	DEF_hist_vars%litr2c_vr	DEF hist vars litr2c vr	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1367
247	case:DEF_hist_vars%litr2nCap_vr	DEF_hist_vars%litr2nCap_vr	DEF hist vars litr2nCap vr	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1389
248	case:DEF_hist_vars%litr2n_vr	DEF_hist_vars%litr2n_vr	DEF hist vars litr2n vr	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1374
249	case:DEF_hist_vars%litr3cCap_vr	DEF_hist_vars%litr3cCap_vr	DEF hist vars litr3cCap vr	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1383
250	case:DEF_hist_vars%litr3c_vr	DEF_hist_vars%litr3c_vr	DEF hist vars litr3c vr	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1368
251	case:DEF_hist_vars%litr3nCap_vr	DEF_hist_vars%litr3nCap_vr	DEF hist vars litr3nCap vr	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1390
252	case:DEF_hist_vars%litr3n_vr	DEF_hist_vars%litr3n_vr	DEF hist vars litr3n vr	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1375
253	case:DEF_hist_vars%livecrootc	DEF_hist_vars%livecrootc	DEF hist vars livecrootc	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:962
254	case:DEF_hist_vars%livecrootcCap	DEF_hist_vars%livecrootcCap	DEF hist vars livecrootcCap	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1296
255	case:DEF_hist_vars%livecrootc_storage	DEF_hist_vars%livecrootc_storage	DEF hist vars livecrootc storage	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:963
256	case:DEF_hist_vars%livecrootc_storageCap	DEF_hist_vars%livecrootc_storageCap	DEF hist vars livecrootc storageCap	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1297
257	case:DEF_hist_vars%livecrootc_xfer	DEF_hist_vars%livecrootc_xfer	DEF hist vars livecrootc xfer	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:964
258	case:DEF_hist_vars%livecrootc_xferCap	DEF_hist_vars%livecrootc_xferCap	DEF hist vars livecrootc xferCap	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1298
259	case:DEF_hist_vars%livecrootn	DEF_hist_vars%livecrootn	DEF hist vars livecrootn	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:983
260	case:DEF_hist_vars%livecrootnCap	DEF_hist_vars%livecrootnCap	DEF hist vars livecrootnCap	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1314
261	case:DEF_hist_vars%livecrootn_storage	DEF_hist_vars%livecrootn_storage	DEF hist vars livecrootn storage	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:984
262	case:DEF_hist_vars%livecrootn_storageCap	DEF_hist_vars%livecrootn_storageCap	DEF hist vars livecrootn storageCap	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1315

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
263	case:DEF_hist_vars%livecrotn_xfer	DEF_hist_vars%livecrootn_xfer	DEF hist vars livecrootn xfer	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:985
264	case:DEF_hist_vars%livecrotn_xferCap	DEF_hist_vars%livecrootn_xferCap	DEF hist vars livecrootn xferCap	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1316
265	case:DEF_hist_vars%livestemc	DEF_hist_vars%livestemc	DEF hist vars livestemc	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:956
266	case:DEF_hist_vars%livestemcCap	DEF_hist_vars%livestemcCap	DEF hist vars livestemcCap	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1290
267	case:DEF_hist_vars%livestemc_storage	DEF_hist_vars%livestemc_storage	DEF hist vars livestemc storage	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:957
268	case:DEF_hist_vars%livestemc_storageCap	DEF_hist_vars%livestemc_storageCap	DEF hist vars livestemc storageCap	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1291
269	case:DEF_hist_vars%livestemc_xfer	DEF_hist_vars%livestemc_xfer	DEF hist vars livestemc xfer	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:958
270	case:DEF_hist_vars%livestemc_xferCap	DEF_hist_vars%livestemc_xferCap	DEF hist vars livestemc xferCap	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1292
271	case:DEF_hist_vars%livestemn	DEF_hist_vars%livestemn	DEF hist vars livestemn	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:977
272	case:DEF_hist_vars%livestemnCap	DEF_hist_vars%livestemnCap	DEF hist vars livestemnCap	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1308
273	case:DEF_hist_vars%livestemn_storage	DEF_hist_vars%livestemn_storage	DEF hist vars livestemn storage	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:978
274	case:DEF_hist_vars%livestemn_storageCap	DEF_hist_vars%livestemn_storageCap	DEF hist vars livestemn storageCap	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1309
275	case:DEF_hist_vars%livestemn_xfer	DEF_hist_vars%livestemn_xfer	DEF hist vars livestemn xfer	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:979
276	case:DEF_hist_vars%livestemn_xferCap	DEF_hist_vars%livestemn_xferCap	DEF hist vars livestemn xferCap	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1310
277	case:DEF_hist_vars%lnfm	DEF_hist_vars%lnfm	DEF hist vars lnfm	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1282
278	case:DEF_hist_vars%m_cwd_c_to_fire	DEF_hist_vars%m_cwd_c_to_fire	DEF hist vars m cwd c to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1276
279	case:DEF_hist_vars%m_cwd_n_to_fire	DEF_hist_vars%m_cwd_n_to_fire	DEF hist vars m cwd n to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1277

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
280	case:DEF_hist_vars%m_deadcrootc_storage_to_fire	DEF_hist_vars%m_deadcrootc_storage_to_fire	DEF hist vars m deadcrootc storage to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1199
281	case:DEF_hist_vars%m_deadcrootc_storage_to_litter_fire	DEF_hist_vars%m_deadcrootc_storage_to_litter_fire	DEF hist vars m deadcrootc storage to litter fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1221
282	case:DEF_hist_vars%m_deadcrootc_to_fire	DEF_hist_vars%m_deadcrootc_to_fire	DEF hist vars m deadcrootc to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1193
283	case:DEF_hist_vars%m_deadcrootc_to_litter_fire	DEF_hist_vars%m_deadcrootc_to_litter_fire	DEF hist vars m deadcrootc to litter fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1215
284	case:DEF_hist_vars%m_deadcrootc_xfer_to_fire	DEF_hist_vars%m_deadcrootc_xfer_to_fire	DEF hist vars m deadcrootc xfer to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1206
285	case:DEF_hist_vars%m_deadcrootc_xfer_to_litter_fire	DEF_hist_vars%m_deadcrootc_xfer_to_litter_fire	DEF hist vars m deadcrootc xfer to litter fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1228
286	case:DEF_hist_vars%m_deadcrootn_storage_to_fire	DEF_hist_vars%m_deadcrootn_storage_to_fire	DEF hist vars m deadcrootn storage to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1241
287	case:DEF_hist_vars%m_deadcrootn_storage_to_litter_fire	DEF_hist_vars%m_deadcrootn_storage_to_litter_fire	DEF hist vars m deadcrootn storage to litter fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1262
288	case:DEF_hist_vars%m_deadcrootn_to_fire	DEF_hist_vars%m_deadcrootn_to_fire	DEF hist vars m deadcrootn to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1235
289	case:DEF_hist_vars%m_deadcrootn_to_litter_fire	DEF_hist_vars%m_deadcrootn_to_litter_fire	DEF hist vars m deadcrootn to litter fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1256
290	case:DEF_hist_vars%m_deadcrootn_xfer_to_fire	DEF_hist_vars%m_deadcrootn_xfer_to_fire	DEF hist vars m deadcrootn xfer to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1247
291	case:DEF_hist_vars%m_deadcrootn_xfer_to_litter_fire	DEF_hist_vars%m_deadcrootn_xfer_to_litter_fire	DEF hist vars m deadcrootn xfer to litter fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1268
292	case:DEF_hist_vars%m_deadstemc_storage_to_fire	DEF_hist_vars%m_deadstemc_storage_to_fire	DEF hist vars m deadstemc storage to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1197
293	case:DEF_hist_vars%m_deadstemc_storage_to_litter_fire	DEF_hist_vars%m_deadstemc_storage_to_litter_fire	DEF hist vars m deadstemc storage to litter fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1219
294	case:DEF_hist_vars%m_deadstemc_to_fire	DEF_hist_vars%m_deadstemc_to_fire	DEF hist vars m deadstemc to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1191

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295	case:DEF_hist_vars%m_deadstemc_to_litter_fire	DEF_hist_vars%m_deadstemc_to_litter_fire	DEF hist vars m deadstemc to litter fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1213
296	case:DEF_hist_vars%m_deadstemc_xfer_to_fire	DEF_hist_vars%m_deadstemc_xfer_to_fire	DEF hist vars m deadstemc xfer to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1204
297	case:DEF_hist_vars%m_deadstemc_xfer_to_litter_fire	DEF_hist_vars%m_deadstemc_xfer_to_litter_fire	DEF hist vars m deadstemc xfer to litter fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1226
298	case:DEF_hist_vars%m_deadstemn_storage_to_fire	DEF_hist_vars%m_deadstemn_storage_to_fire	DEF hist vars m deadstemn storage to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1239
299	case:DEF_hist_vars%m_deadstemn_storage_to_litter_fire	DEF_hist_vars%m_deadstemn_storage_to_litter_fire	DEF hist vars m deadstemn storage to litter fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1260
300	case:DEF_hist_vars%m_deadstemn_to_fire	DEF_hist_vars%m_deadstemn_to_fire	DEF hist vars m deadstemn to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1233
301	case:DEF_hist_vars%m_deadstemn_to_litter_fire	DEF_hist_vars%m_deadstemn_to_litter_fire	DEF hist vars m deadstemn to litter fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1254
302	case:DEF_hist_vars%m_deadstemn_xfer_to_fire	DEF_hist_vars%m_deadstemn_xfer_to_fire	DEF hist vars m deadstemn xfer to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1245
303	case:DEF_hist_vars%m_deadstemn_xfer_to_litter_fire	DEF_hist_vars%m_deadstemn_xfer_to_litter_fire	DEF hist vars m deadstemn xfer to litter fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1266
304	case:DEF_hist_vars%m_frootc_storage_to_fire	DEF_hist_vars%m_frootc_storage_to_fire	DEF hist vars m frootc storage to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1195
305	case:DEF_hist_vars%m_frootc_storage_to_litter_fire	DEF_hist_vars%m_frootc_storage_to_litter_fire	DEF hist vars m frootc storage to litter fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1217
306	case:DEF_hist_vars%m_frootc_to_fire	DEF_hist_vars%m_frootc_to_fire	DEF hist vars m frootc to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1189
307	case:DEF_hist_vars%m_frootc_to_litter_fire	DEF_hist_vars%m_frootc_to_litter_fire	DEF hist vars m frootc to litter fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1211
308	case:DEF_hist_vars%m_frootc_xfer_to_fire	DEF_hist_vars%m_frootc_xfer_to_fire	DEF hist vars m frootc xfer to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1202
309	case:DEF_hist_vars%m_frootc_xfer_to_litter_fire	DEF_hist_vars%m_frootc_xfer_to_litter_fire	DEF hist vars m frootc xfer to litter fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1224
310	case:DEF_hist_vars%m_frootn_storage_to_fire	DEF_hist_vars%m_frootn_storage_to_fire	DEF hist vars m frootn storage to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1237
311	case:DEF_hist_vars%m_frootn_storage_to_litter_fire	DEF_hist_vars%m_frootn_storage_to_litter_fire	DEF hist vars m frootn storage to litter fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1258

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
312	case:DEF_hist_vars%m_frootn_to_fire	DEF_hist_vars%m_frootn_to_fire	DEF hist vars m frootn to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1231
313	case:DEF_hist_vars%m_frootn_to_litter_fire	DEF_hist_vars%m_frootn_to_litter_fire	DEF hist vars m frootn to litter fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1252
314	case:DEF_hist_vars%m_frootn_xfer_to_fire	DEF_hist_vars%m_frootn_xfer_to_fire	DEF hist vars m frootn xfer to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1243
315	case:DEF_hist_vars%m_frootn_xfer_to_litter_fire	DEF_hist_vars%m_frootn_xfer_to_litter_fire	DEF hist vars m frootn xfer to litter fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1264
316	case:DEF_hist_vars%m_gresp_storage_to_fire	DEF_hist_vars%m_gresp_storage_to_fire	DEF hist vars m gresp storage to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1200
317	case:DEF_hist_vars%m_gresp_storage_to_litter_fire	DEF_hist_vars%m_gresp_storage_to_litter_fire	DEF hist vars m gresp storage to litter fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1222
318	case:DEF_hist_vars%m_gresp_xfer_to_fire	DEF_hist_vars%m_gresp_xfer_to_fire	DEF hist vars m gresp xfer to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1207
319	case:DEF_hist_vars%m_gresp_xfer_to_litter_fire	DEF_hist_vars%m_gresp_xfer_to_litter_fire	DEF hist vars m gresp xfer to litter fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1229
320	case:DEF_hist_vars%m_leafc_storage_to_fire	DEF_hist_vars%m_leafc_storage_to_fire	DEF hist vars m leafc storage to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1194
321	case:DEF_hist_vars%m_leafc_storage_to_litter_fire	DEF_hist_vars%m_leafc_storage_to_litter_fire	DEF hist vars m leafc storage to litter fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1216
322	case:DEF_hist_vars%m_leafc_to_fire	DEF_hist_vars%m_leafc_to_fire	DEF hist vars m leafc to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1188
323	case:DEF_hist_vars%m_leafc_to_litter_fire	DEF_hist_vars%m_leafc_to_litter_fire	DEF hist vars m leafc to litter fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1210
324	case:DEF_hist_vars%m_leafc_xfer_to_fire	DEF_hist_vars%m_leafc_xfer_to_fire	DEF hist vars m leafc xfer to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1201
325	case:DEF_hist_vars%m_leafc_xfer_to_litter_fire	DEF_hist_vars%m_leafc_xfer_to_litter_fire	DEF hist vars m leafc xfer to litter fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1223
326	case:DEF_hist_vars%m_leafn_storage_to_fire	DEF_hist_vars%m_leafn_storage_to_fire	DEF hist vars m leafn storage to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1236
327	case:DEF_hist_vars%m_leafn_storage_to_litter_fire	DEF_hist_vars%m_leafn_storage_to_litter_fire	DEF hist vars m leafn storage to litter fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1257
328	case:DEF_hist_vars%m_leafn_to_fire	DEF_hist_vars%m_leafn_to_fire	DEF hist vars m leafn to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1230

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
329	case:DEF_hist_vars%m_leafn_to_litter_fire	DEF_hist_vars%m_leafn_to_litter_fire	DEF hist vars m leafn to litter fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1251
330	case:DEF_hist_vars%m_leafn_xfer_to_fire	DEF_hist_vars%m_leafn_xfer_to_fire	DEF hist vars m leafn xfer to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1242
331	case:DEF_hist_vars%m_leafn_xfer_to_litter_fire	DEF_hist_vars%m_leafn_xfer_to_litter_fire	DEF hist vars m leafn xfer to litter fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1263
332	case:DEF_hist_vars%m_litr1_c_to_fire	DEF_hist_vars%m_litr1_c_to_fire	DEF hist vars m litr1 c to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1270
333	case:DEF_hist_vars%m_litr1_n_to_fire	DEF_hist_vars%m_litr1_n_to_fire	DEF hist vars m litr1 n to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1271
334	case:DEF_hist_vars%m_litr2_c_to_fire	DEF_hist_vars%m_litr2_c_to_fire	DEF hist vars m litr2 c to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1272
335	case:DEF_hist_vars%m_litr2_n_to_fire	DEF_hist_vars%m_litr2_n_to_fire	DEF hist vars m litr2 n to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1273
336	case:DEF_hist_vars%m_litr3_c_to_fire	DEF_hist_vars%m_litr3_c_to_fire	DEF hist vars m litr3 c to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1274
337	case:DEF_hist_vars%m_litr3_n_to_fire	DEF_hist_vars%m_litr3_n_to_fire	DEF hist vars m litr3 n to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1275
338	case:DEF_hist_vars%m_livecrootc_storage_to_fire	DEF_hist_vars%m_livecrootc_storage_to_fire	DEF hist vars m livecrootc storage to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1198
339	case:DEF_hist_vars%m_livecrootc_storage_to_litter_fire	DEF_hist_vars%m_livecrootc_storage_to_litter_fire	DEF hist vars m livecrootc storage to litter fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1220
340	case:DEF_hist_vars%m_livecrootc_to_deadcrootc_fire	DEF_hist_vars%m_livecrootc_to_deadcrootc_fire	DEF hist vars m livecrootc to deadcrootc fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1209
341	case:DEF_hist_vars%m_livecrootc_to_fire	DEF_hist_vars%m_livecrootc_to_fire	DEF hist vars m livecrootc to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1192
342	case:DEF_hist_vars%m_livecrootc_to_litter_fire	DEF_hist_vars%m_livecrootc_to_litter_fire	DEF hist vars m livecrootc to litter fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1214
343	case:DEF_hist_vars%m_livecrootc_xfer_to_fire	DEF_hist_vars%m_livecrootc_xfer_to_fire	DEF hist vars m livecrootc xfer to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1205
344	case:DEF_hist_vars%m_livecrootc_xfer_to_litter_fire	DEF_hist_vars%m_livecrootc_xfer_to_litter_fire	DEF hist vars m livecrootc xfer to litter fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1227
345	case:DEF_hist_vars%m_livecrootn_storage_to_fire	DEF_hist_vars%m_livecrootn_storage_to_fire	DEF hist vars m livecrootn storage to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1240

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
346	case:DEF_hist_vars%m_livecrootn_storage_to_litter_fire	DEF_hist_vars%m_livecrootn_storage_to_litter_fire	DEF hist vars m livecrootn storage to litter fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1261
347	case:DEF_hist_vars%m_livecrootn_to_deadcrootn_fire	DEF_hist_vars%m_livecrootn_to_deadcrootn_fire	DEF hist vars m livecrootn to deadcrootn fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1249
348	case:DEF_hist_vars%m_livecrootn_to_fire	DEF_hist_vars%m_livecrootn_to_fire	DEF hist vars m livecrootn to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1234
349	case:DEF_hist_vars%m_livecrootn_to_litter_fire	DEF_hist_vars%m_livecrootn_to_litter_fire	DEF hist vars m livecrootn to litter fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1255
350	case:DEF_hist_vars%m_livecrootn_xfer_to_fire	DEF_hist_vars%m_livecrootn_xfer_to_fire	DEF hist vars m livecrootn xfer to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1246
351	case:DEF_hist_vars%m_livecrootn_xfer_to_litter_fire	DEF_hist_vars%m_livecrootn_xfer_to_litter_fire	DEF hist vars m livecrootn xfer to litter fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1267
352	case:DEF_hist_vars%m_livestemc_storage_to_fire	DEF_hist_vars%m_livestemc_storage_to_fire	DEF hist vars m livestemc storage to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1196
353	case:DEF_hist_vars%m_livestemc_storage_to_litter_fire	DEF_hist_vars%m_livestemc_storage_to_litter_fire	DEF hist vars m livestemc storage to litter fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1218
354	case:DEF_hist_vars%m_livestemc_to_deadstemc_fire	DEF_hist_vars%m_livestemc_to_deadstemc_fire	DEF hist vars m livestemc to deadstemc fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1208
355	case:DEF_hist_vars%m_livestemc_to_fire	DEF_hist_vars%m_livestemc_to_fire	DEF hist vars m livestemc to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1190
356	case:DEF_hist_vars%m_livestemc_to_litter_fire	DEF_hist_vars%m_livestemc_to_litter_fire	DEF hist vars m livestemc to litter fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1212
357	case:DEF_hist_vars%m_livestemc_xfer_to_fire	DEF_hist_vars%m_livestemc_xfer_to_fire	DEF hist vars m livestemc xfer to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1203
358	case:DEF_hist_vars%m_livestemc_xfer_to_litter_fire	DEF_hist_vars%m_livestemc_xfer_to_litter_fire	DEF hist vars m livestemc xfer to litter fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1225
359	case:DEF_hist_vars%m_livestemn_storage_to_fire	DEF_hist_vars%m_livestemn_storage_to_fire	DEF hist vars m livestemn storage to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1238
360	case:DEF_hist_vars%m_livestemn_storage_to_litter_fire	DEF_hist_vars%m_livestemn_storage_to_litter_fire	DEF hist vars m livestemn storage to litter fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1259
361	case:DEF_hist_vars%m_livestemn_to_deadstemn_fire	DEF_hist_vars%m_livestemn_to_deadstemn_fire	DEF hist vars m livestemn to deadstemn fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1248

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
362	case:DEF_hist_vars%m_livestemn_to_fire	DEF_hist_vars%m_livestemn_to_fire	DEF hist vars m livestemn to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1232
363	case:DEF_hist_vars%m_livestemn_to_litter_fire	DEF_hist_vars%m_livestemn_to_litter_fire	DEF hist vars m livestemn to litter fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1253
364	case:DEF_hist_vars%m_livestemn_xfer_to_fire	DEF_hist_vars%m_livestemn_xfer_to_fire	DEF hist vars m livestemn xfer to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1244
365	case:DEF_hist_vars%m_livestemn_xfer_to_litter_fire	DEF_hist_vars%m_livestemn_xfer_to_litter_fire	DEF hist vars m livestemn xfer to litter fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1265
366	case:DEF_hist_vars%m_retransn_to_fire	DEF_hist_vars%m_retransn_to_fire	DEF hist vars m retransn to fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1250
367	case:DEF_hist_vars%m_retransn_to_litter_fire	DEF_hist_vars%m_retransn_to_litter_fire	DEF hist vars m retransn to litter fire	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1269
368	case:DEF_hist_vars%manunitro	DEF_hist_vars%manunitro	DEF hist vars manunitro	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1143
369	case:DEF_hist_vars%meta	DEF_hist_vars%meta	DEF hist vars meta	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:935
370	case:DEF_hist_vars%ndep_to_sminn	DEF_hist_vars%ndep_to_sminn	DEF hist vars ndep to sminn	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1170
371	case:DEF_hist_vars%netflw	DEF_hist_vars%netflw	DEF hist vars netflw	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1451
372	case:DEF_hist_vars%nfire	DEF_hist_vars%nfire	DEF hist vars nfire	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1177
373	case:DEF_hist_vars%npp_c3arcgrass	DEF_hist_vars%npp_c3arcgrass	DEF hist vars npp c3arcgrass	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1065
374	case:DEF_hist_vars%npp_c3grass	DEF_hist_vars%npp_c3grass	DEF hist vars npp c3grass	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1066
375	case:DEF_hist_vars%npp_c4grass	DEF_hist_vars%npp_c4grass	DEF hist vars npp c4grass	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1067
376	case:DEF_hist_vars%npp_dbfboreal	DEF_hist_vars%npp_dbfboreal	DEF hist vars npp dbfboreal	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1061
377	case:DEF_hist_vars%npp_dbftemp	DEF_hist_vars%npp_dbftemp	DEF hist vars npp dbftemp	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1060
378	case:DEF_hist_vars%npp_dbftrop	DEF_hist_vars%npp_dbftrop	DEF hist vars npp dbftrop	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1059

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379	case:DEF_hist_vars%npp_dbsboreal	DEF_hist_vars%npp_dbsboreal	DEF hist vars npp dbsboreal	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1064
380	case:DEF_hist_vars%npp_dbstemp	DEF_hist_vars%npp_dbstemp	DEF hist vars npp dbstemp	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1063
381	case:DEF_hist_vars%npp_dnfboreal	DEF_hist_vars%npp_dnfboreal	DEF hist vars npp dnfboreal	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1056
382	case:DEF_hist_vars%npp_ebftemp	DEF_hist_vars%npp_ebftemp	DEF hist vars npp ebftemp	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1058
383	case:DEF_hist_vars%npp_ebftrop	DEF_hist_vars%npp_ebftrop	DEF hist vars npp ebftrop	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1057
384	case:DEF_hist_vars%npp_ebstemp	DEF_hist_vars%npp_ebstemp	DEF hist vars npp ebstemp	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1062
385	case:DEF_hist_vars%npp_enfboreal	DEF_hist_vars%npp_enfboreal	DEF hist vars npp enfboreal	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1055
386	case:DEF_hist_vars%npp_enftemp	DEF_hist_vars%npp_enftemp	DEF hist vars npp enftemp	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1054
387	case:DEF_hist_vars%npptoleafc_c3arcgrass	DEF_hist_vars%npptoleafc_c3arcgrass	DEF hist vars npptoleafc c3arcgrass	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1079
388	case:DEF_hist_vars%npptoleafc_c3grass	DEF_hist_vars%npptoleafc_c3grass	DEF hist vars npptoleafc c3grass	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1080
389	case:DEF_hist_vars%npptoleafc_c4grass	DEF_hist_vars%npptoleafc_c4grass	DEF hist vars npptoleafc c4grass	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1081
390	case:DEF_hist_vars%npptoleafc_dbfboreal	DEF_hist_vars%npptoleafc_dbfboreal	DEF hist vars npptoleafc dbfboreal	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1075
391	case:DEF_hist_vars%npptoleafc_dbftemp	DEF_hist_vars%npptoleafc_dbftemp	DEF hist vars npptoleafc dbftemp	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1074
392	case:DEF_hist_vars%npptoleafc_dbftrop	DEF_hist_vars%npptoleafc_dbftrop	DEF hist vars npptoleafc dbftrop	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1073
393	case:DEF_hist_vars%npptoleafc_dbsboreal	DEF_hist_vars%npptoleafc_dbsboreal	DEF hist vars npptoleafc dbsboreal	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1078
394	case:DEF_hist_vars%npptoleafc_dbstemp	DEF_hist_vars%npptoleafc_dbstemp	DEF hist vars npptoleafc dbstemp	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1077
395	case:DEF_hist_vars%npptoleafc_dnfboreal	DEF_hist_vars%npptoleafc_dnfboreal	DEF hist vars npptoleafc dnfboreal	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1070

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
396	case:DEF_hist_vars%npptoleafc_ebftemp	DEF_hist_vars%npptoleafc_ebftemp	DEF hist vars npptoleafc ebftemp	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1072
397	case:DEF_hist_vars%npptoleafc_ebftrop	DEF_hist_vars%npptoleafc_ebftrop	DEF hist vars npptoleafc ebftrop	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1071
398	case:DEF_hist_vars%npptoleafc_ebstemp	DEF_hist_vars%npptoleafc_ebstemp	DEF hist vars npptoleafc ebstemp	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1076
399	case:DEF_hist_vars%npptoleafc_enfboreal	DEF_hist_vars%npptoleafc_enfboreal	DEF hist vars npptoleafc enfboreal	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1069
400	case:DEF_hist_vars%npptoleafc_enftemp	DEF_hist_vars%npptoleafc_enftemp	DEF hist vars npptoleafc enftemp	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1068
401	case:DEF_hist_vars%o3uptakesha	DEF_hist_vars%o3uptakesha	DEF hist vars o3uptakesha	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:948
402	case:DEF_hist_vars%o3uptakesun	DEF_hist_vars%o3uptakesun	DEF hist vars o3uptakesun	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:947
403	case:DEF_hist_vars%olrg	DEF_hist_vars%olrg	DEF hist vars olrg	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:873
404	case:DEF_hist_vars%pdcorn	DEF_hist_vars%pdcorn	DEF hist vars pdcorn	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1135
405	case:DEF_hist_vars%pdcotton	DEF_hist_vars%pdcotton	DEF hist vars pdcotton	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1139
406	case:DEF_hist_vars%pdrice1	DEF_hist_vars%pdrice1	DEF hist vars pdrice1	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1140
407	case:DEF_hist_vars%pdrice2	DEF_hist_vars%pdrice2	DEF hist vars pdrice2	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1141
408	case:DEF_hist_vars%pdsoybean	DEF_hist_vars%pdsoybean	DEF hist vars pdsoybean	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1138
409	case:DEF_hist_vars%pdsugarcane	DEF_hist_vars%pdsugarcane	DEF hist vars pdsugarcane	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1142
410	case:DEF_hist_vars%pdswheat	DEF_hist_vars%pdswheat	DEF hist vars pdswheat	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1136
411	case:DEF_hist_vars%pdwwheat	DEF_hist_vars%pdwwheat	DEF hist vars pdwwheat	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1137
412	case:DEF_hist_vars%peatf	DEF_hist_vars%peatf	DEF hist vars peatf	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1280

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
413	case:DEF_hist_vars%pft_fire_closs	DEF_hist_vars%pft_fire_closs	DEF hist vars pft fire closs	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1180
414	case:DEF_hist_vars%pft_fire_nloss	DEF_hist_vars%pft_fire_nloss	DEF hist vars pft fire nloss	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1181
415	case:DEF_hist_vars%plantdate_irrigated_cotton	DEF_hist_vars%plantdate_irrigated_cotton	DEF hist vars plantdate irrigated cotton	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1101
416	case:DEF_hist_vars%plantdate_irrigated_rice	DEF_hist_vars%plantdate_irrigated_rice	DEF hist vars plantdate irrigated rice	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1103
417	case:DEF_hist_vars%plantdate_irrigated_spwheat	DEF_hist_vars%plantdate_irrigated_spwheat	DEF hist vars plantdate irrigated spwheat	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1095
418	case:DEF_hist_vars%plantdate_irrigated_sugarcane	DEF_hist_vars%plantdate_irrigated_sugarcane	DEF hist vars plantdate irrigated sugarcane	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1105
419	case:DEF_hist_vars%plantdate_irrigated_temp_corn	DEF_hist_vars%plantdate_irrigated_temp_corn	DEF hist vars plantdate irrigated temp corn	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1093
420	case:DEF_hist_vars%plantdate_irrigated_temp_soybean	DEF_hist_vars%plantdate_irrigated_temp_soybean	DEF hist vars plantdate irrigated temp soybean	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1099
421	case:DEF_hist_vars%plantdate_irrigated_trop_corn	DEF_hist_vars%plantdate_irrigated_trop_corn	DEF hist vars plantdate irrigated trop corn	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1107
422	case:DEF_hist_vars%plantdate_irrigated_trop_soybean	DEF_hist_vars%plantdate_irrigated_trop_soybean	DEF hist vars plantdate irrigated trop soybean	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1109
423	case:DEF_hist_vars%plantdate_irrigated_wtwheat	DEF_hist_vars%plantdate_irrigated_wtwheat	DEF hist vars plantdate irrigated wtwheat	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1097
424	case:DEF_hist_vars%plantdate_rainfed_cotton	DEF_hist_vars%plantdate_rainfed_cotton	DEF hist vars plantdate rainfed cotton	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1100
425	case:DEF_hist_vars%plantdate_rainfed_rice	DEF_hist_vars%plantdate_rainfed_rice	DEF hist vars plantdate rainfed rice	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1102
426	case:DEF_hist_vars%plantdate_rainfed_spwheat	DEF_hist_vars%plantdate_rainfed_spwheat	DEF hist vars plantdate rainfed spwheat	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1094
427	case:DEF_hist_vars%plantdate_rainfed_sugarcane	DEF_hist_vars%plantdate_rainfed_sugarcane	DEF hist vars plantdate rainfed sugarcane	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1104
428	case:DEF_hist_vars%plantdate_rainfed_temp_corn	DEF_hist_vars%plantdate_rainfed_temp_corn	DEF hist vars plantdate rainfed temp corn	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1092
429	case:DEF_hist_vars%plantdate_rainfed_temp_soybean	DEF_hist_vars%plantdate_rainfed_temp_soybean	DEF hist vars plantdate rainfed temp soybean	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1098

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
430	case:DEF_hist_vars%plantdate_rainfed_trop_corn	DEF_hist_vars%plantdate_rainfed_trop_corn	DEF hist vars plantdate rainfed trop corn	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1106
431	case:DEF_hist_vars%plantdate_rainfed_trop_soybean	DEF_hist_vars%plantdate_rainfed_trop_soybean	DEF hist vars plantdate rainfed trop soybean	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1108
432	case:DEF_hist_vars%plantdate_rainfed_wtwheat	DEF_hist_vars%plantdate_rainfed_wtwheat	DEF hist vars plantdate rainfed wtwheat	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1096
433	case:DEF_hist_vars%plantdate_unmanagedcrop	DEF_hist_vars%plantdate_unmanagedcrop	DEF hist vars plantdate unmanagedcrop	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1110
434	case:DEF_hist_vars%qcharge	DEF_hist_vars%qcharge	DEF hist vars qcharge	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:895
435	case:DEF_hist_vars%qdrip	DEF_hist_vars%qdrip	DEF hist vars qdrip	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:887
436	case:DEF_hist_vars%qinfl	DEF_hist_vars%qinfl	DEF hist vars qinfl	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:886
437	case:DEF_hist_vars%qintr	DEF_hist_vars%qintr	DEF hist vars qintr	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:885
438	case:DEF_hist_vars%qlayer	DEF_hist_vars%qlayer	DEF hist vars qlayer	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1328
439	case:DEF_hist_vars%qref	DEF_hist_vars%qref	DEF hist vars qref	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:919
440	case:DEF_hist_vars%qresv_in	DEF_hist_vars%qresv_in	DEF hist vars qresv in	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1444
441	case:DEF_hist_vars%qresv_out	DEF_hist_vars%qresv_out	DEF hist vars qresv out	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1445
442	case:DEF_hist_vars%qstar	DEF_hist_vars%qstar	DEF hist vars qstar	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1401
443	case:DEF_hist_vars%reflectance_out	DEF_hist_vars%reflectance_out	DEF hist vars reflectance out	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:911
444	case:DEF_hist_vars%reservoirriver_demand	DEF_hist_vars%reservoirriver_demand	DEF hist vars reservoirriver demand	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1167
445	case:DEF_hist_vars%reservoirriver_supply	DEF_hist_vars%reservoirriver_supply	DEF hist vars reservoirriver supply	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1168
446	case:DEF_hist_vars%respc	DEF_hist_vars%respc	DEF hist vars respc	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:894

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
447	case:DEF_hist_vars%retrasn	DEF_hist_vars%retrasn	DEF hist vars retrasn	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:992
448	case:DEF_hist_vars%rib	DEF_hist_vars%rib	DEF hist vars rib	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1403
449	case:DEF_hist_vars%riv_height	DEF_hist_vars%riv_height	DEF hist vars riv height	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1436
450	case:DEF_hist_vars%riv_veloct	DEF_hist_vars%riv_veloct	DEF hist vars riv veloct	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1437
451	case:DEF_hist_vars%rnet	DEF_hist_vars%rnet	DEF hist vars rnet	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:874
452	case:DEF_hist_vars%rnof	DEF_hist_vars%rnof	DEF hist vars rnof	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:881
453	case:DEF_hist_vars%rootr	DEF_hist_vars%rootr	DEF hist vars rootr	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1334
454	case:DEF_hist_vars%rss	DEF_hist_vars%rss	DEF hist vars rss	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:916
455	case:DEF_hist_vars%rstfacsha	DEF_hist_vars%rstfacsha	DEF hist vars rstfacsha	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1331
456	case:DEF_hist_vars%rstfacsun	DEF_hist_vars%rstfacsun	DEF hist vars rstfacsun	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1330
457	case:DEF_hist_vars%rsub	DEF_hist_vars%rsub	DEF hist vars rsub	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:880
458	case:DEF_hist_vars%rsur	DEF_hist_vars%rsur	DEF hist vars rsur	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:877
459	case:DEF_hist_vars%rsur_ie	DEF_hist_vars%rsur_ie	DEF hist vars rsur ie	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:879
460	case:DEF_hist_vars%rsur_se	DEF_hist_vars%rsur_se	DEF hist vars rsur se	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:878
461	case:DEF_hist_vars%sabg	DEF_hist_vars%sabg	DEF hist vars sabg	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:872
462	case:DEF_hist_vars%sabvsha	DEF_hist_vars%sabvsha	DEF hist vars sabvsha	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:871
463	case:DEF_hist_vars%sabvsun	DEF_hist_vars%sabvsun	DEF hist vars sabvsun	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:870

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
464	case:DEF_hist_vars%sai	DEF_hist_vars%sai	DEF hist vars sai	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:908
465	case:DEF_hist_vars%scv	DEF_hist_vars%scv	DEF hist vars scv	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:899
466	case:DEF_hist_vars%sedcon	DEF_hist_vars%sedcon	DEF hist vars sedcon	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1447
467	case:DEF_hist_vars%sedinp	DEF_hist_vars%sedinp	DEF hist vars sedinp	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1450
468	case:DEF_hist_vars%sedlayer	DEF_hist_vars%sedlayer	DEF hist vars sedlayer	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1452
469	case:DEF_hist_vars%sedout	DEF_hist_vars%sedout	DEF hist vars sedout	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1448
470	case:DEF_hist_vars%sensors	DEF_hist_vars%sensors	DEF hist vars sensors	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1455
471	case:DEF_hist_vars%shearvel	DEF_hist_vars%shearvel	DEF hist vars shearvel	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1453
472	case:DEF_hist_vars%sigf	DEF_hist_vars%sigf	DEF hist vars sigf	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:903
473	case:DEF_hist_vars%sminn_vr	DEF_hist_vars%sminn_vr	DEF hist vars sminn vr	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1396
474	case:DEF_hist_vars%snowdp	DEF_hist_vars%snowdp	DEF hist vars snowdp	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:900
475	case:DEF_hist_vars%soil1cCap_vr	DEF_hist_vars%soil1cCap_vr	DEF hist vars soil1cCap vr	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1384
476	case:DEF_hist_vars%soil1c_vr	DEF_hist_vars%soil1c_vr	DEF hist vars soil1c vr	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1369
477	case:DEF_hist_vars%soil1nCap_vr	DEF_hist_vars%soil1nCap_vr	DEF hist vars soil1nCap vr	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1391
478	case:DEF_hist_vars%soil1n_vr	DEF_hist_vars%soil1n_vr	DEF hist vars soil1n vr	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1376
479	case:DEF_hist_vars%soil2cCap_vr	DEF_hist_vars%soil2cCap_vr	DEF hist vars soil2cCap vr	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1385
480	case:DEF_hist_vars%soil2c_vr	DEF_hist_vars%soil2c_vr	DEF hist vars soil2c vr	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1370

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
481	case:DEF_hist_vars%soil2nCap_vr	DEF_hist_vars%soil2nCap_vr	DEF hist vars soil2nCap vr	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1392
482	case:DEF_hist_vars%soil2n_vr	DEF_hist_vars%soil2n_vr	DEF hist vars soil2n vr	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1377
483	case:DEF_hist_vars%soil3cCap_vr	DEF_hist_vars%soil3cCap_vr	DEF hist vars soil3cCap vr	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1386
484	case:DEF_hist_vars%soil3c_vr	DEF_hist_vars%soil3c_vr	DEF hist vars soil3c vr	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1371
485	case:DEF_hist_vars%soil3nCap_vr	DEF_hist_vars%soil3nCap_vr	DEF hist vars soil3nCap vr	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1393
486	case:DEF_hist_vars%soil3n_vr	DEF_hist_vars%soil3n_vr	DEF hist vars soil3n vr	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1378
487	case:DEF_hist_vars%sol_dif_ln_hires	DEF_hist_vars%sol_dif_ln_hires	DEF hist vars sol dif ln hires	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1430
488	case:DEF_hist_vars%sol_dir_ln_hires	DEF_hist_vars%sol_dir_ln_hires	DEF hist vars sol dir ln hires	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1429
489	case:DEF_hist_vars%solnd	DEF_hist_vars%solnd	DEF hist vars solnd	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1413
490	case:DEF_hist_vars%solndln	DEF_hist_vars%solndln	DEF hist vars solndln	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1422
491	case:DEF_hist_vars%solni	DEF_hist_vars%solni	DEF hist vars solni	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1414
492	case:DEF_hist_vars%solniln	DEF_hist_vars%solniln	DEF hist vars solniln	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1423
493	case:DEF_hist_vars%solvd	DEF_hist_vars%solvd	DEF hist vars solvd	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1411
494	case:DEF_hist_vars%solvdln	DEF_hist_vars%solvdln	DEF hist vars solvdln	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1420
495	case:DEF_hist_vars%solvi	DEF_hist_vars%solvi	DEF hist vars solvi	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1412
496	case:DEF_hist_vars%solviln	DEF_hist_vars%solviln	DEF hist vars solviln	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1421
497	case:DEF_hist_vars%somc_fire	DEF_hist_vars%somc_fire	DEF hist vars somc fire	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1184

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
498	case:DEF_hist_vars%somfire	DEF_hist_vars%somfire	DEF hist vars somfire	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1186
499	case:DEF_hist_vars%sr	DEF_hist_vars%sr	DEF hist vars sr	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1410
500	case:DEF_hist_vars%sr_dif_ln_hires	DEF_hist_vars%sr_dif_ln_hires	DEF hist vars sr dif ln hires	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1432
501	case:DEF_hist_vars%sr_dir_ln_hires	DEF_hist_vars%sr_dir_ln_hires	DEF hist vars sr dir ln hires	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1431
502	case:DEF_hist_vars%srnd	DEF_hist_vars%srnd	DEF hist vars srnd	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1417
503	case:DEF_hist_vars%srndln	DEF_hist_vars%srndln	DEF hist vars srndln	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1426
504	case:DEF_hist_vars%srni	DEF_hist_vars%srni	DEF hist vars srni	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1418
505	case:DEF_hist_vars%srniln	DEF_hist_vars%srniln	DEF hist vars srniln	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1427
506	case:DEF_hist_vars%srvd	DEF_hist_vars%srvd	DEF hist vars srvd	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1415
507	case:DEF_hist_vars%srvdln	DEF_hist_vars%srvdln	DEF hist vars srvdln	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1424
508	case:DEF_hist_vars%srvi	DEF_hist_vars%srvi	DEF hist vars srvi	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1416
509	case:DEF_hist_vars%srviln	DEF_hist_vars%srviln	DEF hist vars srviln	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1425
510	case:DEF_hist_vars%sum_deficit_irrig	DEF_hist_vars%sum_deficit_irrig	DEF hist vars sum deficit irrig	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1162
511	case:DEF_hist_vars%sum_irrig	DEF_hist_vars%sum_irrig	DEF hist vars sum irrig	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1161
512	case:DEF_hist_vars%sum_irrig_count	DEF_hist_vars%sum_irrig_count	DEF hist vars sum irrig count	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1163
513	case:DEF_hist_vars%t2m_wmo	DEF_hist_vars%t2m_wmo	DEF hist vars t2m wmo	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:918
514	case:DEF_hist_vars%t_grnd	DEF_hist_vars%t_grnd	DEF hist vars t grnd	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:896

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
515	case:DEF_hist_vars%t_lake	DEF_hist_vars%t_lake	DEF hist vars t lake	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1346
516	case:DEF_hist_vars%t_roof	DEF_hist_vars%t_roof	DEF hist vars t roof	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:939
517	case:DEF_hist_vars%t_room	DEF_hist_vars%t_room	DEF hist vars t room	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:937
518	case:DEF_hist_vars%t_scalar	DEF_hist_vars%t_scalar	DEF hist vars t scalar	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1320
519	case:DEF_hist_vars%t_soisno	DEF_hist_vars%t_soisno	DEF hist vars t soisno	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1323
520	case:DEF_hist_vars%t_wall	DEF_hist_vars%t_wall	DEF hist vars t wall	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:940
521	case:DEF_hist_vars%tafu	DEF_hist_vars%tafu	DEF hist vars tafu	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:938
522	case:DEF_hist_vars%taux	DEF_hist_vars%taux	DEF hist vars taux	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:859
523	case:DEF_hist_vars%tauy	DEF_hist_vars%tauy	DEF hist vars tauy	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:860
524	case:DEF_hist_vars%tleaf	DEF_hist_vars%tleaf	DEF hist vars tleaf	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:897
525	case:DEF_hist_vars%totcolc	DEF_hist_vars%totcolc	DEF hist vars totcolc	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1005
526	case:DEF_hist_vars%totcoln	DEF_hist_vars%totcoln	DEF hist vars totcoln	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1010
527	case:DEF_hist_vars%totcwdc	DEF_hist_vars%totcwdc	DEF hist vars totcwdc	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1003
528	case:DEF_hist_vars%totcwdn	DEF_hist_vars%totcwdn	DEF hist vars totcwdn	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1008
529	case:DEF_hist_vars%totfire	DEF_hist_vars%totfire	DEF hist vars totfire	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1187
530	case:DEF_hist_vars%totlitc	DEF_hist_vars%totlitc	DEF hist vars totlitc	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1002
531	case:DEF_hist_vars%totlitn	DEF_hist_vars%totlitn	DEF hist vars totlitn	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1007

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
532	case:DEF_hist_vars%totsoiln_vr	DEF_hist_vars%totsoiln_vr	DEF hist vars totsoiln vr	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1011
533	case:DEF_hist_vars%totsomc	DEF_hist_vars%totsomc	DEF hist vars totsomc	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1004
534	case:DEF_hist_vars%totsomn	DEF_hist_vars%totsomn	DEF hist vars totsomn	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1009
535	case:DEF_hist_vars%totvegc	DEF_hist_vars%totvegc	DEF hist vars totvegc	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1001
536	case:DEF_hist_vars%totvegn	DEF_hist_vars%totvegn	DEF hist vars totvegn	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1006
537	case:DEF_hist_vars%trad	DEF_hist_vars%trad	DEF hist vars trad	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:915
538	case:DEF_hist_vars%transmittance_out	DEF_hist_vars%transmittance_out	DEF hist vars transmittance out	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:912
539	case:DEF_hist_vars%tref	DEF_hist_vars%tref	DEF hist vars tref	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:917
540	case:DEF_hist_vars%tstar	DEF_hist_vars%tstar	DEF hist vars tstar	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1400
541	case:DEF_hist_vars%us10m	DEF_hist_vars%us10m	DEF hist vars us10m	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1407
542	case:DEF_hist_vars%ustar	DEF_hist_vars%ustar	DEF hist vars ustar	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1398
543	case:DEF_hist_vars%ustar2	DEF_hist_vars%ustar2	DEF hist vars ustar2	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1399
544	case:DEF_hist_vars%vegwp	DEF_hist_vars%vegwp	DEF hist vars vegwp	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1335
545	case:DEF_hist_vars%vehc	DEF_hist_vars%vehc	DEF hist vars vehc	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:936
546	case:DEF_hist_vars%veloc_hru	DEF_hist_vars%veloc_hru	DEF hist vars veloc hru	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1442
547	case:DEF_hist_vars%vf	DEF_hist_vars%vf	DEF hist vars vf	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1086
548	case:DEF_hist_vars%volresv	DEF_hist_vars%volresv	DEF hist vars volresv	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1443

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
549	case:DEF_hist_vars%vs10m	DEF_hist_vars%vs10m	DEF hist vars vs10m	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1408
550	case:DEF_hist_vars%w_scalar	DEF_hist_vars%w_scalar	DEF hist vars w scalar	.false.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1321
551	case:DEF_hist_vars%wa	DEF_hist_vars%wa	DEF hist vars wa	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1342
552	case:DEF_hist_vars%wa_inst	DEF_hist_vars%wa_inst	DEF hist vars wa inst	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1343
553	case:DEF_hist_vars%wat	DEF_hist_vars%wat	DEF hist vars wat	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:888
554	case:DEF_hist_vars%wat_inst	DEF_hist_vars%wat_inst	DEF hist vars wat inst	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:889
555	case:DEF_hist_vars%waterstorage	DEF_hist_vars%waterstorage	DEF hist vars waterstorage	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1164
556	case:DEF_hist_vars%wdsrf	DEF_hist_vars%wdsrf	DEF hist vars wdsrf	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1339
557	case:DEF_hist_vars%wdsrf_hru	DEF_hist_vars%wdsrf_hru	DEF hist vars wdsrf hru	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1441
558	case:DEF_hist_vars%wdsrf_inst	DEF_hist_vars%wdsrf_inst	DEF hist vars wdsrf inst	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1340
559	case:DEF_hist_vars%wetwat	DEF_hist_vars%wetwat	DEF hist vars wetwat	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:890
560	case:DEF_hist_vars%wetwat_inst	DEF_hist_vars%wetwat_inst	DEF hist vars wetwat inst	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:891
561	case:DEF_hist_vars%wetzwt	DEF_hist_vars%wetzwt	DEF hist vars wetzwt	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:892
562	case:DEF_hist_vars%wfc	DEF_hist_vars%wfc	DEF hist vars wfc	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1337
563	case:DEF_hist_vars%wice_soisno	DEF_hist_vars%wice_soisno	DEF hist vars wice soisno	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1325
564	case:DEF_hist_vars%wliq_soisno	DEF_hist_vars%wliq_soisno	DEF hist vars wliq soisno	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1324
565	case:DEF_hist_vars%xerr	DEF_hist_vars%xerr	DEF hist vars xerr	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:875

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
566	case:DEF_hist_vars%xsubs_bsn	DEF_hist_vars%xsubs_bsn	DEF hist vars xsubs bsn	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1434
567	case:DEF_hist_vars%xsubs_hru	DEF_hist_vars%xsubs_hru	DEF hist vars xsubs hru	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1435
568	case:DEF_hist_vars%xwsub	DEF_hist_vars%xwsub	DEF hist vars xwsub	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:883
569	case:DEF_hist_vars%xwsur	DEF_hist_vars%xwsur	DEF hist vars xwsur	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:882
570	case:DEF_hist_vars%xy_frl	DEF_hist_vars%xy_frl	DEF hist vars xy frl	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:852
571	case:DEF_hist_vars%xy_hpbl	DEF_hist_vars%xy_hpbl	DEF hist vars xy hpbl	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:857
572	case:DEF_hist_vars%xy_ozone	DEF_hist_vars%xy_ozone	DEF hist vars xy ozone	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:856
573	case:DEF_hist_vars%xy_pbot	DEF_hist_vars%xy_pbot	DEF hist vars xy pbot	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:851
574	case:DEF_hist_vars%xy_prc	DEF_hist_vars%xy_prc	DEF hist vars xy prc	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:849
575	case:DEF_hist_vars%xy_prl	DEF_hist_vars%xy_prl	DEF hist vars xy prl	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:850
576	case:DEF_hist_vars%xy_q	DEF_hist_vars%xy_q	DEF hist vars xy q	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:848
577	case:DEF_hist_vars%xy_rain	DEF_hist_vars%xy_rain	DEF hist vars xy rain	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:854
578	case:DEF_hist_vars%xy_snow	DEF_hist_vars%xy_snow	DEF hist vars xy snow	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:855
579	case:DEF_hist_vars%xy_solarin	DEF_hist_vars%xy_solarin	DEF hist vars xy solarin	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:853
580	case:DEF_hist_vars%xy_t	DEF_hist_vars%xy_t	DEF hist vars xy t	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:847
581	case:DEF_hist_vars%xy_us	DEF_hist_vars%xy_us	DEF hist vars xy us	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:845
582	case:DEF_hist_vars%xy_vs	DEF_hist_vars%xy_vs	DEF hist vars xy vs	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:846

#	参数 ID	原始键	中文名	默认值	单位	作用域	可见性	写入目标	默认提供方	源码位置
583	case:DEF_hist_vars%z0m	DEF_hist_vars%z0m	DEF hist vars z0m	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:914
584	case:DEF_hist_vars%zerr	DEF_hist_vars%zerr	DEF hist vars zerr	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:876
585	case:DEF_hist_vars%zol	DEF_hist_vars%zol	DEF hist vars zol	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1402
586	case:DEF_hist_vars%zwt	DEF_hist_vars%zwt	DEF hist vars zwt	.true.		case-scalar	editable-common	history.nml	MOD_Namelist.F90 declaration	MOD_Namelist.F90:1341

附录 C：CLI 命令参考

C.1 命令索引

下表直接来自 `colm-cli --help`（与程序内 USAGE 文本一致）。用法 列原样保留命令与位置参数；所有命令共享的选项（`--host/--root/--model-root` 等）见 C.2。

#	用法	用途
1	`colm-cli scan --dir <Sitedata 目录> [--forcing-dir <Forcing 目录>]`	列出目录下的站点；--quick 跳过强迫场，只读站点文件
2	`colm-cli site-new --out <site.nc> --lon <度> --lat <度> [--landtype N] [--crop 1] [--overwrite 1]`	建一份站点文件：经纬度必给，其余从 rawdata 抽或用 标称假设。--landtype 不给就不写，让 CoLM 回落
3	`colm-cli site-pfts <site.nc> [--crop 1] [--landtype N]`	
4	`colm-cli case-land-classes <case-dir> # land classes present in a gridded case's landdata (JSON)`	输出 PFT/PC 站点的非零 PFT 类型与归一化比例
5	`colm-cli new --site <site.nc> --out <dir> [--name N] [--start Y-M-D] [--end Y-M-D]`	在 ../Forcing/ 下找，那两套约定只覆盖 PLUMBER2 与 Urban-PLUMBER 城市站点由文件形状自动识别。两个目录都可选：预抽表盖住的 21 个 Urban-PLUMBER 站不给也能跑，表外的站点才要 --rawdata
6	`colm-cli run <case-dir> --kernel <dir> [--stream 1] [--force 1] [--ranks N]`	--model-version 用一个工作区（修改版）自己编出的程序跑；跑完写 out/<名>/produced-by.json --force 不读取或写入指纹，三段全部重跑 --stage 只运行指定阶段；与 --force 合用时强制重跑该阶段 --record-input-fingerprints 1 记录文件指纹（非驱动输入/程序/输出；默认关闭；驱动数据始终跳过） --stream 把子进程每一行原样转发出来（GUI 用；终端下嫌吵） --ranks >1：Fortran 内核用 MPI 启动；Rust 引擎只在远程由 COLM_MPIEXEC（srun/mpiexec）启动，默认 1 --preprocessors 选 mksrfdata/mkinidata 的实现，--engine 选 colm 主循环； 两者默认都是 rust。Rust 主循环覆盖 SinglePoint + LCT 土面 patch， 跑不了的算例会在第一段之前被预检拒绝，并提示改用 --engine fortran HYPERSPECTRAL PFT/PC：--highres-params 包含 fsds/、 leaf_optical_properties/ 和 water_params.txt；--soil-hyper-albedo-dir 包含 colm_soil_albedo_400nm.nc ... colm_soil_albedo_2500nm.nc
7	`colm-cli model-list|prune|get|create|draft|save-draft|freeze|migrate-case|rollback|make-run|attach-network`	create JSON: {name,note,parent:{id,revision},workspace?}; save-draft: ModelDraft + --expected-generation N freeze: --model-id ID --asset-root DIR [--keep-draft 1]; migrate-case/rollback: prune JSON: {keep:[{id,revision}],dry_run?}: drop unused old official revisions make-run: --out DIR --operation-id ID; attach-network: --model-id ID --expected-generation N + hybrid-install flags

#	用法	用途
8	`colm-cli model-code-ls|model-code-file --model-root DIR --version ID --revision N [--dir D | --path P]`	只读：冻结版本的代码（修改版按冻结的那次提交读），列目录或读一个文件
9	`colm-cli model-changes --model-root DIR --version ID --revision N [--against ID --against-revision N]`	只读：代码相对官方源码改了的文件，参数/物理方案/过程文件/AI 网络相对父版本（或 --against）的不同，来源研究
10	`colm-cli study-save-version <study-dir> --member best|ID --name NAME --model-root DIR [--note TEXT]`	
11	`colm-cli version-check --name N --kernel <dir> [--engine rust|fortran]`	
12	`colm-cli produced-by <case-dir>`	
13	`colm-cli fingerprint-policy`	查询驱动数据永久排除策略，无需读取算例或数据
14	`colm-cli full-input-identity <case-dir>`	只读核验配置与非驱动输入的 SHA256 内容身份（JSON；驱动数据始终跳过）
15	`colm-cli metrics <case-dir> --obs <Flux.nc> [--spinup N] [--from UNIX] [--to UNIX]`	
16	`colm-cli hybrid-install <case-dir> --model <x.onnx|x.mlp.json> --slot pft|land_class`	把训练好的模型装到算例上：拷进 models/、写 hybrid.toml（含 sha256）。 transform: identity|clamp|sigmoid|softplus；见 docs/design-hybrid.md
17	`colm-cli hybrid-check <case-dir> --kernel <dir>`	加载模型、取特征并推理，打印各插槽的行数与特征/输出范围（JSON），不模拟。 需要算例已做完前处理（mkinidata）；单点与空间算例都支持
18	`colm-cli hybrid-climate <case-dir> --kernel <dir>`	在运行时段内累积强迫，给每个 patch 算气候特征（clim_tair、clim_tair_amplitude、clim_prec、clim_swdown、clim_vpd），写到 /hybrid_climate/；不模拟
19	`colm-cli hybrid-fit --studies <dir,dir,...> --network <net.json> --kernel <dir> --out <x.mlp.json>`	两步训练的第二步：读各站调优 Study 的最优参数，在基础算例上取逐行特征，拟合网络； 两个以上 Study 时做留一站交叉验证，并与“其余站均值”基准比较；不如基准时警告， 加 --require-gate 1 时报错且不写模型。net.json 与 Study 的 hybrid 段同格式
20	`colm-cli hybrid-studies <dir> [--depth N]`	列出目录下（默认最多 4 层）的调优 Study：状态、站点、率定的参数，供两步法挑选
21	`colm-cli hybrid-info <case-dir>`	以 JSON 列出算例的混合模型配置与模型校验结果（没有配置时 installed 为 false）
22	`colm-cli hybrid-remove <case-dir>`	删掉 hybrid.toml 及它引用的 models/ 下的模型与标准化文件
23	`colm-cli evaluation-catalog <case-dir> --obs <Flux.nc>`	列出全部支持的评估变量及当前算例/观测是否可用

#	用法	用途
24	`colm-cli evaluation-plan <case-dir> --obs <Flux.nc> --kernel <dir>`	不依赖 history，按 case.nml + 内核预览可评估目标
25	`colm-cli ws-create|ws-note|ws-list|ws-status|ws-search|ws-read|ws-symbols|ws-patch|ws-ls|ws-file|ws-write|ws-snapshot|ws-put|ws-put-dir|ws-rm|ws-revert|ws-build-engine|ws-build-kernel|`	开发工作区：助手改源码、编译、测试、对照、登记实验内核（用法见 ws_cmd.rs 头部）
26	`colm-cli history-subset <case-dir> --out <dir> [--vars f_a,f_b] [--from YYYY-MM] [--to YYYY-MM] [--compress N] [--list 1]`	把 history 裁成只含所选变量与月份的压缩小文件（远程取回结果用，R5）
27	`colm-cli history-catalog <case-dir>`	
28	`colm-cli history-map <case-dir> --var <name> --index <step>`	格点变量某一时间步的整张图（大网格隔行隔列抽稀）
29	`colm-cli history-repos --history-root <dir> [--desktop-path P] [--upstream-path P]`	双仓库注册表：路径/身份/默认分支/同步记录/历史覆盖（只读本地 git）
30	`colm-cli history-sync --history-root <dir> --repository desktop|colm_upstream|both [--offline]`	初始化/更新应用管理的历史缓存；--offline 绝不 fetch/clone
31	`colm-cli history-track --history-root <dir> --repository selected|desktop|colm_upstream|both`	变量定位 + 修改历史 + 未提交 diff + vendor/上游对应关系（只读本地 git）
32	`colm-cli history-commit --history-root <dir> --repository desktop|colm_upstream --sha <40 hex>`	单个 commit 的详情与相关代码 diff
33	`colm-cli series <case-dir> --vars f_rnet,f_fsena [--from UNIX] [--to UNIX]`	
34	`colm-cli study-parameters`	输出可采样专家参数元数据
35	`colm-cli remote-mkdir --host <服务器> --path <现有父目录> --name <新目录名>`	独占创建一个文件夹，不覆盖已有路径、不创建父目录
36	`colm-cli remote-history-cleanup --host <服务器> --root <工作目录> --job <作业> --operation <运行> --expected-path <目录> [--whole-case 1] [--delete 1]`	
37	`colm-cli remote-history-cleanup --host <服务器> --root <工作目录> --`	

#	用法	用途
	`project 1 --expected-path <项目目录> [--delete 1]`	
38	`colm-cli remote-read <运行目录> --command history-catalog|series|energy-summary [--vars …] [--from …] [--to …] [--max-points N]`	
39	`colm-cli remote-engine-progress --host <服务器> --root <工作目录>`	
40	`colm-cli remote-load --host <服务器>`	只读：登录节点负载与内存、各分区占用、排队与自己的作业 默认预检；只清理身份已核实且已完成的独占服务器运行目录
41	`colm-cli study-preflight <case-root> --spec study.json`	只校验 Study，不创建目录
42	`colm-cli study-create <case-root> --spec study.json`	仅站点模式：创建不确定性分析/参数调优 Study，写采样设计
43	`colm-cli study-status <study-dir>`	输出 Study manifest 与成员状态
44	`colm-cli study-run <study-dir> --kernel <dir> [--stream 1] [--preprocessors rust|fortran]`	串行运行尚未完成的成员算例
45	`colm-cli study-export <study-dir> --out <dir>`	导出 manifest、samples、status、report.md/html
46	`colm-cli study-apply-preview <study-dir> --member <id|best>`	只读预览最佳/指定候选会改哪些参数
47	`colm-cli study-pause|study-resume|study-cancel <study-dir>`	请求暂停派发、恢复派发或取消尚未开始任务
48	`colm-cli study-finalize-cancel <study-dir> --pid <pid>`	GUI 杀死对应调度进程后持久化取消终态
49	`colm-cli study-finalize-idle-cancel <study-dir>`	GUI 无活动登记时，确认调度器已退出后持久化取消终态
50	`colm-cli study-retry <study-dir> [--include-review 1]`	把失败成员（可选含待复核成员）重新放回队列
51	`colm-cli study-apply <study-dir> --member <id|best> --out <dir> [--name N]`	将指定成员参数写入新的独立算例
52	`colm-cli study-result <study-dir> --path <result.json>`	
53	`colm-cli study-verified-side <study-dir>`	只读返回单侧研究的版本、成绩与已核对的完整输入证据（schema 1 JSON）
54	`colm-cli study-verify-comparison <study-dir> [--member <id>[,<id>...]]`	只读核验本地研究证据，供远程比较检查使用（schema 1 JSON）； --member 让指定成员也按 best/baseline 的标准核验原始输出 读取并校验单个成员结果
55	`colm-cli all --site <site.nc> --out <dir> --kernel <dir> [--obs <Flux.nc>] [--name N]`	

#	用法	用途
56	`colm-cli observation-table-probe <obs.csv|txt|tsv> [--json 1]`	探测可选验证数据，自动识别站点/时间/评估变量
57	`colm-cli observation-table-convert <obs.csv|txt|tsv> <Observation-dir>`	输出 _Flux.nc；已有同名文件默认拒绝，--overwrite 1 才覆盖；缺 QC 时有限值=0/缺失=1
58	`colm-cli forcing-probe <met.nc> [--json 1]`	
59	`colm-cli forcing-table-probe <forcing.csv|txt|tsv> [--json 1]`	自动识别分隔符、站点/时间/经纬度列与变量候选
60	`colm-cli forcing-table-convert <src.csv|txt|tsv> <Forcing-dir>`	按站点拆分到 .colm-tabular，缺失整行时间补成缺测值
61	`colm-cli netcdf-probe <data.nc> [--json 1]`	探测一份强迫场文件：八个槽位各猜到了什么变量， 猜不到就是 null；三个观测高度缺失时也是 null， 不是 NaN —— GUI 前处理页据此决定问不问用户
62	`colm-cli mesh-new --out <mesh.nc> --nlon N --nlat N [--grid-kind latlon|unstructured]`	生成 GRIDBASED landmask 或 int64 UNSTRUCTURED elmindex；预检也接受既有 int32 网格；无 bbox/SHP 时为全球
63	`colm-cli spatial-preflight --grid-kind latlon|unstructured|catchment --input <mesh.nc>`	在启动 CoLM 前校验空间文件字段并记录 sha256
64	`colm-cli spatial-new --grid-kind latlon|unstructured|catchment --mesh <mesh.nc>`	空间算例暂不支持参数调优和不确定性分析 创建不含 SITE_* 的空间算例；latlon 需要 dlon/dlat
65	`colm-cli forcing-convert <src.nc> <dst.nc> [--slot N=name:units[+extra] ...] [--height V,T,Q]`	供 GUI 走 sidecar 调用；没给 --slot 的槽位走自动匹配
66	`colm-cli forcing-gap-probe <src.nc> [--slot N=name:units[+extra] ...]`	只诊断缺口与时区，不修改源文件
67	`colm-cli forcing-repair <src.nc> <dst.nc> [--slot N=name:units[+extra] ...]`	写一份带逐时 QC 的修复中间文件，原文件不动
68	`colm-cli era5land-download <dst-dir> --lat LAT --lon LON --start YYYY-MM-DD --end YYYY-MM-DD`	用本机 Python cdsapi 和 ~/.cdsapirc 下载对应 0.1° 格点

C.2 完整 `--help` 原文

```
usage:
  colm-cli scan    --dir <Sitedata 目录> [--forcing-dir <Forcing 目录>]
                   [--out sites.json] [--quick 1]
                   # 列出目录下的站点；--quick 跳过强迫场，只读站点文件
  colm-cli site-new --out <site.nc> --lon <度> --lat <度> [--landtype N] [--crop 1] [--overwrite 1]
                   [--rawdata <dir>]
                   [--mode igbp|usgs|pft|pc|urban|urban-igbp|urban-usgs|urban-pft|urban-pc] [--json 1]
                   # 建一份站点文件：经纬度必给，其余从 rawdata 抽或用
                   # 标称假设。--landtype 不给就不写，让 CoLM 回落
```

```
  colm-cli site-pfts <site.nc> [--crop 1] [--landtype N]
  colm-cli case-land-classes <case-dir>     # land classes present in a gridded case's landdata (JSON)
                   # 输出 PFT/PC 站点的非零 PFT 类型与归一化比例
  colm-cli new     --site <site.nc> --out <dir> [--name N] [--start Y-M-D] [--end Y-M-D]
                   [--met <Met.nc>]   # 前处理转出来的强迫场；不给就按命名约定
                                      # 在 ../Forcing/ 下找，那两套约定只覆盖
                                      # PLUMBER2 与 Urban-PLUMBER
                   [--spinup-years N] [--spinup-repeat N]   (默认 1 年 x 1 遍)
                   [--mode igbp|usgs|pft|pc|urban|urban-igbp|urban-usgs|urban-pft|urban-pc]
                   [--crop 1]   # CROP 内核：按 croptyp/pctcrop 审核农田站点
                   [--rawdata <dir>] [--runtime <dir>]
                   # 城市站点由文件形状自动识别。两个目录都可选：预抽表盖住的
                   # 21 个 Urban-PLUMBER 站不给也能跑，表外的站点才要 --rawdata
  colm-cli run     <case-dir> --kernel <dir> [--stream 1] [--force 1] [--ranks N]
                   [--stage mksrfdata|mkinidata|colm] [--preprocessors rust|fortran]
                   [--engine rust|fortran] [--record-input-fingerprints 0|1]
                   [--highres-params <dir>] [--soil-hyper-albedo-dir <dir>]
                   [--model-version official|<workspace>] [--workspace-root <dir>]
                   [--model-id ID --model-revision N --model-root DIR]
                   # --model-version 用一个工作区（修改版）自己编出的程序跑；跑完写 out/<名>/produced-by.json
                   # --force 不读取或写入指纹，三段全部重跑
                   # --stage 只运行指定阶段；与 --force 合用时强制重跑该阶段
                   # --record-input-fingerprints 1 记录文件指纹（非驱动输入/程序/输出；默认关闭；驱动数据始终跳
过）
                   # --stream 把子进程每一行原样转发出来（GUI 用；终端下嫌吵）
                   # --ranks >1：Fortran 内核用 MPI 启动；Rust 引擎只在远程由 COLM_MPIEXEC（srun/mpiexec）启
动，默认 1
                   # --preprocessors 选 mksrfdata/mkinidata 的实现，--engine 选 colm 主循环；
                   # 两者默认都是 rust。Rust 主循环覆盖 SinglePoint + LCT 土面 patch，
                   # 跑不了的算例会在第一段之前被预检拒绝，并提示改用 --engine fortran
                   # HYPERSPECTRAL PFT/PC：--highres-params 包含 fsds/、
                   # leaf_optical_properties/ 和 water_params.txt；--soil-hyper-albedo-dir
                   # 包含 colm_soil_albedo_400nm.nc ... colm_soil_albedo_2500nm.nc
  colm-cli model-list|prune|get|create|draft|save-draft|freeze|migrate-case|rollback|make-run|attach-ne
twork
                   --model-root DIR [--model-id ID --model-revision N] [--json JSON|-]
                   # create JSON: {name,note,parent:{id,revision},workspace?}; save-draft: ModelDraft +
--expected-generation N
                   # freeze: --model-id ID --asset-root DIR [--keep-draft 1]; migrate-case/rollback: <c
ase-dir>
                   # prune JSON: {keep:[{id,revision}],dry_run?}: drop unused old official revisions
                   # make-run: <case-dir> --out DIR --operation-id ID; attach-network: --model-id ID --
expected-generation N + hybrid-install flags
  colm-cli model-code-ls|model-code-file --model-root DIR --version ID --revision N [--dir D | --path
P]
                   # 只读：冻结版本的代码（修改版按冻结的那次提交读），列目录或读一个文件
  colm-cli model-changes --model-root DIR --version ID --revision N [--against ID --against-revision N]
                   # 只读：代码相对官方源码改了的文件，参数/物理方案/过程文件/AI 网络相对父版本（或 --against）的不
同，来源研究
  colm-cli study-save-version <study-dir> --member best|ID --name NAME --model-root DIR [--note TEXT]
  colm-cli version-check --name N --kernel <dir> [--engine rust|fortran]
                   [--preprocessors rust|fortran] [--root <dir>]
  colm-cli produced-by <case-dir>
  colm-cli fingerprint-policy
                   # 查询驱动数据永久排除策略，无需读取算例或数据
  colm-cli full-input-identity <case-dir>
                   # 只读核验配置与非驱动输入的 SHA256 内容身份（JSON；驱动数据始终跳过）
  colm-cli metrics <case-dir> --obs <Flux.nc> [--spinup N] [--from UNIX] [--to UNIX]
                   [--json 1] [--corrected 1]
                   --corrected: 拿能量闭合订正后的观测比（Qle_cor / Qh_cor）
                   --summary-only: 只返回指标，不携带绘图用配对点
                   --obs-label-position: 观测标签=区间起点还是终点（缺省按文件属性识别）
```

```
                    --history-freq: 只作交叉核对；实际输出频率由 history 标签反推，不符时按 history 并告警
                    --min-coverage: 日/月/年聚合要求的最小有效观测覆盖率（缺省 0.5）
                    --pairs-var: 只评估指定观测变量；可重复给出
                    --max-points: 配对点保极值降采样上限（指标仍用完整样本）
  colm-cli hybrid-install <case-dir> --model <x.onnx|x.mlp.json> --slot pft|land_class
                    --features a,b,c --output NAME[:lo:hi[:transform]] [--output ...]
                    [--normalize <norm.json>] [--force 1]
                    # 把训练好的模型装到算例上：拷进 models/、写 hybrid.toml（含 sha256）。
                    # transform: identity|clamp|sigmoid|softplus；见 docs/design-hybrid.md
  colm-cli hybrid-check <case-dir> --kernel <dir>
                    # 加载模型、取特征并推理，打印各插槽的行数与特征/输出范围（JSON），不模拟。
                    # 需要算例已做完前处理（mkinidata）；单点与空间算例都支持
  colm-cli hybrid-climate <case-dir> --kernel <dir>
                    # 在运行时段内累积强迫，给每个 patch 算气候特征（clim_tair、clim_tair_amplitude、
                    # clim_prec、clim_swdown、clim_vpd），写到 <case>/hybrid_climate/；不模拟
  colm-cli hybrid-fit --studies <dir,dir,...> --network <net.json> --kernel <dir> --out <x.mlp.json>
                    [--weight pftfrac] [--epochs N] [--learning-rate X] [--ridge X] [--seed N] [--requir
e-gate 1]
                    # 两步训练的第二步：读各站调优 Study 的最优参数，在基础算例上取逐行特征，拟合网络；
                    # 两个以上 Study 时做留一站交叉验证，并与“其余站均值”基准比较；不如基准时警告，
                    # 加 --require-gate 1 时报错且不写模型。net.json 与 Study 的 hybrid 段同格式
  colm-cli hybrid-studies <dir> [--depth N]
                    # 列出目录下（默认最多 4 层）的调优 Study：状态、站点、率定的参数，供两步法挑选
  colm-cli hybrid-info <case-dir>
                    # 以 JSON 列出算例的混合模型配置与模型校验结果（没有配置时 installed 为 false）
  colm-cli hybrid-remove <case-dir>
                    # 删掉 hybrid.toml 及它引用的 models/ 下的模型与标准化文件
  colm-cli evaluation-catalog <case-dir> --obs <Flux.nc>
                    # 列出全部支持的评估变量及当前算例/观测是否可用
  colm-cli evaluation-plan <case-dir> --obs <Flux.nc> --kernel <dir>
                    # 不依赖 history，按 case.nml + 内核预览可评估目标
  colm-cli ws-create|ws-note|ws-list|ws-status|ws-search|ws-read|ws-symbols|ws-patch|ws-ls|ws-file|ws-w
rite|ws-snapshot|ws-put|ws-put-dir|ws-rm|ws-revert|ws-build-engine|ws-build-kernel|
                     ws-test|ws-run|ws-compare|ws-parity|ws-regress|ws-kernels|ws-export|ws-delete ...
                     # 开发工作区：助手改源码、编译、测试、对照、登记实验内核（用法见 ws_cmd.rs 头部）
  colm-cli history-subset <case-dir> --out <dir> [--vars f_a,f_b] [--from YYYY-MM] [--to YYYY-MM] [--co
mpress N] [--list 1]
                     # 把 history 裁成只含所选变量与月份的压缩小文件（远程取回结果用，R5）
  colm-cli history-catalog <case-dir>
  colm-cli history-map <case-dir> --var <name> --index <step>
                    # 格点变量某一时间步的整张图（大网格隔行隔列抽稀）
  colm-cli history-repos --history-root <dir> [--desktop-path P] [--upstream-path P]
                    # 双仓库注册表：路径/身份/默认分支/同步记录/历史覆盖（只读本地 git）
  colm-cli history-sync --history-root <dir> --repository desktop|colm_upstream|both [--offline]
                    # 初始化/更新应用管理的历史缓存；--offline 绝不 fetch/clone
  colm-cli history-track --history-root <dir> --repository selected|desktop|colm_upstream|both
                    --symbol <symbol> [--source <dir>] [--path P] [--ref R] [--limit N] [--offset N]
                    [--model-json JSON] [--desktop-path P] [--upstream-path P]
                    # 变量定位 + 修改历史 + 未提交 diff + vendor/上游对应关系（只读本地 git）
  colm-cli history-commit --history-root <dir> --repository desktop|colm_upstream --sha <40 hex>
                    [--symbol S] [--path P] [--desktop-path P] [--upstream-path P]
                    # 单个 commit 的详情与相关代码 diff
  colm-cli series  <case-dir> --vars f_rnet,f_fsena [--from UNIX] [--to UNIX]
                    [--max-points N] [--out series.json]
  colm-cli study-parameters
                    # 输出可采样专家参数元数据
  colm-cli remote-mkdir --host <服务器> --path <现有父目录> --name <新目录名>
                    # 独占创建一个文件夹，不覆盖已有路径、不创建父目录
  colm-cli remote-history-cleanup --host <服务器> --root <工作目录> --job <作业> --operation <运行> --expec
ted-path <目录> [--whole-case 1] [--delete 1]
  colm-cli remote-history-cleanup --host <服务器> --root <工作目录> --project 1 --expected-path <项目目录>
[--delete 1]
```

```
  colm-cli remote-read <运行目录> --command history-catalog|series|energy-summary [--vars …] [--from …]
[--to …] [--max-points N]
  colm-cli remote-engine-progress --host <服务器> --root <工作目录>
  colm-cli remote-load --host <服务器>
                  # 只读：登录节点负载与内存、各分区占用、排队与自己的作业
                  # 默认预检；只清理身份已核实且已完成的独占服务器运行目录
  colm-cli study-preflight <case-root> --spec study.json
                  # 只校验 Study，不创建目录
  colm-cli study-create <case-root> --spec study.json
                  # 仅站点模式：创建不确定性分析/参数调优 Study，写采样设计
  colm-cli study-status <study-dir>
                  # 输出 Study manifest 与成员状态
  colm-cli study-run <study-dir> --kernel <dir> [--stream 1] [--preprocessors rust|fortran]
                  [--engine rust|fortran]
                  # 串行运行尚未完成的成员算例
  colm-cli study-export <study-dir> --out <dir>
                  # 导出 manifest、samples、status、report.md/html
  colm-cli study-apply-preview <study-dir> --member <id|best>
                  # 只读预览最佳/指定候选会改哪些参数
  colm-cli study-pause|study-resume|study-cancel <study-dir>
                  # 请求暂停派发、恢复派发或取消尚未开始任务
  colm-cli study-finalize-cancel <study-dir> --pid <pid>
                  # GUI 杀死对应调度进程后持久化取消终态
  colm-cli study-finalize-idle-cancel <study-dir>
                  # GUI 无活动登记时，确认调度器已退出后持久化取消终态
  colm-cli study-retry <study-dir> [--include-review 1]
                  # 把失败成员（可选含待复核成员）重新放回队列
  colm-cli study-apply <study-dir> --member <id|best> --out <dir> [--name N]
                  # 将指定成员参数写入新的独立算例
  colm-cli study-result <study-dir> --path <result.json>
  colm-cli study-verified-side <study-dir>
                  # 只读返回单侧研究的版本、成绩与已核对的完整输入证据（schema 1 JSON）
  colm-cli study-verify-comparison <study-dir> [--member <id>[,<id>...]]
                  # 只读核验本地研究证据，供远程比较检查使用（schema 1 JSON）；
                  # --member 让指定成员也按 best/baseline 的标准核验原始输出
                  # 读取并校验单个成员结果
  colm-cli all     --site <site.nc> --out <dir> --kernel <dir> [--obs <Flux.nc>] [--name N]
                  [--start Y-M-D] [--end Y-M-D] [--spinup N]
                  [--preprocessors rust|fortran] [--engine rust|fortran]
  colm-cli observation-table-probe <obs.csv|txt|tsv> [--json 1]
                           # 探测可选验证数据，自动识别站点/时间/评估变量
  colm-cli observation-table-convert <obs.csv|txt|tsv> <Observation-dir>
                           --time-column COLUMN [--site-column COLUMN|--site-name NAME]
                           [--variable VAR=column[:qc_column] ...] [--json 1] [--overwrite 1]
                           # 输出 <site>_Flux.nc；已有同名文件默认拒绝，--overwrite 1 才覆盖；缺 QC 时有限值=0/
缺失=1
  colm-cli forcing-probe   <met.nc> [--json 1]
  colm-cli forcing-table-probe <forcing.csv|txt|tsv> [--json 1]
                           # 自动识别分隔符、站点/时间/经纬度列与变量候选
  colm-cli forcing-table-convert <src.csv|txt|tsv> <Forcing-dir>
                           --time COLUMN [--site COLUMN]
                           [--lat-column COLUMN --lon-column COLUMN]
                           [--landtype-column COLUMN] [--offset-column COLUMN]
                           [--land-cover-scheme IGBP|USGS|PFT|PC|Urban]
                           [--lat LAT --lon LON] [--utc-offset HOURS]
                           [--step-seconds N] [--height V,T,Q]
                           --slot N=column:units[+extra] ... [--json 1]
                           # 按站点拆分到 .colm-tabular，缺失整行时间补成缺测值
  colm-cli netcdf-probe    <data.nc> [--json 1]
                           # 探测一份强迫场文件：八个槽位各猜到了什么变量，
                           # 猜不到就是 null；三个观测高度缺失时也是 null，
                           # 不是 NaN —— GUI 前处理页据此决定问不问用户
```

```
  colm-cli mesh-new --out <mesh.nc> --nlon N --nlat N [--grid-kind latlon|unstructured]
                    [--west W --east E --south S --north N | --shp basin.shp]
                    [--non-ocean-mask mask.nc --non-ocean-var non_ocean_mask]
                    # 生成 GRIDBASED landmask 或 int64 UNSTRUCTURED elmindex；预检也接受既有 int32 网格；无
bbox/SHP 时为全球
  colm-cli spatial-preflight --grid-kind latlon|unstructured|catchment --input <mesh.nc>
                    [--out manifest.json]
                    # 在启动 CoLM 前校验空间文件字段并记录 sha256
  colm-cli spatial-new --grid-kind latlon|unstructured|catchment --mesh <mesh.nc>
                    # 空间算例暂不支持参数调优和不确定性分析
                    --out <case-dir> --forcing <forcing.nml> --rawdata <dir> --runtime <dir>
                    --start YYYY-MM-DD --end YYYY-MM-DD --timestep <seconds>
                    [--name N] [--dlon degrees --dlat degrees] [--mode igbp|usgs|pft|pc]
                    # 创建不含 SITE_* 的空间算例；latlon 需要 dlon/dlat
  colm-cli forcing-convert <src.nc> <dst.nc> [--slot N=name:units[+extra] ...] [--height V,T,Q]
                           # 供 GUI 走 sidecar 调用；没给 --slot 的槽位走自动匹配
  colm-cli forcing-gap-probe <src.nc> [--slot N=name:units[+extra] ...]
                           [--short-gap N] [--utc-offset HOURS] [--lat LAT --lon LON] [--json 1]
                           # 只诊断缺口与时区，不修改源文件
  colm-cli forcing-repair <src.nc> <dst.nc> [--slot N=name:units[+extra] ...]
                           [--short-gap N] [--utc-offset HOURS] [--lat LAT --lon LON]
                           [--era5 FILE_OR_DIR] [--min-overlap N] [--json 1]
                           # 写一份带逐时 QC 的修复中间文件，原文件不动
  colm-cli era5land-download <dst-dir> --lat LAT --lon LON --start YYYY-MM-DD --end YYYY-MM-DD
                           # 用本机 Python cdsapi 和 ~/.cdsapirc 下载对应 0.1° 格点
```